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**Hong Kong
Mathematics
Olympiad
Past Paper
Solution (Volume 3)**

香港數學競賽
歷屆比賽題解（第三輯）

2008-2022



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Hong Kong Mathematics Olympiad (2008 / 2009)

Heat Event (Individual) Solution

香港數學競賽 (2008 / 2009)

初賽項目(個人) 題解

$$1. \quad x = 0.\dot{2}\dot{3} + 0.00\dot{2}\dot{3} + 0.0000\dot{2}\dot{3} + 0.000000\dot{2}\dot{3} + \dots$$

$$= \frac{23}{99} + \frac{23}{9900} + \frac{23}{990000} + \dots$$

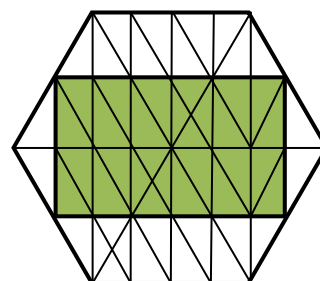
$$= \frac{23}{99} \left(1 + \frac{1}{100} + \frac{1}{10000} + \dots \right)$$

$$= \frac{23}{99} \left(\frac{1}{1 - \frac{1}{100}} \right)$$

$$= \frac{2300}{9801}$$

$$2. \quad \text{Ratio} = \frac{24}{48} = \frac{1}{q}$$

$$\therefore q = 2$$



$$3. \quad 16\sin^4 \theta^\circ - 16(1 - \sin^2 \theta^\circ) - 5 = 0$$

$$16\sin^4 \theta^\circ + 16\sin^2 \theta^\circ - 21 = 0$$

$$\therefore \sin^2 \theta^\circ = \frac{-16 \pm 40}{32}$$

$$= -\frac{56}{32} \text{ (rejected) or } \frac{3}{4}$$

$$\sin \theta^\circ = \frac{\sqrt{3}}{2} \text{ or } -\frac{\sqrt{3}}{2} \text{ (rejected)}$$

$$\theta = 60$$

$$4. \quad 4518 = 2(2008) + 502$$

$$2008 = 4(502)$$

$$\therefore \gcd(2008, 4518) = 502 \text{ and } 502 \text{ has 4 factors}$$

$$\therefore m = 4$$

$$\begin{aligned}
 5. \quad x^2 &= 7 - (y-3)^2 \\
 x^2 + 5y &= -y^2 - 11y - 2 \\
 &= -\left(y - \frac{11}{2}\right)^2 + \left(\frac{11}{2}\right)^2 - 2 \\
 &= -\left(y - \frac{11}{2}\right)^2 + \frac{113}{4} \\
 \therefore k &= \max(x^2 + 5y) = \frac{113}{4}
 \end{aligned}$$

$$6. \quad \text{Note that } f_2(x) = -\frac{1-x}{x}, \quad f_3(x) = x, \quad f_4(x) = f_1(x). \text{ So}$$

$$\begin{aligned}
 f_{2009}(2008) &= f_1(f_{2008}(2008)) \\
 &= f_1(f_1(2008)) \\
 &= f_2(2008) \\
 &= -\frac{1-2008}{2008} \\
 &= \frac{2007}{2008}
 \end{aligned}$$

$$7. \quad c = \text{the area of an equilateral triangle formed inside the hexagon}$$

$$c = \frac{6 \times 6 \sin 60^\circ}{2} = 9\sqrt{3}$$

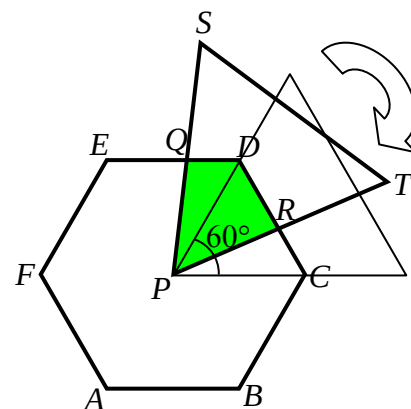
[proof of $\triangle QDP \cong \triangle RCP$:

$$PD = PC$$

$$\angle QPD = \angle RPC$$

$$\angle QDP = \angle RCP$$

$$\therefore \triangle QDP \cong \triangle RCP \text{ (ASA)]}$$



$$\begin{aligned}
 8. \quad &\text{Unit digit of } 7^1 = 7 \\
 &\text{Unit digit of } 7^2 = 9 \\
 &\text{Unit digit of } 7^3 = 3 \\
 &\text{Unit digit of } 7^4 = 1 \\
 &\text{Unit digit of } 7^5 = 7 \dots \\
 &\text{Thus unit digit of } 7^{2009} = 7^1 = 7
 \end{aligned}$$

$$9. \quad \begin{cases} a - 7b = 2 & \dots\dots (1) \\ \log_{2b} a = 2 & \dots\dots (2) \end{cases}$$

$$\text{From (2), we have } \log a = \log(2b)^2 \Rightarrow a = 4b^2 \dots\dots (3)$$

Substituting (3) into (1) gives

$$4b^2 - 7b - 2 = 0$$

$$b = \frac{7 \pm \sqrt{81}}{8}$$

$$= 2 \text{ or } -\frac{1}{4} \text{ (rejected)}$$

$$\therefore a = 16 \text{ and } a \times b = 32$$

10. $40 = b + c + d$

$$80 = e + i + c + k$$

$$b + c + k = d + h$$

$$a + b = i + c + d$$

$$e = f$$

$\therefore$ Area of rectangle

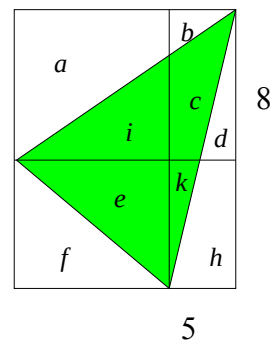
$$= a + b + c + d + e + f + k + h + i$$

$$= 2i + 3c + d + 2e + 2k + b$$

$$= b + c + d + 2(i + e + k + c)$$

$$= 40 + 80 \times 2$$

$$= 200$$



Hong Kong Mathematics Olympiad (2008 / 2009)

Heat Event (Group) Solution

香港數學競賽 (2008 / 2009)

初賽項目(團體)題解

$$\begin{aligned}
 1. \quad & \frac{1}{a(a+1)} + \frac{1}{(a+1)(a+2)} + \cdots + \frac{1}{2008 \times 2009} = \frac{272}{30135} \\
 & \frac{1}{a} - \frac{1}{a+1} + \frac{1}{a+1} - \frac{1}{a+2} + \cdots + \frac{1}{2008} - \frac{1}{2009} = \frac{1}{a} - \frac{1}{2009} \\
 & \frac{1}{a} - \frac{1}{2009} = \frac{272}{30135} \\
 & \frac{1}{a} = \frac{272}{30135} + \frac{1}{2009} \\
 & a = 105
 \end{aligned}$$

$$\begin{aligned}
 2. \quad & x = 1 + \sqrt{2} \\
 & (x-1)^2 = 2 \\
 & x^2 - 2x - 1 = 0 \\
 & x^5 - 2x^4 + 3x^3 - 4x^2 - 10x - 6 \\
 & = (x^3 + 4x + 4)(x^2 - 2x - 1) + 2x - 2 \\
 & = 0 + 2(1 + \sqrt{2}) - 2 \\
 & = 2\sqrt{2}
 \end{aligned}$$

3. Note that both p and q should be positive numbers.

$$\frac{2}{p} + \frac{1}{q} = 1$$

$$p + 2q = pq$$

$$(p-2)(q-1) = 2$$

$\therefore (p, q) = (1, -1)$ (rejected) or $(0, 0)$ (rejected) or $(3, 3)$ or $(4, 2)$, and when $p \geq 5$, q is not an integer. Therefore, the maximum value of $p \times q = 3 \times 3 = 9$.

4. Trivially, $x = 0, 180$

Hence we want to solve $\cos 7x^\circ = \cos 5x^\circ$ for $0 < x < 180$.

$$\cos 7x^\circ = \cos 5x^\circ$$

$$\cos 7x^\circ = \cos(360k - 5x^\circ) \text{ for } k = 1, 2, 3, 4, 5$$

$$\therefore x = 30, 60, 90, 120, 150$$

$$\therefore r = 7$$

5. As x, y, z are all positive integers, $z - \sqrt{28} = z - 2\sqrt{7} = x + y - 2\sqrt{xy}$

$$\therefore (x, y) = (1, 7) \text{ or } (7, 1) \Rightarrow z = 8 \text{ and } x + y + z = 16$$

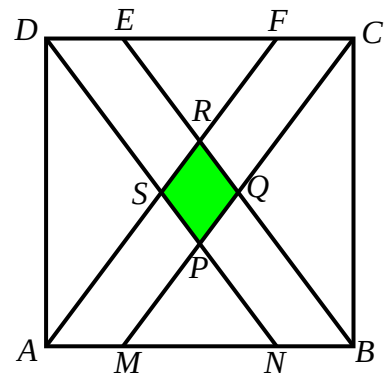
6. area of $\triangle EFR = \triangle NMP = 2c$
 area of $\triangle QBC = \triangle SAD = 3$
 area of $DEBN = \text{area of } FCMA = 4$
 area of $ABCD$

$$= 4^2 = 16$$

$$= \triangle QBC + \triangle SAD + \triangle EFR + \triangle NMP + DEBN + FCMA - c$$

$$= 6 + 8 + 3c$$

$$\therefore c = \frac{2}{3}$$



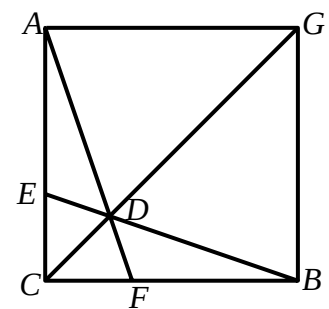
7. Note that x should be negative and $2x + 8$ should equal to 0, otherwise, one side will give an odd number while the other side gives an even number. Therefore $x = -4$

8. Construct a square by adding G in which CDG is a straight line.
 $\triangle DCF \sim \triangle DGA$

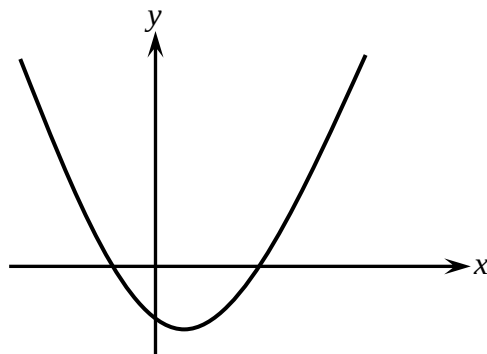
$$\therefore \frac{CD}{DG} = \frac{FD}{DA} = \frac{CF}{AG} = \frac{6}{14}$$

$$CG = 14\sqrt{2}$$

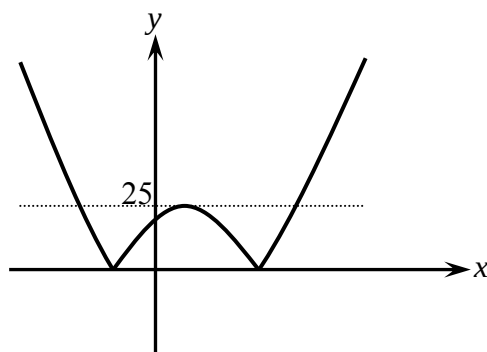
$$CD = \frac{21\sqrt{2}}{5}$$



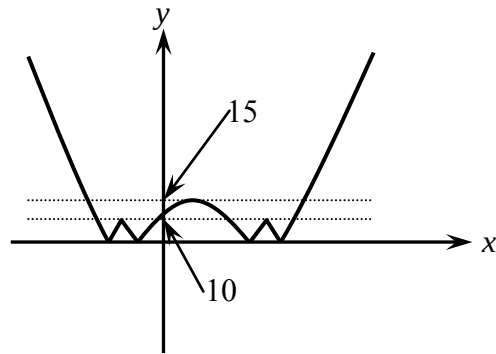
9. The graph of $y = x^2 - 6x - 16$ is:



Thus the graph of $y = |x^2 - 6x - 16|$ is



Therefore the graph of $y = ||x^2 - 6x - 16| - 10|$ is

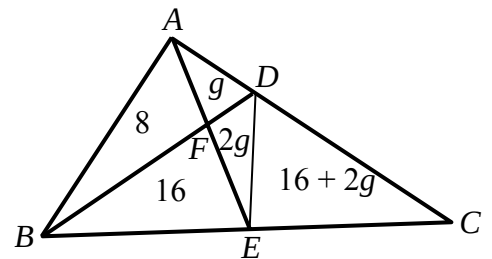


From the graph, the only one value of y which will give 6 distinct roots is $y = 10$.

10. Since $AE : AF = 1 : 3$, we have area of $\triangle ABF$: area of $\triangle BEF = 1 : 2$, and area of $\triangle ADF$: area of $\triangle DEF = 1 : 2$, thus we have the diagram on the right.

$$16 + 2g + 2g + g = 24$$

$$g = 1.6$$



Hong Kong Mathematics Olympiad (2008 / 2009)

Final Event (Individual) Solution

香港數學競賽 (2008 / 2009)

決賽項目(個人)題解

Event 1

1. $\because a, b, c, d$ are roots of $x^4 - 15x^2 + 56 = 0$

$$x^4 - 15x^2 + 56 = 0$$

$$\Rightarrow (x^2)^2 - 15(x^2) + 56 = 0$$

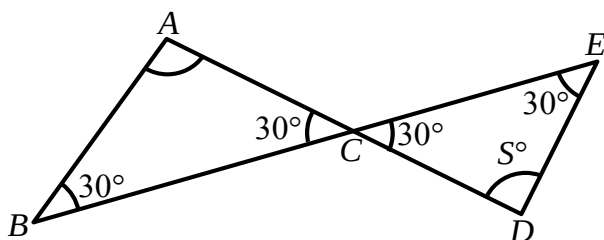
$$\Rightarrow a^2, b^2, c^2, d^2 \text{ are roots of } t^2 - 15t + 56 = 0 \dots (*)$$

Without loss of generality, let $a = -\sqrt{\alpha}$, $b = \sqrt{\alpha}$, $c = \sqrt{\beta}$, $d = -\sqrt{\beta}$ where α, β are roots of (*)

$$\therefore a^2 + b^2 + c^2 + d^2 = 2(\alpha + \beta)$$

$$= 2(15) = 30$$

2.



$$\angle BAC = 180^\circ - 30^\circ - 30^\circ$$

$$= 120^\circ$$

$$\angle ADE = \angle BAC$$

$$= 120^\circ$$

$$\therefore S = 120$$

3. $F = 1 + 2 + 2^2 + 2^3 + \dots + 2^s = 2^{121} - 1$

$$\therefore T = \sqrt{\frac{\log 2^{121}}{\log 2}} = \sqrt{121} = 11$$

4. $U = \frac{f(11)}{(11-1)f(11-3)}$

$$= \frac{f(11)}{10f(8)}$$

$$= \frac{10 \times 9 \times 8 f(8)}{10 \times f(8)}$$

$$= 9 \times 8$$

$$= 72$$

Event 2

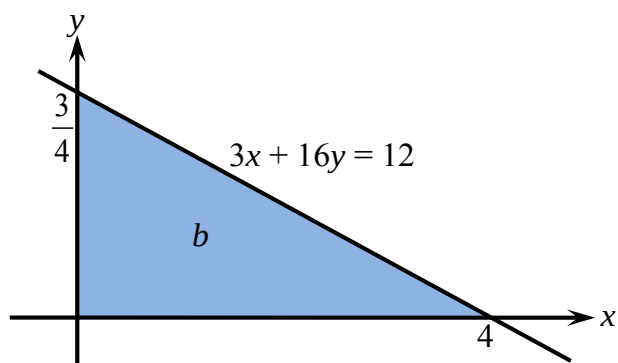
1. $0 < \sqrt{3} - \sqrt{2} < 1$

$\therefore 0 < (\sqrt{3} - \sqrt{2})^{2009} < 1$

$\therefore [(\sqrt{3} - \sqrt{2})^{2009}] = 0$

$\therefore a = 16$

2. $b = \frac{1}{2}(4)\left(\frac{3}{4}\right) = \frac{3}{2}$ or $1\frac{1}{2}$ or 1.5



3. $x - \frac{1}{x} = 3$

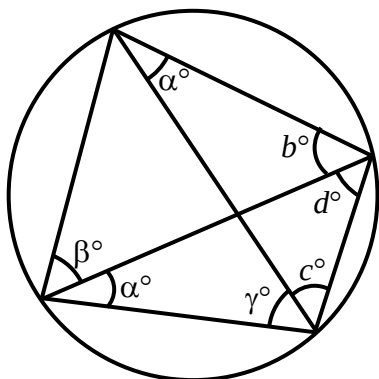
$$\left(x - \frac{1}{x}\right)^3 = x^3 - 3x^2\left(\frac{1}{x}\right) + 3x\left(\frac{1}{x}\right)^2 - \frac{1}{x^3}$$

$$= \left(x^3 - \frac{1}{x^3}\right) - 3\left(x - \frac{1}{x}\right)$$

$$27 = x^3 - \frac{1}{x^3} - 3(3)$$

$\therefore x^3 - \frac{1}{x^3} = 36$

4. $\alpha = 36, b = 59, c = 43$
 $\alpha^\circ + \beta^\circ + b^\circ + d^\circ = 180^\circ$ (Opp $\angle$ s of cyclic quad.)
 $\therefore d = 42$



Event 3

$$1. \quad \frac{4}{\sqrt{6}+\sqrt{2}} - \frac{1}{\sqrt{3}+\sqrt{2}} = \frac{4(\sqrt{6}-\sqrt{2})}{(\sqrt{6}+\sqrt{2})(\sqrt{6}-\sqrt{2})} - \frac{\sqrt{3}-\sqrt{2}}{(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2})}$$

$$= \sqrt{6}-\sqrt{2}-\sqrt{3}+\sqrt{2}$$

$$= \sqrt{6}-\sqrt{3}$$

$$\therefore m = 6 - 3 = 3$$

$$2. \quad A : B = 3 : 2 \quad A : C = n : 1$$

$$\frac{a}{b} = \frac{c}{d}$$

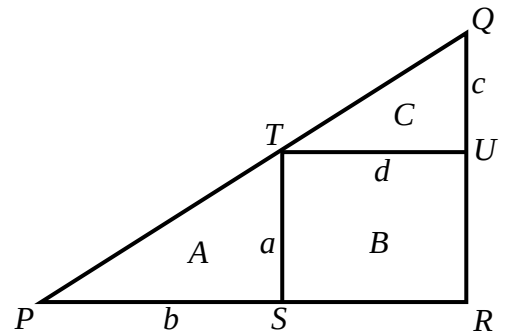
$$A = \frac{ab}{2}, \quad C = \frac{cd}{2}, \quad B = ad$$

$$\frac{A}{B} = \frac{\frac{ab}{2}}{d} = \frac{b}{2d} = \frac{3}{2}$$

$$\therefore b = 3d \Rightarrow a = 3c$$

$$\frac{A}{C} = \frac{\frac{ab}{2}}{\frac{cd}{2}} = \frac{ab}{cd} = \frac{(3c)(3d)}{cd} = 9$$

$$\therefore n = 9$$



$$3. \quad \text{Let } a = x_1, \quad b = x_2 \text{ be the roots of } (x+c)(x+d) = -1 \text{ where } c = x_3 \text{ and } d = x_4$$

$$\text{Then } (x+c)(x+d)+1=(x-a)(x-b) \dots\dots (1)$$

Substitute $x = -c$ in (1) :

$$1 = (-c-a)(-c-b) = (a+c)(b+c)$$

Substitute $x = -d$ in (1) :

$$1 = (-d-a)(-d-b) = (a+d)(b+d)$$

$$\therefore p = (a+c)(b+c) + (a+d)(b+d) = 1+1 = 2$$

$$4. \quad q \geq 1000, \quad 7|q, \quad 3|q-1 \text{ and } 4|q-1 \text{ and } 5|q-1 \Rightarrow 60|q-1 \dots\dots (*)$$

From (*) and $q > 1000$, we have q may equal to 1021, 1081, 1141, ...

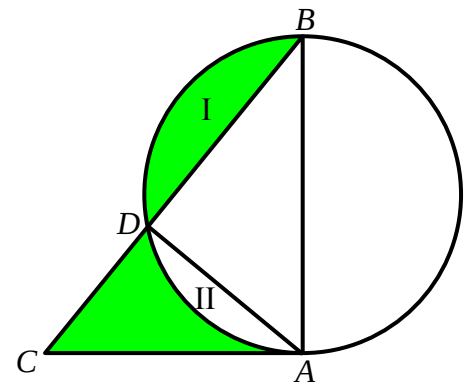
but $7 \nmid 1021$ and $7 \nmid 1081$, $7|1141$,

$$\therefore q = 1141$$

Event 4

$$\begin{aligned}
 1. \quad & x_0^3 + 2x_0^2 + 2 \\
 &= x_0(x_0^2 + x_0) + x_0^2 + 2 \\
 &= x_0 + x_0^2 + 2 \\
 &= 1 + 2 = 3
 \end{aligned}$$

$$\begin{aligned}
 2. \quad & I = II \\
 & \text{Shaded area} \\
 &= \left(\frac{1}{2} \times 3 \times 3\right) \times \frac{1}{2} \\
 &= \frac{9}{4} \text{ or } 2\frac{1}{4}
 \end{aligned}$$



$$\begin{aligned}
 3. \quad & \therefore \log 1 = 0 \\
 & \therefore p = 4\left(\frac{9}{4}\right)(1) = 9
 \end{aligned}$$

$$\begin{aligned}
 4. \quad & (x-3)^2 + (y-3)^2 = 1 \\
 & \therefore k = \frac{y}{x-3} \\
 & \therefore (x-3)^2 + (k(x-3)-3)^2 = 1 \\
 \Rightarrow & (x-3)^2 + k^2(x-3)^2 - 6k(x-3) + 9 = 1 \\
 \Rightarrow & (k^2+1)(x-3)^2 - 6k(x-3) + 8 = 0 \\
 & \therefore x-3 \in \mathbf{R} \\
 & \therefore \Delta \geq 0 \\
 & \therefore (-6k)^2 - 4(k^2+1)(8) \geq 0 \\
 \Rightarrow & 36k^2 - 32k^2 - 32 \geq 0 \\
 \Rightarrow & 4k^2 \geq 32 \\
 \Rightarrow & k^2 \geq 8 \\
 & \therefore q = 8
 \end{aligned}$$

Event 1

1. By triangular inequality,

Case 1: $a = 1$

$$1 \leq 2 \leq 2 \Rightarrow b = 2$$

Case 2: $a = 2$

$$(i) \quad 2 \leq 2 \leq 2 \Rightarrow b = 2 \text{ or } 3$$

$$(ii) \quad 2 \leq 2 \leq 3$$

$$\therefore q = 3$$

2. Case 1: $x \geq 0$

$$\Rightarrow x - \frac{4}{x} = 3 \Rightarrow x^2 - 3x - 4 = 0$$

$$\Rightarrow x = 4 \text{ or } x = -1 \text{ (rejected)}$$

Case 2: $x < 0$

$$\Rightarrow -x - \frac{4}{x} = -3 \Rightarrow x^2 - 3x + 4 = 0 \text{ which has no real root}$$

$$\therefore k = 1$$

3. $x = 7 + y$

$$\therefore \frac{\sqrt{7+y}}{\sqrt{y}} - \frac{\sqrt{y}}{\sqrt{7+y}} = \frac{7}{12}$$

$$\text{Let } M = \sqrt{\frac{7+y}{y}}, \text{ then}$$

$$M - \frac{1}{M} = \frac{7}{12}$$

$$\Rightarrow 12M^2 - 7M - 12 = 0$$

$$\Rightarrow M = \frac{4}{3} \text{ or } -\frac{3}{4} \text{ (rejected)}$$

$$\therefore \frac{7+y}{y} = \frac{16}{9}$$

$$\Rightarrow 63 + 9y = 16y$$

$$\Rightarrow y = 9$$

$$\therefore x = 16$$

$$\therefore w = 16 + 9 = 25$$

$$4. \left| x - \frac{1}{2} \right| + \sqrt{y^2 - 1} = 0$$

$$\Rightarrow x = \frac{1}{2} \text{ and } y^2 - 1 = 0$$

$$\therefore x = \frac{1}{2} \text{ and } |y| = \sqrt{y^2} = 1$$

$$\therefore p = |x| + |y| = 1 + \frac{1}{2} = \frac{3}{2} \text{ or } 1.5$$

Event 2

$$1. \cos \theta = -\frac{12}{13}$$

$$\sin \theta = -\frac{5}{13}$$

$$\therefore A = -\frac{12}{13} - \frac{5}{13} = -\frac{17}{13} \text{ or } -1\frac{4}{13}$$

$$2. \text{ Let } x = 10 + \sqrt{10 + \sqrt{10 + \sqrt{10 + \dots}}}$$

$$\Rightarrow (x - 10)^2 = x$$

$$\Rightarrow x^2 - 21x + 100 = 0$$

$$\Rightarrow x = \frac{21 + \sqrt{41}}{2} \text{ or } \frac{21 - \sqrt{41}}{2} \text{ (rejected)}$$

$$\therefore B = \left\lfloor \frac{21 + \sqrt{41}}{2} \right\rfloor = 13$$

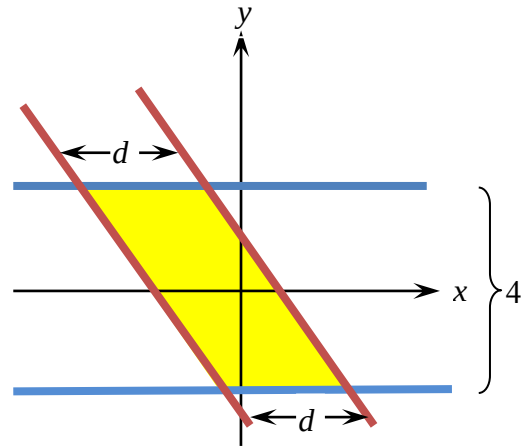
$$\begin{aligned}
 3. \quad (1 \oplus 2) \oplus 3 &= (1 \times 2 + 10) \oplus 3 \\
 &= 12 \oplus 3 \\
 &= 12 \times 3 + 10 \\
 &= 46
 \end{aligned}$$

$$4. \quad d = \frac{5}{2} - (-5) = \frac{15}{2}$$

$\therefore$ Area of the bounded region

$$= \frac{15}{2} \times 4 = 30$$

$$\therefore D = 30$$



Event 3

$$\begin{aligned}
 1. \quad &5 \times 2008 \times 15 < 2008 \times 80 < 6 \times 2008 \times 15 \\
 &5 \times 2009 \times 25 < 2009 \times 130 < 6 \times 2009 \times 25 \\
 &5 \times 2010 \times 35 < 2010 \times 180 < 6 \times 2010 \times 35 \\
 \therefore \quad &5 < \frac{2008 \times 80 + 2009 \times 130 + 2010 \times 180}{2008 \times 15 + 2009 \times 25 + 2010 \times 35} < 6 \\
 \therefore \quad &A = 5
 \end{aligned}$$

$$\begin{aligned}
 2. \quad &\underbrace{99 \dots 9}_{2009} \times \underbrace{99 \dots 9}_{2009} + \underbrace{199 \dots 9}_{2009} \\
 &= \underbrace{99 \dots 9}_{2009} \times \underbrace{99 \dots 9}_{2009} + \underbrace{99 \dots 9}_{2009} + \underbrace{100 \dots 0}_{2009} \\
 &= \underbrace{99 \dots 9}_{2009} \times (\underbrace{99 \dots 9}_{2009} + 1) + \underbrace{100 \dots 0}_{2009} \\
 &= \underbrace{99 \dots 9}_{2009} \times \underbrace{100 \dots 0}_{2009} + \underbrace{100 \dots 0}_{2009} \\
 &= \underbrace{100 \dots 0}_{2009} \times (\underbrace{99 \dots 9}_{2009} + 1) \\
 &= \underbrace{100 \dots 0}_{2009 \times 2}
 \end{aligned}$$

$$\therefore R = 2009 \times 2 = 4018$$

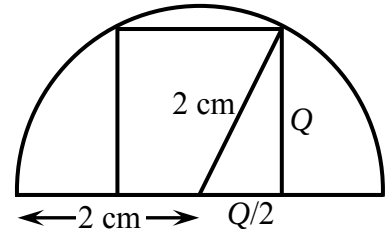
3.

$$Q^2 + \frac{Q^2}{4} = 4$$

$$\Rightarrow 5Q^2 = 16$$

$$\Rightarrow Q = \sqrt{\frac{16}{5}} \text{ or } -\sqrt{\frac{16}{5}} \text{ (rejected)}$$

$$\therefore Q = \frac{4}{\sqrt{5}} \text{ or } \frac{4\sqrt{5}}{5}$$



4.

$$OJ = 4 \text{ cm}$$

$$\Rightarrow \cos \theta = \frac{2}{4}$$

$$\therefore \theta = 60^\circ$$

$$\frac{IJ}{2} = \tan 60^\circ \Rightarrow IJ = 2\sqrt{3} \text{ cm}$$

Area of sector OBJ

$$= \frac{1}{2}(4)^2 \left(\frac{\pi}{3} \right)$$

$$= \frac{1}{2}(16) \left(\frac{3}{3} \right) = 8 \text{ cm}^2$$

Area of sector IKB

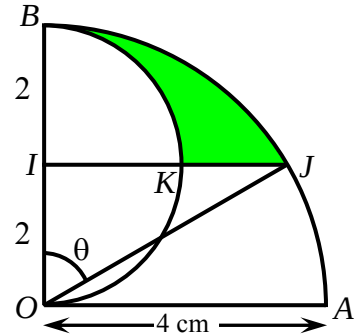
$$= \frac{1}{4}(\pi)(2)^2 = 3 \text{ cm}^2$$

Shaded region area

$$= 8 - 3 - \text{area of } \triangle IJO$$

$$= 5 - \frac{2\sqrt{3} \times 2}{2} = (5 - 2\sqrt{3}) \text{ cm}^2$$

$$\therefore T = 5 - 2\sqrt{3}$$



Event 4

$$1. \quad \sqrt{3-2P} = 2 - \sqrt{1-2P}$$

$$\therefore 3-2P = 4 - 4\sqrt{1-2P} + (1-2P)$$

$$\Rightarrow \frac{1}{2} = \sqrt{1-2P}$$

$$\Rightarrow \frac{1}{4} = 1-2P$$

$$\Rightarrow P = \frac{3}{8}$$

2.

$$\text{Area of } \triangle ABC = \frac{1}{2}(1)(1) = \frac{1}{2} \text{ cm}^2$$

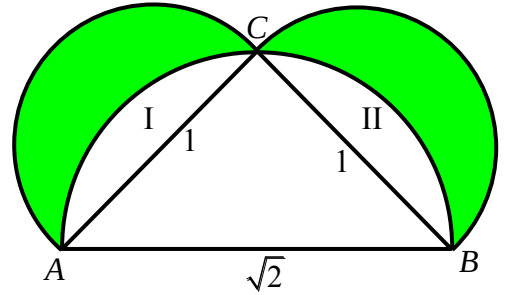
$$\text{Area of the largest semicircle} = \frac{\pi \left(\frac{\sqrt{2}}{2} \right)^2}{2} = \frac{\pi}{4} \text{ cm}^2$$

$$\text{Area of I \& II} = \left(\frac{\pi}{4} - \frac{1}{2} \right) \text{ cm}^2$$

Area of the two small semi-circles

$$= \frac{\pi \left(\frac{1}{2} \right)^2}{2} + \frac{\pi \left(\frac{1}{2} \right)^2}{2} = \frac{\pi}{4} \text{ cm}^2$$

$$\text{Area of shaded region} = \frac{\pi}{4} - \left(\frac{\pi}{4} - \frac{1}{2} \right) = \frac{1}{2} \text{ cm}^2 \text{ or } 0.5 \text{ cm}^2$$



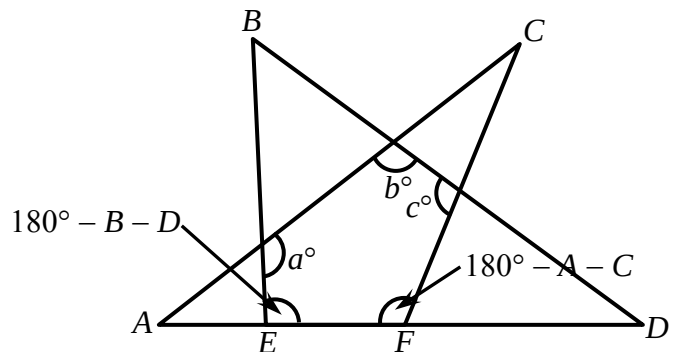
3.

$$\angle E = 180^\circ - B - D$$

$$\angle F = 180^\circ - A - C$$

$$\therefore (a^\circ + b^\circ + c^\circ) + (180^\circ - B - D) + (180^\circ - A - C) = 540^\circ$$

$$\begin{aligned} a^\circ + b^\circ + c^\circ &= 180^\circ + (B + D + A + C) \\ &= 180^\circ + 140^\circ \\ &= 320^\circ \end{aligned}$$



4.

$$\log_{2+\sqrt{2^2-1}} (2 - \sqrt{2^2-1})(2 + \sqrt{2^2-1}) = \log_{2+\sqrt{2^2-1}} 1 = 0$$

$$\therefore \log_{2+\sqrt{2^2-1}} (2 - \sqrt{2^2-1}) + \log_{2+\sqrt{2^2-1}} (2 + \sqrt{2^2-1}) = 0$$

$$\Rightarrow \log_{2+\sqrt{2^2-1}} (2 - \sqrt{2^2-1}) = -1$$

Hong Kong Mathematics Olympiad (2009 / 2010)

Heat Event (Individual) Solution

香港數學競賽 (2009 / 2010)

初賽項目(個人) 題解

1. Put the 8 balls in a single row. Observe that there are 7 gaps between the balls.
What we need to do is to put 2 sticks into any two of these gaps, so as to divide the balls into 3 groups.
The following diagrams show one possible division.



The three boxes contain 2, 5 and 1 balls respectively.

The number of ways is equivalent to the number of choosing 2 gaps for placing the sticks among the 7 gaps.

The number of ways is $C_2^7 = \frac{7 \times 6}{2!} = 21$

2. If α, β are the roots of $x^2 - x - 1 = 0$, then $\alpha + \beta = 1$, $\alpha\beta = -1$.

$$\alpha^2 = \alpha + 1$$

$$\alpha^6 = (\alpha^2)^3 = (\alpha + 1)^3 = \alpha^3 + 3\alpha^2 + 3\alpha + 1$$

$$= \alpha(\alpha^2) + 3(\alpha + 1) + 3\alpha + 1$$

$$= \alpha(\alpha + 1) + 6\alpha + 4$$

$$= \alpha^2 + 7\alpha + 4$$

$$= (\alpha + 1) + 7\alpha + 4 = 8\alpha + 5$$

$$\begin{aligned} \therefore \alpha^6 + 8\beta &= 8\alpha + 5 + 8\beta \\ &= 8(\alpha + \beta) + 5 \\ &= 8 + 5 = 13 \end{aligned}$$

3.
$$\begin{aligned} a &= \frac{1}{5 \times 10} + \frac{1}{10 \times 15} + \frac{1}{15 \times 20} + \cdots + \frac{1}{100 \times 105} \\ &= \frac{1}{25} \left(\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{20 \times 21} \right) \\ &= \frac{1}{25} \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{20} - \frac{1}{21} \right) \\ &= \frac{1}{25} \left(1 - \frac{1}{21} \right) \\ &= \frac{1}{25} \times \frac{20}{21} = \frac{4}{105} \end{aligned}$$

4. Let $x = -a$, where $a > 0$, then $y + z = a + 3$ (1)

$$y^3 + z^3 = a^3 + 3 \text{ (2)}$$

From (2): $(y + z)^3 - 3yz(y + z) = a^3 + 3$

$$\therefore (a + 3)^3 - a^3 - 3 = 3yz(a + 3)$$

$$\begin{aligned} \Rightarrow yz &= \frac{a^3 + 9a^2 + 27a + 27 - a^3 - 3}{3(a + 3)} \\ &= \frac{9a^2 + 27a + 24}{3(a + 3)} \\ &= \frac{3a^2 + 9a + 8}{a + 3} \\ &= 3a + \frac{8}{a + 3} \text{ (3)} \end{aligned}$$

Since yz is an integer $\Rightarrow a = 1$ or 5

$$\therefore (y - z)^2 = (y + z)^2 - 4yz$$

When $a = 1$, $x = -1$, $y + z = 4$ from (1) and $yz = 5$ from (3)

$$\therefore (y - z)^2 = 4^2 - 4 \times 5 = -4 < 0, \text{ which is impossible, hence rejected.}$$

When $a = 5$, $y + z = 8$ and $yz = 16$

Solving for y and z gives $x = -5$, $y = 4$, $z = 4$

$$\therefore y = 4$$

5. $a^{\frac{1}{2}} = b$, $c^{\frac{3}{4}} = d \Rightarrow a = b^2$ and $c = d^{\frac{4}{3}}$

Substitute them into $a - c = 9$

$$b^2 - d^{\frac{4}{3}} = 9$$

$$(b + d^{\frac{2}{3}})(b - d^{\frac{2}{3}}) = 9$$

$$\Rightarrow b + d^{\frac{2}{3}} = 3, b - d^{\frac{2}{3}} = 3 \text{ (no solution, rejected) or } b + d^{\frac{2}{3}} = 9, b - d^{\frac{2}{3}} = 1$$

$$\Rightarrow b = 5 \text{ and } d^{\frac{2}{3}} = 4$$

$$\Rightarrow b = 5, d = 8$$

$$\Rightarrow b - d = -3$$

6. Let $x = \sin A$, $y = \sin B$, then $\sqrt{1 - y^2} = \cos B$, $\sqrt{1 - x^2} = \cos A$

The equation becomes $\sin A \cos B + \cos A \sin B = 1$

$$\therefore \sin(A + B) = 1$$

$$\Rightarrow A + B = 90^\circ \Rightarrow B = 90^\circ - A$$

$$\Rightarrow x^2 + y^2 = \sin^2 A + \sin^2 B$$

$$= \sin^2 A + \sin^2(90^\circ - A)$$

$$= \sin^2 A + \cos^2 A$$

$$= 1$$

7. Draw a perpendicular line from B onto AD .

$$\tan \beta = \frac{12}{12} = 1; \tan(\alpha + \beta) = \frac{12}{12-7} = \frac{12}{5}$$

$$\begin{aligned} \tan \alpha &= \tan[(\alpha + \beta) - \beta] \\ &= \frac{\tan(\alpha + \beta) - \tan \beta}{1 + \tan(\alpha + \beta) \tan \beta} \\ &= \frac{\frac{12}{5} - 1}{1 + \frac{12}{5}} \\ &= \frac{7}{17} \end{aligned}$$

$$\therefore \sin \alpha = \frac{7}{\sqrt{17^2 + 7^2}} = \frac{7}{13\sqrt{2}}; \sin \beta = \frac{1}{\sqrt{2}}$$

$$\therefore \frac{\sin \alpha}{\sin \beta} = \frac{7}{13}$$

8. Let $x = 7k$, $z = 4k$ and $x + y + z = 180$

$$\Rightarrow y = 180 - 11k$$

$$\therefore x \geq y \geq z \Rightarrow 7k \geq 180 - 11k \geq 4k$$

$$\therefore 18k \geq 180 \text{ and } 180 \geq 15k$$

$$\Rightarrow 12 \geq k \geq 10$$

$$\Rightarrow 84 \geq x \geq 70$$

$$\therefore m = 84, n = 70$$

$$\therefore m + n = 154$$

9. When $n = 3$, there are two possible arrangements: 1, 2, 3 or 1, 3, 2.

When $n = 4$, there are two possible arrangements: 1, 2, 4, 3 or 1, 3, 4, 2.

Deductively, for any $n \geq 3$, there are two possible arrangements:

1, 2, 4, 6, 8, ... , largest even integer, largest odd integer, ... , 7, 5, 3 or

1, 3, 5, 7, ... , largest odd integer, largest even integer, ... , 6, 4, 2.

$\therefore$ The answer is 2.

10. Let $f(n) = \frac{n^2}{2010}$, where n is an integer from 1 to 2010.

$$f(n+1) - f(n) = \frac{2n+1}{2010}$$

$$f(n+1) - f(n) < 1 \Leftrightarrow \frac{2n+1}{2010} < 1 \Leftrightarrow n < 1004.5$$

$$\text{Also } f(1005) = \frac{1005^2}{2010} = \frac{1005}{2} = 502.5$$

$\lfloor f(1) \rfloor = 0$, $\lfloor f(2) \rfloor = 0$, ..., $\lfloor f(1005) \rfloor = 502$, the sequence contains 503 different integers.

On the other hand, when $n > 1005$, $f(n+1) - f(n) > 1$

All numbers in the sequence $\lfloor f(1006) \rfloor$, ..., $\lfloor f(2010) \rfloor$ are different, total 1005 numbers.

Thus the number of distinct values is 1508.

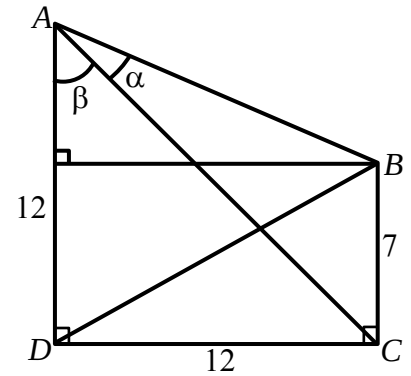


Figure 1

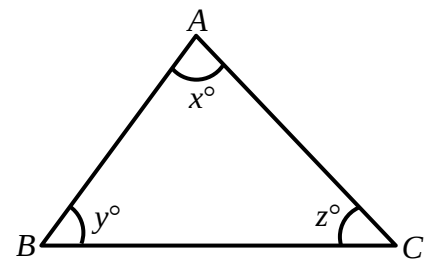


Figure 2

Hong Kong Mathematics Olympiad (2009 / 2010)

Heat Event (Group) Solution

香港數學競賽 (2009 / 2010)

初賽項目(團體) 題解

1. There is no common factor for 7, 9, 11 and the LCM of them is 693.

504 is divisible by 7 and 9.

504504 is divisible by 693.

$$504504 - 693 = 503811$$

$$503811 - 693 = 503118$$

The three-digit number is 118.

2. Sum of odd digits – sum of even digits = multiples of 11

$$n(0+9) - n(2+0) = 11m, \text{ where } m \text{ is an integer.}$$

$$7n = 11m \Rightarrow \text{smallest } n \text{ is } 11$$

3. Let $\angle ACB = y^\circ$, then $\angle ABC = y^\circ + 40^\circ$

$$\angle BAC = 180^\circ - y^\circ - y^\circ - 40^\circ \quad (\angle \text{ sum of } \triangle ABC)$$

$$= 140^\circ - 2y^\circ$$

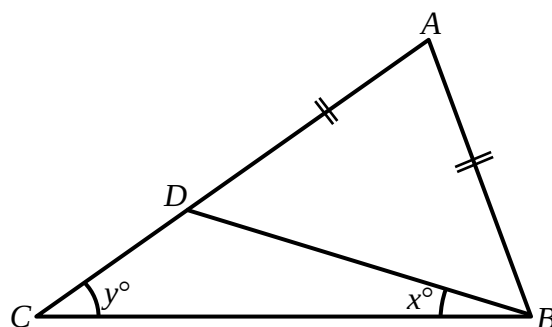
$$\angle ADB = \angle ABD = \frac{180^\circ - (140^\circ - 2y^\circ)}{2} \quad (\text{base } \angle \text{s, isos. } \triangle)$$

$$= 20^\circ + y^\circ$$

$$x^\circ = \angle CBD = \angle ADB - \angle ACB \quad (\text{ext } \angle \text{ of } \triangle BCD)$$

$$= 20^\circ + y^\circ - y^\circ = 20^\circ$$

$$\therefore x = 20$$



4. Suppose AC and BD intersect at K .

$$S_{BCD} = \frac{10 \times 12}{2} = 60$$

$$= S_{CDK} + S_{BCK}$$

$$= 35 + S_{BCK}$$

$$\Rightarrow S_{BCK} = 25$$

$\triangle BCK$ and $\triangle DCK$ have the same height but different bases.

$$BK : KD = S_{BCK} : S_{DCK} = 25 : 35 = 5 : 7$$

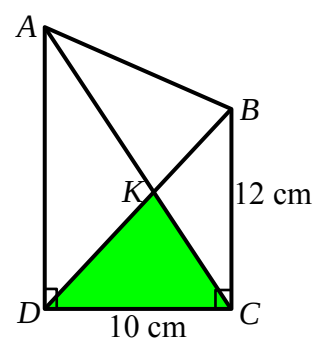
$$\Rightarrow \text{Let } BK = 5t, \quad DK = 7t$$

$$\triangle BCK \sim \triangle DAK \quad (\text{equiangular})$$

$$\Rightarrow S_{BCK} : S_{DAK} = BK^2 : DK^2 = 5^2 : 7^2 = 25 : 49$$

$\triangle ABK$ and $\triangle ADK$ have the same height but different bases.

$$S_{ABK} : S_{ADK} = BK : KD = 5 : 7 \Rightarrow S_{ABCD} = 35 + 25 + 49 + 35 = 144 \text{ cm}^2$$



5. Favourable outcomes = {123, 124, 125, 126, 234, 235, 236, 134, 345, 346, 145, 245, 456, 156, 256, 356}

There are 16 favourable outcomes

$$\therefore \text{Probability} = \frac{16}{C_3^6} = \frac{4}{5} = 0.8$$

$$\begin{aligned} 6. \quad f(500) &= |500-1| + |500-2| + \dots + |500-1000| \\ &= (499 + 498 + \dots + 1) \times 2 + 500 \\ &= 250\,000 \end{aligned}$$

Let n be an integer where $1 \leq n \leq 500$ and $x \leq n$,

$$|x-n| + |x-(1001-n)| = n-x + 1001-n-x = 1001-2x \geq 1001-2n$$

$$\therefore |500-n| + |500-(1001-n)| = 500-n + 501-n = 1001-2n$$

For $1 \leq n < x \leq 500$, $|x-n| + |x-(1001-n)| = x-n + 1001-n-x = 1001-2n$

$$\begin{aligned} \text{If } x \leq 500, \quad f(x) - f(500) &= \sum_{n=1}^{1000} |x-n| - \sum_{n=1}^{1000} |500-n| \\ &= \sum_{n=1}^{500} [|x-n| + |x-(1001-n)|] - \sum_{n=1}^{500} [|500-n| + |500-(1001-n)|] \\ &\geq \sum_{n=1}^{500} [1001-2n - (1001-2n)] = 0 \end{aligned}$$

$$\begin{aligned} f(1001-x) &= |1001-x-1| + |1001-x-2| + \dots + |1001-x-1000| \\ &= |1000-x| + |999-x| + \dots + |1-x| \\ &= |x-1| + |x-2| + \dots + |x-1000| = f(x) \end{aligned}$$

$\therefore f(x) \geq f(500) = 250\,000$ for all real values of x .

$$7. \quad \frac{1}{2010} < \frac{m}{n} < \frac{1}{2009} \Rightarrow 2010 > \frac{n}{m} > 2009 \Rightarrow 2010m > n > 2009m$$

$\therefore m, n$ are positive integer,

$\therefore$ It is equivalent to find the minimum value of m .

When $m = 1$, $2010 > n > 2009$, there is no solution to n .

When $m = 2$, $4020 > n > 4018$, $\Rightarrow n = 4019$.

8.

Number of digits	smallest, ... , largest	Number of lucky numbers	subtotal
1	7	1	1
2	16, 25, ... , 61, 70	7	7
3	106, 115, ... , 160	7	28
	205, 214, ... , 250	6	
	304, 313, ... , 340	5	
	$\vdots$	$\vdots$	
	700	1	
4	1006, 1015, ... , 1060	7	84
	1105, 1114, ... , 1150	6	
	1204, ... , 1240	5	
	$\vdots$	$\vdots$	
	1600	1	
	2005, ... , 2050	6	
	$\vdots$	$\vdots$	
	2500	1	
	3004, ... , 3040	5	
	$\vdots$	$\vdots$	
	3400	1	
	4XYZ	4 + 3 + 2 + 1	
	5XYZ	3 + 2 + 1	
	6XYZ	2 + 1	
	7000	1	
5	100XY	7	
	10105	1	

From the above table, $a_{64} = 1600 \Rightarrow a_{128} = 10105$

9.

$$(x+2y)(x-2y) = 4$$

$$x^2 - 4y^2 = 4$$

$$x^2 = 4y^2 + 4$$

$$T = |x| - |y| = \sqrt{4(y^2 + 1)} - |y|$$

$$T + |y| = \sqrt{4(y^2 + 1)}$$

$$T^2 + y^2 + 2|y|T = 4(y^2 + 1)$$

$$3|y|^2 - 2|y|T + (4 - T^2) = 0$$

$$\Delta = 4[T^2 - 3(4 - T^2)] \geq 0$$

$$4T^2 - 12 \geq 0$$

$$T \geq \sqrt{3}$$

The minimum value of $|x| - |y|$ is $\sqrt{3}$.

10. Let $AP = PQ = QB = BC = t$, let $AQ = y$

$\angle AQP = x^\circ$ (base $\angle$ s, isos. Δ)

$$AQ = y = 2t \cos x^\circ = y + t - t$$

$$= AC - AP = CP$$

$\angle BPQ = \angle PBQ$ (base $\angle$ s, isos. Δ)

$$= \frac{x^\circ}{2} \quad (\text{ext } \angle \text{ of } \Delta BPQ)$$

$\angle QPC = 2x^\circ$ (ext $\angle$ of ΔAPQ)

$$\angle BPC = \angle QPC - \angle BPQ = 2x^\circ - \frac{x^\circ}{2} = \frac{3x^\circ}{2}$$

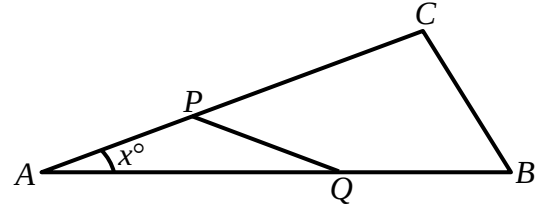
$$\angle ABC = \angle ACB = 90^\circ - \frac{x^\circ}{2} \quad (\angle \text{ sum of } \Delta ABC)$$

$$\angle CBP = \angle ABC - \angle PBQ = 90^\circ - \frac{x^\circ}{2} - \frac{x^\circ}{2} = 90^\circ - x^\circ$$

$$\frac{CP}{\sin \angle CBP} = \frac{BC}{\sin \angle BPC}$$

$$\frac{2t \cos x^\circ}{\sin(90^\circ - x^\circ)} = \frac{t}{\sin \frac{3x^\circ}{2}}$$

$$\sin \frac{3x^\circ}{2} = \frac{1}{2} \Rightarrow x = 20$$



Hong Kong Mathematics Olympiad (2009 / 2010)

Final Event (Individual) Solution

香港數學競賽 (2009 / 2010)

決賽項目(個人) 題解

Event 1

1. Total surface area before being glued together is 84 units^2 . Placing the cube with volume 8 on top of the cube with volume 27 can remove a maximum of $2 \times 4 = 8 \text{ units}^2$. Also placing the unit cube so that one face adjoins the cube with volume 27 and another adjoins the cube with volume 8 can remove up to 4 units^2 . Therefore, $a = 84 - 8 - 4 = 72 \text{ units}^2$ is left.

2. $f(x) = -x^2 + 10x + 9 = -(x-5)^2 + 34 \dots\dots\dots (1)$

Thus maximum value is 34

For minimum value, either substitute $x = 2$ or $x = 8$ into (1) gives $f(2) = f(8) = 25$.

Therefore $b = 9$.

3.
$$\begin{aligned} 70 &= p^2q + q^2p + p + q \\ &= (pq+1)(p+q) \\ &= (b+1)(p+q) \end{aligned}$$

Thus $p+q=7$ and since $c = p^2 + q^2$, we have

$$\begin{aligned} c &= p^2 + q^2 \\ &= (p+q)^2 - 2pq \\ &= 7^2 - 2(9) \\ &= 31 \end{aligned}$$

4. Mean = 64 and number of rows = 31, thus

$$\text{Number of seats} = 31 \times 64 = 1984$$

Event 2

1. By trial and error, a , p and q are respectively 2, 17 and 19. Thus $a = 2$.

2. $b^2 + h^2 = b(a+h) + ah = b(2+h) + 2h$

Therefore $b^2 - (2+h)b + (h^2 - 2h) = 0$

As it has a real solution, we have $(2+h)^2 - 4(h^2 - 2h) \geq 0$, also,

$$(2+h)^2 - 4(h^2 - 2h)$$

$$= h^2 + 4h + 4 - 4h^2 + 8h$$

$$= 4 + 12h - 3h^2$$

$$= 4 - 3(h^2 - 4h + 4) + 12$$

$$= 16 - 3(h-2)^2 \leq 16$$

Since $16 - 3(h-2)^2$ has to be a perfect square, $h = 4$ or $h = 2$, but $b < h$ and consequently $b = 2$.

3. The number of combinations of different pairs of squares chosen

$$= {}^5C_2 {}^5C_1 {}^5C_1$$

$$= 10 \times 5 \times 5$$

$$= 250$$

4. Let $\frac{1}{x} + \frac{1}{2} = a + b$, where a is an integer and $0 \leq b < 1$.

$$\text{Then } \left\lfloor \frac{1}{x} + \frac{1}{2} \right\rfloor = a \text{ and } \frac{1}{x} - \left\lfloor \frac{1}{x} + \frac{1}{2} \right\rfloor = a + b - \frac{1}{2} - a = b - \frac{1}{2}.$$

$$\text{Since } 0 \leq b < 1, -\frac{1}{2} \leq b - \frac{1}{2} < \frac{1}{2} \Rightarrow \left| b - \frac{1}{2} \right| \leq \frac{1}{2}$$

Consequently,

$$f(x) = c \left| \frac{1}{x} - \left\lfloor \frac{1}{x} + \frac{1}{2} \right\rfloor \right|$$

$$= 250 \times \left| b - \frac{1}{2} \right|$$

$$\leq 250 \times \frac{1}{2} = 125$$

$$\therefore d = 125$$

Event 3

1. As $15147 = 3^4 \times 11 \times 17$, we have $a = 3$.

2. As $x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right) = 3^3 - 3 \times 3 = 18$, we have $b = 18$.

3. Since c is odd, $f(c) = c + 5$ and $f(f(c)) = \frac{c+5}{2}$.

If $\frac{c+5}{2}$ is even, then $18 = f\left(\frac{c+5}{2}\right) = \frac{c+5}{4}$, i.e. $c+5 = 72$. Hence $c = 67$.

If $\frac{c+5}{2}$ is odd, then $18 = f\left(\frac{c+5}{2}\right) = \frac{c+5}{2} + 5$, i.e. $\frac{c+15}{2} = 18$. Hence $c = 21$.

Hence the least value of c is 21.

4. If $f\left(\frac{x}{3}\right) = x^2 + x + 1$, we have $f\left(\frac{x}{3}\right) = 9\left(\frac{x}{3}\right)^2 + 3\left(\frac{x}{3}\right) + 1$, thus replacing $\frac{x}{3}$ by y , we have

$$f(y) = 9y^2 + 3y + 1 \text{ and } f(3y) = 81y^2 + 9y + 1.$$

Therefore, $f(3y) = 21 \Rightarrow 81y^2 + 9y + 1 = 21 \Rightarrow 81y^2 + 9y - 20 = 0 \dots\dots\dots (*)$

Therefore, the sum of all roots of (*) is $-\frac{9}{81} = -\frac{1}{9}$, i.e. $d = -\frac{1}{9}$.

Event 4

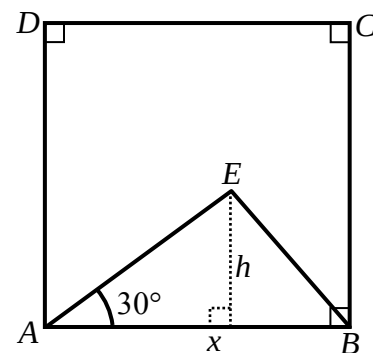
1. Let the length of the square $ABCD$ be x and the height of $\triangle ABE$ be h .

We have $\frac{\frac{x \cdot h}{2}}{x^2} = \frac{h}{2x} = \frac{1}{6}$, thus $\frac{h}{x} = \frac{1}{3}$, therefore

$$a = \frac{AE}{AB} = \frac{\frac{h}{\sin 30^\circ}}{x} = \frac{2h}{x} = \frac{2}{3}$$

$$\therefore a = \frac{2}{3}$$

2.
$$b = \frac{\log 8^{\frac{2}{3}} + \log 27^{\frac{2}{3}} + \log 125^{\frac{2}{3}}}{\log 9 + \log 25 + \log 2 - \log 15}$$
$$= \frac{2(\log 2 + \log 3 + \log 5)}{\log 3 + \log 2 + \log 5}$$
$$= 2$$
$$\therefore b = 2$$

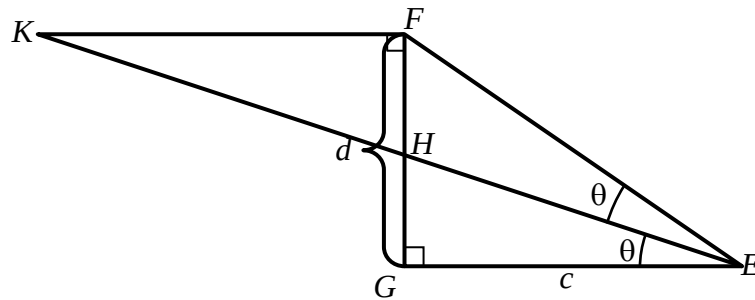


3. By using the formula $1^3 + 2^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$, we have

$$\begin{aligned}
 & 1^3 + 2^3 + \dots + 2009^3 + 2010^3 \\
 &= \frac{1}{4}(2010)^2(2011)^2 \\
 &= 1005^2 \times 2011^2 \\
 &= (4p+1)^2(4q+3)^2 \quad \text{where } p \text{ and } q \text{ are positive integers} \\
 &= (16p^2 + 8p + 1)(16q^2 + 24q + 9)
 \end{aligned}$$

Thus, when it is divided by 4, the remainder is 1, therefore $c = 1$.

4. Referring to the figure below:



Construct a point K on EH produced so that $FK \parallel GE$.

$$\angle FKE = \angle FEK = \theta \quad (\text{by construction})$$

$$\text{Also } \triangle FHK \sim \triangle GHE \quad (\text{by construction})$$

$$\text{Thus } \frac{FH}{GH} = \frac{FK}{GE} \quad (\text{corr. sides, } \sim \Delta s)$$

Therefore

$$\frac{5}{4} = \frac{FK}{1} \Rightarrow FK = \frac{5}{4}$$

$$\text{Consequently, } FE = FK = \frac{5}{4}$$

$$\begin{aligned}
 \therefore d &= \sqrt{\left(\frac{5}{4}\right)^2 - 1^2} \\
 &= \sqrt{\frac{25}{16} - 1} \\
 &= \sqrt{\frac{9}{16}} \\
 &= \frac{3}{4}
 \end{aligned}$$

Hong Kong Mathematics Olympiad (2009 / 2010)

Final Event (Group) Solution

香港數學競賽 (2009 / 2010)

決賽項目(團體)題解

Event 1

1. Let $S = \sin^2 1^\circ + \sin^2 2^\circ + \cdots + \sin^2 89^\circ$, we have $S = \cos^2 89^\circ + \cos^2 88^\circ + \cdots + \cos^2 1^\circ$.
Thus $2S = (\sin^2 1^\circ + \cos^2 1^\circ) + (\sin^2 2^\circ + \cos^2 2^\circ) + \cdots + (\sin^2 89^\circ + \cos^2 89^\circ) = \underbrace{1+1+\cdots+1}_{89 \text{ terms}} = 89$

Therefore $S = \frac{89}{2} = 44.5$

2. $\frac{x}{y} = \frac{x+z+z+2y+x}{2z-x+2x-z+y} = \frac{2(x+y+z)}{x+y+z} = 2$

3. Let $a = 2^x - 4$ and $b = 4^x - 2$, then

$$\begin{aligned} & (4^x + 2^x - 6)^3 \\ &= [(4^x - 2) + (2^x - 4)]^3 \\ &= (a + b)^3 \end{aligned}$$

Thus the given equation becomes

$$(a+b)^3 = a^3 + b^3$$

$$3ab(a+b) = 0$$

Therefore, we have three possibilities

(i) $a = 0 \Rightarrow 2^x = 4 \Rightarrow x = 2$

(ii) $b = 0 \Rightarrow 4^x = 2 \Rightarrow x = \frac{1}{2}$

(iii) $a + b = 0 \Rightarrow 2^x + 4^x = 6 \Rightarrow x = 1$

Consequently, the sum of all real roots x of the equation is $2 + 1 + \frac{1}{2} = \frac{7}{2}$.

4. Make a point G on CD such that $BD \parallel EG$.
 $AB \perp CD \Rightarrow EG \perp CD \Rightarrow \angle DGE = \angle BDC = 90^\circ$.
 Since $\angle BFD = \angle EFG$ and $BF = EF$, we have

$$\triangle BDF \cong \triangle EGF$$

$$\Rightarrow BD = EG \text{ and } DF = FG$$

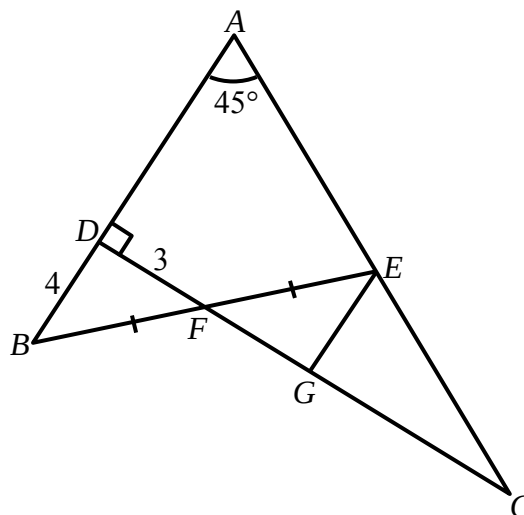
$$\Rightarrow EG = 4 \text{ and } FG = 3$$

Besides, $\triangle ACD$ is an isosceles triangle, then $\triangle EGC$ is also an isosceles triangle and hence

$$CG = EG = 4$$

$$\Rightarrow CD = 3 + 3 + 4 = 10$$

$$\Rightarrow n = AD = 10$$



Event 2

$$\begin{aligned} 1. \quad p &= 2^{10}(2-1) - 2^9 - 2^8 - \dots - 2^2 + 2 \\ &= 2^9(2-1) - 2^8 - 2^7 - \dots - 2^2 + 2 \\ &= 2^8(2-1) - 2^7 - 2^6 - \dots - 2^2 + 2 \\ &= \dots \\ &= 2^2(2-1) + 2 = 6 \end{aligned}$$

$$2. \quad \text{From } x + \frac{1}{y} = y + \frac{1}{z}, \text{ we have } x - y = \frac{1}{z} - \frac{1}{y} = \frac{y-z}{yz}, \text{ thus } yz = \frac{y-z}{x-y}.$$

$$\text{Similarly, } xy = \frac{x-y}{z-x} \text{ and } xz = \frac{z-x}{y-z}, \text{ therefore}$$

$$\begin{aligned} x^2 y^2 z^2 &= (xy)(yz)(zx) \\ &= \left(\frac{x-y}{z-x} \right) \left(\frac{y-z}{x-y} \right) \left(\frac{z-x}{y-z} \right) \\ &= 1 \end{aligned}$$

$$3. \quad \text{Let } f(x) = x \cdot 3^x - 3^{18}$$

$$\text{We have } f(15) = 15 \times 3^{15} - 3^{18} = 3^{15}(15 - 3^3) < 0$$

$$\text{and } f(16) = 16 \times 3^{16} - 3^{18} = 3^{16}(16 - 3^2) > 0$$

$$\text{Thus } k = 15$$

$$4. \quad \text{The answer is } 8 + 94(5) = 478.$$

Event 3

1. Because both 101^{303} and 303^{101} are odd integers, $101^{303} + 303^{101}$ is an even integer. Therefore, the smallest prime factor is 2.

2. Let $\frac{1}{s} = \frac{1}{1980} + \frac{1}{1981} + \dots + \frac{1}{2009}$

Then $\frac{30}{2009} < \frac{1}{s} < \frac{30}{1980}$

$$\Rightarrow \frac{1980}{30} < s < \frac{2009}{30}$$

$$\Rightarrow 66 < s < 66.97$$

Therefore n is 66.

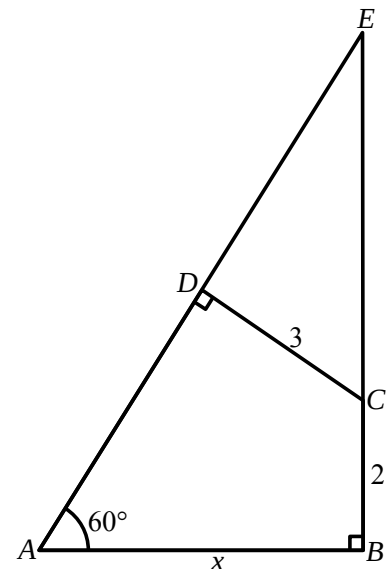
3. Make a point E as shown in the figure.

In $\triangle ABE$, $\angle E = 30^\circ$. In $\triangle CDE$,

$$\frac{CD}{CE} = \sin 30^\circ \Rightarrow CE = 2 \times CD = 6 \Rightarrow BE = 8$$

In $\triangle ABE$,

$$\tan 60^\circ = \frac{BE}{AB} \Rightarrow x = \frac{8}{\sqrt{3}} = \frac{8\sqrt{3}}{3}$$



4. Let two of these roots be $2 + \alpha$ and $2 + \beta$, since $f(2+x) = f(2-x)$ for every x implies that $f(2+\alpha) = f(2-\alpha) = 0$ and $f(2+\beta) = f(2-\beta) = 0$, where $\alpha \neq 0$ and $\beta \neq 0$. Thus the four roots are $2 + \alpha$, $2 + \beta$, $2 - \alpha$ and $2 - \beta$. Consequently, sum of roots $= 2 + \alpha + 2 - \alpha + 2 + \beta + 2 - \beta = 8$.

Event 4

1. The product of the two roots is given by $\frac{a}{a-1}$ which is also an integer. Because

$$\frac{a}{a-1} = 1 + \frac{1}{a-1}$$

that is, $\frac{1}{a-1}$ must be an integer. The only possibility will be $a = 2$.

The product of the two roots is therefore 2. Thus, the two positive integer roots must be 1 and 2. Finally, $m = 1 + 2 = 3$.

2. $y = \sqrt{x^2 - 2x + 2} + \sqrt{x^2 - 10x + 34}$

On squaring both sides, gives

$$\left(y - \sqrt{x^2 - 2x + 2}\right)^2 = x^2 - 10x + 34$$

$$y^2 - 2y\sqrt{x^2 - 2x + 2} + x^2 - 2x + 2 = x^2 - 10x + 34$$

$$y^2 + 8x - 32 = 2y\sqrt{x^2 - 2x + 2}$$

Squaring both sides again, we have

$$64x^2 + y^4 - 64y^2 + 1024 + 16x(y^2 - 32) = 4y^2(x^2 - 2x + 2)$$

$$(64 - 4y^2)x^2 + (24y^2 - 512)x + (y^4 - 72y^2 + 1024) = 0$$

Since x is real, we have $\Delta \geq 0$.

$$(24y^2 - 512)^2 - 4(64 - 4y^2)(y^4 - 72y^2 + 1024) \geq 0$$

$$(128 - 6y^2)^2 + (y^2 - 16)(y^4 - 72y^2 + 1024) \geq 0$$

$$36y^4 - 1536y^2 + 16384 + (y^6 - 72y^4 + 1024y^2 - 16y^4 + 1152y^2 - 16384) \geq 0$$

$$y^6 - 52y^4 + 640y^2 \geq 0$$

$$y^2(y^2 - 32)(y^2 - 20) \geq 0$$

Therefore $y^2 \leq 20$ or $y^2 \geq 32$

When $y^2 \geq 32$,

$$\sqrt{x^2 - 2x + 2} + \sqrt{x^2 - 10x + 34} \geq \sqrt{32}$$

$$x^2 - 10x + 34 \geq 32 + x^2 - 2x + 2 - 2\sqrt{32}\sqrt{x^2 - 2x + 2}$$

$$\sqrt{32}\sqrt{x^2 - 2x + 2} \geq 4x$$

$$32(x^2 - 2x + 2) \geq 16x^2$$

$$x^2 - 4x + 4 \geq 0$$

$$(x - 4)^2 \geq 0$$

which is always true.

When $y^2 \leq 20$,

$$\begin{aligned}
\sqrt{x^2 - 2x + 2} + \sqrt{x^2 - 10x + 34} &\leq \sqrt{20} \\
x^2 - 10x + 34 &\leq 20 + x^2 - 2x + 2 - 2\sqrt{20}\sqrt{x^2 - 2x + 2} \\
\sqrt{20}\sqrt{x^2 - 2x + 2} &\leq 4x - 6 \\
20(x^2 - 2x + 2) &\leq 16x^2 - 48x + 36 \\
x^2 + 2x + 1 &\leq 0 \\
(x+1)^2 &\leq 0
\end{aligned}$$

which is only true when $x = -1$.

But when $x = -1$, $y = 4\sqrt{5} > \sqrt{20}$, so this case is rejected.

$\therefore$ the minimum value of $y = \sqrt{32} = 4\sqrt{2}$.

3. Rewrite the equation as $5^A 2^B = 500^C = 5^3 2^2$, which implies $A = 3C$ and $B = 2C$.
 Since the greatest common divisor of A , B and C is 1, $C = 1$.
 Therefore $A + B + C = 6$.

4. Draw BE .

Then $BE^2 = AB^2 - AE^2 = 64 - 16 = 48$, and

$BF : FC = 3 : 1 \Rightarrow BF = 3x$, $FC = x \Rightarrow BC = 4x$.

Since $\triangle BEF \sim \triangle BCE$, we have

$$\frac{BE}{BC} = \frac{BF}{BE}$$

$$\Rightarrow BE^2 = BC \times BF$$

$$\Rightarrow 48 = (4x)(3x)$$

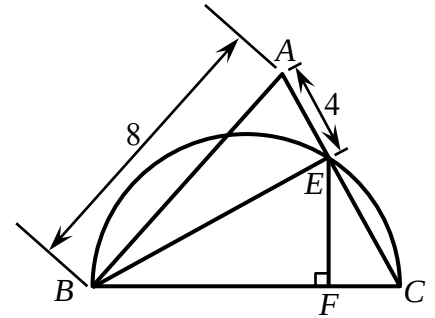
$$\Rightarrow x = 2$$

$$\Rightarrow BC = 8$$

Hence,

$$\triangle CBE \cong \triangle ABE$$

$$\Rightarrow EC = AE = 4$$



Hong Kong Mathematics Olympiad (2010 / 2011)

Heat Event (Individual) Solution

香港數學競賽 (2010 / 2011)

初賽項目(個人) 題解

$$\begin{aligned}
 1. \quad & 2^{2011} \\
 &= (2^6)^{334} \times 2^7 \\
 & (2^6)^{334} \equiv (-1)^{334} \pmod{13} \\
 & (2^6)^{334} \equiv 1 \pmod{13} \\
 & 2^7 \equiv 11 \pmod{13} \\
 \therefore 2^{2011} &\equiv 11 \pmod{13} \\
 \therefore \text{The remainder is } &11.
 \end{aligned}$$

$$\begin{aligned}
 2. \quad & \text{Let } x = \sin \theta, \text{ then } y = \cos \theta, \quad 0 \leq x^2, y^2 \leq 1, \\
 & \text{i.e. } -1 \leq x, y \leq 1
 \end{aligned}$$

$$\begin{aligned}
 & 2x + 5y^2 \\
 &= 2\sin \theta + 5(\cos \theta)^2 \\
 &= 5(1 - \sin^2 \theta) + 2\sin \theta \\
 &= 5 - 5\sin^2 \theta + 2\sin \theta \\
 &= -5 \left[\sin^2 \theta - \frac{2}{5} \sin \theta - 1 \right] \\
 &= -5 \left[\left(\sin \theta - \frac{1}{5} \right)^2 - \frac{26}{25} \right] \\
 &= -5 \left(\sin \theta - \frac{1}{5} \right)^2 + \frac{26}{5}
 \end{aligned}$$

Alternatively, since $y^2 = 1 - x^2$, we have

$$\begin{aligned}
 & 2x + 5y^2 \\
 &= 2x + 5(1 - x^2) \\
 &= 5 + 2x - 5x^2 \\
 &= -5 \left(x^2 - \frac{2}{5}x - 1 \right) \\
 &= -5 \left[\left(x - \frac{1}{5} \right)^2 - \frac{26}{25} \right] \\
 &= -5 \left(x - \frac{1}{5} \right)^2 + \frac{26}{5}
 \end{aligned}$$

Therefore, the maximum value is $\frac{26}{5}$.

$$3. \quad (a+b)^2 = \sqrt{2011} + \sqrt{2010}, \text{ gives } a^2 + 2ab + b^2 = \sqrt{2011} + \sqrt{2010} \dots\dots (1)$$

$$(a-b)^2 = \sqrt{2011} - \sqrt{2010}, \text{ gives } a^2 - 2ab + b^2 = \sqrt{2011} - \sqrt{2010} \dots\dots (2)$$

$$(1) - (2): \quad 4ab = 2\sqrt{2010},$$

$$\text{Therefore } ab = \frac{1}{2}\sqrt{2010}.$$

4. Let the altitudes of the three sides of ΔABC , AB , BC and CA be respectively CF , AD and BE , and $CF : AD : BE = 3 : 4 : 5$

$\therefore CF = 3t$, $AD = 4t$, $BE = 5t$, where t is a constant.

$$\therefore \frac{1}{2}AB \times CF = \frac{1}{2}BC \times AD = \frac{1}{2}CA \times BE$$

$$\therefore \frac{1}{2}AB \times 3t = \frac{1}{2}BC \times 4t = \frac{1}{2}CA \times 5t$$

$$\text{i.e. } 3AB = 4BC = 5CA$$

$$\frac{AB}{20} = \frac{BC}{15} = \frac{CA}{12}$$

Since the $\text{gcd}(20, 15, 12) = 1$, the minimum value of AB is 20.

5. According to the question,

$$\begin{cases} x - 12 = m^2 \\ x + 19 = n^2 \end{cases}, \text{ where } m, n \text{ are integers.}$$

$$\therefore n^2 - m^2 = 31$$

$$(n - m)(n + m) = 31$$

Since 31 is a prime number, we have

$$\begin{cases} n - m = 1 \\ n + m = 31 \end{cases}$$

that is, $n = 16$, $m = 15$

$$\text{Therefore } x = 16^2 - 19 = 237$$

6. There are 5 different ways:

$$A \rightarrow B \rightarrow \begin{cases} A \rightarrow \begin{cases} B \rightarrow C \rightarrow A \\ C \rightarrow B \rightarrow A \end{cases} \\ C \rightarrow \begin{cases} A \rightarrow \begin{cases} B \rightarrow A \\ C \rightarrow A \end{cases} \\ B \rightarrow C \rightarrow A \end{cases} \end{cases}$$

If the positions of B and C are interchanged, we have another 5 ways.

Therefore, there are 10 ways altogether.

$$\begin{aligned}
7. \quad & \sqrt{7 - \sqrt{12} - \sqrt{13 - 2\sqrt{12}}} \\
&= \sqrt{7 - \sqrt{12} - (\sqrt{12} - 1)} \\
&= \sqrt{8 - 2\sqrt{12}} \\
&= \sqrt{6} - \sqrt{2}
\end{aligned}$$

8. Let class A used k \$100 notes and the number of each type of raffle tickets bought be a , b , c and d respectively. Thus $10a + 15b + 25c + 40d = 100k$
i.e., $2a + 3b + 5c + 8d = 20k$, from which we see that b and c are either both even or both odd
If b and c are both even, we have $b = c = 10$, $a = d = 5$ and $k = 6.5$ (since k is not an integer, so reject the answer);
If b and c are both odd, then $b = c = 5$, $a = d = 10$ and $k = 7$.
Therefore class A has used 7 \$100 notes to buy the raffle tickets.

9. Let the length and width of the rectangle be x and y , where $x > y$.

$$\text{Since } xy - 2(x + y) = 9,$$

$$xy - 2x - 2y = 9$$

$$xy - 2x - 2y + 4 = 9 + 4$$

$$(x - 2)(y - 2) = 13$$

$$x - 2 = 13 \text{ and } y - 2 = 1$$

$$\text{i.e. } x = 15 \text{ and } y = 3$$

$$\text{Therefore, the perimeter of the rectangle} = 2(3 + 15) = 36$$

10. Let the sides of $\triangle ABD$ be x units.

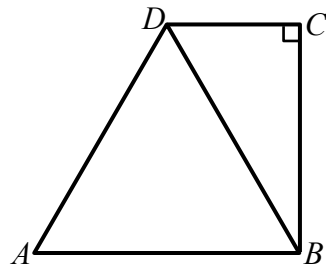
$$BC = \sqrt{x^2 - \left(\frac{x}{2}\right)^2} = \frac{x}{2}\sqrt{3} \quad (\text{or } BC = x \sin 60^\circ = \frac{\sqrt{3}}{2}x)$$

$$\therefore \frac{1}{2}x \left(\frac{x}{2}\sqrt{3} \right) = 16\sqrt{3}$$

$$\therefore x = 8$$

$$\therefore CD = \frac{x}{2} \text{ units and } AB = x \text{ units}$$

$$\therefore \text{The area of trapezium } ABCD = \frac{1}{2} \left(\frac{x}{2} + x \right) \left(\frac{x}{2}\sqrt{3} \right) = \frac{1}{2} \left(\frac{8}{2} + 8 \right) \left(\frac{8}{2}\sqrt{3} \right) = 24\sqrt{3}$$



Alternatively, since $CD = \frac{AB}{2}$, area of $\triangle BCD = \frac{1}{2} \times$ area of $\triangle ABD$

$\therefore$ The area of trapezium $ABCD = \frac{3}{2} \times$ area of $\triangle ABD = \frac{3}{2} \times 16\sqrt{3} = 24\sqrt{3}$

Hong Kong Mathematics Olympiad (2010 / 2011)
Heat Event (Group) Solution
香港數學競賽 (2010 / 2011)
初賽項目(團體) 題解

1. Because a, b, c, d and e are even,
 $(1000 - a), (1000 - b), (1000 - c), (1000 - d)$ and $(1000 - e)$ are even numbers.
Also $24^2 = 2 \times (-2) \times 4 \times 6 \times (-6)$,
Therefore $(1000 - a), (1000 - b), (1000 - c), (1000 - d)$ and $(1000 - e)$ are equal to $2, (-2), 4, 6$ and (-6) .
Given that $a > b > c > d > e$, thus
 $1000 - a = -6, \quad \therefore a = 1006$
 $1000 - b = -2, \quad \therefore b = 1002$
 $1000 - c = 2, \quad \therefore c = 998$
 $1000 - d = 4, \quad \therefore d = 996$
 $1000 - e = 6, \quad \therefore e = 994$

2. The maximum value of $a + b$ is $9 + 9 = 18$.
At the instant, the maximum remainder $R_{\overline{ab}}$ is 17 , but when $a + b = 18$, $\overline{ab}$ must be 99 , and when $\overline{ab} \div (a + b) = 99 \div 18$, the remainder is 9 , thus rejecting it.
Consider the case when the remainder $R_{\overline{ab}}$ is 16 , the possible values of $a + b$ can be 18 (rejected) or 17 .
When $a + b = 17$, $\overline{ab}$ can possibly be 98 or 89 .
The remainder when $\overline{ab} \div (a + b) = 98 \div 17$ is 13 and the remainder when $\overline{ab} \div (a + b) = 89 \div 17$ is 4 , therefore, reject the answer.
Consider the case when the remainder $R_{\overline{ab}}$ is 15 , the possible values of $a + b$ can be 18 (rejected), 17 (rejected) or 16 .
When $a + b = 16$, $\overline{ab}$ can possibly be $97, 88$ or 79 , and the remainder when $\overline{ab} \div (a + b) = 79 \div 16$ is 15 , therefore the maximum value of $R_{\overline{ab}}$ is 15 .

3. Because $a + b = 2011, c - a = 2010$ gives $b + c = 4021$;
If the value of $a + b + c$ attains its maximum, the value of a also attains its maximum.
Because $a + b = 2011$ and $a < b$, we have the maximum value of a is 1005 ,
Thus the possible maximum value of $a + b + c = 1005 + 4021 = 5026$.

$$4. \quad n^4 - 18n^2 + 49$$

$$= (n^2 - 7)^2 - (2n)^2$$

$$= (n^2 - 7 + 2n)(n^2 - 7 - 2n)$$

Since it is prime, then

$$n^2 - 7 - 2n = 1$$

$$(n - 4)(n + 2) = 0$$

$$n = 4 \text{ or } n = -2 \text{ (rejected)}$$

$$5. \quad \text{Consider } f\left(\frac{1}{2011}\right) + f\left(\frac{2010}{2011}\right) = \frac{\frac{1}{4^{2011}}}{\frac{1}{4^{2011}} + 2} + \frac{\frac{2010}{4^{2011}}}{\frac{2010}{4^{2011}} + 2} = \frac{\frac{1}{4^{2011}}\left(\frac{2010}{4^{2011}} + 2\right) + \frac{2010}{4^{2011}}\left(\frac{1}{4^{2011}} + 2\right)}{\left(\frac{1}{4^{2011}} + 2\right)\left(\frac{2010}{4^{2011}} + 2\right)}$$

$$= \frac{4 + 2 \times \frac{1}{4^{2011}} + 4 + 2 \times \frac{2010}{4^{2011}}}{4 + 2 \times \frac{1}{4^{2011}} + 2 \times \frac{2010}{4^{2011}} + 4} = 1$$

$$\begin{aligned} \therefore f\left(\frac{1}{2011}\right) + f\left(\frac{2}{2011}\right) + f\left(\frac{3}{2011}\right) + \dots + f\left(\frac{2009}{2011}\right) + f\left(\frac{2010}{2011}\right) \\ = f\left(\frac{1}{2011}\right) + f\left(\frac{2010}{2011}\right) + f\left(\frac{2}{2011}\right) + f\left(\frac{2009}{2011}\right) + \dots + f\left(\frac{1005}{2011}\right) + f\left(\frac{1006}{2011}\right) \\ = 1005 \end{aligned}$$

6. According to the question, $\angle ABC = 90^\circ$. Let $AB = x$ and $BC = y$, we have $a = AB + BC > AC = 6$.
Meanwhile, $x + y = a$ and $x^2 + y^2 = 36$ gives $xy = \frac{a^2 - 36}{2}$, which means that x and y are the two

real roots of the equation $z^2 - az + \frac{a^2 - 36}{2} = 0$

Therefore $\Delta = a^2 - 4(1)\left(\frac{a^2 - 36}{2}\right) \geq 0$, i.e. $a \leq 6\sqrt{2}$.

$\therefore$ The maximum value of a is $6\sqrt{2}$.

7. The maximum value of k is the number of factors of 5 in 2011!.

$$\begin{aligned} k_{\max} &= \left\lfloor \frac{2011}{5} \right\rfloor + \left\lfloor \frac{2011}{25} \right\rfloor + \left\lfloor \frac{2011}{125} \right\rfloor + \left\lfloor \frac{2011}{625} \right\rfloor \\ &= 402 + 80 + 16 + 3 \\ &= 501 \end{aligned}$$

$$S = 1 + 2 + 3 + \dots + 501$$

$$\begin{aligned} \therefore &= \frac{501 \cdot 502}{2} \\ &= 125751 \end{aligned}$$

8. $ac + bd + ad + bc = 2011$

$$(a + b) \times (c + d) = 2011$$

Because 2011 is prime, we have $(a + b) \times (c + d) = 2011 \times 1$, which gives

$$a + b + c + d = (a + b) + (c + d) = 2011 + 1 = 2012$$

9. Construct a circumscribed circle of $\triangle ABC$. Let the BD produced intersect the circle at K .

Join AK , CK . $\angle AKB = \angle ACB = 18^\circ$. Given that $\angle BDA = 36^\circ$, we have $\angle DAK = 18^\circ$, i.e. $\angle DAK = \angle DKA$, thus $DA = DK$.

Also, $\angle CKB = \angle CAB = 27^\circ$, and $\angle BDC = 54^\circ$, thus $\angle DCK = 54^\circ - \angle DKC = 27^\circ$, i.e. $DC = DK$.

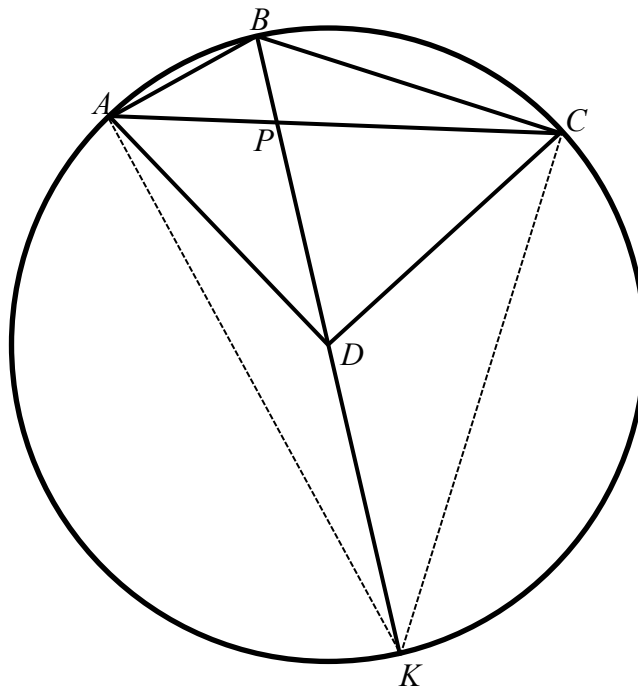
Therefore D is the centre of the circumscribed circle of $\triangle ABC$.

$$\angle ADC = \angle ADB + \angle BDC = 36^\circ + 54^\circ = 90^\circ$$

$$\therefore \angle CPB = \angle PDC + \angle PCD$$

$$= 54^\circ + 45^\circ$$

$$= 99^\circ$$



10. Let $AM = x$, and the radius of the in-circle = r .

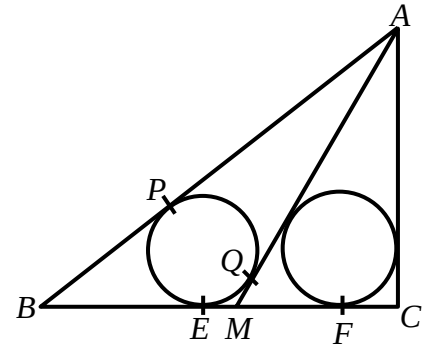
We have, $\frac{1}{2}r(3+4+5+2x) = \text{area of } \triangle ABC$

$$\frac{1}{2}r(12+2x) = \frac{1}{2} \times 3 \times 4$$

$$r(6+x) = 6 \quad (1)$$

Let E and F be the points of contact between the 2 circles and BC as shown in the figure.

Also let the inscribed circle of $\triangle ABM$ touch AB and AM at P and Q respectively.



$$2BE = BP + BE$$

$$= AB - AP + BM - EM$$

$$= AB + BM - (AP + EM)$$

$$= AB + BM - (AQ + QM)$$

$$= AB + BM - x$$

$$BE = \frac{1}{2}(AB + BM - x)$$

Similarly, $CF = \frac{1}{2}(AC + CM - x)$

$$BE + CF = \frac{1}{2}(AB + AC + BC - 2x)$$

$$r \left(\cot \frac{\angle B}{2} + \cot \frac{\angle C}{2} \right) = \frac{1}{2}(3+4+5-2x)$$

Since $\tan \angle B = \frac{3}{4}$, then $\tan \frac{\angle B}{2} = \frac{1}{3}$ and $\tan \frac{\angle C}{2} = \tan \frac{90^\circ}{2} = 1$.

We have

$$r(3+1) = \frac{1}{2}(12-2x)$$

$$4r = 6 - x \quad (2)$$

From (1) and (2), we have

$$(6-x)(6+x) = 24$$

$$x^2 = 12$$

$$x = 2\sqrt{3}$$

$$AM = 2\sqrt{3}$$

Hong Kong Mathematics Olympiad (2010 / 2011)

Final Event (Individual) Solution

香港數學競賽 (2010 / 2011)

決賽項目(個人) 題解

Event 1

1. As $a+b+c=36$, $\frac{(2a+1)+(2b+2)+(2c+3)+2}{4} = \frac{2(a+b+c)+1+2+3+2}{4} = \frac{80}{4} = 20$
2. As $P=20$, divide 20112011 by 20 in succession, we have $f=11$, $e=d=0$, $c=14$, $b=5$ and $a=6$. Therefore, $Q=a+b+c+d+e+f=6+5+14+11=36$
3. As the unit digit of 8^1 is 8, 8^2 is 4, 8^3 is 2, 8^4 is 6, and these unit digits are repeating themselves thereafter.
Likewise, the unit digit of $(7^{10})^1$ is 9, $(7^{10})^2$ is 1, and these unit digits are repeating themselves thereafter.
Also, the unit digit of 6^n is 6, where n is any positive integer; and the unit digit of 5^m is 5, where m is any positive integer.
Since $Q=36$, which is a multiple of 4, we have the unit digit of $8^Q + 7^{10Q} + 6^{100Q} + 5^{1000Q} = 6 + 1 + 6 + 5 = 8$, therefore R is 8.
4. $S = (R-1)! = 7! = 5040$.

Event 2

1. Solving the system of equations gives $x = \frac{5P-13}{2}$ and $y = \frac{13-3P}{2}$. Thus $\frac{5P-13}{2} \geq 1$ and $\frac{13-3P}{2} \geq 1$, i.e. $3 \leq P \leq \frac{11}{3}$. Therefore $P=3$.

2. Given $x + y = P$, $x^2 + y^2 = Q$ and $x^3 + y^3 = P^2$, we have

$$\begin{aligned} P^2 &= x^3 + y^3 \\ &= (x + y)^3 - 3xy(x + y) \\ &= P^3 - 3xyP \end{aligned}$$

Substitute $P = 3$ gives

$$3^2 = 3^3 - 3xy(3)$$

$$1 = 3 - xy$$

$$xy = 2$$

$$\text{Also } Q = x^2 + y^2 = (x + y)^2 - 2xy = P^2 - 2xy = 9 - 2xy \dots\dots\dots (1)$$

$$\text{Put } xy = 2 \text{ into (1) gives } Q = 9 - 2(2) = 5.$$

3. As a and b are roots of the equation $x^2 - Qx + R = 0$, we have $a + b = Q$ and $ab = R$.

Thus $a + b = 5$ and a, b are distinct primes imply that a and b are 2 and 3, and thus $ab = R = (2)(3) = 6$.

4. Note that $\frac{1}{S(S-1)} + \frac{1}{(S+1)S} + \dots + \frac{1}{(S+20)(S+19)} = 1 - \frac{1}{R}$ gives

$$\begin{aligned} \frac{1}{S-1} - \frac{1}{S} + \frac{1}{S} - \frac{1}{S+1} + \dots + \frac{1}{S+19} - \frac{1}{S+20} &= 1 - \frac{1}{R} = \frac{5}{6} \\ \frac{1}{S-1} - \frac{1}{S+20} &= \frac{5}{6} \end{aligned}$$

$$5S^2 + 95S - 226 = 0$$

Therefore

$$\begin{aligned} S &= \frac{-95 \pm \sqrt{95^2 - 4(5)(-226)}}{2(5)} \\ &= \frac{-95 \pm \sqrt{13545}}{10} \\ &= \frac{-95 \pm 3\sqrt{1505}}{10} \end{aligned}$$

Event 3

1. Since the coefficient for x^2 is 1 and coefficient for x is even, both roots are integers only happens when $\Delta = [2(P+1)]^2 - 4(P^2 - P - 14) = 4 \times (\text{perfect square})$, i.e. $3P + 15$ is a perfect square.

The least such prime P is 7.

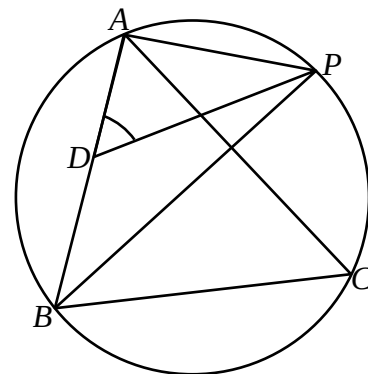
2. We have $2x^3 + 5x^2 + 24x + 11 = (2x+1)(x^2 + 2x + 11)$ and $x^3 + Px - 22 = x^3 + 7x - 22 = (x-2)(x^2 + 2x + 11)$, thus $a = 2$ and $b = 11$, and $Q = a + b = 2 + 11 = 13$.
3. We have $R^3 + 4R^2 + (Q - 93)R + 14Q + 10 = R^3 + 4R^2 - 80R + 192 = (R + 12)(R - 4)^2$. Thus, $R = 5$.

4. As $\angle APB = \angle ACB = \angle ADP$ and $\triangle APB \sim \triangle ADP$. It follows that

$$\frac{AB}{AP} = \frac{AP}{AD}$$

and hence, $(AP)^2 = (AB)(AD) = 5(AD)^2$. Thus, $AP = \sqrt{5}(AD)$.

Therefore $S = \frac{PB}{PD} = \frac{AP}{AD} = \sqrt{5}$.



Event 4

1. As $-\sqrt{A^2 + B^2} \leq A \sin x \pm B \cos x \leq \sqrt{A^2 + B^2}$, we have $a = \sqrt{1^2 + (\sqrt{3})^2} = 2$.

2. We have

$$\begin{aligned} |b - y| &= b + y - 2 \\ b + y &= |b - y| + 2 > 0 \\ |b + y| &= b + y \\ b + y &= b + 2 \\ y &= 2 \\ |b - 2| &= b + 2 - 2 \\ b &= 1 \end{aligned}$$

3. It is clear that either $|x - y|^{2010} = 0$ and $|z - x|^{2011} = 1$ or $|x - y|^{2010} = 1$ and $|z - x|^{2011} = 0$.

In either case, $|y - z| = 1$ and $|x - y| + |z - x| = 1$. That is, $c = 1 + 1 = 2$.

4. Note that $\frac{FG}{GE} = \frac{OB}{BC} = \frac{FE}{EH}$.

As $FE = d$, $FG = d - 1$. Therefore, $d - 1 = \frac{d}{2}$. That is, $d = 2$.

Hong Kong Mathematics Olympiad (2010 / 2011)

Final Event (Group) Solution

香港數學競賽 (2010 / 2011)

決賽項目(團體)題解

Event 1

1. As $\triangle ABC \sim \triangle DBA$, $\frac{\sqrt{1+a^2}}{1} = \frac{5}{\sqrt{1+a^2}}$. That is, $1+a^2=5$, therefore $a=2$.

2. Note that

$$\begin{aligned} b &= 1 - \frac{1}{1 - \frac{1}{1 - \frac{1}{1+2}}} \\ &= 1 - \frac{1}{1 - \frac{1}{\frac{2}{3}}} \\ &= 1 - \frac{1}{-\frac{1}{2}} \\ &= 1 + 2 = 3 \end{aligned}$$

3. Divide the equality by xyz gives $\frac{2}{z} = \frac{3}{x} = \frac{5}{y}$, which means $x : y : z = 3 : 5 : 2$. That is

$$c = \frac{x+3y-3z}{x+3y-6z} = \frac{3+3(5)-3(2)}{3+3(5)-6(2)} = 2$$

4. We have $\log_{\frac{1}{4}}(2x+1) < \log_{\frac{1}{2}}(x-1)$
 $\log_{\frac{1}{2}}(2x+1) < 2\log_{\frac{1}{2}}(x-1)$
 $\log_{\frac{1}{2}}(2x+1) < \log_{\frac{1}{2}}(x-1)^2$

Now, $x-1 > 0$, $2x+1 > 0$ and $2x+1 > (x-1)^2$. That is, $x > 1$, $x > -\frac{1}{2}$ and $x^2 - 2x + 1 < 2x + 1$.

Hence, $1 < x < 4$ and $x = 3$.

Event 2

1. Note that the area of $ABCD = AB \times 4$ as it is a parallelogram. Since

$$AB = \frac{5}{\sin 30^\circ} = \frac{5}{\frac{1}{2}} = 10$$

The area of $ABCD$ is 40 sq. units.

2. If we take 1 as the first number, then the other number should be 100, so there is only one choice; if we take the number 2 as the first, then the second number can either be 99 or 100, which means we have two choices; likewise, if we take the number 3 first, then the second number can be 98, 99 or 100, that means we have three choices; and so on. If repetition is allowed, the total number of ways is

$$\frac{(1+50)50}{2} \times 2 = 2550$$

3. As $\triangle AEO$ is an isosceles triangle, $\angle AOE = 20^\circ$.

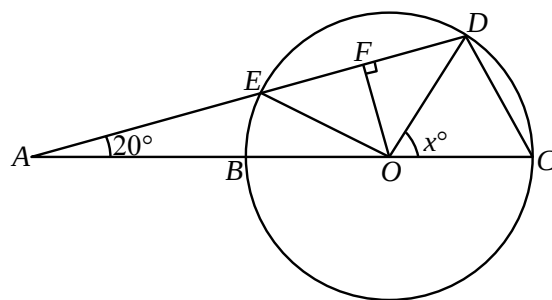
Bisect ED at F .

Join O and F .

Then $\triangle EOF \cong \triangle DOF$.

That is, $\angle EOF = 70^\circ - 20^\circ = 50^\circ$ and $\angle DOF = 50^\circ$.

Thus, $x = 180 - 100 - 20 = 60$



4. Let $\frac{1}{x} + \frac{2}{y} + \frac{3}{z} = 0$ (1)

$$\frac{1}{x} - \frac{6}{y} - \frac{5}{z} = 0 \text{ (2)}$$

$$(1) - (2) \text{ gives } \frac{8}{y} + \frac{8}{z} = 0 \Rightarrow \frac{y}{z} = -1$$

$$(1) \times 3 + (2) \text{ gives } \frac{4}{x} + \frac{4}{z} = 0 \Rightarrow \frac{z}{x} = -1$$

$$(1) \times 5 + (2) \times 3 \text{ gives } \frac{8}{x} - \frac{8}{y} = 0 \Rightarrow \frac{x}{y} = 1$$

$$\text{Thus } P = \frac{x}{y} + \frac{y}{z} + \frac{z}{x} = 1 + (-1) + (-1) = -1$$

Event 3

1. $a^2 + 100a = a(a + 100)$. If $a^2 + 100a$ is a prime, then the only factors of it are 1 and itself. Therefore, a must be 1.

2. As 1 is a root of $x^2 + ax + 2 = 0$, we have $1 + a + 2 = 0 \Rightarrow a = -3$.
Also, a and b are roots of $x^2 + 5x + c = 0$, which means $a + b = -5$ and $ab = c$.
Thus, $b = -2$ and $c = 6$.
Therefore $a + b + c = -3 - 2 + 6 = 1$.

$$\begin{aligned}
 3. \quad & \sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = \frac{10}{3} \\
 \Rightarrow & \frac{x+y}{\sqrt{xy}} = \frac{10}{3} \\
 \Rightarrow & \frac{(\sqrt{x} + \sqrt{y})^2 - 2\sqrt{xy}}{\sqrt{xy}} = \frac{10}{3} \\
 \Rightarrow & \frac{1 - 2\sqrt{xy}}{\sqrt{xy}} = \frac{10}{3} \\
 \Rightarrow & \sqrt{xy} = \frac{3}{16} \\
 \Rightarrow & \sqrt{x} = \frac{1}{4} \text{ and } \sqrt{y} = \frac{3}{4} \\
 \text{Therefore } & y - x = \frac{9}{16} - \frac{1}{16} = \frac{8}{16} = \frac{1}{2}
 \end{aligned}$$

4. Among the numbers, 7 must be part of P_1 .

The factors of 10 are 2 and 5, therefore 10 should be in the numerator, 2 and 5 should be in the denominator.

$9 = 3 \times 3$, therefore 9 should be in the numerator, one of the 3 should be in the denominator, and the other 3 will group with other numbers in the denominator.

$8 = 2 \times 4$, therefore 8 should be in the numerator while 4 should be in the denominator, also the number 2 will group with the number 3 above to form the number 6.

Thus, the minimum value of $\frac{P_1}{P_2} = \frac{7 \times 8 \times 9 \times 10}{1 \times 2 \times 3 \times 4 \times 5 \times 6} = 7$.

Event 4

1. Let $x = 2010$, then

$$\begin{aligned} P &= 2\sqrt[4]{(x-3)(x-1)(x+1)(x+3)+10x^2-9}-4000 \\ &= 2\sqrt[4]{x^4-10x^2+9+10x^2-9}-4000 \\ &= 2(x-2000) \\ &= 20 \end{aligned}$$

2. For $9x^2 + nx + 1$ to be a square, $\Delta = n^2 - 4 \cdot 9 = n^2 - 36 = 0$ implies $n = 6$.

For $4y^2 + 12y + m$ to be a square, $\Delta = 12^2 - 4 \cdot 4m = 144 - 16m = 0$ implies $m = 9$.

Therefore $\frac{n}{m} = \frac{6}{9} = \frac{2}{3}$.

3. $\frac{47}{5} \left(\frac{4}{47} + \frac{n}{141} \right) = \frac{47}{5} \left(\frac{n+12}{141} \right) = \frac{n+12}{15} = k$, where k is a positive integer, then $n = 15k - 12 = 15t + 3$, where $t = k - 1$. Therefore $r = 3$.

4. Area of $\triangle ABF = \text{Area of } \triangle BCE = \frac{1}{2} \text{Area of } ABCD$.

Thus, Area of $\triangle ABF = b + x + d$ and Area of $\triangle BCE = a + x + c$
 $\therefore b + d = a + c$

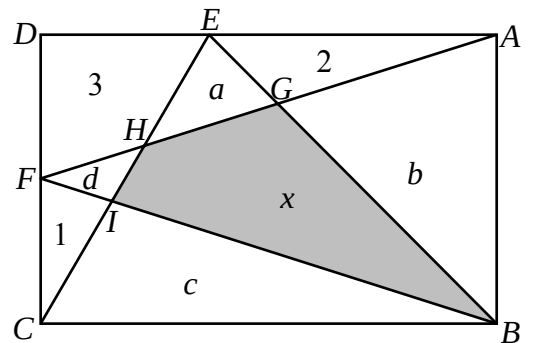
Also,

$$\begin{aligned} &\text{Area of } \triangle ADF + \text{Area of } \triangle BCF \\ &= \text{Area of } \triangle ABF \\ &= \frac{1}{2} \text{Area of } ABCD \end{aligned}$$

$$2 + a + 3 + 1 + c = b + x + d$$

$$6 + a + c = x + b + d$$

$$x = 6$$



Hong Kong Mathematics Olympiad (2011 / 2012)

Heat Event (Individual) Solution

香港數學競賽 (2011 / 2012)

初賽項目(個人) 題解

1. Since the unit digits of $1^2, 2^2, 3^2, \dots, 10^2$ are 1, 4, 9, 6, 5, 6, 9, 4, 1, 0 respectively, the sum of every 10 terms form a group and the unit digit of each sum is the unit digit of $1+4+9+6+5+6+9+4+1+0$, which is 5.

Hence, the unit digit of $1^2 + 2^2 + 3^2 + 4^2 + \dots + 20122010^2 = 5 \times 1 = 5$.

Also, the unit digit of $20122011^2 + 20122012^2 = 1 + 4 = 5$.

Since $5 + 5 = 10$, we have, the unit digit of $1^2 + 2^2 + 3^2 + 4^2 + \dots + 20122012^2 = 0$.

Finally, the unit digit of $2^2 + 3^2 + 4^2 + \dots + 20122012^2 = 10 - 1 = 9$.

2. Let $a = 2A, b = 2B, c = 2C$, where A, B and C are positive integers. The equation becomes $2A + 2B + 2C = 2012 \Rightarrow A + B + C = 1006$.

The answer $= {}_{1005}C_2 = \frac{1005 \times 1004}{2} = 504510$.

3. $\left(\frac{b+1}{a-5}\right)\left(\frac{b-3}{a+3}\right) = -1$ and $(a+3)^2 + (b-3)^2 = (a-5)^2 + (b+1)^2$

From the second equation, $b = 2a - 1$.

Thus $(2a - 1 + 1)(2a - 1 - 3) = -(a - 5)(a + 3) \Rightarrow 2a(2a - 4) + a^2 - 2a - 15 = 0$

$\Rightarrow 5(a^2 - 2a - 3) = 0 \Rightarrow a = 3$ or -1 (rejected). Thus $b = 2(3) - 1 = 5$.

The answer $= 3 + 5 = 8$.

4. $2^{20} \times 25^{12} = 2^{20} \times 5^{24} = 2^{20} \times 5^{20} \times 5^4 = 10^{20} \times 625$ is a 23-digit number.

5. $\log_4 N = \frac{1}{1 - \frac{1}{3}} = \frac{3}{2}$

$$N = 4^{\frac{3}{2}} = 8$$

6. Let a and b be the roots of the equation $x^2 - 19x + m = 0$, then $a + b = 19$.

Since a and b are distinct prime numbers, therefore a and b can only take the values 2 and 17.

Thus, $\frac{a}{b} + \frac{b}{a} = \frac{2}{17} + \frac{17}{2} = \frac{293}{34}$.

7. According to the question,

$$\begin{aligned}
 P &\geq \frac{(a+b)+(a+c)+(b+c)}{3} \\
 &= \frac{2}{3}(a+b+c) \\
 &= \frac{2}{3}(9) = 6
 \end{aligned}$$

Therefore the minimum value of P is 6 .

$$\begin{aligned}
 8. \quad (k^2 - 4)x^2 - (14k + 4)x + 48 &= 0 \\
 [(k+2)x - 6][(k-2)x - 8] &= 0 \\
 x = \frac{6}{k+2} \text{ or } x = \frac{8}{k-2}
 \end{aligned}$$

For the roots are distinct positive integral solutions, then the value of $k = 4$.

$$\begin{aligned}
 9. \quad x^3 - y^3 &= 2189 \\
 (x-y)(x^2 + xy + y^2) &= 11 \times 199 \\
 \begin{cases} x-y=11 \\ x^2 + xy + y^2 = 199 \end{cases} \\
 \Rightarrow \begin{cases} x^2 - 2xy + y^2 = 121 \\ x^2 + xy + y^2 = 199 \end{cases} \\
 \Rightarrow 3xy = 78 \Rightarrow xy = 26 \Rightarrow (x+y)^2 = 225 \\
 \Rightarrow x+y = 15 \Rightarrow x = 13, y = 2
 \end{aligned}$$

$$\begin{aligned}
 10. \text{ Since } AB^2 + BC^2 &= 21^2 + 20^2 \\
 &= 841 \\
 &= 29^2 \\
 &= AC^2
 \end{aligned}$$

$$\therefore \angle ABC = 90^\circ$$

Let $B = (0, 0)$, $A(0, 21)$, $C = (20, 0)$, $D = (10, 0)$ and $E(0, 7)$.

$$\text{Equation of } AD \text{ is } \frac{x}{10} + \frac{y}{21} = 1 \Rightarrow 21x + 10y = 210$$

$$\text{Equation of } CE \text{ is } \frac{x}{20} + \frac{y}{7} = 1 \Rightarrow 7x + 20y = 140$$

$$\text{Coordinates of } F = \left(8, \frac{21}{5}\right).$$

$$BF^2 = (8)^2 + \left(\frac{21}{5}\right)^2 = 64 + \frac{441}{25} = \frac{2041}{25} \quad \left(\text{or } 81\frac{16}{25} \text{ or } 81.64\right)$$

Hong Kong Mathematics Olympiad (2011 / 2012)
Heat Event (Group) Solution
香港數學競賽 (2011 / 2012)
初賽項目(團體) 題解

1. Let $x = y - 1$ and $z = y + 1$.

$$\begin{aligned} & \frac{y}{x} + \frac{z}{x} + \frac{x}{y} + \frac{z}{y} + \frac{x}{z} + \frac{y}{z} \\ &= \frac{y}{y-1} + \frac{y+1}{y-1} + \frac{y-1}{y} + \frac{y+1}{y} + \frac{y-1}{y+1} + \frac{y}{y+1} \\ &= \frac{2y+1}{y-1} + \frac{2y}{y} + \frac{2y-1}{y+1} \\ &= 2 + \frac{(2y+1)(y+1) + (2y-1)(y-1)}{(y-1)(y+1)} \\ &= 2 + \frac{2y^2 + 3y + 1 + 2y^2 - 3y + 1}{y^2 - 1} \\ &= 2 + \frac{4y^2 + 2}{y^2 - 1} \\ &= 2 + 4 + \frac{6}{y^2 - 1} \end{aligned}$$

Since $2 + 4 + \frac{6}{y^2 - 1}$ is an integer, $\frac{6}{y^2 - 1}$ must be an integer, i.e., $(y^2 - 1)$ must be a factor of 6, therefore $y = 2$ and $x + y + z = y - 1 + y + y + 1 = 3y = 6$.

2. Because $x - 2012 > 0$, therefore $\sqrt{(5 - x)^2} = x - 5$

$$\begin{aligned} \sqrt{x - 2012} + x - 5 &= x \\ \sqrt{x - 2012} &= 5 \\ x - 2012 &= 25 \\ \therefore x &= 2037 \end{aligned}$$

3.
$$\begin{aligned} & \sqrt{2^2 + 2^{1008} + 2^{2012}} \\ &= \sqrt{(2)^2 + 2 \times 2 \times 2^{1006} + (2^{1006})^2} \\ &= \sqrt{(2 + 2^{1006})^2} \\ &= 2 + 2^{1006} \end{aligned}$$

$$\begin{aligned}
4. \quad & \frac{1}{\sqrt{2012} + \sqrt{2011}} + \frac{1}{\sqrt{2011} + \sqrt{2010}} + \cdots + \frac{1}{\sqrt{3} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{1}} \\
&= \frac{1}{\sqrt{2012} + \sqrt{2011}} \times \frac{\sqrt{2012} - \sqrt{2011}}{\sqrt{2012} - \sqrt{2011}} + \cdots + \frac{1}{\sqrt{3} + \sqrt{2}} \times \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{1}} \times \frac{\sqrt{2} - \sqrt{1}}{\sqrt{2} - \sqrt{1}} \\
&= \sqrt{2012} - \sqrt{2011} + \sqrt{2011} - \sqrt{2010} + \cdots + \sqrt{3} - \sqrt{2} + \sqrt{2} - \sqrt{1} \\
&= \sqrt{2012} - 1
\end{aligned}$$

$$\begin{aligned}
5. \quad & x^2 + y^2 - 10x - 6y + 2046 \\
&= x^2 - 10x + 25 + y^2 - 6y + 9 + 2012 \\
&= (x-5)^2 + (y-3)^2 + 2012 \\
&\geq 2012
\end{aligned}$$

Then, the minimum value = 2012.

6. According to the question, $\angle DAB = 90^\circ$. Let E be the midpoint of BD and $BE = x$, then $AE = BE$ and $BD = 2x$.

Meanwhile, $\triangle ABE \sim \triangle CBA$, therefore

$$\begin{aligned}
& \frac{AB}{BE} = \frac{BC}{AB} \\
& \frac{12}{x} = \frac{2x-2}{12} = \frac{x-1}{6} \\
& x^2 - x - 72 = 0 \\
& (x-9)(x+8) = 0 \\
& x = 9
\end{aligned}$$

Thus $BC = 16$.

7. As $a^x = b^y = c^z = 30^w$, we have $a^{\frac{1}{w}} = 30^{\frac{1}{x}}$, $b^{\frac{1}{w}} = 30^{\frac{1}{y}}$, $c^{\frac{1}{w}} = 30^{\frac{1}{z}}$

Therefore

$$\begin{aligned}
& \left(a^{\frac{1}{w}}\right)\left(b^{\frac{1}{w}}\right)\left(c^{\frac{1}{w}}\right) = 30^{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}} \\
& (abc)^{\frac{1}{w}} = 30^{\frac{1}{w}} \\
& abc = 30
\end{aligned}$$

Since $30 = 2 \times 3 \times 5$ and 2, 3 and 5 are co-prime, therefore $30 = 2 \times 3 \times 5$ is the only feasible factorisation, i.e., $a = 2$, $b = 3$ and $c = 5$, thus $a + b + c = 2 + 3 + 5 = 10$.

8. Let α, β be the roots of the equation $x^2 + px + q = 0$, then $\alpha + \beta = -p$ and $\alpha\beta = q$,

$$\begin{aligned} 60 &= p + q = -(\alpha + \beta) + \alpha\beta \\ &= -\alpha - \beta + \alpha\beta \\ &= (1 - \alpha)(1 - \beta) - 1 \\ &= (\alpha - 1)(\beta - 1) - 1 \end{aligned}$$

$$\therefore (\alpha - 1)(\beta - 1) = 61$$

$$\Rightarrow \alpha, \beta = 2, 62 \text{ or } \alpha, \beta = 0, -60$$

$$\text{Therefore } q = 2 \times 62 = 124.$$

$$\begin{aligned} 9. \quad & \sin^2 1^\circ + \sin^2 2^\circ + \sin^2 3^\circ + \dots + \sin^2 44^\circ + \sin^2 45^\circ + \sin^2 46^\circ + \dots + \sin^2 89^\circ \\ &= \cos^2 89^\circ + \cos^2 88^\circ + \dots + \cos^2 46^\circ + \sin^2 45^\circ + \sin^2 46^\circ + \dots + \sin^2 88^\circ + \sin^2 89^\circ \\ &= \sin^2 89^\circ + \cos^2 89^\circ + \sin^2 88^\circ + \cos^2 88^\circ + \dots + \sin^2 46^\circ + \cos^2 46^\circ + \sin^2 45^\circ \\ &= \underbrace{1 + 1 + \dots + 1}_{44 \text{ terms}} + \left(\frac{\sqrt{2}}{2}\right)^2 = 44 \frac{1}{2} \end{aligned}$$

Thus

$$\begin{aligned} & \sin^2 1^\circ + \sin^2 2^\circ + \sin^2 3^\circ + \dots + \sin^2 359^\circ + \sin^2 360^\circ \\ &= 4 \times (\sin^2 1^\circ + \sin^2 2^\circ + \sin^2 3^\circ + \dots + \sin^2 89^\circ) + \sin^2 90^\circ + \sin^2 180^\circ + \sin^2 270^\circ + \sin^2 360^\circ \\ &= 4 \times \left(44 \frac{1}{2}\right) + 2 = 180 \end{aligned}$$

10. Let there be altogether n people.

$$\text{Originally, the total number of hand-shakes} = \frac{n(n-1)}{2}$$

$$\text{Apart from Steven, the total number of hand-shakes of } (n-1) \text{ people} = \frac{(n-1)(n-2)}{2}$$

$$\therefore \frac{(n-1)(n-2)}{2} < 60 \leq \frac{n(n-1)}{2}$$

$$\therefore (n-1)(n-2) < 120 \leq n(n-1)$$

Since $11 \times 10 < 120 \leq 12 \times 11$, we have $n = 12$.

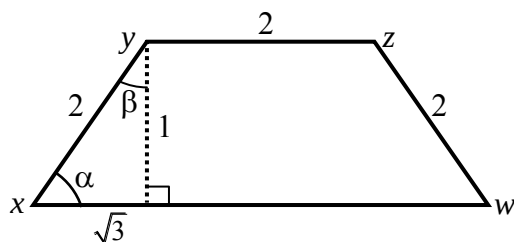
$$\text{Therefore, apart from Steven, the total number of hand-shakes of the other 11 people} = \frac{11 \times 10}{2} = 55$$

$$\text{Thus, the number of people that Steven knows} = 60 - 55 = 5.$$

Event 1

$$\begin{aligned}
 1. \quad & x^4 + 6x^3 + 12x^2 + 9x + 2 \\
 &= (x+1)(x+2)(x^2 + 3x + 1) \\
 &= (x+1)(x+2)\left(x + \frac{3+\sqrt{5}}{2}\right)\left(x + \frac{3-\sqrt{5}}{2}\right) \\
 \therefore \quad & A = 1^2 + 2^2 + \left(\frac{3+\sqrt{5}}{2}\right)^2 + \left(\frac{3-\sqrt{5}}{2}\right)^2 \\
 &= 1 + 4 + 7 = 12
 \end{aligned}$$

2. Note that



$$\beta = 150^\circ - 90^\circ = 60^\circ$$

$$\begin{aligned}
 \text{Area of } xyzw &= \left[2 + (2 + 2\sqrt{3})\right] \times \frac{1}{2} \\
 &= 2 + \sqrt{3}
 \end{aligned}$$

$$\Rightarrow a = 2, b = 3 \Rightarrow B = 2^2 3^3 = 108$$

$$\begin{aligned}
 3. \quad C &= (1 + 2 + 2^2)(1 + 3 + 3^2 + 3^3) \\
 &= 280
 \end{aligned}$$

$$\begin{aligned}
 4. \quad D &= \left\lfloor \frac{280}{5} \right\rfloor + \left\lfloor \frac{280}{5^2} \right\rfloor + \left\lfloor \frac{280}{5^3} \right\rfloor \\
 &= 56 + 11 + 2 = 69
 \end{aligned}$$

Event 2

1. Let $u^2 = x^2 + 9x + 21 \geq 0$, we have

$$u^2 - 8 = 2\sqrt{u^2} = 2u$$

$$\Rightarrow u^2 - 2u - 8 = 0$$

$$\Rightarrow (u - 4)(u + 2) = 0$$

$$\Rightarrow u = 4$$

$$\therefore x^2 + 9x + 21 = u^2 = 16$$

$$\Rightarrow x^2 + 9x + 5 = 0$$

$$\Rightarrow P = 5$$

2. Note that $f(x) + f(1-x) = 1$, so we have

$$Q = \left[f\left(\frac{1}{25}\right) + f\left(\frac{24}{25}\right) \right] + \cdots + \left[f\left(\frac{12}{25}\right) + f\left(\frac{13}{25}\right) \right] \\ = 12$$

3. $X = \sqrt{100 \cdot 102 \cdot 103 \cdot 105 + 9}$, the unit digit can only be 3 as $100 \cdot 102 \cdot 103 \cdot 105$ is a multiple of 100. Alternatively,

$$x(x+2)(x+3)(x+5) + 9 \\ = (x^2 + 2x)(x^2 + 8x + 15) + 9 \\ = x^4 + 10x^3 + 31x^2 + 30x + 9 \\ = (x^2 + 5x + 3)^2$$

$$\text{Putting } x = 100 \text{ gives } x = 100^2 + 500 + 3 \Rightarrow R = 3.$$

4. $\log_{2012}(\sqrt{2012} x^{\log_{2012} x}) = \log_{2012} x^3$

$$\Rightarrow \frac{1}{2} + (\log_{2012} x)^2 = 3 \log_{2012} x$$

Putting $y = \log_{2012} x$. The above equation becomes

$$y^2 - 3y + \frac{1}{2} = 0$$

$$\Rightarrow y_1 + y_2 = 3$$

$$\Rightarrow \log_{2012} x_1 + \log_{2012} x_2 = 3$$

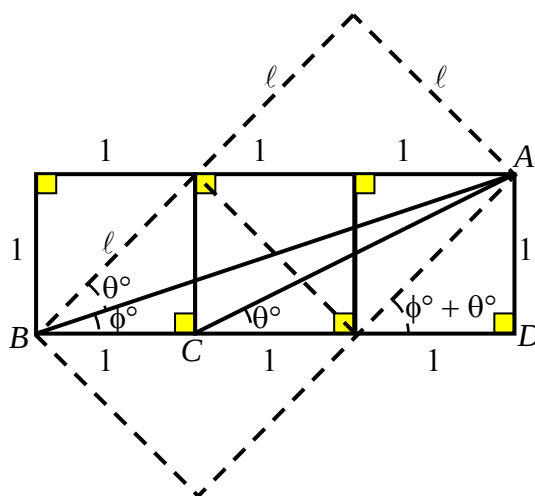
$$\Rightarrow \log_{2012} x_1 x_2 = 3$$

$$\Rightarrow x_1 x_2 = 2012^3 = (2000 + 12)^3$$

The last 3 digits are 7, 2 and 8 $\Rightarrow S = 17$

Event 3

1. From the figure below, $\alpha^\circ = \phi^\circ + \theta^\circ = 45^\circ$. Therefore $\alpha = 45$



$$2. \quad \sin A = \frac{36}{45} = \frac{4}{5}, \quad \cos A = \frac{3}{5}, \quad \sin B = \frac{12}{13}, \quad \cos B = \frac{5}{13}.$$

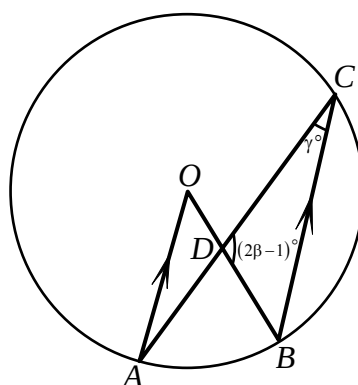
$$\Rightarrow \sin C = \sin(180^\circ - A - B) = \sin(A + B)$$

$$= \sin A \cos B + \cos A \sin B$$

$$= \frac{56}{65}$$

$$\Rightarrow \beta = 56$$

$$3. \quad 2\gamma + \gamma = 180 - (2\beta - 1) \Rightarrow 3\gamma = 69 \Rightarrow \gamma = 23.$$



$$4. \quad \text{Coefficient of } x^\gamma = x^{23} \text{ is } C_{23}^{2012} a^{23} b^{2012-23}$$

$$\text{Coefficient of } x^{\gamma+1} = x^{24} \text{ is } C_{24}^{2012} a^{24} b^{2012-24}$$

$$\Rightarrow C_{23}^{2012} a^{23} b^{2012-23} = C_{24}^{2012} a^{24} b^{2012-24}$$

$$\therefore \frac{a}{b} = \frac{C_{23}^{2012}}{C_{24}^{2012}} = \frac{24}{1989} = \frac{8}{663}$$

$$\therefore \delta = 671$$

Event 4

$$\begin{aligned} 1. \quad & \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(A+1) \cdot (A+3)} \\ &= \frac{1}{2} \left[\left(1 - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{A+1} - \frac{1}{A+3}\right) \right] \\ &= \frac{1}{2} \left(\frac{A+2}{A+3} \right) = \frac{12}{25} \\ \Rightarrow \quad & A = 22 \end{aligned}$$

$$2. \quad xy = x + y + 22 \Rightarrow xy - x - y + 1 = 23 \Rightarrow (x-1)(y-1) = 23, \text{ which is a prime.}$$

Since $x-1$ and $y-1$ are natural numbers with $x-1 > y-1$, it follows that $x-1 = 23$ and $y-1 = 1 \Rightarrow x = 24$ and $y = 2 \Rightarrow B = 12$.

$$\begin{aligned} 3. \quad & f(4) = f(2 \cdot 2) = f(2) \cdot f(2) = 4 \\ \therefore \quad & f \text{ is increasing, } f(2) < f(3) < f(4). \\ \Rightarrow \quad & f(3) = 3 \\ \Rightarrow \quad & f(B) = f(12) = f(3 \cdot 4) = f(3)f(4) = 12. \end{aligned}$$

$$\begin{aligned} 4. \quad & 2401 \cdot 7^C = 7^4 \cdot 7^{12} = (7^4)^4 = (2400+1)^4 \\ &= 2400^4 + 4 \times 2400^3 + 6 \times 2400^2 + 4 \times 2400 + 1 \\ &\text{The last three digits are determined by } 4 \times 2400 + 1, \text{ which is } 601. \\ \therefore \quad & D = 7 \end{aligned}$$

Event 1

$$\begin{aligned}
 1. \quad & 2011^{2011} \\
 & \equiv 11^{11} \pmod{100} \\
 & \equiv (11)^2 (11)^2 (11)^2 (11)^2 (11)^2 (11) \\
 & \equiv (21)(21)(21)(21)(21)(11) \pmod{100} \\
 & \equiv (41)(41)(21)(11) \pmod{100} \\
 & \equiv 81 \cdot 31 \equiv 11 \pmod{100}
 \end{aligned}$$

$\therefore$ The answer is 1.

$$\begin{aligned}
 2. \quad & a_2 = a_1 + 1, \quad a_4 = a_3 + 1, \dots, \quad a_{100} = a_{99} + 1 \\
 \Rightarrow & 2P = (a_1 + 1) + a_2 + \dots + (a_{99} + 1) + a_{100} \\
 & = 2012 + 50 \\
 \Rightarrow & P = 1031
 \end{aligned}$$

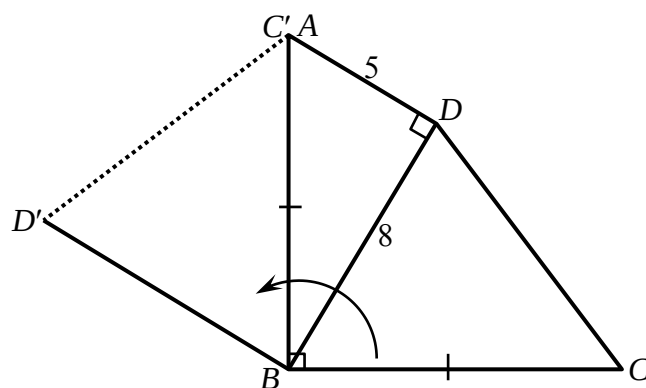
$$3. \quad \left\lfloor \frac{90}{5} \right\rfloor + \left\lfloor \frac{90}{5^2} \right\rfloor = 21$$

4. Rotate $\triangle BCD$ anti-clockwise about B so that BC coincides with AB .

Then $AD'BD$ is a trapezium with area

$$\frac{1}{2}(AD + BD') \cdot 8 = 52$$

$$\text{So, the area of } \triangle BCD = 52 - \frac{1}{2} \cdot 5 \cdot 8 = 32$$



Event 2

1. Note that $\tan(90^\circ - \theta)\tan \theta = 1$.

$$\begin{aligned} & 2 \times \tan 1^\circ \times \tan 2^\circ \times \tan 3^\circ \times \cdots \times \tan 87^\circ \times \tan 88^\circ \times \tan 89^\circ \\ &= 2(\tan 1^\circ \times \tan 89^\circ) \times \cdots \times (\tan 44^\circ \times \tan 46^\circ) \times \tan 45^\circ \\ &= 2 \end{aligned}$$

$$2. \quad (x^2 - 3x + 2)^2 - 3(x^2 - 3x) - 4 = (x^2 - 3x + 2)^2 - 3(x^2 - 3x + 2) + 2 = 0$$

$$\Rightarrow x^2 - 3x + 2 = 1 \text{ or } x^2 - 3x + 2 = 2$$

$$\Rightarrow x^2 - 3x + 1 = 0 \text{ or } x^2 - 3x = 0$$

$$\Rightarrow x = \frac{3 \pm \sqrt{5}}{2} \text{ or } x = 0 \text{ or } x = 3$$

$$\Rightarrow K = 2$$

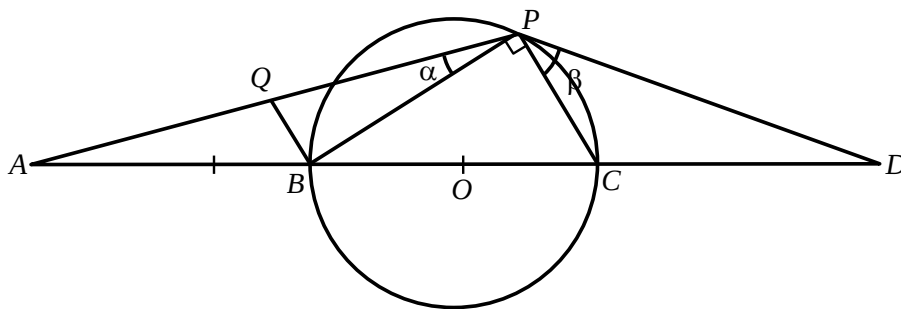
$$3. \text{ For } x < 2, \quad |x - 2| + |x - 47| = 2 - x + 47 - x = 49 - 2x > 45$$

$$\text{For } 2 \leq x \leq 47, \quad |x - 2| + |x - 47| = x - 2 + 47 - x = 45$$

$$\text{For } x > 47, \quad |x - 2| + |x - 47| = x - 2 + x - 47 = 2x - 49 > 2(47) - 49 = 45$$

$$\therefore \ell = 45$$

4.



Choose a point Q on AP so that $QB \perp PB$.

$$\Rightarrow QB \parallel PC \text{ and } QB = \frac{1}{2}PC$$

$$\therefore \tan \alpha = \frac{QB}{PB} = \frac{\frac{1}{2}PC}{PB} = \frac{PC}{2PB}$$

$$\text{And similarly, } \tan \beta = \frac{PB}{2PC} \Rightarrow \tan \alpha \tan \beta = \frac{1}{4}$$

Event 3

1. Note that $x + y = 5$ and $xy = 1$.

$$\begin{aligned}\text{So } 192z &= (x^2 + y^2)^2 - 2x^2y^2 + (x + y)^4 \\ &= [(x + y)^2 - 2xy]^2 - 2(xy)^2 + (x + y)^4 \\ &= (5^2 - 2)^2 - 2 + 5^4 = 1152 \\ \Rightarrow z &= 6\end{aligned}$$

2. $c_1 + c_2 + c_3 + c_4 + c_5 = 360^\circ$

$$\angle A = 180^\circ - c_5 - c_1$$

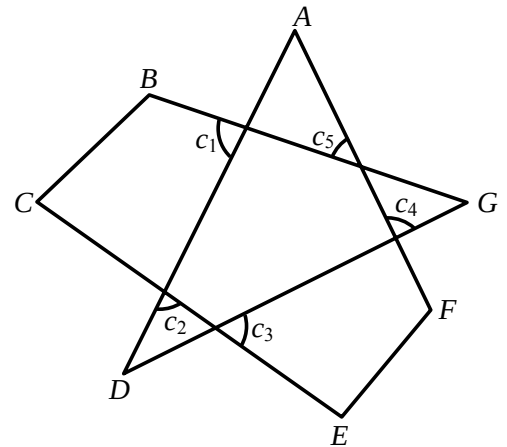
$$\angle B + \angle C = 360^\circ - c_1 - c_2$$

$$\angle D = 180^\circ - c_2 - c_3$$

$$\angle E + \angle F = 360^\circ - c_3 - c_4$$

$$\angle G = 180^\circ - c_4 - c_5$$

$$\begin{aligned}\Rightarrow \angle A + \dots + \angle G &= 3 \cdot 180^\circ + 2 \cdot 360^\circ - 2(c_1 + \dots + c_5) \\ &= 540^\circ\end{aligned}$$



3. $f(112) = 991$, $f(f(112)) = 998$, $f(f(f(112))) = 998, \dots$

$$\therefore D = 998$$

4. $F_1(7) = 49$

$$F_2(7) = 16 + 81 = 97$$

$$F_3(7) = 81 + 49 = 130$$

$$F_4(7) = 1 + 9 + 0 = 10$$

$$F_5(7) = 1 = \dots = F_{2012}(7)$$

Event 4

1. Extend BA and CE to meet at F .

Then $\angle FBE = \angle ACF$

$$\Rightarrow \triangle ABD \cong \triangle ACF$$

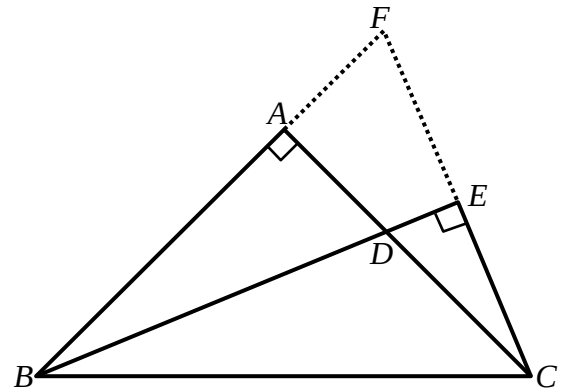
$$\Rightarrow CF = BD$$

$\therefore BE \perp CF$, BE is the bisector of $\angle FBC$

$\therefore BE$ bisects CF . E is the mid-point of CF .

$$\therefore 2CE = FC = BD$$

$$\Rightarrow P = 2$$



2. Either $3Q - |1 - 2Q| = 2$ or $3Q - |1 - 2Q| = -2$

For the first case,

$$3Q - |1 - 2Q| = 2$$

$$\Rightarrow |1 - 2Q| = 3Q - 2$$

$$\Rightarrow 3Q - 2 \geq 0 \Rightarrow Q \geq \frac{2}{3}$$

$$\Rightarrow 1 - 2Q = 3Q - 2 \quad \text{or} \quad 1 - 2Q = -(3Q - 2)$$

$$\Rightarrow Q = \frac{3}{5} \text{ (rejected) or } Q = 1$$

For the second case,

$$3Q - |1 - 2Q| = -2$$

$$\Rightarrow |1 - 2Q| = 3Q + 2$$

$$\Rightarrow 3Q + 2 \geq 0 \Rightarrow Q \geq -\frac{2}{3}$$

$$\Rightarrow 1 - 2Q = 3Q + 2 \quad \text{or} \quad 1 - 2Q = -(3Q + 2)$$

$$\Rightarrow Q = -\frac{1}{5} \text{ (rejected) or } Q = -3 \text{ (rejected)}$$

$$3. \quad \frac{1}{1+x+xy+xyz} = \frac{t}{t+xt+xyt+xyzt} = \frac{t}{1+t+xt+xyt}$$

$$\text{Similarly, } \frac{1}{1+y+yz+yzt} = \frac{tx}{1+t+xt+xyt} \quad \text{and} \quad \frac{1}{1+z+zt+ztz} = \frac{txy}{1+t+xt+xyt}$$

$$\therefore R = 1$$

4. Suppose that $x_i \leq x_{i+1}$, $i = 1, 2, 3, 4$.

$$\begin{aligned}
 1 &= \frac{1}{x_2 x_3 x_4 x_5} + \frac{1}{x_1 x_2 x_4 x_5} + \frac{1}{x_1 x_3 x_4 x_5} + \frac{1}{x_1 x_2 x_3 x_5} + \frac{1}{x_1 x_2 x_3 x_4} \\
 &\leq \frac{1}{x_4 x_5} + \frac{1}{x_4 x_5} + \frac{1}{x_4 x_5} + \frac{1}{x_5} + \frac{1}{x_4} \\
 &\leq \frac{3 + x_4 + x_5}{x_4 x_5}
 \end{aligned}$$

$$\therefore x_4 x_5 \leq 3 + x_4 + x_5$$

$$\therefore (x_4 - 1)(x_5 - 1) \leq 4$$

If $x_4 = 1$, then $x_1 = x_2 = x_3 = x_4 = 1 \Rightarrow 4 + x_5 = x_5$ which is impossible

$$\therefore x_4 > 1, \text{ then } x_5 - 1 \leq 4 \Rightarrow x_5 \leq 5$$

For $x_5 = 5$, then $x_4 = 2$, $\therefore x_1 = x_2 = x_3 = 1$

$\therefore$ The maximum value of x_5 is 5.

Hong Kong Mathematics Olympiad (2012 / 2013)

Heat Event (Individual) Solution

香港數學競賽 (2012 / 2013)

初賽項目(個人) 題解

1. $\sqrt{94 - 2\sqrt{2013}}$

$$= \sqrt{(33 + 61) - 2\sqrt{33 \times 61}}$$

$$= \sqrt{(\sqrt{61} - \sqrt{33})^2}$$

$$= \sqrt{61} - \sqrt{33}$$

2. Let the length of the sides of the parallelogram be a and b , we have

$$ab \sin 60^\circ = 178 \times \frac{1}{2} \times 1 \times 1 \times \sin 60^\circ$$

$$ab = 89 = 1 \times 89$$

As a and b must be integers, the maximum value of $P = 2 \times (1 + 89) = 180$.

3. Let $AD = x$

Given that $AC = 3a$, thus

$$x = 3a \tan 30^\circ = \frac{3a}{\sqrt{3}} = \sqrt{3}a.$$

$$\text{Therefore } \tan \theta = \frac{a}{\sqrt{3}a} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = 30^\circ$$

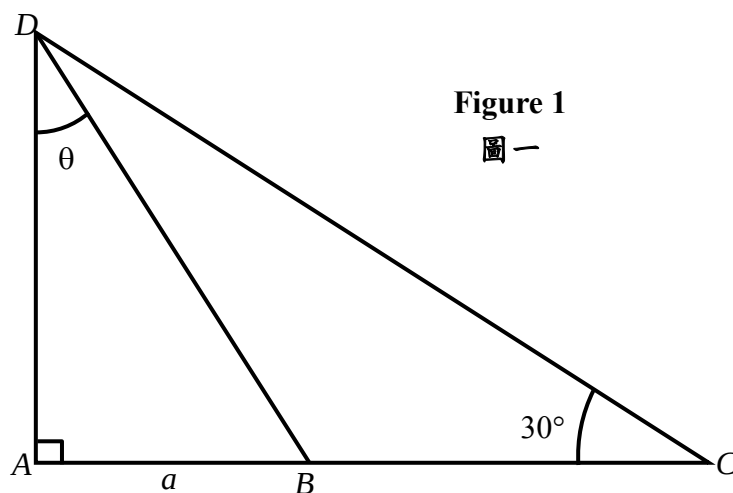


Figure 1
圖一

4. Trial and error. First, find a value of y which satisfies $2^y > 399$. The first one should be $y = 9$, which gives $2^9 = 512 > 399$, but $512 - 399 = 113$, which is not a squared number, i.e. there is no x for $x^2 = 113$. The next number is $y = 10$, which gives $2^{10} = 1024 > 399$, and this time, $x^2 = 1024 - 399 = 625$, therefore $x = 25$.

Alternatively, for all positive integers k , $2^{2k+1} \equiv 2^{2k} \cdot 2 \equiv 4^k \cdot 2 \equiv (-1)^k \cdot 2 \equiv 2 \text{ or } 3 \pmod{5}$

However $x^2 \equiv 0, 1 \text{ or } 4 \pmod{5}$, thus y is an even number. Let $y = 2k$, where k is a positive number, then

$x^2 + 399 = 2^y \Rightarrow 2^{2k} - x^2 = 399$, i.e. $(2^k + x)(2^k - x) = 399$. As $399 = 3 \times 7 \times 19$, there are four possibilities:

(1) $2^k + x = 399$ and $2^k - x = 1 \Rightarrow 2^{k+1} = 400$, no solution;

(2) $2^k + x = 133$ and $2^k - x = 3 \Rightarrow 2^{k+1} = 136$, no solution;

(3) $2^k + x = 57$ and $2^k - x = 7 \Rightarrow 2^{k+1} = 64 \Rightarrow k = 5 \Rightarrow y = 10$, thus $x^2 + 399 = 2^{10} = 1024 \Rightarrow x^2 = 1024 - 399 = 625 \Rightarrow x = 25$;

(4) $2^k + x = 21$ and $2^k - x = 19 \Rightarrow 2^{k+1} = 40$, no solution.

$$\begin{aligned} 5. \quad y &= (x+1)(x+2)(x+3)(x+4) + 2013 \\ &= [(x+2)(x+3)][(x+1)(x+4)] + 2013 \\ &= [x^2 + 5x + 6][x^2 + 5x + 4] + 2013 \end{aligned}$$

$$\text{Let } t = x^2 + 5x + 5$$

$$\begin{aligned} y &= (t-1)(t+1) + 2013 \\ &= t^2 + 2012 \end{aligned}$$

As $t^2 \geq 0$, the minimum value of y is 2012.

6. Let α be the size of the angle selected. Then $0^\circ < \alpha < 180^\circ$.

$$2013^\circ < (n-2) \times 180^\circ < 2013^\circ + 180^\circ$$

$$11\frac{11}{60} < n-2 < 12\frac{11}{60}$$

$$13\frac{11}{60} < n < 14\frac{11}{60}$$

Thus $n = 14$.

7. Join CD and BE . Then $\angle CDB = \angle BEC = 90^\circ$.
Let $CD = a, BE = b$.

Then $AD = CD = a, AE = BE = b$.

$$AB = \frac{BE}{\sin 45^\circ} = \sqrt{2}b \text{ and } AC = \frac{CD}{\sin 45^\circ} = \sqrt{2}a.$$

$$\text{Area of } \triangle ABC = \frac{1}{2}(\sqrt{2}b)(\sqrt{2}a)\sin 45^\circ = \frac{1}{\sqrt{2}}ab.$$

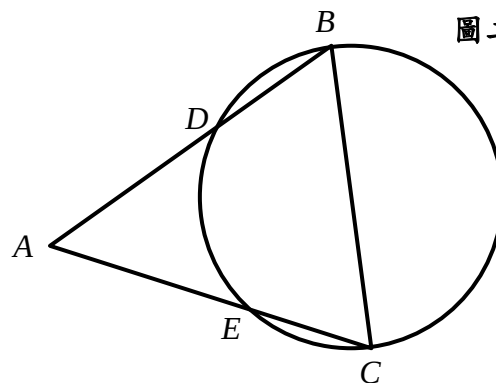
$$\text{Area of } \triangle ADE = \frac{1}{2}(a)(b)\sin 45^\circ = \frac{1}{2\sqrt{2}}ab$$

$$\text{Area of } BCED = \frac{1}{\sqrt{2}}ab - \frac{1}{2\sqrt{2}}ab = \frac{1}{2\sqrt{2}}ab.$$

The answer = 1.

Figure 2

圖二



8. Let $y = \sqrt{31+x}$ (1)
Also, $x = \sqrt{31-y}$ (2)

$$(1)^2 - (2)^2 :$$

$$y^2 - x^2 = x + y$$

$$(y+x)(y-x) = x+y$$

$$y-x=1 \text{ or } y+x=0 \text{ (rejected)}$$

$$\sqrt{31+x} - x = 1$$

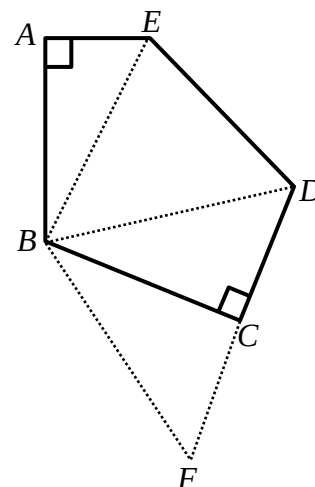
$$31+x = (x+1)^2$$

$$x^2 + x - 30 = 0$$

$$(x+6)(x-5) = 0$$

$$\therefore x = -6 \text{ (rejected) or } x = 5$$

9. Extend DC to point F so that $CF = AE$.
Join BE , BD and BF , then $DF = DE = 3$.
Also $\triangle BCF \cong \triangle BAE$ (SAS)
which gives $BF = BE$
Since $BD = BD$
we have $\triangle BED \cong \triangle BFD$ (SSS)
 $\therefore$ area of $\triangle BED$ = area of $\triangle BFD$
 $\therefore$ Area of the pentagon = area of quadrilateral $BEDF$
 $= 2 \times \text{area of } \triangle BED$
 $= 2 \times \frac{1}{2} \times 3 \times 3$
 $= 9$



10. Let $t = a + b$.

$$t^2 = (a + b)^2 \text{ , } t^2 = a^2 + b^2 + 2ab \text{ , } t^2 = t + 2ab \text{ , gives } ab = \frac{t^2 - t}{2}$$

$$0 \leq (a - b)^2 = a^2 + b^2 - 2ab = (a + b)^2 - 4ab = 2t - t^2 = t(2 - t)$$

Solving it gives $0 \leq t \leq 2$.

Therefore the maximum value of $a + b$ is 2 .

Hong Kong Mathematics Olympiad (2012 / 2013)
Heat Event (Group) Solution
香港數學競賽 (2012 / 2013)
初賽項目(團體) 題解

1. Let a and b be the roots of $x^2 - (m+2)x + 4m = 0$, then

$$\begin{cases} a+b = m+2 \\ ab = 4m \end{cases}$$

$$a+b = \frac{ab}{4} + 2$$

$$ab - 4a - 4b + 8 = 0$$

$$(a-4)(b-4) = 8$$

Without loss of generality, let $a \geq b$ and the length of the third side be c , then $a-4 \geq b-4$.

From $(a-4)(b-4) = 8 = 8 \times 1 = 4 \times 2$, we have

$$(1) \quad \begin{cases} a-4=8 \\ b-4=1 \end{cases} \text{ gives } \begin{cases} a=12 \\ b=5 \end{cases}$$

$$\Rightarrow c = \sqrt{12^2 + 5^2} = 13 \quad \text{or} \quad c = \sqrt{12^2 - 5^2} = \sqrt{119} \text{ (rejected)}$$

$$(2) \quad \begin{cases} a-4=4 \\ b-4=2 \end{cases} \text{ gives } \begin{cases} a=8 \\ b=6 \end{cases}$$

$$\Rightarrow c = \sqrt{8^2 + 6^2} = 10 \quad \text{or} \quad c = \sqrt{8^2 - 6^2} = 2\sqrt{7} \text{ (rejected)}$$

$\therefore$ The required answer is 13.

2.

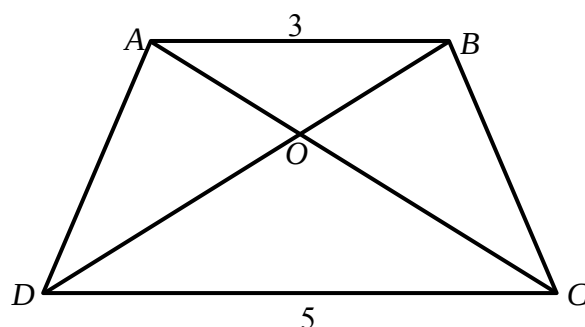


Figure 1

圖一

Since the ratio of areas of two triangles with the same height = the ratio of the length of their bases, we have

$$\frac{\text{Area of } \triangle AOD}{\text{Area of } \triangle AOB} = \frac{OD}{OB} = \frac{5}{3} \text{ (i.e. Area of } \triangle AOD = 45). \text{ Meanwhile, as } \triangle AOB \sim \triangle COD,$$

$$\frac{\text{Area of } \triangle COD}{\text{Area of } \triangle AOB} = \frac{5^2}{3^2} = \frac{25}{9} \text{ (i.e. Area of } \triangle COD = 75).$$

Also, area of $\triangle ADB$ = area of $\triangle ACB$, thus

$$\text{Area of } \triangle ABD = \text{Area of } \triangle AOB + \text{Area of } \triangle AOD$$

$$\text{and } \text{Area of } \triangle ABC = \text{Area of } \triangle AOB + \text{Area of } \triangle BOC$$

$$\text{we have } \text{Area of } \triangle AOD = \text{Area of } \triangle BOC$$

Therefore

$$\begin{aligned} \text{Area of } ABCD &= \text{Area of } \triangle AOB + \text{Area of } \triangle BOC + \text{Area of } \triangle COD + \text{Area of } \triangle AOD \\ &= \text{Area of } \triangle AOB + 2 \times \text{Area of } \triangle AOD + \text{Area of } \triangle COD \\ &= 10 + (2 \times 45) + 75 \\ &= 192 \end{aligned}$$

$$3. \text{ Let } x^2 - xy + y^2 = k.$$

$$\text{Then } 2xy = (x^2 + xy + y^2) - (x^2 - xy - y^2) = 2013 - k.$$

$$k = x^2 - xy + y^2 = x^2 + 2xy + y^2 - 3xy = (x + y)^2 - 3xy$$

$$k = (x + y)^2 - 3\left(\frac{2013 - k}{2}\right) \geq \frac{3k - 6039}{2}$$

$$2k \geq 3k - 6039$$

$$k \leq 6039$$

The maximum value of k is 6039.

(This happens when $y = -x$, i.e. $x^2 + x(-x) + (-x)^2 = 2013$. We may set $x = \sqrt{2013}$ and $y = -\sqrt{2013}$.)

$$4. \text{ Given that } \alpha, \beta \text{ are roots of } x^2 + 2013x + 5 = 0, \text{ we have}$$

$$\alpha^2 + 2011\alpha + 3 = -2\alpha - 2$$

$$\beta^2 + 2015\beta + 7 = 2\beta + 2$$

Then ,

$$(\alpha^2 + 2011\alpha + 3)(\beta^2 + 2015\beta + 7)$$

$$= (-2\alpha - 2)(2\beta + 2)$$

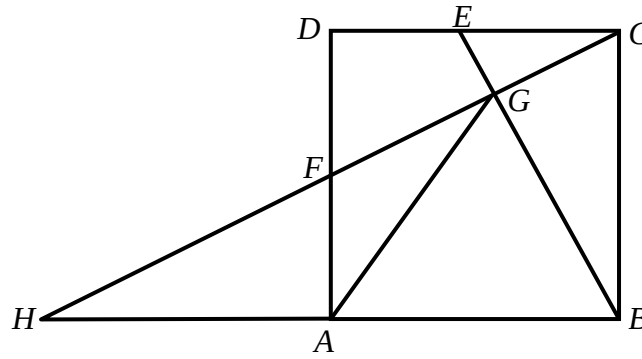
$$= -4[\alpha\beta + (\alpha + \beta) + 1]$$

$$= -4(5 - 2013 + 1)$$

$$= 8028$$

$$\therefore \text{ The value } = 8028$$

5.



As $\triangle BCE \cong \triangle CDF$ (SAS)

we have $\angle CBE = \angle DCF$

and $\angle BEC = \angle CFD$

In $\triangle CEG$, $\angle CGE = 180^\circ - \angle DCF - \angle BEC = 180^\circ - \angle DCF - \angle CFD = 90^\circ$

CF and BA produced to intersect at H , $\angle HGB = \angle CGE = 90^\circ$.

Because $\triangle CDF \cong \triangle HAF$ (ASA), which gives $HA = CD$, thus $HA = AB$.

Therefore A is the centre of the circle HGB , HA , AB and AG are radii of the circle HGB , therefore $AG = 10$.

6. From $x^2 + ax + 2b = 0$, $\Delta = a^2 - 8b \geq 0$, i.e. $a^2 \geq 8b$

From $x^2 + 2bx + a = 0$, $\Delta = 4b^2 - 4a \geq 0$, i.e. $4b^2 \geq 4a$ and $b^2 \geq a$

$\therefore a^4 \geq 64b^2 \geq 64a$

Since $a > 0$, i.e. $a^3 \geq 64$ and $a \geq 4$

Also $4b^2 \geq 4a \geq 16$, gives $b \geq 2$

Therefore, the minimum value of $a + b = 4 + 2 = 6$.

7. Let $f(x) = x^3 - 12x^2 + 47x - 60 = 0$, since $f(3) = (3)^3 - 12(3)^2 + 47(3) - 60 = 0$, by factor theorem,

$x - 3$ is a factor of $f(x)$, therefore $f(x) = x^3 - 12x^2 + 47x - 60 = (x - 3)(x - 4)(x - 5)$, which gives

the length of the three sides of the triangle to be 3, 4 and 5, thus the triangle is a right-angled triangle,

and its area $= \frac{1}{2} \times 3 \times 4 = 6$ square units.

Alternatively, let the length of the three sides of the triangle be $t - d$, t and $t + d$, where d is the common difference, then $t - d + t + t + d = 3t = 12$, i.e. $t = 4$. Also

$$t(t-d)(t+d) = 60$$

$$t(t^2 - d^2) = 60$$

$$4(16 - d^2) = 60$$

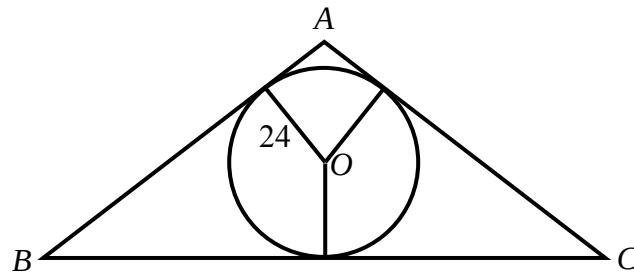
$$16 - d^2 = 15$$

$$d = 1$$

Therefore, the length of the three sides of $\triangle ABC$ are $t - 1 = 3$, $t = 4$ and $t + 1 = 5$.

This means $\triangle ABC$ is a right-angled triangle and its area = $\frac{1}{2} \times 3 \times 4 = 6$ square units.

8. Let $AB = x$. Then the perimeter of $\triangle ABC = 240 + 2x$. From the figure below, if the centre of the circle is O , we have



Area of $\triangle ABC$ = area of $\triangle AOB$ + area of $\triangle BOC$ + area of $\triangle COA$

$$= \frac{1}{2}x(24) + \frac{1}{2}(240)(24) + \frac{1}{2}x(24)$$

$$= \frac{1}{2}(240 + 2x)(24)$$

$$\text{Thus } \frac{1}{2}(240)\sqrt{x^2 - 120^2} = \frac{1}{2}(240 + 2x)(24)$$

$$5\sqrt{x^2 - 120^2} = (x + 120)$$

$$25(x + 120)(x - 120) = (x + 120)^2$$

$$25(x - 120) = x + 120$$

$$25x - 3000 = x + 120$$

$$24x = 3120$$

$$x = 130$$

9. In the set of integers : $1, 2, 3, \dots, 2011, 2012, 2013$, there are 671 integers which give a remainder of 1 when each of them is divided by 3.

Out of these 671 integers, the sum of any two of them has a remainder of 2 when it is divided by 3, while the difference between them is always a multiple of 3, i.e., the sum of any two numbers taken out from these 671 integers is not a multiple of the difference between the two numbers.

If 672 integers are taken out, then in the following 671 groups of numbers

$$(1, 2, 3), (4, 5, 6), (7, 8, 9), \dots, (2011, 2012, 2013)$$

There must be two numbers taken out from the same group.

- (1) If these two numbers are next to each other, then the sum of these two numbers must be a multiple of the difference between them (which is 1);
- (2) If these two numbers are not next to each other, then the sum and the difference of these two numbers are even, which means their sum must be a multiple of their difference

Therefore, at most one can take out 671 integers.

10. In (ii), replacing n by $n-1$, we have $f(1) + f(2) + \dots + f(n-1) = (n-1)^2 f(n-1)$ (iii)

Subtracting (iii) from (ii) gives $f(n) = n^2 f(n) - (n-1)^2 f(n-1)$

On rearranging, $f(n) = \frac{(n-1)^2}{n^2-1} f(n-1) = \frac{n-1}{n+1} f(n-1)$, thus

$$f(2012) = \frac{2011}{2013} \times \frac{2010}{2012} \times \frac{2009}{2011} \times \dots \times \frac{2}{4} \times \frac{1}{3} \times f(1) = \frac{2}{2013 \times 2012} \times 2012 = \frac{2}{2013}$$

Hong Kong Mathematics Olympiad (2012 / 2013)

Final Event (Individual) Solution

香港數學競賽 (2012 / 2013)

決賽項目(個人)題解

Event 1

1. $a = (4+3+2+1)(4+3+2+1) = 100$

Reason: horizontally: each row has $4+3+2+1$ rectangles.

vertically: each column has $4+3+2+1$ rectangles.

2. Since 7 divides 111111, it suffices to work out the last four 1's, i.e. 1111.

Since $1111 = 158 \cdot 7 + 5$, we have $1111 \equiv 5 \pmod{7}$ and hence, $b = 5$.

3. $c \equiv 3^{100} + 4^{50} + 5^{25} \equiv 0^{100} + 1^{50} + (25^{12} \cdot 5) \equiv 0 + 1 + 1 \cdot 2 \equiv 0 \pmod{3}$

4. $x = -1$, $y = 1$ and $z = 0$. So d is 2.

Event 2

1. Note that $f(x) = x^2 - (6)x + (9) = (x-3)^2$. Hence, $a = 3$.

2. It is clear that $b = 6x^2$. Hence, $2b - 3 = (x+3)^3 - x^3 = 9x^2 + 27x + 27$, which is the same as $0 = 3x^2 - 27x - 30 = x^2 - 9x - 10 = (x-10)(x+1)$ and $x = 10$. Therefore, $b = 6x^2 = 600$.

3. Working in congruence modulo 3, we have

$$\begin{array}{cccccccccccc} 3 & 5 & 8 & 13 & 21 & 34 & 55 & 89 & ||| & 144 & 233 \end{array}$$

$$\begin{array}{cccccccccccc} 0 & 2 & 2 & 1 & 0 & 1 & 1 & 2 & ||| & 0 & 2 \pmod{3} \end{array}$$

This sequence has period length 8. Since $600 = 8 \cdot 75$, $f(600) \equiv 2 \pmod{3}$ and $c = 2$.

4. Since $\angle X + \angle Y + \angle Z = 180^\circ$ and $2 \cdot \angle X = 6 \cdot \angle Z$, it follows that $3 \cdot \angle Z + \angle Y + \angle Z = 180^\circ$ or

$\angle Y = 180^\circ - 4 \cdot \angle Z$. Since $\angle Z \leq \angle Y$, we have $\angle Z \leq 180^\circ - 4 \cdot \angle Z$ and hence, $\angle Z \leq \frac{180^\circ}{5} = 36^\circ$.

Therefore, $d = 36$.

Event 3

1. Since $(7+4\sqrt{3})^{\frac{1}{2}} - (7-4\sqrt{3})^{\frac{1}{2}} = \sqrt{(2+\sqrt{3})^2} - \sqrt{(2-\sqrt{3})^2} = 2+\sqrt{3} - (2-\sqrt{3}) = 2\sqrt{3}$, we have $a=2$.

2. $b = F[3, f(4)] = F[3, 2] = 4 + 3 = 7$

3. Note that $392 - b = 392 - 7 = 385 = 5 \cdot 7 \cdot 11$. Hence, the two-digit numbers are 11, 35, 55 and 77 and $c=4$.

4. $y = 3 + \frac{1}{x^2 + x + 1}$ and $x^2 + x + 1 = (x + \frac{1}{2})^2 + \frac{3}{4}$.

So the least denominator value is $3/4$. Thus, $d = 3 + \frac{4}{3} = 4\frac{1}{3}$.

Event 4

1. $f(1) = f(1 \cdot 1) = f(1) \cdot f(1) = f(1)^2$. Therefore $f(1) = 1$ or $f(1) = 0$. However, if $f(1) = 0$, then $f(0) = f(1 \cdot 0) = f(1) \cdot f(0) = 0 \cdot f(0) = 0$. Therefore $f(1) = 1$.

2. $F(4) = 2$, $F(5) = \frac{2 \times 1 + 1}{1} = 3$ and $b = F(6) = \frac{2 \times 3 + 1}{1} = 7$.

3. Because the sum of the four roots of the equation is equal to the coefficient of x^3 ($= 0$), thus $1 + 2 + 3 + \text{the fourth root} = 0$, which means that the fourth root is -6 . We have $(x-1)(x-2)(x-3)(x+6) = x^4 - 25x^2 + 60x - 36 = x^4 + rx^2 + sx + t$, i.e. $c = r + t = -25 - 36 = -61$.

4. $x^2y + yx^2 + x + y = xy(x+y) + x + y = (x+y)(xy+1) = (x+y)(6+1) = 7(x+y)$
i.e. $x+y=9$. Therefore, $x^2 + y^2 = (x+y)^2 - 2xy = 81 - 12 = 69$.

Hong Kong Mathematics Olympiad (2012 / 2013)

Final Event (Group) Solution

香港數學競賽 (2012 / 2013)

決賽項目(團體) 題解

Event 1

1. The unit digit of $2^{14} + 1$ is 5, the rest are odd numbers. Therefore the answer is 5. Inside which the recurrence of powers of 2 is

2^1	Unit Digit = 2
2^2	Unit Digit = 4
2^3	Unit Digit = 8
2^4	Unit Digit = 6

Therefore the unit digit of 2^{14} is 4. Meanwhile, in the calculation process, it is only necessary to consider each unit digit only. 5 times the unit digit of any odd number is still 5, and an odd number multiplied by an odd number is still an odd number.

2. Original expression > 16 , $(0.6 \times 20) = 16$, $12 > 1$

Original expression < 16 , $(0.4 \times 20) = 16$, $8 = 2$

Therefore the integral part is 1.

3. It is feasible to use $\{1, 2, 6\}$, $\{1, 4, 7\}$, $\{2, 6, 7\}$ and $\{2, 4, 6\}$

Each group can generate 6 numbers, altogether 24 numbers.

As long as the sum of all digits in a number is a multiple of 3, the number is divisible by 3. Note that $1 + 2 + 6 = 9$ which is a multiple of 3, therefore 126, 162, 216, 261, 612 and 621 are all multiples of 3.

3. The same applies to other 3 groups.

4. Apparently E is 5. The condition that $ABCD$ is divisible by 4 is CD is divisible by 4, because B , D and F are even numbers, thus C cannot be even. Therefore, possible values of CD are 12, 16, 32, 36. On the other hand, $A + B + C$ must be a multiple of 3 (a sufficient condition for ABC to be divisible by 3), $A + B + C + D + E + F$ is also a multiple of 3 ($ABCDEF$ is divisible by 6 means that it is an even number and is divisible by 3). Therefore, $D + E + F = D + 5 + F$ is a multiple of 3. This implies D is either 2 or 6. If D is 2, $2 + 5 + F$ cannot be a multiple of 3, thus D must be 6, which implies F must be 4.

Together $ABCDEF = A2C654$, A and C can both be 1 or 3. Consequently, the maximum value of A is 3.

Event 2

1. Note that $4^3 + 4^r + 4^4 = 2^6 + 2^{2r} + 2^8 = (2^4 + 2^r)^2$ if $r = 1$.

$$\begin{array}{ll}
2. & B_1 \quad G_1 \quad _ \quad _ \quad _ \quad _ & 2! \times 2! \\
& G_1 \quad B_1 \quad G_1 \quad _ \quad _ \quad _ & 2! \times 2! \times 2 \\
& _ \quad G_1 \quad B_1 \quad G_1 \quad _ \quad _ & 2! \times 2! \times 2 \\
& _ \quad _ \quad G_1 \quad B_1 \quad G_1 \quad _ & 2! \times 2! \times 2 \\
& _ \quad _ \quad _ \quad G_1 \quad B_1 \quad G_1 & 2! \times 2! \times 2 \\
& _ \quad _ \quad _ \quad _ \quad B_1 \quad G_1 & 2! \times 2!
\end{array}$$

Therefore, $s = 40$.

3. Consider $y = \frac{x+a}{x^2 + \frac{1}{2}}$, then $yx^2 - x + \left(\frac{y}{2} - a\right) = 0$ with $\Delta = (-1)^2 - 4y\left(\frac{y}{2} - a\right) \geq 0$, which can be rewritten as $2y^2 - 4ay - 1 \leq 0$.

Since y varies between $\frac{1}{2}$ and -1 , we have $(y - (-1))\left(y - \frac{1}{2}\right) \leq 0$ or $2y^2 + y - 1 \leq 0$.

It follows that $a = -\frac{1}{4}$ and hence, $t = f(0) = -\frac{1}{2}$.

4. Let x_1, x_2 be roots of $ax^2 - \sqrt{2} \cdot bx + a = 0$. Then $x_1 + x_2 = \frac{\sqrt{2}b}{a}$ and $x_1 x_2 = \frac{a}{a} = 1$.

Also, since $(x_1 - x_2)^2 = 2$, it follows that $b^2 = 3a^2$. Thus, $CD = \frac{b}{2} = \frac{a\sqrt{3}}{2}$, D the mid-point of AC . And

$$\cos(\angle BCA) = \frac{\sqrt{3}}{2} \text{ or } \angle BCA = 30^\circ. \text{ Therefore, } u = 120.$$

Event 3

1. $43 = m^2 - n^2 = (m - n)(m + n)$ is a prime implies that $m = 22$ and $n = 21$. Thus,

$$m^3 - n^3 = (m - n)(m^2 + mn + n^2) = 1387.$$

2. If -1 appears m times, 1 appears n times and 2 appears r times, then $m + n + r = 10$ and $-m + n + 2r = 11$. It follows that $2m = r - 1 \geq 0$ and $2n = 21 - 3r - 1 \geq 0$ and $1 \leq r \leq 7$ ($r \geq 1$).

$$\text{Now, } x_1^2 + x_2^2 + \dots + x_{10}^2 = m + n + 4r = 3r + 10 \leq 31.$$

3. Note that

$$f(3) = a^3 + b^3 = (a + b)((a + b)^2 - 3ab)$$

and

$$f(3) = [f(1)]^3 + f(1) = (a + b)^3 + (a + b) = (a + b)((a + b)^2 + 1).$$

$$\text{Thus, } ab = -\frac{1}{3}.$$

4. Take E to be the point of intersection of CD with the circle. Then $\triangle DOE \cong \triangle COE$. Thus $\triangle DOA \cong \triangle COB$.

$$\text{Hence, } \frac{6}{4} = \frac{BC}{6} \text{ and } BC = 9.$$

Event 4

1. Let the multiplicand be x , and the multiplier be y , then

$$3.6 \times 10^{18} < x < 3.7 \times 10^{18}$$

$$3.4 \times 10^{14} < y < 3.5 \times 10^{14}$$

$$3.6 \times 3.4 \times 10^{32} < xy < 3.7 \times 3.5 \times 10^{32}$$

$$1.244 \times 10^{33} < xy < 1.259 \times 10^{33}$$

Therefore $P = XY$ is a 34 digits number.

2. The original equation can be simplified as $\frac{x+2013}{2013x} = \left(\frac{7}{4} - \frac{1}{4}\right) \times \frac{1}{4} = \frac{3}{8}$.

Therefore the answer is $1872 + 48 \times \frac{8}{3} = 2000$.

3. Let the number be N , then $N+1$ is a multiple of 10, 9, 8, 7, 6, 5, 4, 3 and 2. Since the least common multiple of 10, 9, 8, 7, 6, 5, 4, 3 and 2 is $2^3 \times 5 \times 3^2 \times 7 = 2520$, we have $N = 2520 - 1 = 2519$.

4. A must be 1 (otherwise, the product will be too large, 2 doesn't work as there is a digit 1 on the left most place).

E must be 5 (only $9 \times 5 = 45$ and $9 \times 0 = 0$). Therefore $ABCED = 111105 \div 9 = 12345$.

The answer = $1 + 2 + 3 + 4 + 5 = 15$.

Hong Kong Mathematics Olympiad (2013 / 2014)

Heat Event (Individual) Solution

香港數學競賽 (2013 / 2014)

初賽項目(個人) 題解

1. We have:

$$\begin{cases} \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}} = \frac{1}{2} \\ \frac{1}{\sqrt{b}} + \frac{1}{\sqrt{c}} = \frac{1}{3} \\ \frac{1}{\sqrt{c}} + \frac{1}{\sqrt{a}} = \frac{1}{5} \end{cases}$$

then, $\frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}} + \frac{1}{\sqrt{c}} = \frac{31}{60}$

We have:

$$\frac{1}{\sqrt{a}} = \frac{11}{60}, \quad \frac{1}{\sqrt{b}} = \frac{19}{60} \text{ and } \frac{1}{\sqrt{c}} = \frac{1}{60}$$

Hence, $\frac{a}{\sqrt{bc}} = \frac{60^2}{11^2} \times \frac{19 \times 1}{60 \times 60} = \frac{19}{121}$.

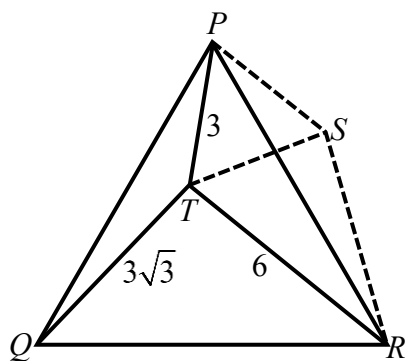
2. $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$

Given that $a = 2014x + 2011$, $b = 2014x + 2013$ and $c = 2014x + 2015$.

Therefore $a - b = -2$, $b - c = -2$ and $c - a = 4$.

$$\Rightarrow a^2 + b^2 + c^2 - ab - bc - ca = 12.$$

3. Construct a point S in the figure such that PTS is an equilateral triangle. Join SR .



From the figure above, consider $\triangle PTQ$ and $\triangle PSR$.

$$PQ = PR \quad (\text{given})$$

$$PT = PS \quad (\text{by construction})$$

$$\angle QPT = 60^\circ - \angle RPT = \angle RPS$$

$$\therefore \triangle PTQ \cong \triangle PSR \text{ (S.A.S.)}$$

$$\therefore SR = QT = 3\sqrt{3} \text{ and since } ST = 3, \text{ we have } ST^2 + SR^2 = 3^2 + (3\sqrt{3})^2 = 36 = 6^2 = TR^2.$$

Therefore by the converse of Pythagoras' Theorem, $\angle TSR = 90^\circ$.

$$\therefore \sin \angle STR = \frac{SR}{RT} = \frac{3\sqrt{3}}{6} = \frac{\sqrt{3}}{2} \Rightarrow \angle STR = 60^\circ$$

$$\therefore \angle PTR = 60^\circ + 60^\circ = 120^\circ.$$

Alternative Solution:

Rotate PT about P by 60° to PT' and $\triangle PT'T$ is equilateral. Since $PQ = PR$, R is the image of Q under the rotation.

$$\therefore \triangle PTQ \cong \triangle PT'R$$

$$\cos \angle T'TR = \frac{3^2 + 6^2 - (3\sqrt{3})^2}{2 \times 3 \times 6} = \frac{1}{2}$$

$$\therefore \angle T'TR = 60^\circ$$

$$\therefore \angle PTR = 60^\circ + 60^\circ = 120^\circ$$

$$\begin{aligned} 4. \quad & \frac{\alpha^2}{\beta^2 + 1} + \frac{\beta^2}{\alpha^2 + 1} \\ &= \frac{\alpha^2}{14\beta} + \frac{\beta^2}{14\alpha} \\ &= \frac{1}{14} \left(\frac{\alpha^3 + \beta^3}{\alpha\beta} \right) \\ &= \frac{(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}{14} \\ &= \frac{14^3 - 3(14)}{14} \\ &= 14^2 - 3 \\ &= 193 \end{aligned}$$

5. Let $\angle ADC = x^\circ$.

$$\therefore \angle ABC = 180^\circ - x^\circ$$

$$AD^2 + DC^2 - 2AD \times DC \times \cos x^\circ$$

$$= BC^2 + AB^2 - 2BC \times AB \times \cos(180^\circ - x^\circ)$$

$$5^2 + 14^2 - 2 \times 5 \times 14 \times \cos x^\circ = 10^2 + 11^2 - 2 \times 10 \times 11 \times (-\cos x^\circ)$$

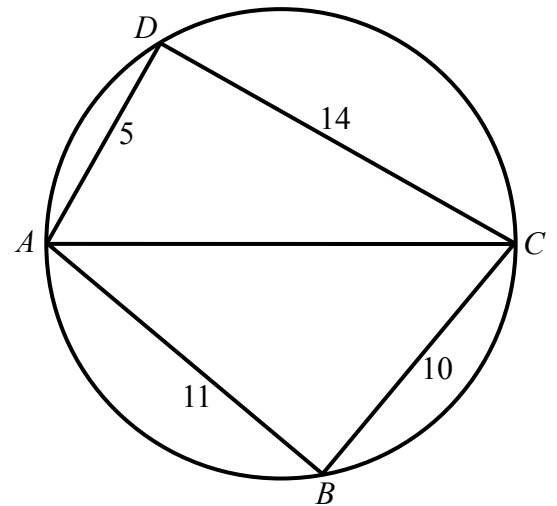
$$221 - 140 \times \cos x^\circ = 221 + 220 \times \cos x^\circ$$

$$\cos x^\circ = 0$$

$$x = 90$$

$\therefore$ The area of the quadrilateral $ABCD$

$$= \frac{1}{2} \times 5 \times 14 + \frac{1}{2} \times 10 \times 11 = 90.$$



6. $n^{2014} - 1 = (n-1)(n^{2013} + n^{2012} + n^{2011} + \dots + n + 1),$

$\therefore (n^{2013} + n^{2012} + n^{2011} + \dots + n + 1)$ is divisible by $(n-1).$

$$(n^{2013} + n^{2012} + n^{2011} + \dots + n + 1) = (n^{2013} - 1) + (n^{2012} - 1) + (n^{2011} - 1) + \dots + (n^2 - 1) + (n - 1) + 2013 + 1$$

and $(n^k - 1)$ is divisible by $(n-1)$, where $k = 1, 2, 3, \dots, 2013.$

$\therefore 2014$ is divisible by $(n-1).$

$$\text{As } 2014 = 2 \times 19 \times 53 = 19 \times 106,$$

Therefore, the maximum value of n is 107.

7. Since $x^3 + x^2 + x + 1 = 0$, $x^3 + x^2 + x + 1 = x^2(x+1) + (x+1) = (x^2+1)(x+1)$ which gives $x = -1$ or $\pm i$.

$$\text{Therefore } x^{-2014} + x^{-2013} + x^{-2012} + \dots + x^{-1} + 1 + x + x^2 + \dots + x^{2012} + x^{2013} + x^{2014} = 1 \text{ or } -1.$$

The answer is 1 or -1.

8. $\overline{xy} + \overline{yx} = 10x + y + 10y + x = 11(x + y)$

Since $2 \leq x + y \leq 18$, when $x + y = 11$, $\overline{xy} + \overline{yx}$ is a square number, i.e. $\overline{xy}$ can be 29, 38, 47, 56, 65, 74, 83 and 92, altogether 8 of them.

9. $4x^2 + 2xy + y^2 + 6z$

$$= (2x - y)^2 + 6(xy + z)$$

$$= (2x - y)^2 + 6\left(xy + \frac{64}{xy}\right)$$

$$= (2x - y)^2 + 6\left(\sqrt{xy} - \frac{8}{\sqrt{xy}}\right)^2 + 6(2)(8)$$

$$\text{For minimum, } 2x - y = \sqrt{xy} - \frac{8}{\sqrt{xy}} = 0$$

$$\begin{aligned}
 x(2x) &= 8 \\
 x &= 2 \\
 (x, y, z) &= (2, 4, 8) \\
 \therefore S &= 14
 \end{aligned}$$

10. Let $\angle A = a^\circ$, $\angle B = b^\circ$, $\angle C = c^\circ$.

Because x is the minimum of $a - b$, $b - c$ and $90 - a$, we have

$$\begin{cases} x \leq a - b \\ x \leq b - c \\ x \leq 90 - a \end{cases}$$

$$\begin{cases} 2x \leq 2a - 2b \\ x \leq b - c \\ 3x \leq 270 - 3a \end{cases}$$

$$6x \leq 270 - (a + b + c) = 270 - 180 = 90$$

$$x \leq 15$$

$\therefore$ The maximum value of x is 15.

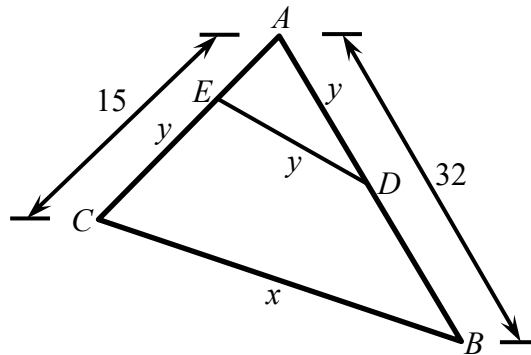
Hong Kong Mathematics Olympiad (2013 / 2014)
Heat Event (Group) Solution
香港數學競賽 (2013 / 2014)
初賽項目(團體) 題解

1. Because $(\sqrt{2014-x^2} - \sqrt{2004-x^2})(\sqrt{2014-x^2} + \sqrt{2004-x^2}) = 2014 - x^2 - 2004 + x^2 = 10$

$$\therefore 2(\sqrt{2014-x^2} + \sqrt{2004-x^2}) = 10$$

Therefore $\sqrt{2014-x^2} + \sqrt{2000-x^2} = 5$.

2. For $\triangle ADE$, the sum of two sides must be greater than the third, thus $y + y > 15 - y$, which leads to $5 < y < 15$.



Since $\triangle DAE$ is an isosceles triangle, we have $\cos A = \frac{\frac{1}{2}AE}{AD} = \frac{15-y}{2y}$.

In $\triangle ABC$, by using cosine formula, we have

$$\begin{aligned} x^2 &= 15^2 + 32^2 - 2 \times 15 \times 32 \times \frac{15-y}{2y} \\ &= 1729 - \frac{2^5 \times 3^2 \times 5^2}{y} \end{aligned}$$

From the above equation, y can take the value of 6, 8, 9, 10 and 12.

However, only $y = 6$ will give a perfect square $x^2 = 529 \Rightarrow x = 23$.

3. $\left(\cos \theta - \frac{7}{13}\right)^2 = \sin^2 \theta$

$$\cos^2 \theta - \frac{14}{13} \cos \theta + \frac{49}{169} = 1 - \cos^2 \theta$$

$$2 \cos^2 \theta - \frac{14}{13} \cos \theta - \frac{120}{169} = 0$$

$$169 \cos^2 \theta - 91 \cos \theta - 60 = 0$$

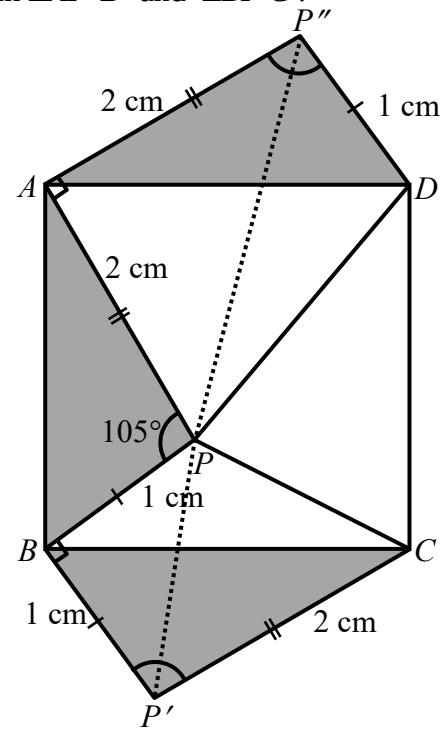
$$(13 \cos \theta + 5)(13 \cos \theta - 12) = 0$$

$$\cos \theta = -\frac{5}{13} \quad \text{or} \quad \cos \theta = \frac{12}{13} \quad (\text{rejected})$$

$$\therefore \cos \theta + \cos^3 \theta + \cos^5 \theta + \dots = \frac{-\frac{5}{13}}{1 - \left(\frac{5}{13}\right)^2} = -\frac{65}{144}$$

4. Rotate $\triangle APB$ about point A and point B respectively to obtain $\triangle AP''D$ and $\triangle BP'C$.

$$\begin{aligned} PP' &= \sqrt{1^2 + 1^2} = \sqrt{2} \text{ cm} \\ \angle CP'P &= 105^\circ - 45^\circ = 60^\circ \\ CP^2 &= 2 + 4 - 2(\sqrt{2})(2)\cos 60^\circ = 6 - 2\sqrt{2} \\ PP'' &= \sqrt{2^2 + 2^2} = \sqrt{8} \text{ cm} \\ \angle DP''P &= 105^\circ - 45^\circ = 60^\circ \\ DP^2 &= 8 + 1 - 2(\sqrt{8})(1)\cos 60^\circ = 9 - 2\sqrt{2} \\ x &= 6 - 2\sqrt{2} + 9 - 2\sqrt{2} = 15 - 4\sqrt{2} \end{aligned}$$



5. $6x + 7 - x^2 = 3y^2 \geq 0$

$$x^2 - 6x - 7 \leq 0$$

$$(x - 7)(x + 1) \leq 0$$

$$-1 \leq x \leq 7$$

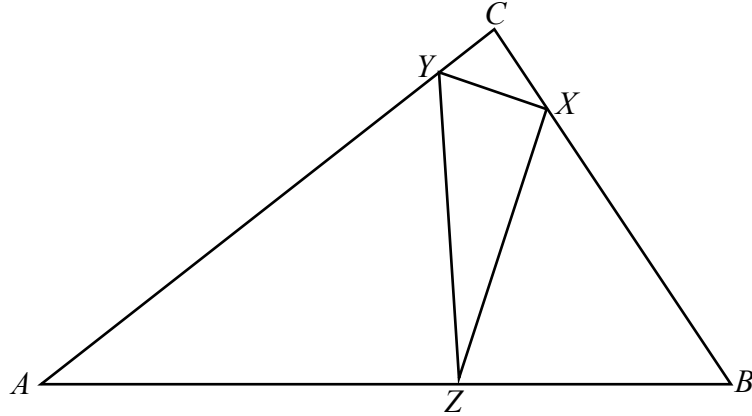
$$x^2 + y^2 = x^2 + \frac{6x + 7 - x^2}{3}$$

$$= \frac{2}{3}x^2 + 2x + \frac{7}{3}$$

$$= \frac{2}{3}\left(x + \frac{3}{2}\right)^2 + \frac{5}{6}$$

$$\therefore \text{Maximum value of } x^2 + y^2 = \frac{2}{3}\left(7 + \frac{3}{2}\right)^2 + \frac{5}{6} = 49$$

6. Let $\angle AZY = \angle BZX = z$, $\angle BXZ = \angle CXY = x$ and $\angle CYX = \angle AYZ = y$.



Consider the sum of interior angles of $\triangle XYZ$.

$$180^\circ - 2x + 180^\circ - 2y + 180^\circ - 2z = 180^\circ$$

$$2(x + y + z) = 360^\circ$$

$$x + y + z = 180^\circ$$

$\therefore \angle A = x$, $\angle B = y$ and $\angle C = z$.

i.e. $\triangle ABC \sim \triangle AYZ \sim \triangle XBZ \sim \triangle XYC$.

Let $AZ = 9a$, $AY = 10a$, $BZ = 6b$, $BX = 10b$, $CX = 9c$ and $CY = 6c$, where a , b and c are constants.

$$\begin{cases} 9a + 6b = 10 \\ 10b + 9c = 6 \\ 10a + 6c = 9 \end{cases}$$

$$\Rightarrow \begin{cases} 9a + 6b = 10 \\ (30a + 18c) - (20b + 18c) = 9 \times 3 - 6 \times 2 \end{cases}$$

$$\Rightarrow \begin{cases} 9a + 6b = 10 \\ 9a - 6b = 4.5 \end{cases}$$

$$\Rightarrow 18a = 14.5$$

$$\Rightarrow AZ = 9a = 7.25 \text{ or } \frac{29}{4}$$

7. Let $a + b + c + d = m$, $a + c = x$ and $b + c = y$.

$$\therefore a + d = m - y \text{ and } b + d = m - x,$$

$$\therefore (a + c)(a + d) = mx - xy = 1 \text{ and } (b + c)(b + d) = my - xy = 1$$

$$\therefore mx - xy = my - xy$$

$$\therefore m(x - y) = 0$$

Since a , b , c and d are four distinct numbers, we have $a + c \neq b + c$, i.e. $x \neq y$, $\therefore m = 0$

$$\text{Also } mx - xy = 1, \therefore xy = -1$$

$$\text{Therefore } (a + c)(b + c) = xy = -1$$

8. $a_{n+2} = 3a_{n+1} - 2a_n \Rightarrow a_{n+2} - 2a_{n+1} = a_{n+1} - 2a_n$
 Thus $a_{2014} - 2a_{2013} = a_{2013} - 2a_{2012} = \dots = a_2 - 2a_1 = 2014 - 2(215) = 1584$.

9. $\therefore \frac{-(-4)}{2(1)} = 2 > 1$

$\therefore y$ attains its minimum value at $\sin x = 1$.

Then,

$$(1)^2 - 4(1) + m = \frac{-8}{3}$$

$$m = \frac{1}{3}$$

Also, for $-1 \leq \sin x \leq 1$,

$$(1)^2 - 4(1) + \frac{1}{3} \leq y \leq (-1)^2 - 4(-1) + \frac{1}{3}$$

$$\frac{-8}{3} \leq y \leq \frac{16}{3}$$

Hence, the minimum value of m^y is $3^{-\frac{16}{3}}$.

10. $\tan\left(\frac{90^\circ}{\tan x}\right) \times \tan(90^\circ \tan x) = 1$

Then,

$$\frac{90^\circ}{\tan x} = 90^\circ + 180^\circ k - 90^\circ \tan x \quad \text{where } k \text{ is any integer.}$$

$$\frac{1}{\tan x} = 1 + 2k - \tan x$$

$$\tan x + \frac{1}{\tan x} = 2k + 1$$

For $1 < \tan x < 3$, the only feasible value for k is 1, because $1\frac{1}{3} < 2k + 1 < 4$ and k must be an integer.

Hence,

$$\tan x + \frac{1}{\tan x} = 2(1) + 1$$

$$\tan^2 x - 3\tan x + 1 = 0$$

$$\tan x = \frac{3 - \sqrt{5}}{2} \text{ (rejected) or } \tan x = \frac{3 + \sqrt{5}}{2} \text{ which is in the range.}$$

Therefore,

$$\tan x = \frac{3 + \sqrt{5}}{2}$$

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Hong Kong Mathematics Olympiad (2013 / 2014)

Final Event (Individual) Solution

香港數學競賽 (2013 / 2014)

決賽項目(個人) 題解

Event 1

$$1. \quad 3 + \alpha + y = 1 + 3 + 1 + 3 + y$$

$$\alpha = 5$$

$$2. \quad 1, 2, 3, 4, 5, 6, 7, 8, 9, 55$$

$$\Rightarrow \beta = 55$$

$$3. \quad 1, 11, 21, 31, 41, 51$$

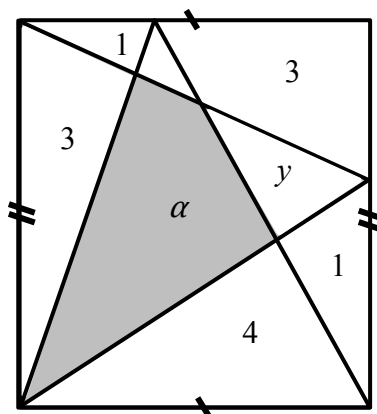
$$\Rightarrow \gamma = 6$$

$$4. \quad \log_2 a + \log_2 b = \log_2 ab > 6$$

$$\Rightarrow ab \geq 2^6$$

Since $a + b \geq 2\sqrt{ab} \geq 2 \cdot 2^3 = 2^4$

So $\delta = 16$



Event 2

$$1. \quad (x + \sqrt{x}) + (x - \sqrt{x}) - 2\sqrt{x + \sqrt{x}}\sqrt{x - \sqrt{x}} = x$$

$$2\sqrt{x + \sqrt{x}}\sqrt{x - \sqrt{x}} = x$$

$$4(x + \sqrt{x})(x - \sqrt{x}) = x^2$$

$$4(x^2 - x) = x^2$$

$$3x^2 - 4x = 0$$

$$\Rightarrow x = 0 \text{ or } x = \frac{4}{3}$$

$$\therefore x = \frac{4}{3}$$

$$2. \quad \beta = 8 \cdot \frac{4}{4/3} = 24$$

$$\begin{aligned}
3. \quad & \sqrt{x} = 4\sqrt{2} + \sqrt{24\gamma} \\
\Rightarrow & x = 32 + 8\sqrt{48\gamma} + 24\gamma \\
\Rightarrow & \gamma = 3
\end{aligned}$$

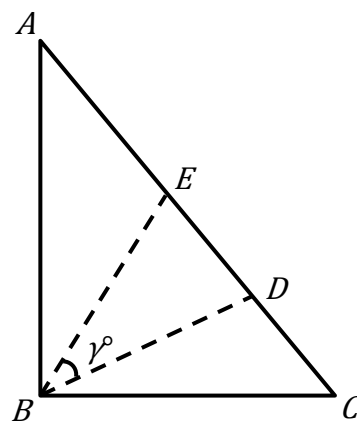
$$\begin{aligned}
4. \quad & \left((3^3)^3\right)^3 \equiv (7^3)^3 \equiv 3^3 \equiv 7 \pmod{10} \\
\therefore & \delta = 7
\end{aligned}$$

Event 3

$$\begin{aligned}
1. \quad & 1\,000\,000 = 10^6 \\
\Rightarrow & \frac{1}{11}(1+2+3+\dots+\alpha) = 6 \\
\Rightarrow & \frac{1}{11} \frac{(\alpha+1)\alpha}{2} = 6 \\
\Rightarrow & \alpha = 11
\end{aligned}$$

$$\begin{aligned}
2. \quad & \beta \left(\frac{1}{2} \left(\frac{1}{1 \cdot 2} - \frac{1}{9 \cdot 10} \right) \right) = 11 \\
\Rightarrow & \beta \left(\frac{22}{90} \right) = 11 \\
\Rightarrow & \beta = 45
\end{aligned}$$

$$\begin{aligned}
3. \quad & \angle ABE = a^\circ \\
& \angle CBD = b^\circ \\
\Rightarrow & \angle DBE = \gamma^\circ \\
\because & AB = AD, \angle BDE = a^\circ + \gamma^\circ \\
\because & CB = CE, \angle BED = b^\circ + \gamma^\circ \\
\text{Since } & a^\circ + b^\circ + \gamma^\circ = 90^\circ, \angle BED + \angle BDE + \gamma^\circ = 180^\circ \\
\Rightarrow & 3\gamma^\circ + a^\circ + b^\circ = 180^\circ \\
\Rightarrow & 2\gamma^\circ = 90^\circ \\
\Rightarrow & \gamma = 45
\end{aligned}$$



4. Let k be the number of 1's among the first 45 terms.

As the n th 1 is immediately followed by n 2s, we may write

$$45 \leq k + (1 + 2 + 3 + \dots + k) = k + \frac{k(k+1)}{2}$$

$$\Rightarrow 8 \leq k$$

For the first 8 1's, there are altogether 44 terms, so we need to tag on one more 1 to the sum. Therefore,

$$\delta = 8 + \frac{8(8+1)}{2} \cdot 2 + 1 = 8 + 72 + 1 = 81$$

Event 4

$$1. \quad \frac{6\sqrt{3}(3\sqrt{3} + 2\sqrt{3})}{(3\sqrt{2})^2 - (2\sqrt{3})^2} = 3\sqrt{\alpha} - 6$$

$$\Rightarrow \sqrt{3}(3\sqrt{2} - 2\sqrt{3}) = 3\sqrt{\alpha} - 6$$

$$\Rightarrow 3\sqrt{6} - 6 = 3\sqrt{\alpha} - 6$$

$$\Rightarrow \alpha = 6$$

$$2. \quad 0 < \frac{n-1}{n} < \frac{\alpha}{7}$$

$$\Rightarrow 1 < \alpha < 7$$

$$\Rightarrow \alpha = 2, 3, 4, 5 \text{ and } 6$$

$$\Rightarrow \beta = 5$$

3. Partition the equilateral triangle into 4 congruent equilateral triangular pieces and partition the regular hexagon into 6 equilateral triangular pieces.

$\Rightarrow$ Individual triangular pieces in the original triangle and hexagon are identical

$\Rightarrow$ Ratio of areas is $6 : 4 = 3 : 2$

$$\Rightarrow \gamma = \text{area of hexagon} = \frac{3}{2} \cdot \beta = \frac{3}{2} \cdot 5 = 7.5 \text{ (square units)}$$

$$\begin{aligned}
4. \quad \delta &= \left(1 + \frac{1}{2}\right) + \left(1 + \frac{1}{4}\right) + \cdots + \left(1 + \frac{1}{64}\right) - 7.5 \\
&= \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{64} - 1.5 \\
&= \left(1 - \frac{1}{64}\right) - 1.5 \\
&= -\frac{1}{64} + \left(-\frac{1}{2}\right) \\
&= -\frac{33}{64}
\end{aligned}$$

Hong Kong Mathematics Olympiad (2013 / 2014)

Final Event (Group) Solution

香港數學競賽 (2013 / 2014)

決賽項目(團體) 題解

Event 1

1. Let b be the base length. Then leg length $= \frac{1}{2}(32 - b)$.

$$\Rightarrow \left[\frac{1}{2}(32 - b) \right]^2 - \left(\frac{b}{2} \right)^2 = 8^2$$

$$\Rightarrow b = 12$$

$$\therefore \text{Area} = \frac{1}{2} \cdot 12 \cdot 8 = 48 \text{ (square units)}$$

$$2. \left(x^3 + \frac{1}{x^3} \right)^2 = x^6 + 2 + \frac{1}{x^6}$$

$$\begin{aligned} f(x) &= \frac{\left[\left(x + \frac{1}{x} \right)^3 \right]^2 - \left[x^3 + \frac{1}{x^3} \right]^2}{\left(x + \frac{1}{x} \right)^3 + \left(x^3 + \frac{1}{x^3} \right)} \\ &= \left(x + \frac{1}{x} \right)^3 - \left(x^3 + \frac{1}{x^3} \right) \\ &= 3 \left(x + \frac{1}{x} \right) \end{aligned}$$

$$\therefore 0 \leq \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)^2 = x + \frac{1}{x} - 2 \quad \text{or} \quad 2 \leq x + \frac{1}{x}$$

$$\Rightarrow f(x) = 3 \left(x + \frac{1}{x} \right) \text{ has a minimum value of 6 with } x = 1.$$

$$3. \quad \underbrace{111\dots1}_{81 \text{ 1's}} = \underbrace{111\dots1}_{9 \text{ 1's}} \cdot \underbrace{10000000001000000000\dots10000000001}_{8 \text{ sets of } 1000000000}$$

$$\therefore 9 \text{ divides both the first and the second factors} \Rightarrow 81 \mid \overline{111\dots1}$$

$$\Rightarrow \text{remainder} = 0$$

4. For $k \geq 2$, $a_k = k^2 a_k - (k-1)^2 a_{k-1}$

$$\Rightarrow (k^2 - 1)a_k = (k-1)^2 a_{k-1}$$

$$\begin{aligned} a_k &= \frac{k-1}{k+1} a_{k-1} \\ &= \frac{(k-1)(k-2)}{(k+1)k} a_{k-2} \\ &= \frac{(k-1)!}{(k+1)(k) \cdots 3} a_1 \\ &= \frac{(k-1)!}{(k+1)!} \\ &= \frac{1}{(k+1)k} \end{aligned}$$

$$\therefore a_{100} = \frac{1}{101 \cdot 100} = \frac{1}{10100}$$

Event 2

1. Remove $\frac{1}{8}$ and $\frac{1}{10}$

$$\Rightarrow \text{Product} = \frac{1}{80}$$

2. When n is even, $S_n = (1-2) + (3-4) + \cdots + (n-1-n) = -\frac{n}{2}$

When n is odd, $S_n = 1 + (-2+3) + (-4+5) + \cdots + (-(n-1)+n) = 1 + \frac{n-1}{2} = \frac{n+1}{2}$

$$\therefore S_{17} + S_{33} + S_{50} = \frac{18}{2} + \frac{34}{2} - \frac{50}{2} = 1$$

3. We may first determine who works on Sunday of the 50th week. This Sunday is $7 \cdot 49 + 1 = 344$ days from the first Sunday.

The person who works on this Sunday is $344 \equiv 2(6)$, which means that B is the person. Therefore A works on the Friday of that week.

4. As $AB = AC = AD$, B , C and D sit on a circle with centre at A and radius AB .

$$\text{So } \alpha^\circ = \angle BDC = \frac{1}{2} \angle ABC = 30^\circ.$$

Event 3

1. The product can be rewritten as

$$\left(1 - \frac{1}{2}\right)\left(1 + \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 + \frac{1}{3}\right) \cdots \left(1 - \frac{1}{10}\right)\left(1 + \frac{1}{10}\right)$$

$$= \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdots \frac{9}{10} \cdot \frac{11}{10} = \frac{11}{20}$$

2. The sum can be rewritten as

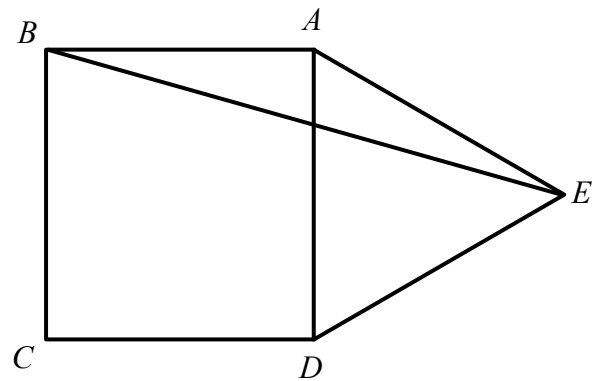
$$\log_{100!} 2 + \log_{100!} 3 + \cdots + \log_{100!} 100$$

$$= \log_{100!} (2 \cdot 3 \cdot \cdots 100) = \log_{100!} 100! = 1$$

3. Since $\angle BAE = 90^\circ + 60^\circ = 150^\circ$ and $BA = AE$,

$$\angle AEB = \frac{180^\circ - 150^\circ}{2} = 15^\circ$$

$$\therefore \alpha = 15$$



- 4.

6	÷	2	=	3
+		×		
1	+	4	=	α
=		=		
7		8		

$$\therefore \alpha = 5$$

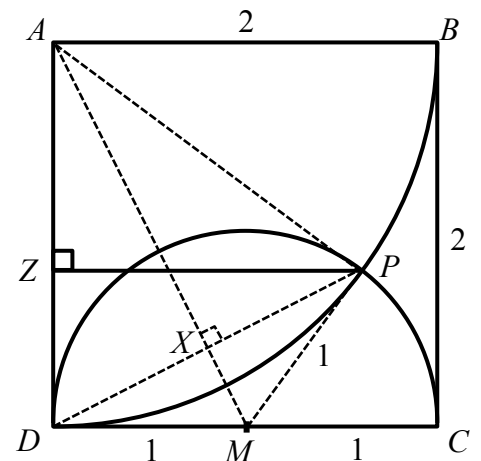
Event 4

1. Note that $ADMP$ is a kite.

$$\triangle ADM \approx \triangle PZD \approx \triangle PXM$$

$$\Rightarrow \frac{PM}{PX} = \frac{AM}{AD} = \frac{\sqrt{5}}{2}$$

$$\therefore PM = 1 \Rightarrow PX = \frac{2}{\sqrt{5}} \text{ and } PD = 2PX = \frac{4}{\sqrt{5}}$$



$$\frac{PZ}{PD} = \frac{PX}{PM} \Rightarrow PZ = PD \cdot \frac{PX}{PM} \Rightarrow PZ = \frac{4}{\sqrt{5}} \cdot \frac{2}{\sqrt{5}} = \frac{8}{5}$$

2. $x + y = \sqrt{5}$ and $xy = 1$

$$x^3y + 2x^2y^2 + xy^3 = xy(x + y)^2 = (\sqrt{5})^2 = 5$$

3. $c = 0$, $b = 6$, $a = 4$ and $d = 2$

4. Put $y = x - 2$ and rewrite the equation as

$$(y + 2)^4 + (y - 2)^4 = 32$$

$$\Rightarrow y^4 + 24y^2 = 0$$

$$\Rightarrow y^2(y^2 + 24) = 0$$

$$\Rightarrow y = 0 \text{ is the only possible real root here (double roots)}$$

$$\therefore x = 2 \text{ (double roots)}$$

$$\Rightarrow \text{Product } 2 \cdot 2 = 4$$

Hong Kong Mathematics Olympiad (2014 / 2015)

Heat Event (Individual) Solution

香港數學競賽 (2014 / 2015)

初賽項目(個人) 題解

1. 一對相異整數可說成為「好」的組合若它們的積是 5 的倍數。那麼從 1 到 2015 之間有多少對這樣的組合？〔答：730639〕

A pair of distinct integers is said to be “nice” pair if their product is a multiple of 5. How many “nice” pairs are there for integers from 1 to 2015?

Solution

Number of multiples of 5 between 0 and 2015 = 403

$$\text{Number of “nice” pairs} = 403(2015 - 403) + \frac{403(403 - 1)}{2} = 730639$$

2. 已知 $(10^{2015})^{-10^2} = 0.\underbrace{000\cdots 01}_{n \text{ 個 } 0}$ ，求 n 的值？〔答：201499〕

Given that $(10^{2015})^{-10^2} = 0.\underbrace{000\cdots 01}_{n \text{ times}}$, find the value of n .

Solution

$$(10^{2015})^{-10^2} = 10^{-201500} = \frac{1}{10^{201500}}, \text{ therefore there are } 201499 \text{ zeros after the integral part.}$$

(Note: $\frac{1}{10^2}$ has 1 zero after the integral part.)

3. 設正 n 邊形的內角為 x° ，其中 x 為整數。問 n 有多少個可能值？〔答：22〕

Let x° be the size of an interior angle of a n -sided regular polygon, where x is an integer, how many possible values of n are there?

Solution

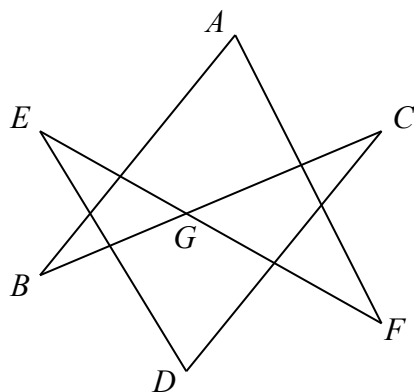
$$\frac{(n-2) \times 180^\circ}{n} = 180^\circ - \frac{360^\circ}{n}$$

n must be a factor of 360

$$360 = 2^3 \times 3^2 \times 5$$

Apart from 1 and 2, n can take $(3+1)(2+1)(1+1) - 2 = 22$

4. 如圖，已知 $\angle EGB = 64^\circ$ ，則 $\angle A + \angle B + \angle C + \angle D + \angle E + \angle F = ?$ [答： 232°]
 As shown in the figure, given that $\angle EGB = 64^\circ$, $\angle A + \angle B + \angle C + \angle D + \angle E + \angle F = ?$



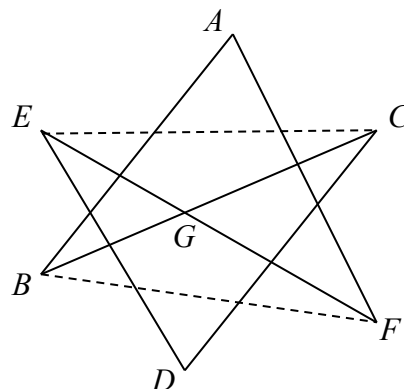
Solution

Join EC and BF .

In $\triangle ABF$, $\angle A + \angle B + \angle F = 180^\circ - \angle EGB$

In $\triangle CDE$, $\angle C + \angle D + \angle E = 180^\circ - \angle EGB$

$$\begin{aligned}\angle A + \angle B + \angle C + \angle D + \angle E + \angle F &= 360^\circ - 2 \times \angle EGB \\ &= 360^\circ - 2 \times 64^\circ \\ &= 232^\circ\end{aligned}$$



5. 已知 $a_1, a_2, \dots, a_n, \dots$ 為一正實數序列，其中 $a_1 = 1$ 及 $a_{n+1} = a_n + \sqrt{a_n} + \frac{1}{4}$ 。求 a_{2015} 的值。

[答：1016064]

It is given that $a_1, a_2, \dots, a_n, \dots$ is a sequence of positive real numbers such that $a_1 = 1$ and

$a_{n+1} = a_n + \sqrt{a_n} + \frac{1}{4}$. Find the value of a_{2015} .

Solution

$$a_{n+1} = a_n + \sqrt{a_n} + \frac{1}{4}$$

$$= \left(\sqrt{a_n} + \frac{1}{2} \right)^2$$

$$\Rightarrow \sqrt{a_{n+1}} = \sqrt{a_n} + \frac{1}{2}$$

$$\sqrt{a_n} = 1 + (n-1) \frac{1}{2}$$

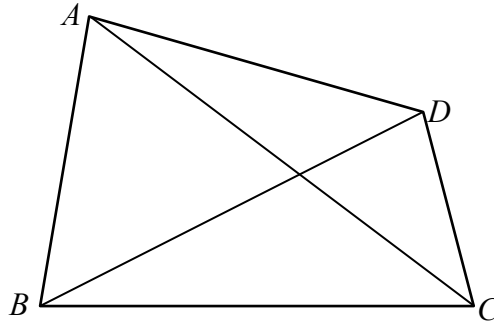
$$= \frac{1}{2} + \frac{1}{2}n$$

$$\therefore a_n = \frac{1}{4}(n+1)^2$$

$$a_{2015} = \frac{1}{4}(2016)^2 = 1016064$$

6. 如圖， $ABCD$ 是一個凸四邊形， $AB+BD+CD=16$ ，求 $ABCD$ 的最大面積。〔答：32〕

As shown in the figure, $ABCD$ is a convex quadrilateral, $AB+BD+CD=16$, find the maximum area of $ABCD$.



Solution

Let $AB = x$, $BD = y$.

Then $CD = 16 - x - y$

Area of $ABCD$

= Area of $\triangle ABD$ + area of $\triangle BCD$

$$\leq \frac{1}{2}xy + \frac{1}{2}y(16 - x - y)$$

$$= \frac{1}{2}xy + 8y - \frac{1}{2}xy - \frac{1}{2}y^2$$

$$= -\frac{1}{2}(y^2 - 16y)$$

$$= -\frac{1}{2}(y - 8)^2 + 32$$

Thus the maximum area of $ABCD = 32$.

(Note: $\angle ABD = \angle BDC = 90^\circ$, $AC = 8\sqrt{2}$, $y = 8$)

7. 設 $x, y, z > 1$, p 是正數及 $\log_x p = 18$, $\log_y p = 21$ 及 $\log_{xyz} p = 9$ 。求 $\log_z p$ 的值。〔答：126〕

Let $x, y, z > 1$, p is a positive number and $\log_x p = 18$, $\log_y p = 21$ and $\log_{xyz} p = 9$. Find the value of $\log_z p$.

Solution

$$(I) \quad \log_x p = 18 \Rightarrow \log_p x = \frac{1}{18}, \quad \log_y p = 21 \Rightarrow \log_p y = \frac{1}{21}, \quad \log_{xyz} p = 9 \Rightarrow \log_p xyz = \frac{1}{9}$$

We have

$$\begin{aligned} \log_p x + \log_p y + \log_p z &= \log_p xyz \\ \frac{1}{18} + \frac{1}{21} + \log_p z &= \frac{1}{9} \\ \log_p z &= \frac{1}{9} - \frac{1}{18} - \frac{1}{21} \\ &= \frac{1}{18} - \frac{1}{21} \\ &= \frac{1}{126} \end{aligned}$$

Therefore, $\log_z p = 126$.

$$(II) \quad p = x^{18}, \quad p = y^{21}, \quad p = (xyz)^9, \text{ we have}$$

$$\begin{aligned} z^9 &= \frac{p}{x^9 y^9} \\ &= \frac{p}{p^{\frac{1}{2}} p^{\frac{3}{7}}} \\ &= p^{1 - \frac{1}{2} - \frac{3}{7}} \\ &= p^{\frac{1}{14}} \end{aligned}$$

$$\therefore p = z^{9 \times 14} = z^{126} \Rightarrow \log_z p = 126$$

8. 求 $\frac{1}{4029} + \frac{2 \times 2014}{2014^2 + 2015^2} + \frac{4 \times 2014^3}{2014^4 + 2015^4} - \frac{8 \times 2014^7}{2014^8 - 2015^8}$ 的值。〔答：1〕

Find the value of $\frac{1}{4029} + \frac{2 \times 2014}{2014^2 + 2015^2} + \frac{4 \times 2014^3}{2014^4 + 2015^4} - \frac{8 \times 2014^7}{2014^8 - 2015^8}$.

Solution

Let $x = 2014$, $y = 2015$. Then the expression

$$\begin{aligned} &\frac{1}{4029} + \frac{2 \times 2014}{2014^2 + 2015^2} + \frac{4 \times 2014^3}{2014^4 + 2015^4} - \frac{8 \times 2014^7}{2014^8 - 2015^8} \\ &= \frac{1}{x+y} + \frac{2x}{x^2+y^2} + \frac{4x^3}{x^4+y^4} - \frac{8x^7}{x^8-y^8} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{x-y} + \frac{1}{x+y} + \frac{2x}{x^2+y^2} + \frac{4x^3}{x^4+y^4} - \frac{8x^7}{x^8+y^8} - \frac{1}{x-y} \\
&= \frac{2x}{x^2-y^2} + \frac{2x}{x^2+y^2} + \frac{4x^3}{x^4+y^4} - \frac{8x^7}{x^8+y^8} - \frac{1}{x-y} \\
&= \frac{4x^3}{x^4-y^4} + \frac{4x^3}{x^4+y^4} - \frac{8x^7}{x^8+y^8} - \frac{1}{x-y} \\
&= \frac{8x^7}{x^8-y^8} - \frac{8x^7}{x^8-y^8} - \frac{1}{x-y} \\
&= \frac{-1}{x-y} \\
&= \frac{-1}{2014-2015} \\
&= 1
\end{aligned}$$

9. 設 x 為實數。求 $\sqrt{x^2-4x+13} + \sqrt{x^2-14x+130}$ 的最小值。〔答: 13〕

Let x be a real number. Find the minimum value of $\sqrt{x^2-4x+13} + \sqrt{x^2-14x+130}$.

Solution

$$\sqrt{x^2-4x+13} + \sqrt{x^2-14x+130} = \sqrt{(x-2)^2+3^2} + \sqrt{(x-7)^2+9^2}$$

Let $A = (2, 3)$, $B = (7, 9)$ and $C = (x, 0)$.

Then $AC + CB = \sqrt{(x-2)^2+3^2} + \sqrt{(x-7)^2+9^2}$.

Reflect A along the x -axis to A' . Then $A' = (2, -3)$

The minimum value of the expression = the shortest distance between A' and B

$$= A'B = \sqrt{(7-2)^2 + (9+3)^2} = \sqrt{5^2 + 12^2} = 13$$

10. 已知 B 、 H 及 I 為圓上的點。 C 是該圓外的一點。 BC 是該圓在點 B 的切線。 HC 和 IC 分別通過該圓於點 D 及 G 。已知 HDC 是 $\angle BCI$ 的角平分線，若 $BC = 12$ ， $DC = 6$ 及 $GC = 9$ ，求 $\frac{\triangle BDH \text{ 的面積}}{\triangle DHIG \text{ 的面積}}$ 。〔答: $\frac{36}{55}$ 〕

Given that B , H and I are points on the circle. C is a point outside the circle. BC is tangent to the circle at B . HC and IC cut the circle at D and G respectively. It is given that HDC is the angle bisector of $\angle BCI$.

If $BC = 12$, $DC = 6$ and $GC = 9$, find $\frac{\text{area of } \triangle BDH}{\text{area of } \triangle DHIG}$.

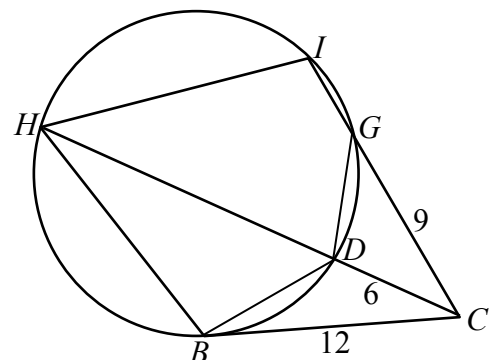
Solution

$$12^2 = 6(6 + DH) = 9(9 + IG)$$

Thus $DH = 18$ and $IG = 7$.

Let $\theta = \angle BCH = \angle HCI$.

Then



$$\begin{aligned}
\frac{\text{area of } \triangle BDH}{\text{area of } DHIG} &= \frac{\frac{1}{2}(12)(6+18)\sin\theta - \frac{1}{2}(12)(6)\sin\theta}{\frac{1}{2}(6+18)(9+7)\sin\theta - \frac{1}{2}(6)(9)\sin\theta} \\
&= \frac{12 \times 24 - 12 \times 6}{24 \times 16 - 6 \times 9} = \frac{36}{55}
\end{aligned}$$

Hong Kong Mathematics Olympiad (2014 / 2015)
Heat Event (Group) Solution
香港數學競賽 (2014 / 2015)
初賽項目(團體) 題解

1. 求 $\frac{1}{1860 \times 1865} + \frac{1}{1865 \times 1870} + \frac{1}{1870 \times 1875} + \dots + \frac{1}{2010 \times 2015}$ 的值。〔答： $\frac{1}{120900}$ 〕

Find the value of $\frac{1}{1860 \times 1865} + \frac{1}{1865 \times 1870} + \frac{1}{1870 \times 1875} + \dots + \frac{1}{2010 \times 2015}$.

Solution

$$\begin{aligned} & \frac{1}{1860 \times 1865} + \frac{1}{1865 \times 1870} + \frac{1}{1870 \times 1875} + \dots + \frac{1}{2010 \times 2015} \\ &= \frac{1}{5} \left(\frac{1}{1860} - \frac{1}{1865} + \frac{1}{1865} - \frac{1}{1870} + \dots + \frac{1}{2010} - \frac{1}{2015} \right) \\ &= \frac{1}{5} \left(\frac{1}{1860} - \frac{1}{2015} \right) \\ &= \frac{1}{5} \left(\frac{2015 - 1860}{1860 \times 2015} \right) = \frac{1}{5} \left(\frac{155}{1860 \times 2015} \right) = \frac{1}{120900}. \end{aligned}$$

2. 已知等邊三角形 ABC 的邊長為 3， P 為三角形內一點。設 PX 、 PY 及 PZ 分別為 P 至三角形三邊 AB 、 BC 及 CA 的垂足，求 $PX + PY + PZ$ 的值。〔答： $\frac{3\sqrt{3}}{2}$ 〕

Given an equilateral triangle ABC with side 3 and P is an interior point of the triangle. Let PX , PY and PZ be the feet of perpendiculars from P to AB , BC and CA respectively, find the value of $PX + PY + PZ$.

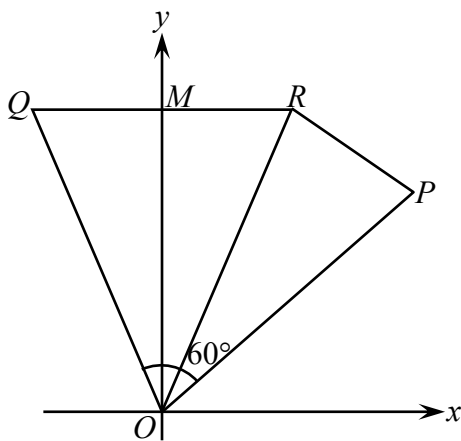
Solution

By area of triangle,

$$\begin{aligned} \frac{1}{2} PX(3) + \frac{1}{2} PY(3) + \frac{1}{2} PZ(3) &= \frac{1}{2} (3)^2 \sin 60^\circ \\ PX + PY + PZ &= 3 \left(\frac{\sqrt{3}}{2} \right) \\ PX + PY + PZ &= \frac{3\sqrt{3}}{2} \end{aligned}$$

3. 點 P 的坐標為 $(\sqrt{3}+1, \sqrt{3}+1)$ 。設點 P 繞原點作逆時針方向 60° 旋轉至點 Q ，接著點 Q 再沿 y 軸反射至點 R 。求 PR^2 的值。〔答：4〕

The coordinates of P are $(\sqrt{3}+1, \sqrt{3}+1)$. P is rotated anticlockwise about the origin through 60° to Q . Q is then reflected along the y -axis to R . Find the value of PR^2 .



Solution

Note that $\angle MOP = 45^\circ$. Thus $\angle QOM = 15^\circ$.

Hence $\angle ROP = \angle QOR = 30^\circ$.

$$OP = OR = \sqrt{(\sqrt{3}+1)^2 + (\sqrt{3}+1)^2} = \sqrt{2}(\sqrt{3}+1)$$

$$\begin{aligned} PR^2 &= OP^2 + OR^2 - 2OP \times OR \cos 30^\circ \\ &= 2OP^2 \left(1 - \frac{\sqrt{3}}{2}\right) \end{aligned}$$

$$\begin{aligned} &= 2OP^2 \left(1 - \frac{\sqrt{3}}{2}\right) \\ &= 4(\sqrt{3}+1)^2 \left(\frac{2-\sqrt{3}}{2}\right) \end{aligned}$$

$$\begin{aligned} &= 4(4+2\sqrt{3}) \left(\frac{2-\sqrt{3}}{2}\right) \\ &= 4(2+\sqrt{3})(2-\sqrt{3}) = 4. \end{aligned}$$

4. 已知 $a^2 + \frac{b^2}{2} + 9 \leq ab - 3b$ ，其中 a, b 為實數，求 ab 的值。〔答：18〕

Given that $a^2 + \frac{b^2}{2} + 9 \leq ab - 3b$, where a, b are real numbers. Find the value of ab .

Solution

For $a^2 + \frac{b^2}{2} + 9 \leq ab - 3b$, then

$$\left(a^2 - ab + \frac{b^2}{4}\right) + \left(\frac{b^2}{4} + 3b + 9\right) \leq 0$$

$$\left(a - \frac{b}{2}\right)^2 + \left(\frac{b}{2} + 3\right)^2 \leq 0$$

$$\begin{cases} a - \frac{b}{2} = 0 \\ \frac{b}{2} + 3 = 0 \end{cases}$$

$$b = -6,$$

$$a = -3.$$

$$ab = 18$$

5. 已知方程 $x^2 + 15x + 58 = 2\sqrt{x^2 + 15x + 66}$ 有兩個實根，求兩根之和。〔答：-15〕

Given that the equation $x^2 + 15x + 58 = 2\sqrt{x^2 + 15x + 66}$ has two real roots, find the sum of roots.

Solution

- (I) Rewrite the equation as $x^2 + 15x + 66 - 2\sqrt{x^2 + 15x + 66} - 8 = 0$

$$\text{Therefore} \quad (\sqrt{x^2 + 15x + 66} + 2)(\sqrt{x^2 + 15x + 66} - 4) = 0$$

$$\Rightarrow \sqrt{x^2 + 15x + 66} = -2 \text{ (rejected)} \quad \text{or} \quad \sqrt{x^2 + 15x + 66} = 4$$

$$\Rightarrow x^2 + 15x + 50 = 0$$

$\therefore$ The sum of root is -15.

- (II) Let $t = \sqrt{x^2 + 15x + 66}$, then the original equation becomes $t^2 - 8 = 2t$

$$\Rightarrow t^2 - 2t - 8 = 0$$

$$(t+2)(t-4) = 0$$

Because $t \geq 0$, $t = 4$, therefore

$$\Rightarrow x^2 + 15x + 50 = 0$$

$\therefore$ The sum of root is -15.

- (III) Rewrite the equation as $x^2 + 15x + 66 - 2\sqrt{x^2 + 15x + 66} + 1 = 9$

$$\text{Therefore} \quad (\sqrt{x^2 + 15x + 66} - 1)^2 = 9$$

$$\Rightarrow \sqrt{x^2+15x+66}-1=-3 \text{ (rejected)} \quad \text{or} \quad \sqrt{x^2+15x+66}-1=3$$

$$\Rightarrow x^2+15x+50=0$$

$\therefore$ The sum of root is -15 .

6. 已知三角形中兩角之和為 n° ，最大角比最小角大 30° ，求 n 的最大值。〔答：140〕

Given that the sum of two angles of a triangle is n° , and the largest angle is 30° more than the smallest angle, find the largest value of n .

Solution

Let the sizes of the three angles of the triangle be a° , b° and c° , where $a \geq b \geq c$. Then $a - c = 30$.

- (1) If $a + b = n$, then $c = 180 - n$. From $a - c = 30$, we have $a - 180 + n = 30$,

$$\text{i.e. } a = 210 - n, \quad b = 2n - 210$$

$$\text{Because } 210 - n \geq 2n - 210 \geq 180 - n$$

$$\therefore 130 \leq n \leq 140$$

- (2) If $a + c = n$, then $b = 180 - n$. From $a - c = 30$, we have $a = \frac{n+30}{2}$ and $c = \frac{n-30}{2}$.

$$\text{Because } \frac{n+30}{2} \geq 180 - n \geq \frac{n-30}{2}, \text{ we have } 110 \leq n \leq 130.$$

- (3) If $b + c = n$, then $a = 180 - n$. From $a - c = 30$, we have $180 - n - c = 30$,

$$\text{i.e. } c = 150 - n \text{ and } b = n - c = 2n - 150.$$

$$\text{Because } 180 - n \geq 2n - 150 \geq 150 - n, \text{ we have } 100 \leq n \leq 110.$$

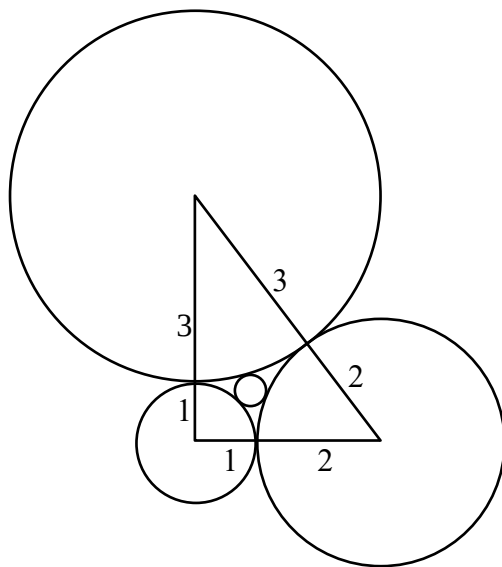
Therefore the maximum value of n is 140.

(Note: in this case, the three angles of the triangle are 40° , 70° and 70°)

7. 四個半徑分別為 1 單位、2 單位、3 單位及 r 單位的圓互相相切如下圖所示。求 r 的值。〔答：

$$r = \frac{6}{23} \text{ 〕}$$

Fours circles with radii 1 unit, 2 units, 3 units and r units are touching each other as shown in the figure. Find the value of r .



Solution

Consider the figure:

$$\cos \alpha = \frac{4^2 + (1+r)^2 - (3+r)^2}{2(4)(1+r)} = \frac{2-r}{2(1+r)}$$

$$\cos \beta = \frac{3^2 + (1+r)^2 - (2+r)^2}{2(3)(1+r)} = \frac{3-r}{3(1+r)}$$

Since $\alpha + \beta = 90^\circ$, $\cos^2 \alpha + \cos^2 \beta = 1$, thus

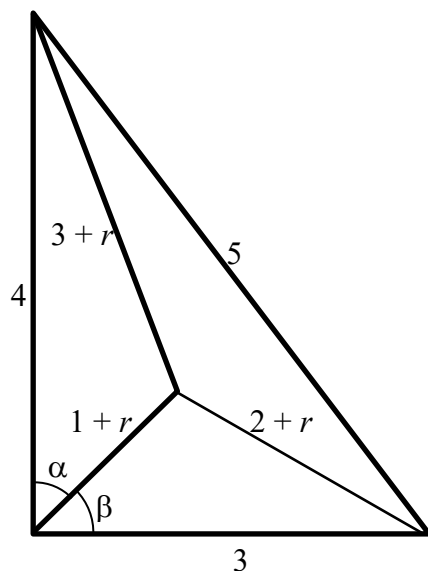
$$\frac{(2-r)^2}{4(1+r)^2} + \frac{(3-r)^2}{9(1+r)^2} = 1$$

$$9(r^2 - 4r + 4) + 4(r^2 - 6r + 9) = 36(r^2 + 2r + 1)$$

$$23r^2 + 132r - 36 = 0$$

$$(23r - 6)(r + 6) = 0$$

$$\therefore r = \frac{6}{23}$$



8. 已知 a, b, x, y 為非零整數，其中 $ax + by = 4$, $ax^2 + by^2 = 22$, $ax^3 + by^3 = 46$ 及 $ax^4 + by^4 = 178$ 。

求 $ax^5 + by^5$ 的值。〔答：454〕

Given that a, b, x, y are non-zero integers, where $ax + by = 4$, $ax^2 + by^2 = 22$, $ax^3 + by^3 = 46$ and $ax^4 + by^4 = 178$, find the value of $ax^5 + by^5$.

Solution

$$ax + by = 4 \quad \dots\dots\dots (1)$$

$$ax^2 + by^2 = 22 \quad \dots\dots\dots (2)$$

$$ax^3 + by^3 = 46 \quad \dots\dots\dots (3)$$

$$ax^4 + by^4 = 178 \quad \dots\dots\dots (4)$$

From (4), we have $ax^4 = 178 - by^4$, therefore $ax^5 = 178x - bxy^4 \dots\dots\dots (5)$

Likewise, we have $by^5 = 178y - ax^4y \dots\dots\dots (6)$

$$(5) + (6) \text{ gives } ax^5 + by^5 = 178(x + y) - xy(ax^3 + by^3) \Rightarrow ax^5 + by^5 = 178(x + y) - 46xy \dots\dots\dots (7)$$

Similarly, from (3), we have $ax^3 = 46 - by^3$, therefore $ax^4 = 46x - bxy^3 \dots\dots\dots (8)$

Likewise, we have $by^4 = 46y - ax^3y \dots\dots\dots (9)$

$$(5) + (6) \text{ gives } ax^4 + by^4 = 46(x + y) - xy(ax^2 + by^2) \Rightarrow 178 = 46(x + y) - 22xy \dots\dots\dots (10)$$

From (2), we have $ax^2 = 22 - by^2$, therefore $ax^3 = 22x - bxy^2 \dots\dots\dots (11)$

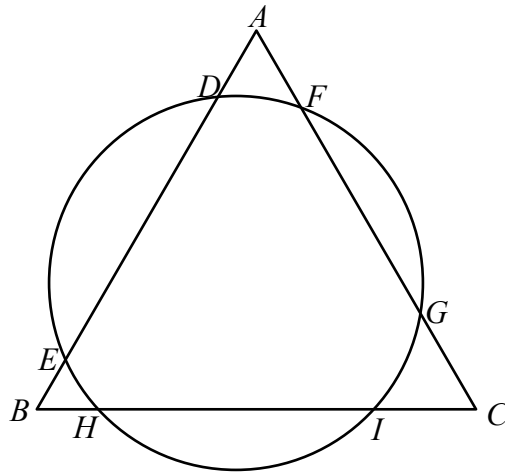
Likewise, we have $by^3 = 22y - ax^2y \dots\dots\dots (12)$

$$(5) + (6) \text{ gives } ax^3 + by^3 = 22(x + y) - xy(ax + by) \Rightarrow 46 = 22(x + y) - 4xy \dots\dots\dots (13)$$

Solving (10) and (13) for $x + y$ and xy , we have $x + y = 1$ and $xy = -6$.

$$\begin{aligned} \text{Therefore } ax^5 + by^5 &= 178(x + y) - 46xy \\ &= 178(1) - 46(-6) \\ &= 454 \end{aligned}$$

9. 已知 $\triangle ABC$ 為等邊三角形， $AF = 2$ 、 $FG = 10$ 、 $GC = 1$ 及 $DE = 5$ 。求 HI 。〔答： $2\sqrt{14}$ 〕
Given that $\triangle ABC$ is equilateral. $AF = 2$, $FG = 10$, $GC = 1$ and $DE = 5$. Find HI .



Solution

Let $AD = a$, $BH = c$, $IC = d$ and $HI = x$.

Then, we have:

$$a(a+5) = 2(2+10)$$

$$a^2 + 5a - 24 = 0$$

$$a = 3 \text{ or } a = -8(\text{rej})$$

$$c(c+x) = 5(10) = 50$$

$$\left(c + \frac{x}{2}\right)^2 = 50 + \left(\frac{x}{2}\right)^2$$

$$d(d+x) = 1(11) = 11$$

$$\left(d + \frac{x}{2}\right)^2 = 11 + \left(\frac{x}{2}\right)^2$$

Then,

$$\left(c + \frac{x}{2}\right)^2 - \left(d + \frac{x}{2}\right)^2 = 50 - 11$$

$$(c-d)(c+d+x) = 39$$

$$(c-d)(13) = 39$$

$$c-d = 3$$

Also,

$$c+d+x = 13$$

$$d = 5 - \frac{x}{2}$$

$$\left(5 - \frac{x}{2}\right)\left(5 + \frac{x}{2}\right) = 11$$

$$14 = \left(\frac{x}{2}\right)^2$$

$$x = 2\sqrt{14}$$

10. 設 A_n 、 B_n 為某二次函數 $C_n: y = n(n-1)x^2 - (2n-1)x + 1$ 圖像的 x 截距，其中 n 為一個大於 1 的整數。求 $A_2B_2 + A_3B_3 + \cdots + A_{2015}B_{2015}$ 的值。〔答： $\frac{2014}{2015}$ 〕

Let A_n , B_n be the x -intercepts of the graph of $C_n: y = n(n-1)x^2 - (2n-1)x + 1$, where n is an integer greater than 1. Find the value of $A_2B_2 + A_3B_3 + \cdots + A_{2015}B_{2015}$.

Solution

Let α_n, β_n be the 2 roots of $0 = n(n-1)x^2 - (2n-1)x + 1$.

Then

$$\begin{aligned} & A_n B_n^2 \\ &= (\alpha_n - \beta_n)^2 \\ &= (\alpha_n + \beta_n)^2 - 4\alpha_n \beta_n \\ &= \frac{(2n-1)^2 - 4n(n-1)}{(n(n-1))^2} \\ &= \frac{1}{(n(n-1))^2} \\ & \quad A_n B_n \\ &= \frac{1}{n(n-1)} \end{aligned}$$

Hence, $A_2B_2 + A_3B_3 + \cdots + A_{2015}B_{2015}$

$$\begin{aligned} &= \frac{1}{2(1)} + \frac{1}{3(2)} + \cdots + \frac{1}{2015(2014)} \\ &= 1 - \frac{1}{2015} \\ &= \frac{2014}{2015} \end{aligned}$$

Hong Kong Mathematics Olympiad (2014 / 2015)

Final Event (Individual) Solution

香港數學競賽 (2014 / 2015)

決賽項目(個人) 題解

Event 1

1. Since each absolute value is at least 0, it follows that they are all 0 and hence, $x = -\sqrt{5}$, $y = \sqrt{5}$, $z = 0$ and $\alpha = 5 + 5 + 0 = 10$.

2. Notice that

$$\begin{aligned} & \underbrace{111111111}_{10 \text{ 1's}} \times \underbrace{999999999}_{10 \text{ 9's}} \\ &= 111111111 \times (1000000000 - 1) \\ &= 111111111 \times 1000000000 - 111111111 \\ &= 1111111110888888889 \\ &\text{which implies that the sum of digits is } 9 + 72 + 9 = 90. \end{aligned}$$

3. Notice that $f(1) = f(1 \times 1) = f(1) \times f(1) = f(1)^2$. Thus, $f(1) = 1$ or $f(1) = 0$. Since $f(1) < 1$, it follows that $f(1) = 0$. Hence, $f(90) = f(90) \times f(1) = f(90) \times 0 = 0$ and $\gamma = f(\beta) + 100 - \beta = 10$.

4. Note that $f(11) = 2 \times f(10) + 1$ and

$$2^{11} = f(11) - f(10) = 2 \times f(10) + 1 - f(10) = f(10) + 1.$$

$$\text{Thus, } \delta = f(10) = 2^{11} - 1 = 2048 - 1 = 2047.$$

Event 2

1. Summing all three equations gives $x + y + z - 2x - 2y - 2z = -7$, which is $-x - y - z = -7$ and $\alpha = 7$.
2. Since 7 divides 111111, we may remove 6 1's from the left at a time and arrive at the last 4 1's, namely 1111, which will have the same remainder when divided by 7. It is easy to compute that the remainder is 5.

3. As $\left[(\beta-2)^{100} + \beta^{50} + (\beta+2)^{25}\right] = \left[3^{100} + 5^{50} + 7^{25}\right]$, we only need to compute the remainder of $(5^{50} + 7^{25}) \div 3$. Since the remainder of 5^{50} divided by 3 is congruent to $(-1)^{50}$ which is 1 and 7^{25} has the same remainder as 1^{25} . Thus, $\gamma = 2$.

4. Denote the unknown root as R . Since we have a polynomial of degree 4 with 4 real roots, the sum of the four roots, namely, $1 + 2 + 4 + R$, is the negative value the coefficient of x^3 (which is 0) divided by the coefficient of x^4 (which is 1). It follows that

$$1 + 2 + 4 + R = 0$$

and the fourth root, $R = -7$. Thus,

$$0 = x^4 + ax^2 + bx + \delta = (x-1)(x-2)(x-4)[x-(-7)]$$

Hence, $\delta = (-1)(-2)(-4)(7) = -56$.

Event 3

1. Note that there are 200 multiples of 5, 142 multiples of 7 and 28 multiples of 35, the number of positive integers in question is $1000 - 200 - 142 + 28 = 686$.

2. $\beta = 1 + (3-2)(3+2) + \dots + (686-685)(686+685) = 1 + 2 + 3 + \dots + 686 = 253641$

3. Note that for any n , its last appearance on the sequence is on the position $1 + 2 + \dots + n = \frac{n(n+1)}{2}$. Since $\frac{62 \times 63}{2} = 1953$ and $\frac{63 \times 64}{2} = 2016$, the 1993rd term is 63. Thus, the remainder of interest $\gamma = 21$.

4. Note that $\triangle BED \sim \triangle BCA$. Thus, $\frac{ED}{CA} = \frac{BE}{BC}$.

Since $BC + ED = 1$, it follows that $BC = 1 - ED$. Further, since $BE = CA$ and $BD = \frac{1}{2}$, we have

$$ED = \frac{CA \times BE}{BC} = \frac{BE^2}{1 - ED} = \frac{\frac{1}{4} - ED^2}{1 - ED}, \text{ which yields } ED = \frac{1}{4} \text{ and } \delta = \frac{\gamma}{4} = \frac{21}{4}$$

Event 4

1. Note that when $2^6 = 64$ divided by 13 has remainder -1 . Thus, $2^{12} = 2^6 \times 2^6$ divided by 13 has remainder $(-1)(-1) = 1$. Since $1000 = 12 \times 83 + 4$, we have $2^{1000} = (2^{12})^{83} \times 16$ divided by 13 has the same remainder as 16 divided by 13, which is 3. Therefore, $\alpha = 3$.

2. Note that

$$(7 + 4\sqrt{3})^{\frac{1}{2}} - (7 - 4\sqrt{3})^{\frac{1}{2}} = \sqrt{(2 + \sqrt{3})^2} - \sqrt{(2 - \sqrt{3})^2} = 2 + \sqrt{3} - (2 - \sqrt{3}) = 2\sqrt{3}.$$

Hence, $\beta = 2$.

3. $\gamma = F(3, f(4)) = F(3, 2) = 4 + 3 = 7$

4. Taking logarithm of the equation yields $(\log_7 x)(\log_7 x) = \log_7 10$. This implies that

$$\log_7 x = \pm \sqrt{\log_7 10}, \text{ or equivalently, } x = 7^{\pm \sqrt{\log_7 10}}. \text{ Hence, } \delta = 1.$$

Hong Kong Mathematics Olympiad (2014 / 2015)

Final Event (Group) Solution

香港數學競賽 (2014 / 2015)

決賽項目(團體)題解

Event 1

$$\begin{aligned} 1. \quad & \left(\frac{1 \times 3 \times 9 + 2 \times 6 \times 18 + \cdots + n \times 3n \times 9n}{1 \times 5 \times 25 + 2 \times 10 \times 50 + \cdots + n \times 5n \times 25n} \right)^{\frac{1}{3}} \\ &= \left(\frac{27(1^3 + 2^3 + \cdots + n^3)}{75(1^3 + 2^3 + \cdots + n^3)} \right)^{\frac{1}{3}} \\ &= \frac{3}{5} \end{aligned}$$

2. The number of that do not solve 1st, 2nd, 3rd, 4th problem are 5, 10, 15 and 20 respectively. Since each team does not solve at least one problem and there are $5 + 10 + 15 + 20 = 50$ questions not solved, it follows that each team does not solve exactly one problem. Hence, the teams that do not solve the first or second must have solved the third and the fourth. Therefore, there are $5 + 10 = 15$ teams.

3. Let $x = 3659893456789325678$ and $y = 342973489379256$, then

$$3.6 \times 10^{18} < x < 3.7 \times 10^{18}$$

$$3.4 \times 10^{14} < y < 3.5 \times 10^{14}$$

$$3.6 \times 3.4 \times 10^{32} < xy < 3.7 \times 3.5 \times 10^{32}$$

$$1.244 \times 10^{33} < xy < 1.259 \times 10^{33}$$

Thus, the product $n = xy$ has 34 digits.

4. The three circles of radii 2, 3 and 10 have their circumferences tangent to each other and their centres at A , B and C . We may form triangle $\triangle ABC$ with $AB = 5$, $AC = 12$ and $BC = 13$. This is a Pythagoras triangle (5, 12, 13), which means that $\triangle ABC$ is a right angle triangle. Now, we form a rectangle of side length 5 and 12 with vertices at A , B , D and C . Draw lines out from D to intersect with the three circles at P , Q and R , respectively. It is clear that $DP = DQ = DR = 15$, which is the radius of the big circle.

Event 2

1. Case 1: All squares in the middle row are Red (RRR).
The first and the third rows should not consist of two Red squares next to each other. The remaining possible patterns are with one Blue (RBR), two Blues (BBR, BRB, RBB), or three Blues (BBB). The number of possible colourings in this case is $1 \times 5^2 = 25$.
- Case 2: Two of the three squares in the middle row is Red (RRB, BRR, RBR).
For RRB and BRR, not the same pattern nor all Reds are in the first row and the third row. The number of possible colourings is $2 \times (8 - 2)^2 = 72$.
For RBR, any pattern can be in the first row and the third row. The number of possible colourings is $1 \times 8^2 = 64$.
- Case 3: One or none of the squares in the middle row is Red (RBB, BRB, BBR, BBB).
Any patterns could be in the first row and the third row. The number of possible colourings in this case is $4 \times 8^2 = 256$.

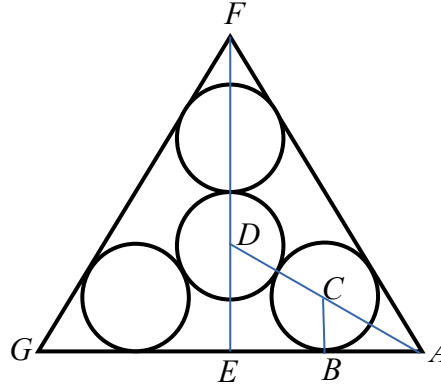
Hence the total number of possible colourings is $25 + 72 + 64 + 256 = 417$.

2. Let the least of the 25 positive integers be a . Then

$$(a + a + 24) \times 25 \div 2 = (a + 12) \times 5 \times 5$$

Since we are looking for a product of three primes, we may take $a = 1$ and get the least sum as $1 + 12 + 5 + 5 = 23$.

3. When $x \leq -3$, we have $-(x+3) - (1-x) = x+1$ and the real root $x = -5$.
When $-3 < x < 1$, we have $(x+3) - (1-x) = x+1$ and the real root $x = -1$.
When $1 \leq x$, we have $(x+3) - (x-1) = x+1$ and the real root $x = 3$. Therefore, the sum of the three roots is -3 .
4. $\triangle AEF$ and $\triangle ABC$ are right triangle with $\angle AFE = \angle BAC = 30^\circ$ and $\angle EAF = \angle ACB = 60^\circ$. $BC = 1$, so that $AC = 2$, and $AB = \sqrt{3}$. $CD = 2r = 1$, so that $AD = 4r$, $AE = 2\sqrt{3}$.



So, the length of side triangle, $s = 2AE = 4\sqrt{3}$ and $EF = 6$, so the area of $\triangle AGF = \frac{6 \times 4\sqrt{3}}{2} = 12\sqrt{3} \text{ cm}^2$.

Therefore, the answer is $12\sqrt{3}$.

Event 3

- Let $x = \sqrt{3+\sqrt{5}} + \sqrt{3-\sqrt{5}}$, then

$$x^2 = (3+\sqrt{5}) + (3-\sqrt{5}) + 2\sqrt{3+\sqrt{5}}\sqrt{3-\sqrt{5}} = 9 + 2\sqrt{9-5} = 13.$$

Since $x \geq 0$, we have $x = \sqrt{13}$.

- If m is odd, $(m+2)^3$ is odd but $p(p+m)+2p$ is even. Thus m is even, and hence $(m+2)^3$ is also even. Then p is also even, thus $p = 2$. By substituting $p = 2$, $2(2+m)+4 = (m+2)^3$ or $m(m^2+6m+10) = 0$; hence $m = 0$.

- Let $a = \left(x - \frac{1}{x}\right)^{\frac{1}{2}}$ and $b = \left(1 - \frac{1}{x}\right)^{\frac{1}{2}}$. Clearly $x \neq 0$, and we have $a - b = \frac{a^2 - b^2}{a + b} = \frac{x - 1}{x} = 1 - \frac{1}{x}$.

Summing this equation with the original equation gives $2a = x - \frac{1}{x} + 1 = a^2 + 1$ or $a^2 - 2a + 1 = 0$ and

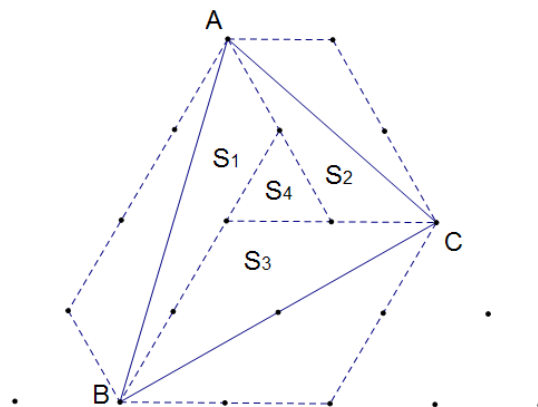
hence $a = 1$. Thus, $x - \frac{1}{x} = 1$ and $x^2 - x - 1 = 0$ and consequently, $x = \frac{1+\sqrt{5}}{2}$ as x is non-negative.

4. Consider

$$S_1 = 3, \quad S_2 = 2,$$

$$S_3 = 4, \quad S_4 = 1.$$

Hence, Area = $1 + 2 + 3 + 4 = 10$.



Event 4

$$1. \quad 1 + (2013^2 - 2^2) - (2012^2 - 3^2) + \dots - (1008^2 - 1007^2)$$

$$= 1 + 2015 \times 2011 - 2015 \times 2009 + \dots - 2015 \times 1$$

$$= 1 + 2015 \times (2011 - 2009 + \dots + 3 - 1)$$

Thus the remainder is 1.

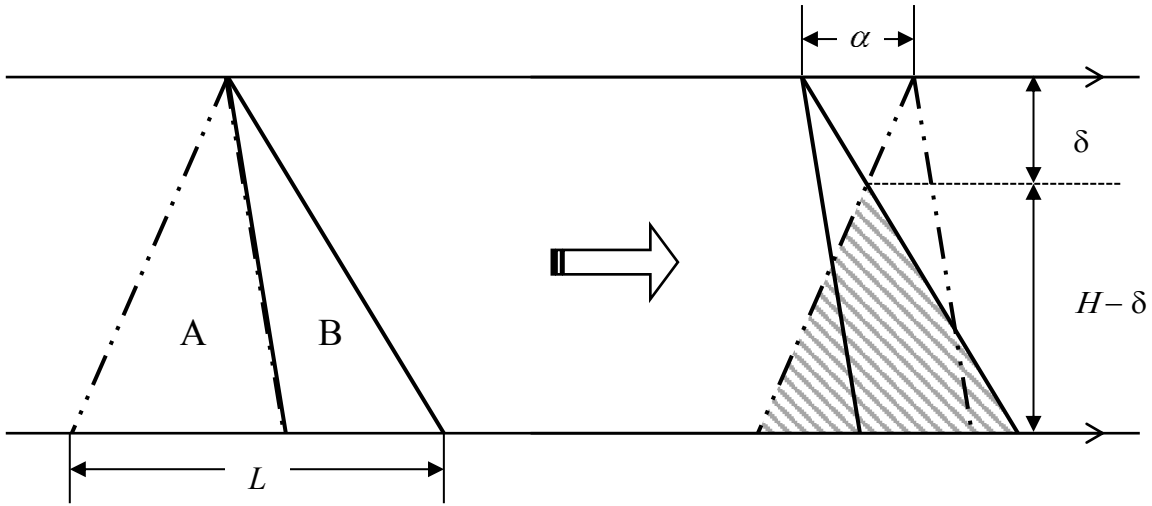
$$2. \quad 625$$

$$3. \quad \text{Note that } x^5 + \frac{1}{x^5} = \left(x^3 + \frac{1}{x^3}\right)\left(x^2 + \frac{1}{x^2}\right) - \left(x + \frac{1}{x}\right), \quad x^3 + \frac{1}{x^3} = \left(x^2 + \frac{1}{x^2}\right)\left(x + \frac{1}{x}\right) - \left(x + \frac{1}{x}\right) \quad \text{and}$$

$$x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)\left(x + \frac{1}{x}\right) - 2 = -1$$

$$\text{Thus, } x^5 + \frac{1}{x^5} = 1.$$

4.



Since the two indicated triangles are similar, we have $\frac{\alpha}{L-\alpha} = \frac{\delta}{H-\delta} \Rightarrow \frac{\alpha}{L} = \frac{\delta}{H}$.

Area of the shaded region is $\frac{1}{2}(H-\delta)(L-\alpha) = \frac{1}{2}HL\left(1-\frac{\delta}{H}\right)\left(1-\frac{\alpha}{L}\right) = \frac{1}{2}HL\left(1-\frac{\alpha}{L}\right)^2$.

Therefore, $\left(1-\frac{\alpha}{L}\right)^2 = \frac{\alpha}{L} \Rightarrow \frac{\alpha}{L} = \frac{3-\sqrt{5}}{2}$.

Hong Kong Mathematics Olympiad (2015 / 2016)

Heat Event (Individual) Solution

香港數學競賽 (2015 / 2016)

初賽項目(個人)題解

甲部

Part A

1. 計算 $0.125^{2016} \times (2^{2017})^3$ 的值。〔答：8〕

Find the value of $0.125^{2016} \times (2^{2017})^3$.

Solution

$$0.125^{2016} \times (2^{2017})^3 = (0.125 \times 2^3)^{2016} \times 2^3 = 8$$

2. 已知方程 $\begin{cases} x_1 + x_2 = x_2 + x_3 = x_3 + x_4 = \cdots = x_{2014} + x_{2015} = x_{2015} + x_{2016} = 1 \\ x_1 + x_2 + x_3 + \cdots + x_{2015} + x_{2016} = x_{2016} \end{cases}$ ，求 x_1 的值？〔答：

-1007〕

Given the equations $\begin{cases} x_1 + x_2 = x_2 + x_3 = x_3 + x_4 = \cdots = x_{2014} + x_{2015} = x_{2015} + x_{2016} = 1 \\ x_1 + x_2 + x_3 + \cdots + x_{2015} + x_{2016} = x_{2016} \end{cases}$, find the value of

x_1 .

Solution

$$x_1 + x_2 + x_3 + \cdots + x_{2015} + x_{2016} = x_{2016}$$

From the first equation,

$$\therefore 1 + 1 + 1 + \cdots + 1 + 1 = x_{2016}$$

$$\therefore x_{2016} = 1008$$

$$x_{2015} + x_{2016} = 1 \quad x_{2014} + x_{2015} = 1 \quad x_1 + x_2 = 1$$

$$\therefore x_{2015} + 1008 = 1, \therefore x_{2014} - 1007 = 1008, \dots, \therefore x_1 + 1008 = 1$$

$$\therefore x_{2015} = -1007 \quad \therefore x_{2014} = 1008 \quad \therefore x_1 = -1007$$

3. 有多少個整數 x 使得 $\sqrt{2016 - \sqrt{x}}$ 為整數？〔答：45〕

How many integers x can there be so that $\sqrt{2016 - \sqrt{x}}$ is an integer?

Solution

Let $k = \sqrt{2016 - \sqrt{x}}$.

$$k^2 = 2016 - \sqrt{x}$$

$$\therefore 0 \leq k \leq \sqrt{2016}$$

$$\therefore 0 \leq k < 45 ,$$

$$\therefore k = 0, 1, 2, \dots, 43, 44$$

Also $k^2 = 2016 - \sqrt{x}$, which gives $x = (2016 - k^2)^2$, where k has 45 values.

Therefore, x has 45 values.

4. 若 x, y 為整數，有多少對 x, y 且滿足 $(x+1)^2 + (y-2)^2 = 50$ 。 [答：12]

If x, y are integers, how many pairs of x, y are there which satisfy the equation

$$(x+1)^2 + (y-2)^2 = 50 .$$

Solution

$$\begin{cases} (x+1)^2 = 1 \\ (y-2)^2 = 49 \end{cases} \quad \therefore \begin{cases} (x+1) = \pm 1 \\ (y-2) = \pm 7 \end{cases} \Rightarrow 4 \text{ pairs}$$

$$\begin{cases} (x+1)^2 = 25 \\ (y-2)^2 = 25 \end{cases} \quad \therefore \begin{cases} (x+1) = \pm 5 \\ (y-2) = \pm 5 \end{cases} \Rightarrow 4 \text{ pairs}$$

$$\begin{cases} (x+1)^2 = 49 \\ (y-2)^2 = 1 \end{cases} \quad \therefore \begin{cases} (x+1) = \pm 7 \\ (y-2) = \pm 1 \end{cases} \Rightarrow 4 \text{ pairs}$$

There are 12 pairs of x, y which will satisfy the equation $(x+1)^2 + (y-2)^2 = 50$.

5. 63 個連續整數的和是 2016，求緊接該 63 個連續整數後的 63 個連續整數的和。 [答：5985]

The sum of 63 consecutive integer is 2016 , find the sum of the next 63 consecutive integers.

Solution

First solution:

Let the 63 consecutive integers be $x, x+1, x+2, \dots, x+61, x+62$ (1)

Then the next 63 consecutive integers are $x+63, x+64, x+65, \dots, x+124, x+125$ (2)

In (2), each integer is exactly 63 larger than its counterpart in (1),

$\therefore$ The sum of all 63 upcoming consecutive integers is $2016 + 63 \times 63 = 5985$.

Second Solution:

Because $\frac{63 \times (1 + 63)}{2} = 2016$, those 63 consecutive integers are 1, 2, 3, ..., 62, 63,

$\therefore$ The sum of all 63 upcoming consecutive integers is $\frac{63 \times (2 \times 64 + 62 \times 1)}{2} = 5985$.

6. 已知 8 個整數的平均數、中位數、分佈域及唯一眾數均為 8。若 A 為該 8 個整數中的最大數，求 A 的最大值。〔答：14〕

Given that the mean, median, range and the unique mode of 8 integers are also 8. If A is the largest integer among those 8 integers, find the maximum value of A .

Solution

1. If the maximum value of A is 16, then the mean value of the 8 integers cannot be 8, therefore remove this possibility;
2. If the maximum value of A is 15, then the smallest one from the 8 integers is 7, arranging these integers in ascending order of magnitude, we have 7, a , b , 8, 8, c , d , 15. If a , b , c and d take their smallest possible value, then we have 7, 7, 7, 8, 8, 8, 8, 15 and their mean is $= \frac{7+7+7+8+8+8+8+15}{8} = 8.5 > 8$, therefore remove this possibility;
3. If the maximum value of A is 14, then the smallest one from the 8 integers is 6, arranging these integers in ascending order of magnitude, we have 6, 6, 6, 8, 8, 8, 8, 14, then their mean, median, range and the only mode are 8.

7. 在整數 1 至 500 之間出現了多少個數字「2」？〔答：200〕

How many '2's are there in the numbers between 1 to 500.

Solution

Numbers	Number of '2'
1 – 100	20
101 – 200	21
201 – 300	119
301 – 400	20
401 – 500	20
Total	200

8. 某數的 16 進制位數是 1140。而同一數字的 a 進制位數是 240，求 a 。〔答：46〕
A number in base 16 is 1140. The same number in base a is 240, what is a ?

Solution

$$16^2 + 16^2 + 4(16) = 2a^2 + 4a$$

$$2a^2 + 4a = 4416$$

$$a^2 + 2a - 2208 = 0$$

$$(a + 48)(a - 46) = 0$$

$$a = -48 \text{ (rejected) or } 46$$

9. P 點的極坐標為 $(6, 240^\circ)$ 。若 P 向右平移 16 單位，求 P 的像與極點之間的距離。〔答: 14〕
The polar coordinates of P are $(6, 240^\circ)$. If P is translated to the right by 16 units, find the distance between its image and the pole.

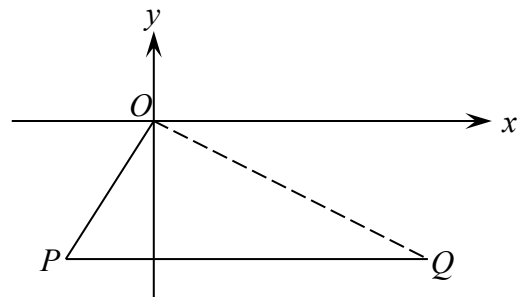
Solution

$$OP = 6$$

$$PQ = 16$$

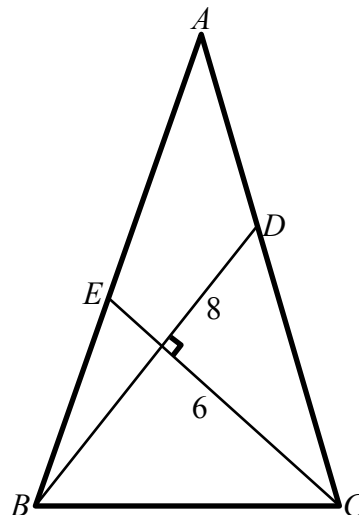
$$\angle OPQ = 60^\circ$$

$$OQ = \sqrt{6^2 + 16^2 - 2(6)(16)\cos 60^\circ} = 14$$



10. 如圖一，在 $\triangle ABC$ 中， BD 和 CE 分別是 AC 和 AB 兩邊上的中線，且 $BD \perp CE$ 。已知 $BD = 8$ ， $CE = 6$ ，求 $\triangle ABC$ 的面積。〔答：32〕

As shown in Figure 1, BD and CE are the medians of the sides AC and AB of $\triangle ABC$ respectively, and $BD \perp CE$. Given that $BD = 8$, $CE = 6$, find the area of $\triangle ABC$.



圖一
Figure 1

Solution

Join ED .

We have

$$\text{Area of } BCDE = \frac{1}{2}BD \cdot CE = 24$$

Since DE is the line joining the midpoints of two sides of $\triangle ABC$, we have

$$\text{Area of } \triangle ABC = \frac{4}{3} \times \text{Area of } BCDE = \frac{4}{3} \times 24 = 32$$

乙部

Part B

11. 已知方程 $100[\log(63x)][\log(32x)]+1=0$ 有兩個相異的實數根 α 及 β ，求 $\alpha\beta$ 的值。〔答：

$$\frac{1}{2016} \text{〕}$$

Given that the equation $100[\log(63x)][\log(32x)]+1=0$ has two distinct real roots α and β . Find the value of $\alpha\beta$.

Solution

$$100[\log(63x)][\log(32x)]+1=0$$

$$100(\log 63 + \log x)(\log 32 + \log x) + 1 = 0$$

$$100(\log x)^2 + 100(\log 63 + \log 32)(\log x) + (100\log 63\log 32 + 1) = 0$$

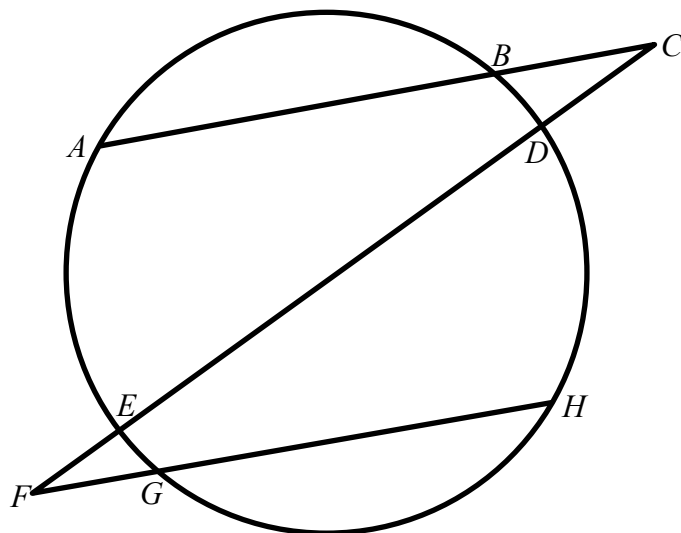
$$100(\log x)^2 + 100(\log 2016)(\log x) + (100\log 63\log 32 + 1) = 0$$

$$\log \alpha + \log \beta = -\log 2016$$

$$\therefore \alpha\beta = \frac{1}{2016}$$

12. 如圖二所示， ABC ， $CDEF$ 及 FGH 皆為直線，且 $ABC \parallel FGH$ 。 $AB = 42$ ， $GH = 40$ ， $EF = 6$ 及 $FG = 8$ 。已知 ABC 與 FGH 之間的最短距離為 41，求 BC 。〔答： $\frac{17}{3}$ 〕

As shown in Figure 2, ABC , $CDEF$ and FGH are straight lines, $ABC \parallel FGH$. $AB = 42$, $GH = 40$, $EF = 6$ and $FG = 8$. Given that the shortest distance between ABC and FGH is 41, find BC .



圖二

Figure 2

Solution

Let r be the radius and x be the shortest distance between the centre and ABC .

$$\text{Then } 21^2 + x^2 = r^2$$

$$\text{and } 20^2 + (41 - x)^2 = r^2.$$

Solving gives $r = 29$ and $x = 20$.

Note also that $(FE)(FD) = (FG)(FH)$.

$$\text{Thus } (6)(6 + ED) = (8)(8 + 40)$$

$$ED = 58$$

As $58 = 2(29)$, ED is the diameter of the circle.

Let O be the centre.

Let M and N be the mid-points of AB and GH respectively.

Then $\triangle OMC \sim \triangle ONF$.

$$\frac{21 + BC}{20 + 8} = \frac{20}{21}$$

$$BC = \frac{17}{3}$$

13. 設 A 、 B 和 C 為三個數字。利用這三個數字組成的三位數有以下性質：

- (a) ACB 可以被 3 整除；
- (b) BAC 可以被 4 整除；
- (c) BCA 可以被 5 整除；及
- (d) CBA 的因數數目為單數。

求三位數 ABC 。〔答：522〕

Let A , B and C be three digits. The number formed by these three digits has the following properties:

- (a) ACB is divisible by 3 ;
- (b) BAC is divisible by 4 ;
- (c) BCA is divisible by 5 ; and
- (d) CBA has an odd number of factors.

Find the 3-digit number ABC .

Solution

From (c), $A = 5$

From (b), $C = 2$ or 6

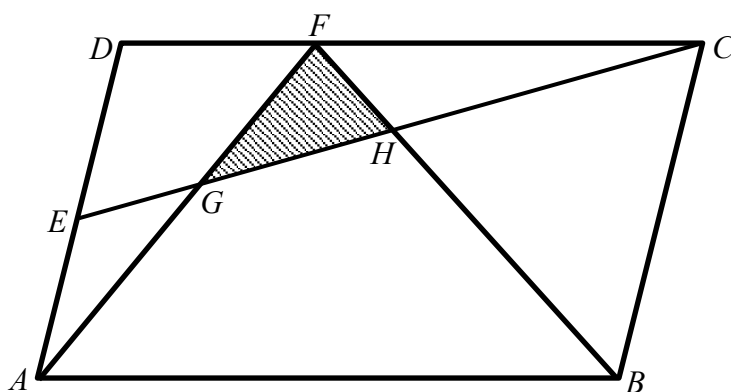
From (a), when $A = 5$, $C = 2$, the number ABC can be 522 , 552 and 582 .

From (a), when $A = 5$, $C = 6$, the number ABC can be 516 , 546 and 576 .

Among all these numbers, only 522 has odd number of factors, therefore $ABC = 522$.

14. 如圖三，在圖中， $ABCD$ 為一平行四邊形。 E 為 AD 的中點及 F 是 DC 上的點且滿足 $DF:FC=1:2$ 。 FA 及 FB 分別相交 EC 於 G 及 H 。 求 $\frac{ABCD \text{ 的面積}}{\Delta FGH \text{ 的面積}}$ 的值。 [答：20]

As shown in Figure 3 , $ABCD$ is a parallelogram. E is the mid-point of AD and F is a point on DC such that $DF:FC=1:2$. FA and FB intersect EC at G and H respectively. Find the value of $\frac{\text{Area of } ABCD}{\text{Area of } \Delta FGH}$.



圖三

Figure 3

Solution

R is a point on BC such that $DC \parallel ER$.

$\triangle PAE \sim \triangle FAD$ and $\triangle EPG \sim \triangle CFG$

$AG : GF = 3 : 2$

$\triangle QBR \sim \triangle FBC$ and $\triangle FHC \cong \triangle QHE$

$AH : HF = 1 : 3$

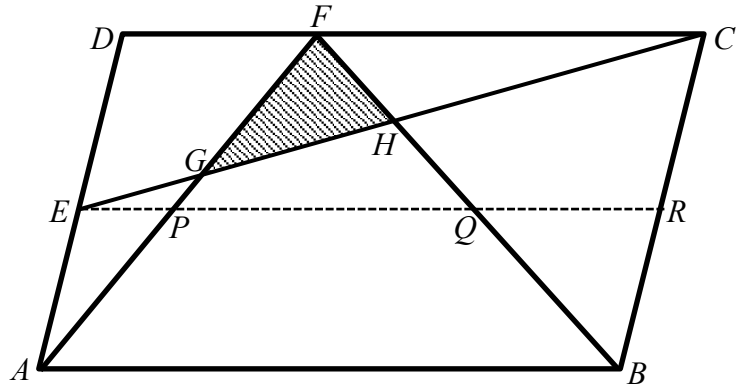
Let x be the area of $\triangle AFH$.

Area of $\triangle FGH = \frac{2}{5}x$

Area of $\triangle AFB = 4x$

Area of $ABCD = 8x$

$$\therefore \frac{\text{Area of } ABCD}{\text{Area of } \triangle FGH} = \frac{8x}{\frac{2}{5}x} = 20$$



15. 已知數列 $\{a_n\}$ 其中 $a_{n+2} = a_{n+1} - a_n$ 。若 $a_2 = -1$ 及 $a_3 = 1$ ，求 a_{2016} 的值。〔答：-1〕

Given a sequence $\{a_n\}$, where $a_{n+2} = a_{n+1} - a_n$. If $a_2 = -1$ and $a_3 = 1$, find the value of a_{2016} .

Solution

From $a_{n+3} = a_{n+2} - a_{n+1} = a_{n+1} - a_n - a_{n+1} = -a_n$ gives $a_{n+6} = -a_{n+3} = a_n$

Which means the sequence $\{a_n\}$ has a period of 6

$$\therefore 2016 \div 6 = 336 \dots 0$$

$$\therefore a_{2016} = a_{2010} = \dots = a_6 = a_5 - a_4 = a_4 - a_3 - a_4 = -a_3 = -1$$

Hong Kong Mathematics Olympiad (2015 / 2016)
Heat Event (Group) Solution
香港數學競賽 (2015 / 2016)
初賽項目(團體) 題解

1. 最初甲瓶裝有 1 公升酒精，乙瓶是空的。

第 1 次將甲瓶全部的酒精倒入乙瓶中，第 2 次將乙瓶酒精的 $\frac{1}{2}$ 倒回甲瓶，

第 3 次將甲瓶酒精的 $\frac{1}{3}$ 倒入乙瓶，第 4 次將乙瓶酒精的 $\frac{1}{4}$ 倒回甲瓶，……，

第 2016 次後，甲瓶還有多少公升酒精？〔答： $\frac{1}{2}$ 〕

At the beginning, there was 1 litre of alcohol in Bottle *A* and bottle *B* is an empty bottle.

First, pour all alcohol from bottle *A* to bottle *B*; second, pour $\frac{1}{2}$ of the alcohol from bottle *B* back to

bottle *A*; third, pour $\frac{1}{3}$ of the alcohol from bottle *A* to bottle *B*; fourth, pour $\frac{1}{4}$ of the alcohol from bottle

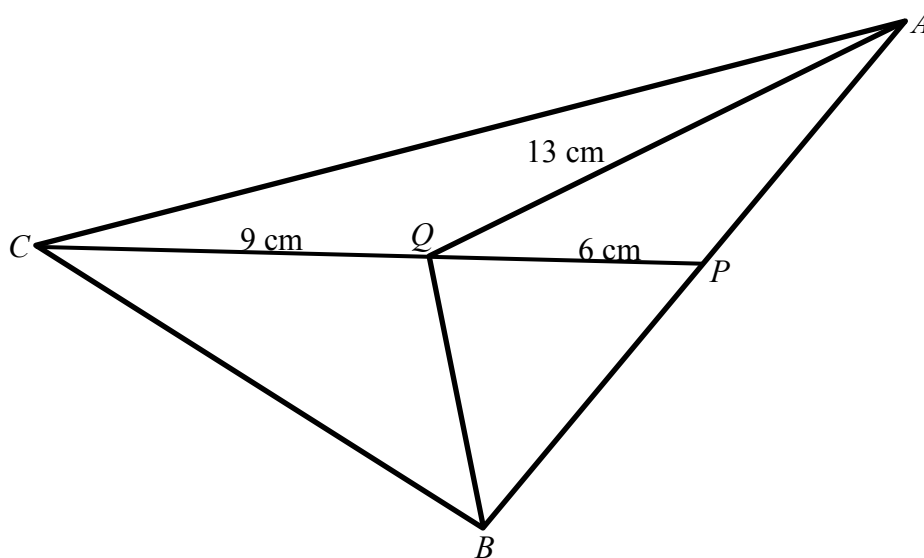
B back to bottle *A*, After the 2016th pouring, how much alcohol was left in bottle *A* ?

Solution

	Bottle <i>A</i> (litre)		Bottle <i>B</i> (litre)
Originally	1		0
After the 1st pouring	0	→	1
After the 2nd pouring	$\frac{1}{2}$	←	$\frac{1}{2}$
After the 3rd pouring	$\frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$	→	$1 - \frac{1}{3} = \frac{2}{3}$
After the 4th pouring	$1 - \frac{1}{2} = \frac{1}{2}$	←	$\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$
After the 5th pouring	$\frac{1}{2} \times \frac{4}{5} = \frac{2}{5}$	→	$1 - \frac{2}{5} = \frac{3}{5}$
After the 6th pouring	$1 - \frac{1}{2} = \frac{1}{2}$	←	$\frac{3}{5} \times \frac{5}{6} = \frac{1}{2}$
After the 7th pouring	$\frac{1}{2} \times \frac{6}{7} = \frac{3}{7}$	→	$1 - \frac{3}{7} = \frac{4}{7}$
After the 8th pouring	$1 - \frac{1}{2} = \frac{1}{2}$	←	$\frac{4}{7} \times \frac{7}{8} = \frac{1}{2}$
...			
After the 2016th pouring	$1 - \frac{1}{2} = \frac{1}{2}$	←	$\frac{1008}{2015} \times \frac{2015}{2016} = \frac{1}{2}$

2. 圖一顯示 $\triangle ABC$ ，*P* 為 *AB* 的中點及 *Q* 是 *CP* 上的一點。已知 $BQ \perp CP$ ， $PQ = 6$ cm、 $CQ = 9$ cm 及 $AQ = 13$ cm。求 $\triangle ABC$ 的面積。〔答：75〕

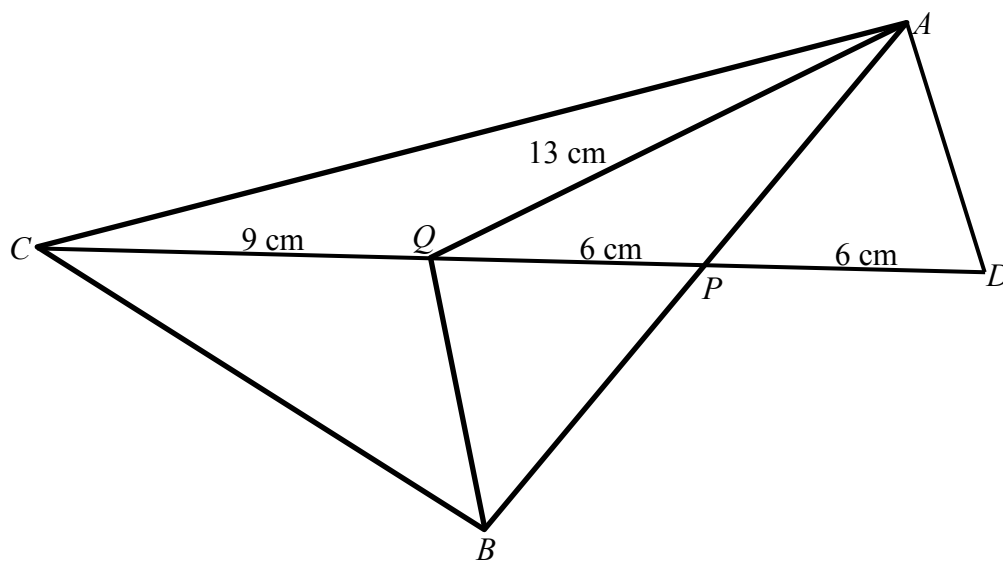
Figure 1 shows $\triangle ABC$, P is the mid-point of AB and Q is a point on CP . It is known that $BQ \perp CP$, $PQ = 6$ cm, $CQ = 9$ cm and $AQ = 13$ cm. Find the area of $\triangle ABC$.



圖一

Figure 1

Solution



Produce QP to D such that $PD = 6$ cm.

$\triangle APD \cong \triangle BPQ$ (SAS)

$\therefore \angle D = 90^\circ$

$\therefore AD = 5$ cm

$\therefore \text{area of } \triangle ABC = (9 + 6) \times 5 = 75 \text{ cm}^2$.

3. 考慮數列 $a_1, a_2, a_3, \dots$ 。定義 $S_n = a_1 + a_2 + \dots + a_n$ 其中 n 為任何整數。若 $S_n = 2 - a_n - \frac{1}{2^{n-1}}$,

求 a_{2016} 的值。〔答： $\frac{63}{2^{2011}}$ 〕

Consider a sequence of numbers $a_1, a_2, a_3, \dots$. Define $S_n = a_1 + a_2 + \dots + a_n$ for any positive integer n . Find the value of a_{2016} if $S_n = 2 - a_n - \frac{1}{2^{n-1}}$.

Solution

$$S_n = 2 - a_n - \frac{1}{2^{n-1}}$$

$$S_n - S_{n-1} = \left(2 - a_n - \frac{1}{2^{n-1}}\right) - \left(2 - a_{n-1} - \frac{1}{2^{n-2}}\right)$$

$$a_n = -a_n + a_{n-1} + \frac{1}{2^{n-1}}$$

$$2a_n = a_{n-1} + \frac{1}{2^{n-1}}$$

$$2^n a_n = 2^{n-1} a_{n-1} + 1$$

$\therefore \{2^n a_n\}$ is an arithmetic sequence with common difference 1.

$$\text{Consider } S_1 = 2 - a_1 - 1$$

$$a_1 = 1 - a_1$$

$$2a_1 = 1$$

$$\therefore 2^{2016} a_{2016} = 2a_1 + 2015(1)$$

$$= 2016$$

$$a_{2016} = \frac{2^5 \cdot 63}{2^{2016}}$$

$$= \frac{63}{2^{2011}}$$

4. 設 x 及 y 為正整數且滿足 $\log x + \log y = \log(2x - y) + 1$, 求 (x, y) 的數量。〔答：6〕

If x and y are positive integers that satisfy $\log x + \log y = \log(2x - y) + 1$, find the number of possible pairs of (x, y) .

Solution

$$\log x + \log y = \log(2x - y) + 1$$

$$\log xy = \log 10(2x - y)$$

$$xy - 20x + 10y = 0$$

$$(x+10)(y-20) = -200$$

$$(x+10)(20-y) = 200$$

$$\therefore x, y \in \mathbf{Z}^+$$

$$x+10 \text{ and } 20-y \in \mathbf{Z}^+$$

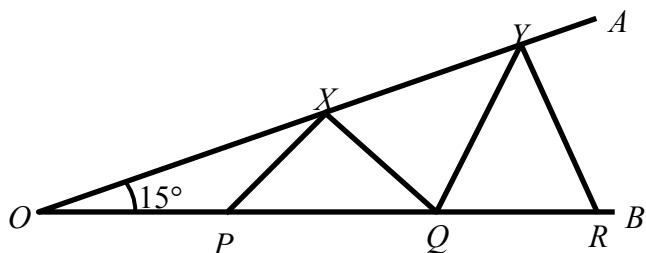
$$0 < 20-y < 20 \text{ and } (20-y) | 200$$

$$20-y = 1, 2, 4, 5, 8, 10$$

$\therefore$ There are 6 pairs of (x, y) .

5. 圖二中， $\angle AOB = 15^\circ$ 。X、Y 是 OA 上的點，P、Q、R 是 OB 上的點使得 $OP = 1$ 及 $OR = 3$ 。若 $s = PX + XQ + QY + YR$ ，求 s 的最小值。〔答： $\sqrt{7}$ 〕

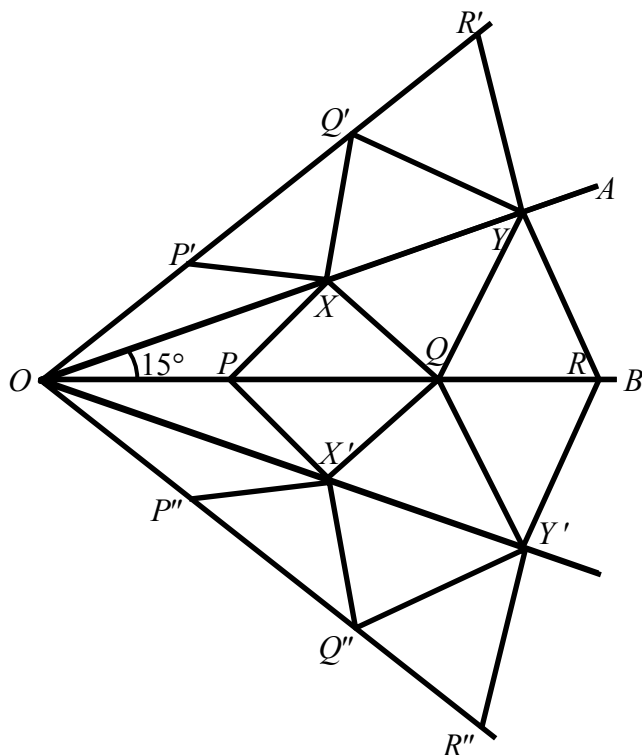
In Figure 2, $\angle AOB = 15^\circ$, X, Y are points on OA, P, Q, R are points on OB such that $OP = 1$ and $OR = 3$. If $s = PX + XQ + QY + YR$, find the least value of s.



圖二

Figure 2

Solution



Reflect the given figure in the line segment OA and then OB as shown.

$$\begin{aligned} s_{\min} &= P'R'' \\ &= \sqrt{1^2 + 3^2 - 2(1)(3)\cos 60^\circ} \\ &= \sqrt{7} \end{aligned}$$

6. 設 $y = px^2 + qx + r$ 為一二次函數。已知

(1) y 的對稱軸為 $x = 2016$

(2) 該函數的圖像通過 x 軸於 A 、 B 兩點，其中 $AB = 4$ 單位

(3) 該函數的圖像通過直線 $y = -10$ 於 C 、 D 兩點，其中 $CD = 16$ 單位

求 q 的值。〔答：672〕

Let $y = px^2 + qx + r$ be a quadratic function. It is known that

(1) The axis of symmetry of y is $x = 2016$.

(2) The curve cuts the x -axis at two points A and B such that $AB = 4$ units.

(3) The curve cuts the line $y = -10$ at two points C and D such that $CD = 16$ units.

Find the value of q .

Solution

The roots of $y = 0$ are 2014 and 2018

$$\therefore y = p(x - 2014)(x - 2018)$$

When $y = -10$, $x = 2008$ or 2024

$$\therefore -10 = p(2024 - 2014)(2024 - 2018)$$

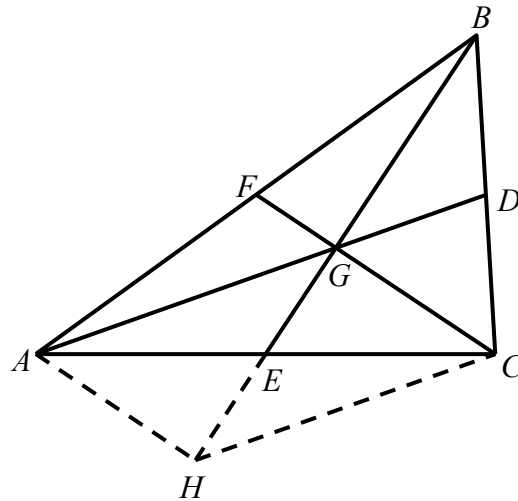
$$\therefore p = -\frac{1}{6}$$

$$\text{Since } 2016 = -\frac{q}{2p}, q = 672.$$

7. 設三角形三條中線的長度為 9、12 及 15。求該三角形的面積。〔答：72〕

The lengths of the three medians of a triangle are 9, 12 and 15. Find the area of the triangle.

Solution



Refer to the above figure, AD , BE and CF are medians of $\triangle ABC$ and their lengths are 15, 12 and 9 respectively. G is the centroid of $\triangle ABC$. Let H be a point on BE produced such that $GE = HE$. Then, $AHCG$ is a parallelogram. (Diagonals bisect each other.)

$$AG = \frac{2}{3}(15) = 10$$

$$GE = \frac{1}{3}(12) = 4$$

$$GH = 2(4) = 8$$

$$AH = CG = \frac{2}{3}(9) = 6$$

Obviously, $\triangle AHG$ is a right angled triangle.

$$\begin{aligned}
& \text{Area of } \triangle ABC \\
&= 6 \times \text{Area of } \triangle AGE \\
&= 6 \times \frac{1}{2} \times \text{Area of } \triangle AHG \\
&= 6 \times \frac{1}{2} \times \frac{6 \times 8}{2} \\
&= 72
\end{aligned}$$

8. 若某正整數的二進位表示有以下特質：

- (1) 有 11 個位，
 - (2) 有六個位是 1，有五個位是零，
- 則稱該數為「好數」。

(例如：2016 是一個「好數」，因為 $2016 = 11111100000_2$ 。)

求所有「好數」的和。〔答：386946〕

If the binary representation of a positive integer has the following properties:

- (1) the number of digits = 11,
- (2) the number of 1's = 6 and the number of 0's = 5,

then the number is said to be a “good number”.

(For example, 2016 is a “good number” as $2016 = 11111100000_2$.)

Find the sum of all “good numbers”.

Solution

The leftmost digit must be 1. Thus there are 5 1's and 5 0's in the remaining ten digits.

The number of “good numbers” = ${}_{10}C_5 = 252$.

We can pair up the 252 “good numbers” in the following way:

$$11111100000 \leftrightarrow 10000011111$$

$$11010101010 \leftrightarrow 10101010101$$

and so on.

The sum of two numbers in each pair = $2(2^{10}) + [2^9 + 2^8 + \dots + 2 + 1] = 3(2^{10}) - 1$.

The answer = $126[3(2^{10}) - 1] = 386946$.

9. 設整數 a 、 b 及 c 為三角形的邊長。已知 $f(x) = x(x-a)(x-b)(x-c)$ ，且 x 為一個大於 a 、 b 及 c 的整數。若 $x = (x-a) + (x-b) + (x-c)$ 及 $f(x) = 900$ ，求該三角形三條垂高的總和。

〔答： $\frac{281}{13}$ 〕

Let the three sides of a triangle are of lengths a , b and c where all of them are integers. Given that $f(x) = x(x-a)(x-b)(x-c)$ where x is an integer of size greater than a , b and c . If $x = (x-a) + (x-b) + (x-c)$ and $f(x) = 900$, find the sum of the lengths of the three altitudes of this triangle.

Solution

$$\therefore x = (x-a) + (x-b) + (x-c)$$

$$\therefore x = \frac{a+b+c}{2} \quad (1)$$

$$\therefore f(x) = 900$$

$$\therefore x(x-a)(x-b)(x-c) = 900$$

$$\sqrt{x(x-a)(x-b)(x-c)} = 30 \quad (2)$$

Equations (1) and (2) together satisfy the conditions of the Heron's Formula, therefore, by simple deduction, we have

$$x(x-a)(x-b)(x-c) = 15 \times 10 \times 3 \times 2$$

Thus a , b and c should be 5, 12 and 13, which means the triangle is a right-angled triangle.

$$\therefore \text{The required sum} = 5 + 12 + \frac{5 \times 12}{13} = \frac{281}{13}$$

10. 求 $\frac{1^4 + 2015^4 + 2016^4}{1^2 + 2015^2 + 2016^2}$ 的值。〔答：4,062,241〕

Find the value of $\frac{1^4 + 2015^4 + 2016^4}{1^2 + 2015^2 + 2016^2}$.

Solution

Let $a = 2015$

$$\begin{aligned} \frac{1^4 + 2015^4 + 2016^4}{1^2 + 2015^2 + 2016^2} &= \frac{1^4 + a^4 + (a+1)^4}{1^2 + a^2 + (a+1)^2} \\ &= \frac{2a^4 + 4a^3 + 6a^2 + 4a + 2}{2a^2 + 2a + 2} \\ &= \frac{a^4 + 2a^3 + 3a^2 + 2a + 1}{a^2 + a + 1} \\ &= \frac{(a^2 + a + 1)^2}{a^2 + a + 1} \\ &= a^2 + a + 1 \\ &= 2015^2 + 2015 + 1 \\ &= (2015)(2016) + 1 \\ &= 4062241 \end{aligned}$$

Hong Kong Mathematics Olympiad (2015 / 2016)

Final Event (Individual) Solution

香港數學競賽 (2015 / 2016)

決賽項目(個人) 題解

Event 1

1. Note that

$$\begin{aligned}\frac{\log a}{\log 5} + \frac{\log a}{\log 3} &= \frac{\log a}{\log 5} \cdot \frac{\log a}{\log 3} \\ (\log 3 + \log 5) \log a &= \log^2 a \\ \log 15 &= \log a \\ a &= 15\end{aligned}$$

2. Note that

$$\begin{aligned}b &= \left(\sqrt{8+\sqrt{15}} + \sqrt{8-\sqrt{15}} \right)^2 \\ &= 8 + \sqrt{15} + 2\sqrt{8+\sqrt{15}} \cdot \sqrt{8-\sqrt{15}} + 8 - \sqrt{15} \\ &= 16 + 2\sqrt{8^2 - 15} \\ &= 30\end{aligned}$$

3. The roots have product equal to 30. There are only two possibilities. 5 and 6 or -5 and -6 . The maximum value of c is 11.

4. Let $d = 100x + 10y + z$ with $z \neq 0$. Then $11 \mid (100x + 10y + z + 100z + 10y + x)$ which is the same as $11 \mid 101x + 20y + 101z$.

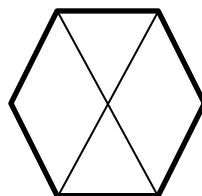
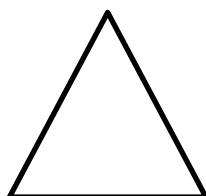
Reducing the coefficients modulo 11 gives $11 \mid 2x - 2y + 2z$.

As 2 and 11 are relatively prime it follows that $11 \mid x + z - y$.

A quick observation for the largest possible value for d is 979.

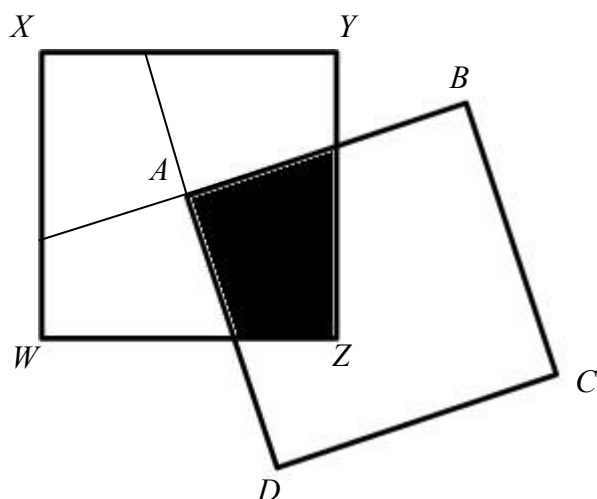
Event 2

1. Note that the figures below have perimeter ratio 1 : 1 and it is clear that the area ratio of the equilateral triangle to the regular hexagon is 4 : 6 or 2 : 3.



$$\begin{aligned}
2. \quad b &= \left(\log_2(a^2) + \log_4\left(\frac{1}{a^2}\right) \right) \times \left(\log_a(2) + \log_{a^2}\left(\frac{1}{2}\right) \right) \\
&= \left(\frac{\log 3^2}{\log 2} + \frac{\log \frac{1}{3^2}}{\log 4} \right) \left(\frac{\log 2}{\log 3} + \frac{\log \frac{1}{2}}{\log 3^2} \right) \\
&= \left(\frac{2\log 3}{\log 2} + \frac{(-2)\log 3}{2\log 2} \right) \left(\frac{\log 2}{\log 3} + \frac{(-1)\log 2}{2\log 3} \right) \\
&= \left(\frac{2\log 3}{\log 2} \right) \left(\frac{\log 2}{\log 3} \right) + \left(\frac{2\log 3}{\log 2} \right) \left(\frac{(-1)\log 2}{2\log 3} \right) + \left(\frac{(-2)\log 3}{2\log 2} \right) \left(\frac{\log 2}{\log 3} \right) + \left(\frac{(-2)\log 3}{2\log 2} \right) \left(\frac{(-1)\log 2}{2\log 3} \right) \\
&= \frac{1}{2}
\end{aligned}$$

3. Note that by extending the sides AD to XY and AB to XW the $XYZW$ is partitioned into four identical quadrilaterals of which one is the shaded region. Hence $c = 4$.



4. Since $456 \times 4 = 76 \times d$ we have $d = 24$.

Event 3

1. Note that $(x-1)f(x) = x^5 - 1$ and

$$\begin{aligned}
f(x^5) &= (x^5)^4 + (x^5)^3 + (x^5)^2 + (x^5) + 1 \\
&= x^{20} + x^{15} + x^{10} + x^5 + 1 \\
&= (x^{20} - 1) + (x^{15} - 1) + (x^{10} - 1) + (x^5 - 1) + 4 + 1
\end{aligned}$$

As $x^5 - 1$ divides $x^{5m} - 1$ for any positive integer m it follows that the remainder r is 5.

2. Since 2, 3 and 5 all divide $n^5 - n$ it follows that 30 divides $n^5 - n$ and the remainder r is $b = 0$.

$$\begin{aligned}
 3. \quad c &= \left(\frac{\sqrt{1+x}}{\sqrt{1+x}-\sqrt{1-x}} + \frac{1-x}{\sqrt{1-x^2}+x-1} \right) \times \left(\sqrt{\frac{1}{x^2}-b^2} - 1 - \frac{1}{x-b} \right) \\
 &= \left(\frac{\sqrt{1+x}}{\sqrt{1+x}-\sqrt{1-x}} + \frac{1-x}{\sqrt{1-x^2}+x-1} \right) \times \left(\sqrt{\frac{1}{x^2}-1} - \frac{1}{x} \right) \\
 &= \frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}-\sqrt{1-x}} \times \frac{\sqrt{1-x^2}-1}{x} \\
 &= \frac{(\sqrt{1+x})^2 - (\sqrt{1-x})^2}{(\sqrt{1+x}-\sqrt{1-x})^2} \times \frac{\sqrt{1-x^2}-1}{x} \\
 &= \frac{2x}{2-2\sqrt{1-x^2}} \times \frac{\sqrt{1-x^2}-1}{x} \\
 &= -1
 \end{aligned}$$

4. Note that the equation can be written as $(x-2y)^2 = xy$. So it follows that

$$x^2 - 5xy + 4y^2 = 0 = (x-y)(x-4y). \text{ This implies that } x-y=0 \text{ or } x-4y=0.$$

As $x=y$ implies that $x-2y < 0$ a contradiction it must be the case that $x-4y=0$ and hence

$$d = \frac{x}{y} = 4.$$

Event 4

$$\begin{aligned}
 1. \quad a &= \log_2 \left(\left(\frac{m^4 n^{-4}}{m^{-1} n} \right)^{-3} \div \left(\frac{m^{-2} n^2}{mn^{-1}} \right)^5 \right) \\
 &= \log_2 \left((m^5 n^{-5})^{-3} \div (m^{-3} n^3)^5 \right) \\
 &= \log_2 (m^{-15} n^{15} \div m^{-15} n^{15}) \\
 &= \log_2 (1) \\
 &= 0
 \end{aligned}$$

2. We compute that

$$1453 - 1108 = 345; 1844 - 1453 = 391; 2281 - 1844 = 437; 437 - 291 = 391 - 345 = 46 = 2 \times 23$$

As $(N_1 d + r) - (N_2 d + r) = d(N_1 - N_2)$ and 2 cannot be the divisor as it divides 1108 but not 1453 the divisor $d = 23$ and hence the remainder $b = 4$.

3. Notice that $c = \sqrt{x^2 - 2x + 1} + \sqrt{x^2 - 6x + 9} = (x-1) + (3-x) = 2$.

4. We are interested in knowing what the fraction $\frac{1+3^2+(3^2)^2+(3^2)^3+(3^2)^4}{1+3+3^2+3^3+3^4} = \frac{(3^2)^{4+1}-1}{3^2-1} \times \frac{3-1}{3^{4+1}-1}$

produces for a remainder. This remainder r is 0 if 3^2-1 and $3^{4+1}-1$ both divide $(3^2)^{4+1}-1$ and they do not share divisors other than $3-1$. This is precisely the case as the exponents 2 and $4+1$ are relatively prime. Thus the remainder r is 0.

Hong Kong Mathematics Olympiad (2015 / 2016)

Final Event (Group) Solution

香港數學競賽 (2015 / 2016)

決賽項目(團體) 題解

Event 1

1. Project Schedule

# of Days	1	2	3	4	5	6	7	8	9	10	11	12
<i>A</i>												
<i>B</i>												
<i>C</i>												

$$\therefore P = 12.$$

2. Both red and green lights blink every 12 seconds, however, all lights blink every 24 seconds. So, both red and yellow lights blink but not the green light at $12 + 24(n-1)$, $n = 1, 2, 3, \dots$. Hence, $Q = 60$.

3. It is easy to see that $f_{4r-2} < f_{4r} < f_{4r-3} < f_{4r-1}$. Since $f_2 = 58$ and $f_{2r} - f_{2r-2} = 1$, it follows that $r = 63 - 58 = 5$ and thus $f_{12} = 63$. Therefore, $f_n \geq 63$, for $n \geq 12$. Also, $f_{12} < f_{11}$. Hence, $n = 11$.

Sequence: 60 58 61 59 62 60 63 61 64 62 65 63 66 64 67

4. Note that $1 = \frac{1}{2}(3^{2^0} - 1)$. Then

$$\begin{aligned}
 T &= (3^{2^0} + 1) \times (3^{2^1} + 1) \times (3^{2^2} + 1) \times \cdots \times (3^{2^{10}} + 1) \\
 &= \frac{1}{2}(3^{2^0} - 1)(3^{2^0} + 1) \times (3^{2^1} + 1) \times (3^{2^2} + 1) \times \cdots \times (3^{2^{10}} + 1) \\
 &= \frac{1}{2}(3^{2^1} - 1)(3^{2^1} + 1) \times (3^{2^2} + 1) \times \cdots \times (3^{2^{10}} + 1) \\
 &= \frac{1}{2}(3^{2^{10}} - 1)(3^{2^{10}} + 1) \\
 &= \frac{1}{2}(3^{2^{11}} - 1)
 \end{aligned}$$

Event 2

1. The number of combinations is $\binom{5}{2} = 10$. As $3+10=4+9=13$, the number of possible sums is 9.

2. Starting with $f_1 = 9$, $f_{3r+1} = 9+r$, $f_{3r-1} = 7+r$ and $f_{3r} = 10+r$.
 For the case $3r+1$, $r=1$ for which $f_{3r+1} < 11$.
 For the case $3r-1$, $r=3$ for which $f_{3r-1} < 11$.
 For the case $3r$, $f_{3r} \geq 11$ for any $r > 0$.
 Therefore, $B = 2 + 3 = 5$.
 Sequence: 9 8 11 10 9 12 11 10 13 12 11 14 13 12 15 14

3. Consider $\mathbf{a} = (a_1, a_2, a_3, a_4, a_5, a_6)$, $f(\mathbf{a}) = a_1 + a_2 + a_3 + a_4 + a_5 + a_6$ and
 $g(\mathbf{a}) = a_1 + 2a_2 + 3a_3 + 4a_4 + 5a_5 + 6a_6$. Under the condition that $f(\mathbf{a}) = 5$ and $g(\mathbf{a}) = 26$, then the solution $\mathbf{a}$ also satisfies $h(a) = 5a_1 + 4a_2 + 3a_3 + 2a_4 + a_5 = 4$ and $a_6 = 5 - a_1 - a_2 - a_3 - a_4 - a_5$.
 Starting with $a_5 = 4$, $(a_1, a_2, a_3, a_4, a_6) = (0, 0, 0, 0, 1)$ is the unique solution.
 For $a_5 = 3$, there is no solution.
 For $a_5 = 2$, $(a_1, a_2, a_3, a_4, a_6) = (0, 0, 0, 1, 2)$ is the unique solution.
 For $a_5 = 1$, $(a_1, a_2, a_3, a_4, a_6) = (0, 0, 1, 0, 3)$ is the unique solution.
 For $a_5 = 0$, $(a_1, a_2, a_3, a_4, a_6) = ((0, 1, 0, 0, 4), (0, 0, 0, 2, 3))$ are solutions.
 So, the number of possible ways is 5, i.e.,
 $a = \{(0, 1, 0, 0, 0, 4), (0, 0, 1, 0, 1, 3), (0, 0, 0, 2, 0, 3), (0, 0, 0, 1, 2, 2), (0, 0, 0, 0, 4, 1)\}$.

4. Note that $\frac{3^d}{f} = \prod_{i=1}^4 a_i = (0.9)^{15} = \frac{3^{30}}{10^{15}}$. Therefore, $d = 30$.

Event 3

1. The distance traveled from point X to point Y through points $A_1, A_2, \dots, A_k$ and then return to point X is equal to twice the number of edges, $2K+2$. Since there are 12 edges in the graph, $K = 2 \times 12 \times 10 \div 60 = 4$ hours.

2. Sequence:

$$a_1 = \frac{4}{5} = \frac{8}{10}$$

$$a_1 \times a_2 = \frac{8 \times 8^2}{10^3} = \frac{512}{1,000}$$

$$a_1 \times a_2 \times a_3 = \frac{8 \times (8^3)^2}{10^7} = \frac{2,097,152}{10,000,000} < 0.3$$

Hence, $L = 3$.

3. Note that $\sqrt[3]{5+\sqrt{x}} + \sqrt[3]{5-\sqrt{x}} = 1$ can be rewritten as $\left(\sqrt[3]{5+\sqrt{x}} + \sqrt[3]{5-\sqrt{x}}\right)^3 = 1$, which gives

$$5 + \sqrt{x} + 3(5 + \sqrt{x})^{\frac{2}{3}}(5 - \sqrt{x})^{\frac{1}{3}} + 3(5 + \sqrt{x})^{\frac{1}{3}}(5 - \sqrt{x})^{\frac{2}{3}} + 5 - \sqrt{x} = 1$$

$$10 + 3(5 + \sqrt{x})^{\frac{1}{3}}(5 - \sqrt{x})^{\frac{1}{3}}\left((5 + \sqrt{x})^{\frac{1}{3}} + (5 - \sqrt{x})^{\frac{1}{3}}\right) = 1$$

Since $(5 + \sqrt{x})^{\frac{1}{3}} + (5 - \sqrt{x})^{\frac{1}{3}} = 1$, it follows that

$$10 + 3(5 + \sqrt{x})^{\frac{1}{3}}(5 - \sqrt{x})^{\frac{1}{3}} = 1 \text{ or equivalently } (25 - x)^{\frac{1}{3}} = -3,$$

which means that $x = 52$.

4. As $a + b + y = 5$, it follows that $a + b = 5 - y$ which implies that $(a + b)^2 = (5 - y)^2$. Further, since

$$ab + by + ay = 3, \text{ it follows that } ab = 3 - by - ay = 3 - (a + b)y = 3 - (5 - y)y.$$

$$\text{Since } 0 \leq (a - b)^2 = (a + b)^2 - 4ab$$

$$= (5 - y)^2 - 4[3 - (5 - y)y]$$

$$= -3y^2 + 10y + 13$$

$$= (13 - 3y)(1 + y)$$

we get $-1 \leq y \leq \frac{13}{3}$.

Indeed, when $a = \frac{1}{3} = b$, we get $y = \frac{13}{3}$.

Event 4

1. Note that the difference the two squares is even implies that the integers a and b must have the same parity. Now, $144 = a^2 - b^2 = (a+b)(a-b) = 144 \times 1 = 72 \times 2$.

Since $a+b=144$ entails that $a-b=1$, which does not meet the requirement, it follows that $d = a+b = 72$.

2. Let $n = 10x + y$, where x and y are digits. Then $n^2 = 100x^2 + 20xy + y^2 = 20z + y^2$.

As the tens digit of n^2 is 7, it follows that the tens digit of y^2 must be odd. The only possibilities are that $y^2 = 16$ or 36. In either case, we have $u = 6$.

3. Note that

$$\begin{aligned}
 & \frac{(4 + \sqrt{15})^{\frac{3}{2}} + (4 - \sqrt{15})^{\frac{3}{2}}}{(6 + \sqrt{35})^{\frac{3}{2}} - (6 - \sqrt{35})^{\frac{3}{2}}} \times \frac{2^{\frac{3}{2}}}{2^{\frac{3}{2}}} \\
 &= \frac{(8 + 2\sqrt{15})^{\frac{3}{2}} + (8 - 2\sqrt{15})^{\frac{3}{2}}}{(12 + 2\sqrt{35})^{\frac{3}{2}} - (12 - 2\sqrt{35})^{\frac{3}{2}}} \\
 &= \frac{(5 + 2\sqrt{15} + 3)^{\frac{3}{2}} + (5 - 2\sqrt{15} + 3)^{\frac{3}{2}}}{(7 + 2\sqrt{35} + 5)^{\frac{3}{2}} - (7 - 2\sqrt{35} + 5)^{\frac{3}{2}}} \\
 &= \frac{(\sqrt{5} + \sqrt{3})^3 + (\sqrt{5} - \sqrt{3})^3}{(\sqrt{7} + \sqrt{5})^3 - (\sqrt{7} - \sqrt{5})^3} \\
 &= \frac{2(5\sqrt{5} + 9\sqrt{5})}{2(21\sqrt{5} + 5\sqrt{5})} \\
 &= \frac{7}{13}
 \end{aligned}$$

4. The above expression for x is the same as $x = 1 + \frac{1}{x}$. Finding x amounts to solving the quadratic equation $x^2 - x - 1 = 0$, which has roots $x = \frac{1+\sqrt{5}}{2}$ or $x = \frac{1-\sqrt{5}}{2}$. As we look for the positive value of x , $x = \frac{1+\sqrt{5}}{2}$.

Hong Kong Mathematics Olympiad (2016 / 2017)

Heat Event (Individual) Solution

香港數學競賽 (2016 / 2017)

初賽項目(個人) 題解

甲部

Part A

1. 已知 $A2017B$ 是一個六位數，且可被 72 整除，求 A 的值。〔答：2〕

Given that $A2017B$ is a 6-digit number which is divisible by 72, find the value of A .

Solution

$A2017B$ is divisible by 72 means it is divisible by 8 and 9 respectively.

$\Rightarrow 17B$ is divisible by 8

$\Rightarrow B = 6$

$\Rightarrow A+2+0+1+7+6$ should be divisible by 9

$\therefore A = 2$

2. 已知 $0 \leq p \leq 1$ ，求 $Q = 3p^2(1-p) + 6p(1-p)^2 + 3(1-p)^3$ 的最大值？〔答：3〕

Given that $0 \leq p \leq 1$, find the greatest value of $Q = 3p^2(1-p) + 6p(1-p)^2 + 3(1-p)^3$.

Solution

$$\begin{aligned} Q &= 3p^2(1-p) + 6p(1-p)^2 + 3(1-p)^3 \\ &= (1-p)(3p^2 + 6p - 6p^2 + 3 - 6p + 3p^2) \\ &= 3(1-p) \end{aligned}$$

3. 已知 $\triangle ABC$ 的三條邊的長是 a 、 b 和 c ，其中 $3 \leq a \leq 5 \leq b \leq 12 \leq c \leq 15$ ，求當 $\triangle ABC$ 的面積最大時，它的周界是多少。〔答：30〕

Given that the three sides of $\triangle ABC$ are of length a , b and c , where $3 \leq a \leq 5 \leq b \leq 12 \leq c \leq 15$, find the perimeter of $\triangle ABC$ when its area attains the maximum value.

Solution

$$\text{Area of } \triangle ABC = \frac{1}{2}ab \sin C \leq \frac{1}{2}(5)(12) \sin C \leq 30.$$

When area of $\triangle ABC$ attains its maximum value, i.e., 30, $a = 5$, $b = 12$ and the angle between these two sides is a right angle (90°).

Therefore $\triangle ABC$ is a right-angled triangle and $c = \sqrt{5^2 + 12^2} = 13$.

Consequently the perimeter of $\triangle ABC$ is $5 + 12 + 13 = 30$.

4. 設 B 及 C 為正整數。求 C 的最小值使得 $B^2 = C + 134$ 。〔答：10〕
Let B and C be positive integers. Find the least value of C satisfying $B^2 = C + 134$.

Solution

$C > 0$. Therefore the closest value of C which will give a perfect square when added to 134 is 10, this will give $C + 134 = 10 + 134 = 144$, this means $B = 12$.

5. 若把一組自然數之和 $1 + 2 + 3 + \cdots + 2015 + 2016 + 2017$ 除以 9，餘數是什麼？〔答：1〕
Determine the remainder of the sum of natural numbers $1 + 2 + 3 + \cdots + 2015 + 2016 + 2017$ divided by 9.

Solution

$$1 + 2 + 3 + \cdots + 2015 + 2016 + 2017 = \frac{2017(1 + 2017)}{2} = 2035153$$

Since $2 + 0 + 3 + 5 + 1 + 5 + 3 = 19$ divided by 9 has remainder 1, the answer is 1.

6. 已知 $a_0 = 2$ ， $a_1 = -1$ 及 $a_{n+1} = 2a_n - a_{n-1}$ ，其中 $n \geq 1$ ，求 a_{2017} 的值。〔答：-6049〕

Given that $a_0 = 2$ ， $a_1 = -1$ and $a_{n+1} = 2a_n - a_{n-1}$ ，where $n \geq 1$ ，determine the value of a_{2017} .

Solution

$a_2 = -2 - 2 = -4$ ， $a_3 = -8 + 1 = -7$. The said sequence is an arithmetic sequence with common difference = -3 . Thus $a_{2017} = -1 + 2016(-3) = -6049$.

7. 設 N 為完全立方數。已知 $N = 161x + 23y$ ，其中 x 和 y 均為正整數。求 $x + y$ 的最小值。〔答：79〕

Let N be a perfect cube number. Given that $N = 161x + 23y$ ，where x and y are positive integers, find the minimum value of $x + y$.

Solution

$$N = 23(7x + y)$$

From the question, N should be $23^3 = 529 \times 23$

$$529 = 7x + y$$

Let $x = t$ ，then $y = 529 - 7t$ ，where $t > 0$ and $529 - 7t > 0$ ，this gives $0 < t < 75.6$.

$x + y = t + 529 - 7t = 529 - 6t$. Therefore it attains the minimum value when $t = 75$ ，and the minimum value of $x + y = 79$

8. 已知 $\textcircled{2}=1\times 2\times 3\times 4$, $\textcircled{3}=2\times 3\times 4\times 5$, $\textcircled{4}=3\times 4\times 5\times 6$, ... 及 $\frac{1}{\textcircled{15}}-\frac{1}{\textcircled{17}}=\frac{1}{\textcircled{17}}\times A$, 求 A 的值。

[答 : $\frac{22}{35}$]

Given that $\textcircled{2}=1\times 2\times 3\times 4$, $\textcircled{3}=2\times 3\times 4\times 5$, $\textcircled{4}=3\times 4\times 5\times 6$, ... and $\frac{1}{\textcircled{15}}-\frac{1}{\textcircled{17}}=\frac{1}{\textcircled{17}}\times A$, find the value of A .

Solution

$$\begin{aligned}\frac{1}{\textcircled{17}}\times A &= \frac{1}{\textcircled{15}} - \frac{1}{\textcircled{17}} \\ A &= \frac{16\times 17\times 18\times 19}{14\times 15\times 16\times 17} - 1 \\ &= \frac{57}{35} - 1 \\ &= \frac{22}{35}\end{aligned}$$

9. 已知 $\sin x \cdot \cos x = 0$ 及 $\sin^3 x - \cos^3 x = 1$, 其中 $90^\circ \leq x < 180^\circ$, 求 x 的值。 [答 : 90°]

Given that $\sin x \cdot \cos x = 0$ and $\sin^3 x - \cos^3 x = 1$, where $90^\circ \leq x < 180^\circ$, find the value of x .

Solution

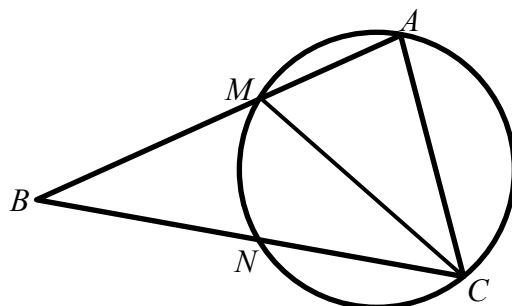
$$\text{Since } \sin^3 x - \cos^3 x = (\sin x - \cos x)(1 + \sin x \cos x)$$

$$\text{we have } \sin x - \cos x = 1$$

$$\text{Since } 90^\circ \leq x < 180^\circ, \text{ we have } x = 90^\circ$$

10. 如圖一， CM 是 $\angle ACB$ 的角平分線，且 $AB = 2AC$ 。已知 $\triangle AMC$ 的外接圓與 BC 相交於 N 。若 $BN = 10$ ，求 AM 的長度。 [答 : 5]

In Figure 1, CM is the angle bisector of $\angle ACB$ and $AB = 2AC$. Given that the circumscribed circle of $\triangle AMC$ intersects BC at N . If $BN = 10$, find the length of AM .



圖一

Figure 1

Solution

$$\text{Let } \angle ACM = \theta \text{ and } \angle AMC = \beta.$$

$$\text{In } \triangle AMC, \text{ by sine formula, } \frac{AM}{AC} = \frac{\sin \theta}{\sin \beta}$$

In $\triangle BMC$, by sine formula, as CM is the angle bisector of $\angle ACB$,

$$\frac{BM}{BC} = \frac{\sin \theta}{\sin(180^\circ - \beta)} = \frac{\sin \theta}{\sin \beta}$$

$$\frac{AM}{AC} = \frac{BM}{BC} \quad \dots (1)$$

Since $BM \times BA = BN \times BC$

$$\frac{BM}{BC} = \frac{BN}{BA} \quad \dots (2)$$

From (1) and (2),

$$\frac{AM}{AC} = \frac{BN}{BA}$$

$$\frac{BN}{AM} = \frac{AB}{AC} = 2$$

$$\therefore BN = 10, AM = 5$$

乙部

Part B

11. 已知 x 為一實數，求 $\sqrt{x(x+3)(x+6)(x+9)+2017}$ 的最小值。〔答：44〕

Given that x is a real number, find the least value of $\sqrt{x(x+3)(x+6)(x+9)+2017}$.

Solution

$$\begin{aligned} & \sqrt{x(x+3)(x+6)(x+9)+2017} \\ &= \sqrt{(x^2+9x)(x^2+9x+18)+2017} \\ &= \sqrt{(x^2+9x)^2+18(x^2+9x)+2017} \\ &= \sqrt{(x^2+9x)^2+18(x^2+9x)+9^2-9^2+2017} \\ &= \sqrt{(x^2+9x+9)^2+1936} \end{aligned}$$

The least value of $\sqrt{x(x+3)(x+6)(x+9)+2017}$ is 44 when $x = \frac{-9 \pm 3\sqrt{5}}{2}$.

12. 已知 $\frac{x}{x^2-5x+1} = \frac{1}{2}$ ，求 $\frac{x^2}{x^4-5x^2+1}$ 的值。〔答： $\frac{1}{42}$ 〕

Given that $\frac{x}{x^2-5x+1} = \frac{1}{2}$, find the value of $\frac{x^2}{x^4-5x^2+1}$.

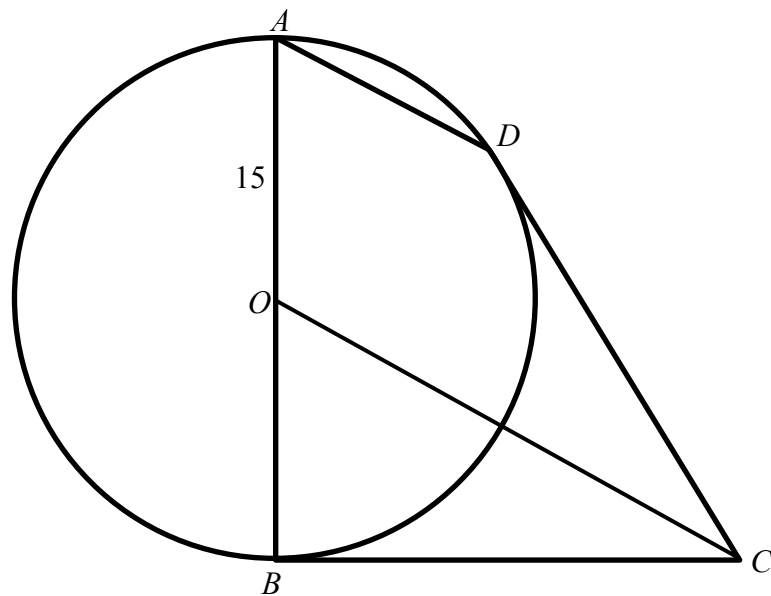
Solution

$$\begin{aligned}\frac{x}{x^2-5x+1} &= \frac{1}{2} \Rightarrow \frac{x^2-5x+1}{x} = 2 \\ x-5+\frac{1}{x} &= 2 \\ x+\frac{1}{x} &= 7 \\ \frac{x^4-5x^2+1}{x^2} &= x^2-5+\frac{1}{x^2} \\ &= \left(x+\frac{1}{x}\right)^2 - 2 - 5 \\ &= 49 - 7 = 42\end{aligned}$$

$$\therefore \frac{x^2}{x^4-5x^2+1} = \frac{1}{42}$$

13. 如圖二， O 是圓 ADB 的圓心。 BC 及 CD 分別是圓 O 在點 B 及 D 的切線， $OC \parallel AD$ ， $OA = 15$ 。若 $AD + OC = 43$ ，求 CD 的長。〔答：20 或 $3\sqrt{11}$ 〕

As shown in Figure 2, O is the centre of the circle ADB . BC and CD are tangents to O at points B and D respectively. $OC \parallel AD$, $OA = 15$. If $AD + OC = 43$, find the length of CD .



圖二
Figure 2

Solution

Since $\triangle ADB \sim \triangle OBC$, $\frac{AD}{OB} = \frac{AB}{OC} \Rightarrow AD \times OC = 450$.

Given that $AD + OC = 43$, this means the lengths of AD and OC are the roots of the quadratic equation $x^2 - 43x + 450 = 0$.
therefore $OC = 25$ or 18 .

$$\therefore CD = \sqrt{25^2 - 15^2} = 20 \text{ or } CD = \sqrt{18^2 - 15^2} = 3\sqrt{11}.$$

14. 若 $a + \log_2 b = a^2 + \log_2 b^3 - 10 = 3$ ，其中 $b > 1$ ，求 b 的值。〔答：16〕

If $a + \log_2 b = a^2 + \log_2 b^3 - 10 = 3$, where $b > 1$, find the value of b .

Solution

$$\begin{cases} a + \log_2 b = 3 \\ a^2 + \log_2 b^3 - 10 = 3 \end{cases} \Rightarrow \begin{cases} \log_2 b = 3 - a \\ a^2 + 3\log_2 b - 13 = 0 \end{cases}$$

We have $a^2 - 3a - 4 = 0$

$$(a - 4)(a + 1) = 0$$

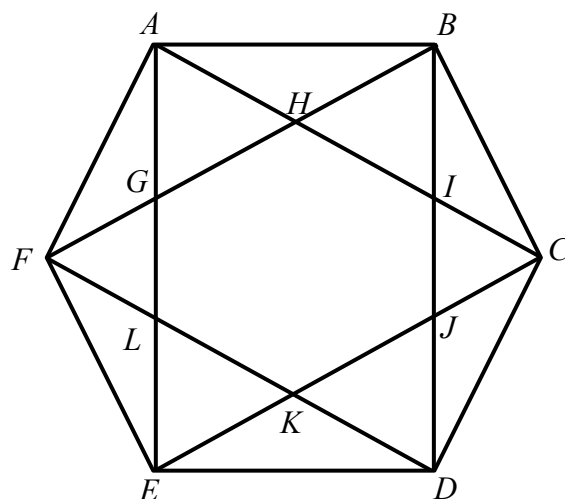
$$a = -1 \text{ or } 4$$

Correspondingly, we have $b = \frac{1}{2}$ or 16.

As $b > 1$, $b = 16$.

15. 在圖三中，已知 $ABCDEF$ 是一個正六邊形，且它的面積是 $90\sqrt{3}$ ，求 GJ 的值。〔答： $4\sqrt{5}$ 〕

In Figure 3, given that $ABCDEF$ is a regular hexagon and its area is $90\sqrt{3}$, find the length of GJ .



圖三

Figure 3

Solution

By finding $\angle IBC = 30^\circ$, we can easily obtain that $AH = HI = IC$.

Then the 3 areas $\triangle ABH$, $\triangle HBI$ and $\triangle IBC$ are equal.

Similarly, we can find all 18 small triangles shown below are equal,

$$\text{We have area } \triangle GOL = \frac{90\sqrt{3}}{18} = 5\sqrt{3}$$

Let GO be x .

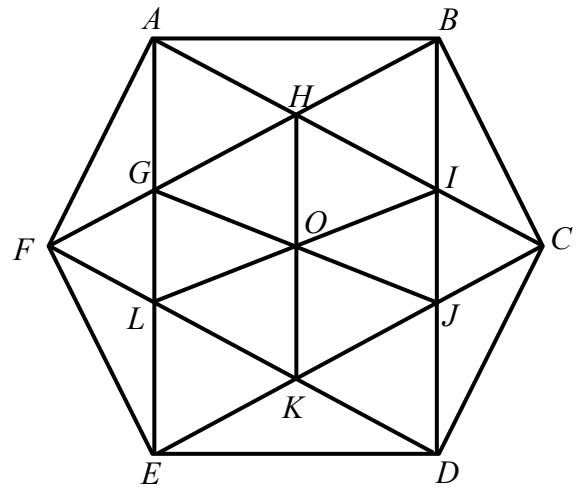
$$\frac{1}{2}x^2 \sin 60^\circ = 5\sqrt{3}$$

$$x = \sqrt{20}$$

Since $GO = OJ$,

Then,

$$GJ = GO + OJ = 2\sqrt{20} = 4\sqrt{5}$$



Hong Kong Mathematics Olympiad (2016 / 2017)
Heat Event (Group) Solution
香港數學競賽 (2016 / 2017)
初賽項目(團體) 題解

1. 設 $\triangle ABC$ 為一個等腰直角三角形，頂點 A 及 B 的坐標分別為 $(-2, 0)$ 及 $(18, 0)$ ，且 C 的坐標是正數。當 $\triangle ABC$ 的面積為最小時，求 C 的坐標。〔答：(8, 10)〕

Suppose that $\triangle ABC$ is an isosceles right-angled triangle with the coordinates of the vertices A and B as $(-2, 0)$ and $(18, 0)$, respectively, and the coordinates of C having positive values. Determine the coordinate of C when the area of $\triangle ABC$ attains its minimum.

Solution

Case (1): AB is the hypotenuse

$AB = 20$, the x -coordinate of $C = 8$, because $\angle CAB = 45^\circ$, and the coordinates of C are positive numbers, therefore, the y -coordinates of C is 10

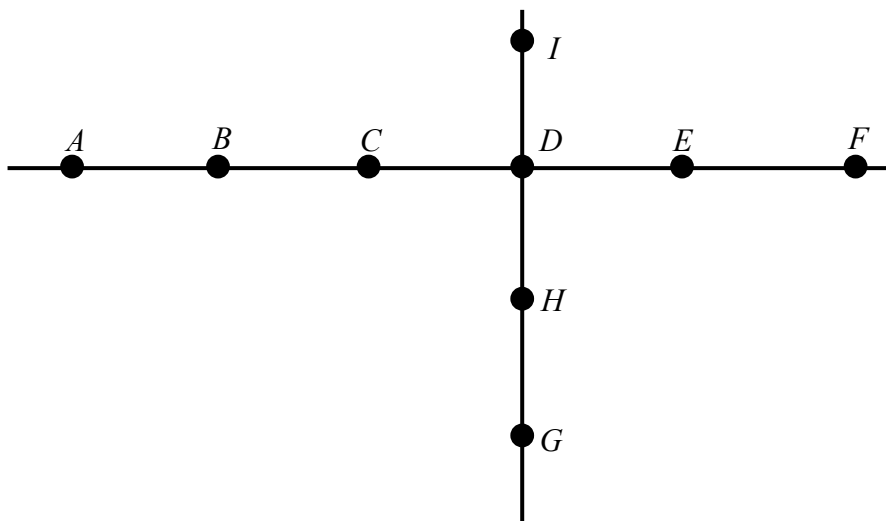
$$\text{The area of } \triangle ABC = \frac{20 \times 10}{2} = 100$$

Case (2): AB is adjacent to the right angle

$$AB = 20, \text{ the area of } \triangle ABC = \frac{20 \times 20}{2} = 200 > 100$$

2. 如圖一所示，點 A, B, C, D 及 E 均在一直線上，點 G, H, D 及 I 在另一直線上。揀選三點，可形成多少個三角形？〔答：60〕

As shown in Figure 1, points A, B, C, D, E and F lie on the same straight line, and points G, H, D , and I lie on another straight line. How many triangles can be made by connecting any three points?



圖一

Figure 1

Solution

$$\begin{aligned} C_2^6 \times 3 + C_2^3 \times 5 &= 15 \times 3 + 3 \times 5 \\ &= 45 + 15 \\ &= 60 \end{aligned}$$

3. 如圖二所示， P 、 Q 分別是正方形 $ABCD$ 的邊 BC 及 CD 上的點。已知 $\triangle PCQ$ 的周界的長等於正方形 $ABCD$ 的周界的長的 $\frac{1}{2}$ ，求 $\angle PAQ$ 。〔答：45 或 45° 〕

As shown in Figure 2, P , Q are points on the sides BC and CD of a square $ABCD$. Given that the perimeter of $\triangle PCQ$ is $\frac{1}{2}$ of that of the square $ABCD$, find $\angle PAQ$.

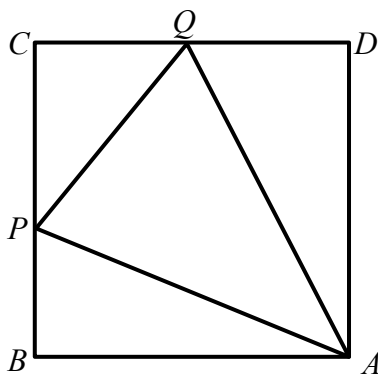


Figure 2

圖二

Solution

Rotate $\triangle ABP$ about A at 90° clockwise so that AB coincides with AD and $\triangle ABP \approx \triangle ADP'$.

Since $\angle ADP' = \angle ABP = 90^\circ$, we have $\angle QDP' = 180^\circ$, i.e.

Q , D and P' lie on the same straight line.

Consider $\triangle AQP'$ and $\triangle AQP$

$$AP' = AP$$

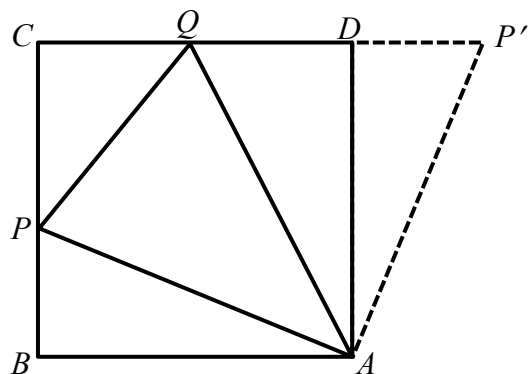
$$AQ = AQ$$

Because the perimeter of $\triangle PCQ$ is $\frac{1}{2}$ of that of the square

$ABCD = BC + CD$, we have $PQ = BP + QD = P'Q$

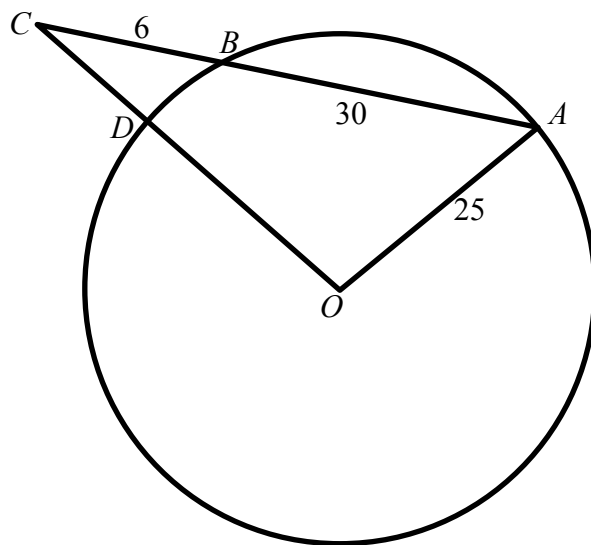
Therefore $\triangle AQP' \approx \triangle AQP$.

Consequently, $\angle PAQ = \angle QAP' = \frac{1}{2} \angle BAD = 45^\circ$.



4. 在圖三中， O 是圓心。弦 AB 及半徑 OD 的延線相交於 C 。已知 $OA=25$ 、 $AB=30$ 及 $BC=6$ 。求 CD 的長。〔答：4〕

In Figure 3, O is the centre of the circle. Chord AB and radius OD are produced to meet at C . Given that $OA=25$, $AB=30$ and $BC=6$. Find the length of CD .



圖三

Figure 3

Solution

Construct a perpendicular OM , where M is a point on AB .

$$AM = 15 \quad (\perp \text{ from centre bisects chord})$$

$$OM = 20 \quad (\text{Pythagoras theorem})$$

$$\begin{aligned} OC &= \sqrt{(15+6)^2 + 20^2} \quad (\text{Pythagoras theorem}) \\ &= 29 \end{aligned}$$

$$\begin{aligned} CD &= OC - OD \\ &= 4 \end{aligned}$$

5. 設 Q 為所有能滿足不等式 $\frac{9p^2}{(\sqrt{3p+1}-1)^2} < 3p+10$ 的整數 p 之和，求 Q 的值。〔答：10〕

Let Q be the sum of all integers p satisfying the inequality $\frac{9p^2}{(\sqrt{3p+1}-1)^2} < 3p+10$, find the value of Q .

Solution

Since $\frac{9p^2}{(\sqrt{3p+1}-1)^2} = (\sqrt{3p+1}+1)^2 = 3p+2+2\sqrt{3p+1}$, we have

$$\frac{9p^2}{(\sqrt{3p+1}-1)^2} < 3p+10$$

$$\Rightarrow 3p+2+2\sqrt{3p+1} < 3p+10$$

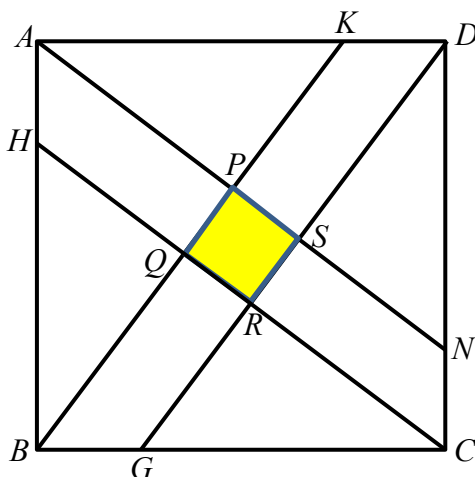
$$\Rightarrow \sqrt{3p+1} < 4$$

Corresponding values of p are 1, 2, 3 and 4.

$$\therefore Q = 1+2+3+4 = 10$$

6. 在圖四中，正方形 $ABCD$ 的邊長為 20。已知 $DK:KA=AH:HB=1:3$ 及 $BK \parallel GD$ ， $HC \parallel AN$ ，求陰影部分 $PQRS$ 的面積。〔答：16〕

In Figure 4, square $ABCD$ has sides of length 20. Given that $DK:KA=AH:HB=1:3$ and $BK \parallel GD$, $HC \parallel AN$, find the area of the shaded region $PQRS$.



圖四

Figure 4

Solution

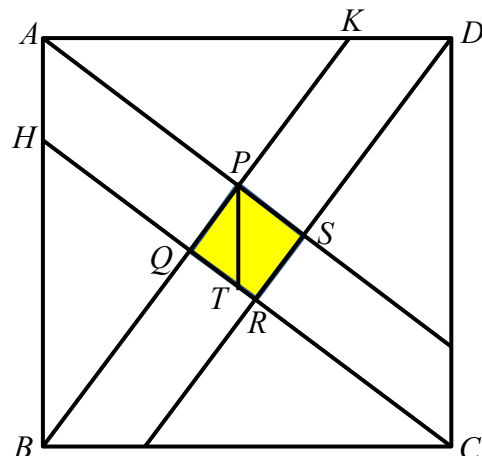
Draw $PT \parallel AB$, which will intersect the line HC at point T , then $\angle ABK = \angle QPT$.

Also, $\angle BAK = \angle PQR = 90^\circ$, therefore $\triangle ABK \sim \triangle QPT$.

$$AB = 20, BK = \sqrt{20^2 + 15^2} = 25, PT = AH = 5,$$

then the required area is

$$PQ^2 = \left(PT \times \frac{AB}{BK} \right)^2 = \left(5 \times \frac{20}{25} \right)^2 = 16$$



7. 已知對於實數 $x_1, x_2, x_3, \dots, x_{2017}$,

$$\sqrt{x_1-1} + \sqrt{x_2-1} + \sqrt{x_3-1} + \dots + \sqrt{x_{2017}-1} = \frac{1}{2}(x_1 + x_2 + x_3 + \dots + x_{2017}).$$

求 $x_1 + x_2 + x_3 + x_4 + \dots + x_{2017}$ 的值。〔答：4034〕

It is given that for real numbers $x_1, x_2, x_3, \dots, x_{2017}$,

$$\sqrt{x_1-1} + \sqrt{x_2-1} + \sqrt{x_3-1} + \dots + \sqrt{x_{2017}-1} = \frac{1}{2}(x_1 + x_2 + x_3 + \dots + x_{2017}).$$

Find the value of $x_1 + x_2 + x_3 + x_4 + \dots + x_{2017}$.

Solution

$$\sqrt{x_1-1} + \sqrt{x_2-1} + \sqrt{x_3-1} + \dots + \sqrt{x_{2017}-1} = \frac{1}{2}(x_1 + x_2 + x_3 + \dots + x_{2017})$$

$$(\sqrt{x_1-1}-1)^2 + (\sqrt{x_2-1}-1)^2 + (\sqrt{x_3-1}-1)^2 + \dots + (\sqrt{x_{2017}-1}-1)^2 = 0$$

$$\sqrt{x_1-1} = \sqrt{x_2-1} = \sqrt{x_3-1} = \dots = \sqrt{x_{2017}-1} = 1$$

$$x_1 = x_2 = x_3 = \dots = x_{2017} = 2$$

$$\begin{aligned} & x_1 + x_2 + x_3 + x_4 + \dots + x_{2017} \\ &= 2 \times 2017 \\ &= 4034 \end{aligned}$$

8. 設正整數 T 能滿足條件： T 的數字的積 $= T^2 - 11T - 23$ 。求該等正整數之和， S ，的值。〔答：13〕

Let positive integers, T , satisfy the condition: the product of the digits of $T = T^2 - 11T - 23$. Find the sum, S , of all such positive integers.

Solution

Let $T = 10^n D_n + 10^{n-1} D_{n-1} + \dots + D_0$, then $D_n \cdot D_{n-1} \cdots D_0 < D_n \cdot 10 \cdots 10 = 10^n D_n \leq T$

$$\therefore T^2 - 11T - 23 < T$$

$$T^2 - 12T - 23 < 0$$

$$6 - \sqrt{59} < T < 6 + \sqrt{59}$$

Possible values of T are 1, 2, 3, ..., 13

T	Product of digits	$T^2 - 11T - 23$	Satisfy
1	1	Negative	×
2	2	Negative	×
3	3	Negative	×
4	4	Negative	×
5	5	Negative	×
6	6	Negative	×
7	7	Negative	×
8	8	Negative	×

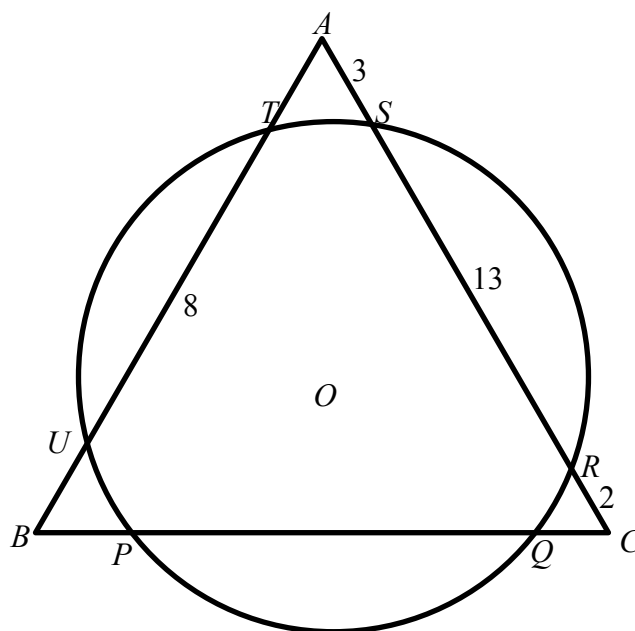
9	9	Negative	×
10	0	Negative	×
11	1	Negative	×
12	2	Negative	×
13	3	3	✓

∴ T can only take the value 13 .

∴ $S = 13$

9. 在圖五中， ABC 是一個等邊三角形且與一圓相交於六點， P 、 Q 、 R 、 S 、 T 及 U 。若 $AS=3$ ， $SR=13$ ， $RC=2$ 及 $UT=8$ ，求 $BP-QC$ 的值。〔答：3〕

In Figure 5, ABC is an equilateral triangle intersecting the circle at six points P , Q , R , S , T and U . If $AS=3$, $SR=13$, $RC=2$ and $UT=8$, find the value of $BP-QC$.



圖五

Figure 5

Solution

Since $AS \times AR = AT \times AU$, we have $AT = 4$, $BU = 6$

Let $BP = x$, $PQ = y$ and $QC = z$.

$$x(x + y) = 6(6 + 8) \dots (1)$$

$$z(y + z) = 2(2 + 13) \dots (2)$$

$$(1) - (2), \quad x^2 - z^2 + xy - yz = 54$$

$$(x - z)(x + z + y) = 54$$

$$x - z = 3$$

10. 已知方程 $a^2x^2 - (4a - 3a^2)x + 2a^2 - a - 21 = 0$ (其中 $a > 0$) 有最少一個整數根，求所有 a 的可能整數值之和。〔答：11〕

It is given that the equation $a^2x^2 - (4a - 3a^2)x + 2a^2 - a - 21 = 0$ (where $a > 0$) has at least one integer root. Find the sum of all possible integral values of a .

Solution

$$a^2x^2 - (4a - 3a^2)x + 2a^2 - a - 21 = 0$$

$$(ax)^2 - (4 - 3a)ax + (2a - 7)(a + 3) = 0$$

$$(ax + 2a - 7)(ax + a + 3) = 0$$

$$x = -1 - \frac{3}{a} \quad \text{or} \quad x = \frac{7}{a} - 2$$

Since the equation has at least one integral root and $a > 0$, $a = 1, 3$ or 7

The sum of all possible integral values of a is 11.

Hong Kong Mathematics Olympiad (2016 / 2017)

Final Event (Individual) Solution

香港數學競賽 (2016 / 2017)

決賽項目(個人) 題解

Event 1

1. Suppose there is a real solution x . Then,

$$\begin{aligned}\frac{1}{(x+2)(x+3)} &= \frac{1}{(x+1)(x+4)} \\ \Rightarrow x^2 + 5x + 6 - x^2 - 5x - 4 &= 0 \\ \Rightarrow 2 &= 0\end{aligned}$$

which leads to contradiction. So, there are no real solutions.

2. Let $f(x) = |x-9| + |x-10|$. For $9 \leq x \leq 10$, $f(x) = x-9 - x+10 = 1$.

For $x < 9$, $f(x) = 9-x - (x-10) = 19-2x > 19-2(9) = 1$.

For $x > 10$, $f(x) = x-9 + x-10 = 2x-19 > 2(10)-19 = 1$.

So, the minimum value of $f(x)$ for real x is 1. The maximum value of $-|x-9| - |10-x| =$ The maximum value of $-f(x) = -1$.

3. $119 = 4x^2 + 4y^2 + 9xy = (2x-2y)^2 + 17xy \geq 17xy$ (Equality holds if $x = y = \sqrt{7}$), which implies that

$$xy \leq \frac{119}{17} = 7.$$

4. Note that $\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 = 9 \Rightarrow x + \frac{1}{x} = 3$ (as $x > 0$). Further,

$$\left(x^2 + \frac{1}{x^2}\right)\left(x + \frac{1}{x}\right) = x^3 + \frac{1}{x^3} + \left(x + \frac{1}{x}\right),$$

which implies that $x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)\left(x^2 + \frac{1}{x^2} - 1\right) = 3(7-1) = 18$.

Event 2

1. Without loss of generality, assume the running direction of the student running with 6km/h is clockwise. After the two students start, they meet at $\frac{10}{16} = \frac{5}{8}$ of the length of track in the clockwise direction the first time. The second time they meet at $\frac{5}{8} + \frac{5}{8} = \frac{2}{8} \pmod{1}$,. Thus, the 8th ($= \frac{\text{LCM}(5,8)}{5}$), time they meet at the starting point again. Hence, $a = 7$.

2. Let x, y be the numbers of red and blue marbles respectively. Then

$$\begin{cases} \frac{x+7}{y} = 3 \\ \frac{x}{y+7} = 2 \end{cases} \text{ and } \begin{cases} x+7 = 3y \\ x = 2y+14 \end{cases} \text{ or } 3y-7 = 2y+14$$

Hence, $y = 21$ and $x = 56$ and the sum is 77.

3. Let the square number be N^2 , where N is a positive integer. Since N^2 is divisible by 77, so is N . The number closest to 1 000 that is divisible by 77 is 1001. Therefore, the square number is $1001^2 = 1\,002\,001$ and its distance from 1 million is 2001.

4. In a month, a technician completes $\frac{1}{d}$ of the project while a labour completes $\frac{1}{y}$ of the project. If y technicians are used, the amount of work done in a fixed duration is $\frac{y}{d}$ to that of $\frac{d}{y}$ by d labours.

Thus

$$\frac{y}{d} = 4 \frac{d}{y} \text{ or } 4d^2 = y^2 \text{ or } 2d = y$$

From $d + y = 2001$, we get $d = 667$.

Event 3

1. Observe that $(1, + (2, - (3)$ gives $a = x_0 + y_0 + z_0 = -15 + 18 - (-5) = 8$.

$$\begin{aligned}
2. \quad 2b &= \sqrt{6+2\sqrt{a}} + \sqrt{6-2\sqrt{a}} \\
&= \sqrt{6+2\sqrt{8}} + \sqrt{6-2\sqrt{8}} \\
&= \sqrt{6+4\sqrt{2}} + \sqrt{6-4\sqrt{2}} \\
&= \left[(2+\sqrt{2})^2 \right]^{\frac{1}{2}} + \left[(2-\sqrt{2})^2 \right]^{\frac{1}{2}} \\
&= 4
\end{aligned}$$

$$\begin{aligned}
3. \quad & \log_{10} b^x > 3 \\
& \Leftrightarrow \log 2^x > \log 1000 \\
& \Leftrightarrow 2^x > 1000 \\
& 2^9 = 512, \quad 2^{10} = 1024
\end{aligned}$$

$$4. \quad d = f(10) = 2^0 + 2^1 + 2^2 + \dots + 2^8 + 2^9 = 1023$$

Event 4

$$\begin{aligned}
1. \quad & \Delta = (a-3)^2 - 4a(a-2) \geq 0 \\
& \Leftrightarrow 3a^2 - 2a - 9 \leq 0 \\
& \Leftrightarrow \frac{1-\sqrt{28}}{3} \leq a \leq \frac{1+\sqrt{8}}{3} \\
& \Leftrightarrow a = -1, 0, 1, 2 \text{ but } a > 0 \Rightarrow a = 1 \text{ or } 2
\end{aligned}$$

$$\begin{aligned}
2. \quad & \text{Write } \log_x y = t, \text{ then } t + \frac{3}{t} = \frac{13}{2} \Rightarrow 2t^2 - 13t + 6 = 0 \Rightarrow t = \frac{1}{2} \text{ or } t = 6 \Rightarrow y = \sqrt{x} \text{ or } y = x^6 \\
& (\text{rejected as } 1 < y < x,.)
\end{aligned}$$

$$\text{Thus, } b = \frac{x+y^4}{x^2+y^2} = \frac{x+x^2}{x^2+x} = 1.$$

$$3. \quad c = \frac{C_3^3 + C_3^4 + C_3^5}{C_3^{12}} = \frac{15}{220} = \frac{3}{44}$$

$$\begin{aligned}
 4. \quad & \sin^4 \theta + \cos^4 \theta \\
 &= (\sin^2 \theta + \cos^2 \theta)^2 - 2\sin^2 \theta \cos^2 \theta \\
 &= 1 - \frac{1}{2} \sin^2 2\theta \\
 &= 1 - \frac{1}{2} (1 - \cos^2 2\theta) \\
 &= \frac{1}{2} + \frac{1}{2} \cos^2 2\theta \\
 &= \frac{1}{2} + \frac{1}{2} \left(\frac{3}{44} \right)^2 \\
 &= \frac{1}{2} \left(1 + \left(\frac{3}{44} \right)^2 \right) \\
 &= \frac{1}{2} \left(\frac{44^2 + 9}{44^2} \right)
 \end{aligned}$$

Event 1

$$\begin{aligned}
 1. \quad & \left(x + \frac{1}{y}\right)\left(y + \frac{1}{z}\right)\left(z + \frac{1}{x}\right) = (xyz) + \frac{1}{xyz} + \left(x + \frac{1}{y}\right) + \left(y + \frac{1}{z}\right) + \left(z + \frac{1}{x}\right) \\
 \Rightarrow & \quad (-1)(-2)(-5) = xyz + \frac{1}{xyz} - 1 - 2 - 5 \\
 \Rightarrow & \quad xyz + \frac{1}{xyz} = -2 \\
 \Rightarrow & \quad \frac{1}{xyz} = -1 \text{ (By solving a quadratic equation of } xyz)
 \end{aligned}$$

$$2. \quad |x - |2x - 1|| = \frac{1}{2} \Leftrightarrow x - |2x - 1| = \pm \frac{1}{2}$$

For $2x - 1 \geq 0$:

$$x - |2x - 1| = \pm \frac{1}{2} \Leftrightarrow x - (2x - 1) = \pm \frac{1}{2} \Leftrightarrow x = \frac{1}{2} \text{ or } \frac{3}{2}$$

$$\text{Also, } 2x - 1 \geq 0 \Leftrightarrow x \geq \frac{1}{2}.$$

$$\text{So, for } 2x - 1 \geq 0, \text{ we have } x = \frac{1}{2} \text{ or } \frac{3}{2}.$$

$$\text{Similarly, for } 2x - 1 < 0, \text{ we have } x = \frac{1}{6}.$$

Therefore, there are 3 distinct real solutions.

$$3. \quad \left(1 - \frac{1}{x}\right)\left(1 - \frac{1}{y}\right) = 1 + \frac{1 - (x + y)}{xy} = 1 - \frac{2}{xy}$$

$$3^2 = (x + y)^2 \Rightarrow x^2 + y^2 = 9 - 2xy$$

$$0 \leq (x - y)^2 = 9 - 2xy - 2xy = 9 - 4xy \text{ (Equality holds when } x = y = \frac{3}{2})$$

$$\Rightarrow -\frac{2}{xy} \leq -\frac{8}{9} \text{ (as } xy > 0)$$

$$\Rightarrow \left(1 - \frac{1}{x}\right)\left(1 - \frac{1}{y}\right) \leq 1 + \left(-\frac{8}{9}\right) = \frac{1}{9} \quad (\text{Equality holds when } x = y = \frac{3}{2})$$

$$4. \quad \left(x^2 + \frac{1}{x^2}\right) = \left(x - \frac{1}{x}\right)^2 + 2 = 11$$

$$\left(x^2 + \frac{1}{x^2}\right)\left(x - \frac{1}{x}\right) = \left(x^3 - \frac{1}{x^3}\right) - \left(x - \frac{1}{x}\right)$$

$$\Rightarrow \left(x^3 - \frac{1}{x^3}\right) = \left(x - \frac{1}{x}\right)\left[\left(x^2 + \frac{1}{x^2}\right) + 1\right] = 3(11 + 1) = 36$$

$$\left(x^2 + \frac{1}{x^2}\right)\left(x^3 - \frac{1}{x^3}\right) = \left(x^5 - \frac{1}{x^5}\right) + \left(x - \frac{1}{x}\right)$$

$$\Rightarrow \left(x^5 - \frac{1}{x^5}\right) = \left(x^2 + \frac{1}{x^2}\right)\left(x^3 - \frac{1}{x^3}\right) - \left(x - \frac{1}{x}\right) = 11(36) - 3 = 393$$

Event 2

$$\begin{aligned} 1. \quad & 6^4 + 2 \times 6^3 + 3 \times 6^2 + 4 \times 6 + 5 \\ &= (3 \times 2)^4 + 2 \times (3 \times 2)^3 + 3 \times (3 \times 2)^2 + [5 \times (3 \times 2) - 1] \\ &= (3 \times 3) \times K + 29 \\ &= 9(K + 3) + 2 \end{aligned}$$

where K is an integer. Therefore, $6^4 + 2 \times 6^3 + 3 \times 6^2 + 4 \times 6 + 5 = 9(K + 3) + 2$. Hence, $A = 2$.

2. A cube has 8 vertices. The greatest number of straight lines being formed is $7 + 6 + 5 + 4 + 3 + 2 + 1 = 28$.

3. As $0 \leq (x - 10)^2 + (y - 10)^2 + (z - 10)^2 = x^2 + y^2 + z^2 - 20(x + y + z) + 300 = C - 300$. The equality holds when $x = y = z = 10$, so the least value of $C = 300$.

4. $(x - 1)^3$ is an increasing function in x . The greatest value of y is 11 when $x = 3$.

Event 3

$$\begin{aligned}
 1. \quad & (ab-1)(bc-1)(ac-1) = (abc)(abc-a-b-c) + (ab+bc+ac-1) \\
 & \Rightarrow abc(ab+bc+ac-1) \\
 & \Rightarrow \text{remainder} = 0.
 \end{aligned}$$

$$\begin{aligned}
 2. \quad & \left(\frac{\sqrt{1+x}}{\sqrt{1+x}-\sqrt{1-x}} + \frac{1-x}{\sqrt{1-x^2}+x-1} \right) \cdot \left(\sqrt{\frac{1}{x^2}-1} - \frac{1}{x} \right) \\
 & = \left(\frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}-\sqrt{1-x}} \right) \cdot \left(\frac{\sqrt{1-x^2}-1}{x} \right) \\
 & = \frac{(\sqrt{1+x})^2 + (\sqrt{1-x})^2}{(\sqrt{1+x}-\sqrt{1-x})^2} \cdot \frac{\sqrt{1-x^2}-1}{x} \\
 & = \frac{2x}{2-2\sqrt{1-x^2}} \cdot \frac{\sqrt{1-x^2}-1}{x} \\
 & = -1
 \end{aligned}$$

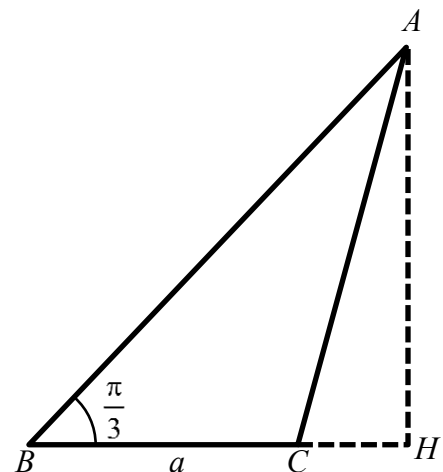
$$\begin{aligned}
 3. \quad & \text{Set } x = y + 2, \text{ then } (y+2)^4 + (y-2)^4 = 2 \cdot 272 \\
 & \Rightarrow 2y^4 + 2 \cdot 24y^2 + 2 \cdot 2^4 = 2 \cdot 272 \\
 & \Rightarrow y^4 + 24y^2 + 2^4 - 272 = 0 \\
 & \Rightarrow y^4 + 24y^2 - 256 = 0 \\
 & y^2 = \frac{-24 \pm \sqrt{24^2 - 4(-256)}}{2} \\
 & \Rightarrow y^2 = 8 \text{ or } -32 \text{ (rejected)} \\
 & \Rightarrow y = \pm 2\sqrt{2} \\
 & \Rightarrow x = \pm 2\sqrt{2} + 2 \\
 & \Rightarrow \text{sum of real roots} = 4
 \end{aligned}$$

$$4. \quad \text{Area of } \triangle ABC = \frac{1}{2}a \cdot AH = \sqrt{3}a^2 \Rightarrow AH = 2\sqrt{3}a$$

$$BH = \frac{AH}{\tan\left(\frac{\pi}{3}\right)} = 2\sqrt{3}a \cdot \frac{1}{\sqrt{3}} = 2a$$

$$\Rightarrow CH = a$$

$$\Rightarrow \tan(\angle C) = -\tan(\angle ACH) = -\frac{AH}{CH} = -2\sqrt{3}.$$



Event 4

1.

日數(days)	甲先生(Mr. A)		乙先生 (Mr. B)	
	倒模 moulding	上色 painting	倒模 moulding	上色 painting
1	3		5	
2	3		5	
3	3		5	
4	3		5	
5	3		5	
6	3		5	
7	3		5	
8	3		5	
9	3		5	
10	3		5	
11		15	5	
12		15	5	
13		15	5	
14		15	5	
15		15	5	
16		15	5	
17		15	5	
18		15	5	
19		15		15

2. $P(\text{Hitting the 6}^{\text{th}} \text{ duck at the 10}^{\text{th}} \text{ shot})$

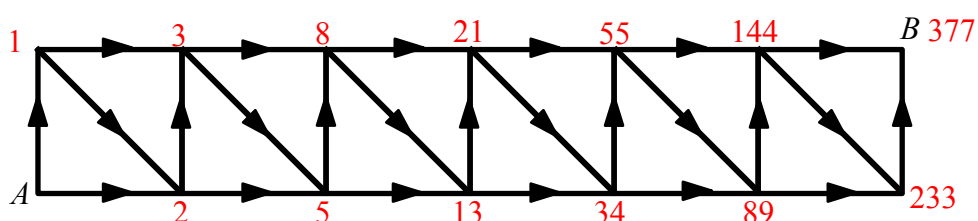
$= P(\text{Hitting 5 ducks in the first 9 shots and hitting a duck in the 10th})$

$= P(\text{Hitting 5 ducks in the first 9 shots}) \times P(\text{Hitting a duck at the 10th})$

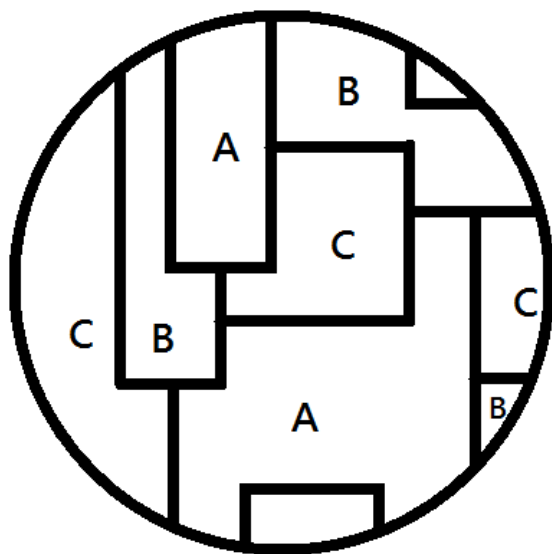
$$= C_5^9 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^4 \times \frac{1}{2}$$

$$= \frac{63}{512}$$

3.



4.



Hong Kong Mathematics Olympiad (2017 / 2018)

Heat Event (Individual) Solution

香港數學競賽 (2017 / 2018)

初賽項目(個人) 題解

甲部

Part A

1. 若 a 及 b 均為實數，求 $a^2 + b^2 + 12a - 8b + 2018$ 的最小值。〔答：1966〕

If a and b are real numbers, find the minimum value of $a^2 + b^2 + 12a - 8b + 2018$.

Solution

$$\begin{aligned} & a^2 + b^2 + 12a - 8b + 2018 \\ &= a^2 + 12a + b^2 - 8b + 2018 \\ &= a^2 + 12a + 36 + b^2 - 8b + 16 + 2018 - 36 - 16 \\ &= (a + 6)^2 + (b - 4)^2 + 1966 \end{aligned}$$

$\therefore$ Minimum value = 1966

2. 設 a 及 k 均為常數。若 $(6x^3 + ax^2 + 7x - 3) \div (2x^2 + kx - 1)$ 的商和餘式分別為 $3x + 5$ 及 $-5x + 2$ ，求 a 的值。〔答：19〕

Let a and k be constants. If the quotient and the remainder of $(6x^3 + ax^2 + 7x - 3) \div (2x^2 + kx - 1)$ are $3x + 5$ and $-5x + 2$ respectively, find the value of a .

Solution

By division algorithm, we have

$$\begin{aligned} 6x^3 + ax^2 + 7x - 3 &= (2x^2 + kx - 1)(3x + 5) + (-5x + 2) \\ &= 6x^3 + (10 + 3k)x^2 + (5k - 8)x - 3 \end{aligned}$$

By comparing the coefficients of x and x^2 on both sides,

$$7 = 5k - 8 \quad \text{and} \quad a = 10 + 3k$$

$\therefore a = 19$

3. 在編制某雜誌中每頁的頁碼時，總共用去了 2,046 個數字，問該雜誌總共有多少頁？（假設該雜誌第一頁的頁碼是 1。）〔答：718〕

In numbering the pages of a magazine, 2,046 digits were used. How many pages are there in the magazine? (Assume the page number of the magazine starts from 1.)

Solution

Page Number	Number of Digits Used
1 – 9	9
10 – 99	90×2
100 – 199	100×3

$$\begin{aligned}\therefore \text{Number of Digits} &= (2046 - 9 - 2 \times 90) \div 3 + 99 \\ &= 718\end{aligned}$$

4. 解 $\log\left(1+\frac{1}{1}\right)+\log\left(1+\frac{1}{2}\right)+\log\left(1+\frac{1}{3}\right)+\dots+\log\left(1+\frac{1}{n}\right)=5$ 。〔答：99999〕

$$\text{Solve } \log\left(1+\frac{1}{1}\right)+\log\left(1+\frac{1}{2}\right)+\log\left(1+\frac{1}{3}\right)+\dots+\log\left(1+\frac{1}{n}\right)=5.$$

Solution

$$\log\left(1+\frac{1}{1}\right)+\log\left(1+\frac{1}{2}\right)+\log\left(1+\frac{1}{3}\right)+\dots+\log\left(1+\frac{1}{n}\right)=5$$

$$\log\left(\frac{2}{1}\right)+\log\left(\frac{3}{2}\right)+\log\left(\frac{4}{3}\right)+\dots+\log\left(\frac{n+1}{n}\right)=5$$

$$\log\left[2\left(\frac{3}{2}\right)\left(\frac{4}{3}\right)\left(\frac{5}{4}\right)\dots\left(\frac{n+1}{n}\right)\right]=\log 100000$$

$$n+1=100000$$

$$n=99999$$

5. 已知 $\frac{1-2^{\frac{-1}{x}}}{2^{\frac{-1}{x}}-2^{\frac{-2}{x}}}=4$ 。求 x 的值。〔答： $x=\frac{1}{2}$ 〕

Given that $\frac{1-2^{\frac{-1}{x}}}{2^{\frac{-1}{x}}-2^{\frac{-2}{x}}}=4$. Find the value of x .

Solution

$$\text{Let } y = 2^{\frac{-1}{x}}.$$

$$\frac{1-y}{y-y^2}=4$$

$$4y^2-5y+1=0$$

$$(4y-1)(y-1)=0$$

$$y=\frac{1}{4} \quad \text{or} \quad 1$$

$$2^{\frac{-1}{x}}=2^{-2} \quad \text{or} \quad 2^{\frac{-1}{x}}=2^0 \text{ (rejected)}$$

$$\frac{-1}{x}=-2$$

$$x=\frac{1}{2}$$

6. 若 x 為有理數，求 x 的值滿足聯立方程 $\begin{cases} y=2x^2-11x+15 \\ y=2x^3-17x^2+16x+35 \end{cases}$ 。〔答： $\frac{5}{2}$ 〕

If x is a rational number, find the value of x satisfying the simultaneous equations

$$\begin{cases} y=2x^2-11x+15 \\ y=2x^3-17x^2+16x+35 \end{cases}.$$

Solution

$$2x^3-17x^2+16x+35=2x^2-11x+15$$

$$2x^3-19x^2+27x+20=0$$

$$(2x-5)(x^2-7x-4)=0$$

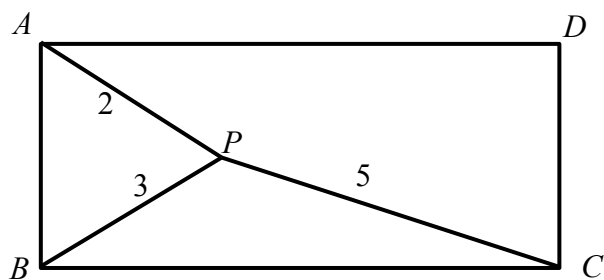
$$x=\frac{5}{2} \quad \text{or} \quad \frac{7\pm\sqrt{65}}{2}$$

$\therefore x$ is a rational number.

$$\therefore x=\frac{5}{2} \quad \text{only}$$

7. 如圖一所示， P 為長方形 $ABCD$ 內的一點，使得 $PA=2$ ， $PB=3$ 及 $PC=5$ 。求 PD 的長度。〔答： $2\sqrt{5}$ 〕

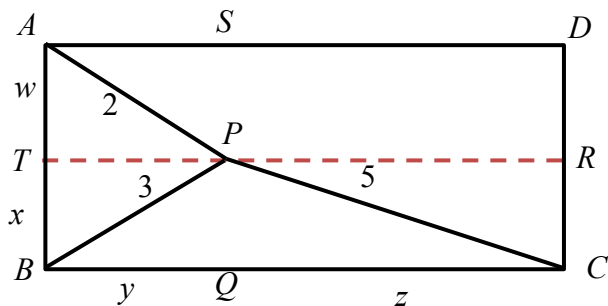
As shown in Figure 1, P is a point inside a rectangle $ABCD$ such that $PA=2$, $PB=3$ and $PC=5$. Find the length of DP .



圖一

Figure 1

Solution



$$x^2 + y^2 = 9 \quad \dots (1)$$

$$w^2 + y^2 = 4 \quad \dots (2)$$

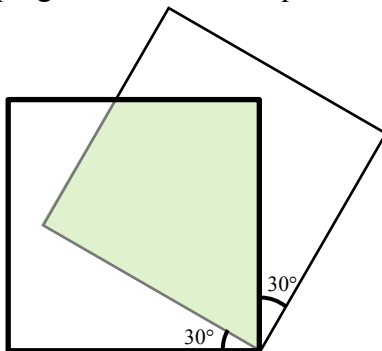
$$x^2 + z^2 = 25 \quad \dots (3)$$

$$(2) + (3) - (1) \quad w^2 + z^2 = 25 + 4 - 9 = 20$$

$$PD = \sqrt{20} = 2\sqrt{5}$$

8. 如圖二所示，兩個邊長為 x cm 的正方形於一角重疊。若兩個正方形的非重疊部分與重疊部分面積的比是 $a:1$ ，求 a 的值。〔答： $2(\sqrt{3}-1)$ 〕

As shown in Figure 2, two squares with side x cm coincide at one corner. If the ratio of the non-overlapping area to the overlapping area of the two squares is $a:1$, find the value of a .



圖二

Figure 2

Solution

Let x be the length of the square.

The overlapping area

$$= x \cdot x \tan 30^\circ \cdot \frac{1}{2} \cdot 2$$

$$= \frac{\sqrt{3}}{3} x^2$$

The non-overlapping area

$$= 2 \left(\frac{3 - \sqrt{3}}{3} \right) x^2$$

The non-overlapping area : the overlapping area

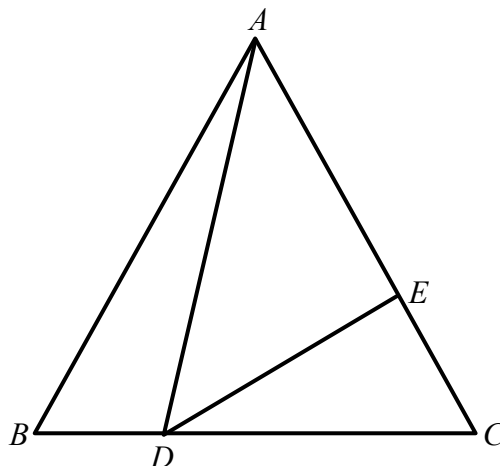
$$= 2 \left(\frac{3 - \sqrt{3}}{3} \right) x^2 : \frac{\sqrt{3}}{3} x^2$$

$$= 2(\sqrt{3} - 1) : 1$$

$$\therefore a = 2(\sqrt{3} - 1)$$

9. 如圖三所示， ABC 是一個等腰三角形，其中 $AB=AC=8$ 及 $BC=4$ 。° D 及 E 分別為 BC 及 AC 上的點使得 $BD=1$ 及 $\angle ABC = \angle ADE$ 。求 AE 的長度。
〔 答: $\frac{61}{8}$ 〕

As shown in Figure 3, ABC is an isosceles triangle with $AB=AC=8$ and $BC=4$. D and E are points lying on BC and AC respectively such that $BD=1$ and $\angle ABC = \angle ADE$. Find the length of AE .



圖三

Figure 3

Solution

$$\angle ADC = \angle ABC + \angle BAD \quad (\text{ext. } \angle \text{ of } \Delta)$$

$$\angle ADE + \angle EDC = \angle ABC + \angle DAB$$

$$\angle EDC = \angle DAB$$

$$\angle ABD = \angle DCE \quad (\text{base } \angle \text{s isos. } \Delta)$$

$$\Delta ABD \sim \Delta DCE$$

$$\frac{CE}{DC} = \frac{BD}{AB}$$

$$CE = \frac{3}{8}$$

$$AE = 8 - \frac{3}{8} = \frac{61}{8}$$

10. PQR 是一個三角形，其中 $PQ=13$ 、 $QR=14$ 及 $PR=15$ 。以 PR 為直徑繪畫出圓 C ， C 相交 QR 於點 T 。求 ΔPTR 的面積。
 [答:54]
 PQR is a triangle with $PQ=13$, $QR=14$ and $PR=15$. The circle C is drawn with diameter PR . C intersects QR at point T . Find the area of ΔPTR .

Solution

$PT \perp QR$ ($\angle$ in the semi-circle)

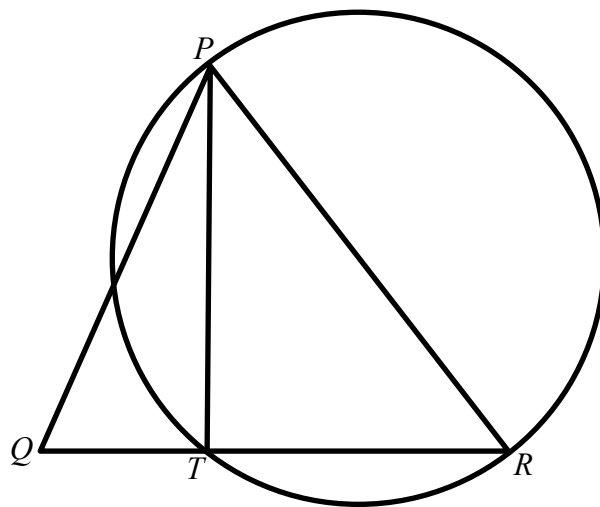
$$s = \frac{13+14+15}{2} = 21$$

$$\text{Area of } \Delta PQR = \sqrt{21(21-13)(21-14)(21-15)} = 84$$

$$PT = \frac{2 \times 84}{14} = 12$$

$$TR = \sqrt{15^2 - 12^2} = 9$$

$$\text{Area of } \Delta PTR = \frac{12 \times 9}{2} = 54$$



乙部

Part B

11. 求 $3^x + 5 + \frac{36}{3^x + 4}$ 的最小值。 [答: 13]

Find the minimum value of $3^x + 5 + \frac{36}{3^x + 4}$.

Solution

$$\begin{aligned} & 3^x + 5 + \frac{36}{3^x + 4} \\ &= \left(3^x + 4 + \frac{36}{3^x + 4} \right) + 1 \\ &\geq 2 \cdot \sqrt{(3^x + 4) \left(\frac{36}{3^x + 4} \right)} + 1 \end{aligned}$$

$$= 13$$

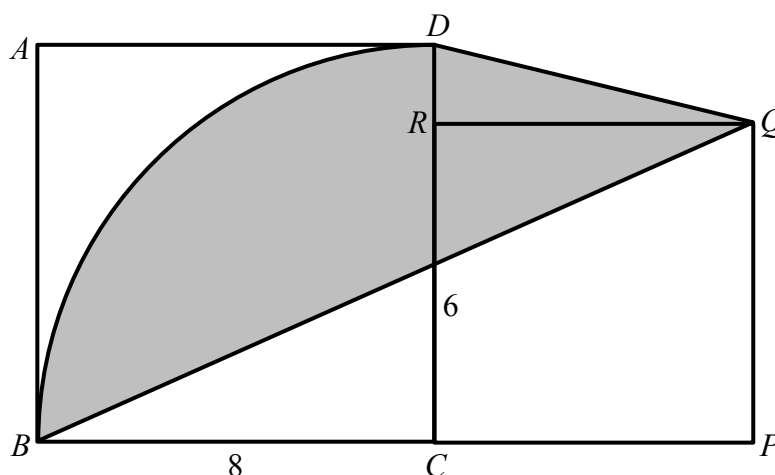
Alternatively,

$$\begin{aligned} & 3^x + 5 + \frac{36}{3^x + 4} \\ &= \left(\sqrt{3^x + 4} - \frac{6}{\sqrt{3^x + 4}} \right)^2 + 1 + 12 \\ &= \left(\sqrt{3^x + 4} - \frac{6}{\sqrt{3^x + 4}} \right)^2 + 13 \end{aligned}$$

$\therefore$ The minimum value is 13 when $x = \log_3 2$.

12. 如圖四所示， $ABCD$ 及 $PQRC$ 為兩個連接的正方形。以 C 為圓心及 BC 為半徑繪畫出弧 BD 。已知 $BC=8$ 及 $RC=6$ 。求弧 BD 及線段 DQ 與 BQ 所圍成的區域的面積。(答案須以 π 表示) [答： 16π]

As shown in Figure 4, two squares $ABCD$ and $PQRC$ are joined together. An arc BD is drawn with centre C and radius BC . Given that $BC=8$ and $RC=6$. Find the area of the region bounded by the arc BD , line segments DQ and BQ . (Express your answer in terms of π)



圖四

Figure 4

Solution

Draw two diagonals BD and QC . Then $BD \parallel QC$.

Therefore, area of $\triangle BDQ$ is equal to area of $\triangle BCD$.

Thus the area of the shaded region = area of the sector BCD

$$\begin{aligned}
 &= \frac{1}{4}(\pi)(BC)^2 \\
 &= \frac{\pi(8)^2}{4} \\
 &= 16\pi
 \end{aligned}$$

13. 一個四位數可以透過把它的所有數字加起來，變成另一個數。例如：1234 可以變成 10，因為 $1+2+3+4=10$ 。究竟從 1998 至 4998（包括此兩個數）有多少個四位數的經上述變換後不可以被 3 整除？〔答：2000〕

A 4-digit number can be transformed into another number by adding its digits. For example, 1234 can be transformed into 10 as $1+2+3+4=10$. How many numbers from 1998 to 4998 inclusive are **NOT** divisible by 3 after the above transformation?

Solution

Number	Transformed number	Remainder when divided by 3
1998	$1+9+9+8=27$	0
1999	$1+9+9+9=28$	1
2000	$2+0+0+0=2$	2
2001	$2+0+0+1=3$	0
2002	$2+0+0+2=4$	1
2003	$2+0+0+3=5$	2

Hence, the required number of transformed number that are divisible by 3 is $(4998-1999+1) \div 3 + 1 = 1001$

$\therefore$ The required number $= 4998 - 1998 + 1 - 1001 = 2000$

14. 對任意實數 x ($x \neq 1$)，定義函數 $f(x) = \frac{x}{1-x}$ 及 $f \circ f(x) = f(f(x))$ 。求

$2017 \underbrace{f \circ f \circ f \circ \dots \circ f}_{2018 \text{ 個 } f}(2018)$ 的值。〔答： $-\frac{2018}{2019}$ 〕

For any real number x ($x \neq 1$), define a function $f(x) = \frac{x}{1-x}$ and $f \circ f(x) = f(f(x))$. Find the value of

$2017 \underbrace{f \circ f \circ f \circ \dots \circ f}_{2018 \text{ s } f}(2018)$.

Solution

$$\begin{aligned}
 & 2017 \underbrace{f \circ f \circ f \circ \dots \circ f}_{2018 \text{ s } f}(2018) \\
 &= 2017 \times \frac{2018}{1-2018^2} \\
 &= 2017 \times \frac{2018}{(1-2018)(1+2018)} \\
 &= 2017 \times \frac{2018}{(1-2018)(1+2018)} \\
 &= -\frac{2018}{2019}
 \end{aligned}$$

15. 設 $N^2 = \overline{abcdefabc}$ 為一個 9 位整數，其中 N 是 4 個相異質數的積及 a, b, c, d, e, f 均為非零數字且滿足 $\overline{def} = 2 \times \overline{abc}$ 。求 N^2 的最小值。〔答：289578289〕

Let $N^2 = \overline{abcdefabc}$ be a nine-digit positive integer, where N is the product of four distinct primes and a, b, c, d, e, f are non-zero digits that satisfy $\overline{def} = 2 \times \overline{abc}$. Find the least value of N^2 .

Solution

Note that $N^2 = \overline{abcdefabc} = (\overline{abc}) \times 1002001 = (\overline{abc}) \times 7^2 \times 11^2 \times 13^2$. This implies that $\overline{abc} = p^2$ for some prime p . It is clear that $100 < \overline{abc} < 1000$ and this means that $10 < p < 23$. Hence, $p = 17$ or 19 and $N^2 = 289578289$.

Hong Kong Mathematics Olympiad (2017 / 2018)
Heat Event (Group) Solution
香港數學競賽 (2017 / 2018)
初賽項目(團體) 題解

1. 設 $f(x)$ 為二次多項式，其中 $f(1) = \frac{1}{2}$ ， $f(2) = \frac{1}{6}$ ， $f(3) = \frac{1}{12}$ 。求 $f(6)$ 的值。〔答： $\frac{4}{3}$ 〕

Let $f(x)$ be a polynomial of degree 2, where $f(1) = \frac{1}{2}$, $f(2) = \frac{1}{6}$, $f(3) = \frac{1}{12}$. Find the value of $f(6)$.

Solution

Let $f(x) = ax^2 + bx + c$, where a, b and c are constants, $a \neq 0$

From the given conditions, we have $a + b + c = \frac{1}{2}$, $4a + 2b + c = \frac{1}{6}$, $9a + 3b + c = \frac{1}{12}$

Solving the three equations, we get $a = \frac{1}{8}$, $b = -\frac{17}{24}$, $c = \frac{13}{12}$

$$\therefore f(x) = \frac{1}{8}x^2 - \frac{17}{24}x + \frac{13}{12}$$

$$\begin{aligned} f(6) &= \frac{1}{8}(6)^2 - \frac{17}{24}(6) + \frac{13}{12} \\ &= \frac{4}{3} \end{aligned}$$

2. 求 $\sqrt{2018 \times 2012 \times 1988 \times 1982 + 8100}$ 。〔答：3999766〕

Evaluate $\sqrt{2018 \times 2012 \times 1988 \times 1982 + 8100}$.

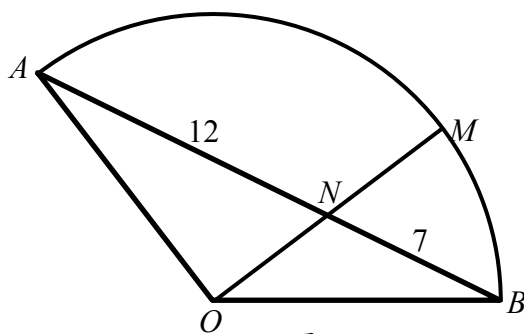
Solution

Let $x = 2000$.

$$\begin{aligned} &\sqrt{2018 \times 2012 \times 1988 \times 1982 + 8100} \\ &= \sqrt{(x+18)(x+12)(x-12)(x-18) + 8100} \\ &= \sqrt{(x^2 - 324)(x^2 - 144) + 8100} \\ &= \sqrt{(x^4 - 468x^2 + (18 \times 12)^2) + 8100} \\ &= \sqrt{x^4 - 468x^2 + (18 \times 13)^2} \\ &= \sqrt{(x^2 - 234)^2} \\ &= x^2 - 234 \\ &= 2000^2 - 234 \\ &= 3999766 \end{aligned}$$

3. 如圖一所示， OAB 是一個以 O 為圓心的圓的扇形。 N 則為半徑 OM 與 AB 的交點。已知 $AN = 12$ ， $BN = 7$ 及 $3ON = 2MN$ 。求 OM 的長。〔答：10〕

As shown in Figure 1, OAB is a sector of a circle with centre O . N is the intersecting point of the radii OM and AB . Given that $AN = 12$, $BN = 7$ and $3ON = 2MN$. Find the length of OM .



圖一

Figure 1

Solution

Let $OM = x$.

Extend the arc AMB to form a circle and extend the radius MO to form a diameter MP . Then, by the intersecting chord theorem,

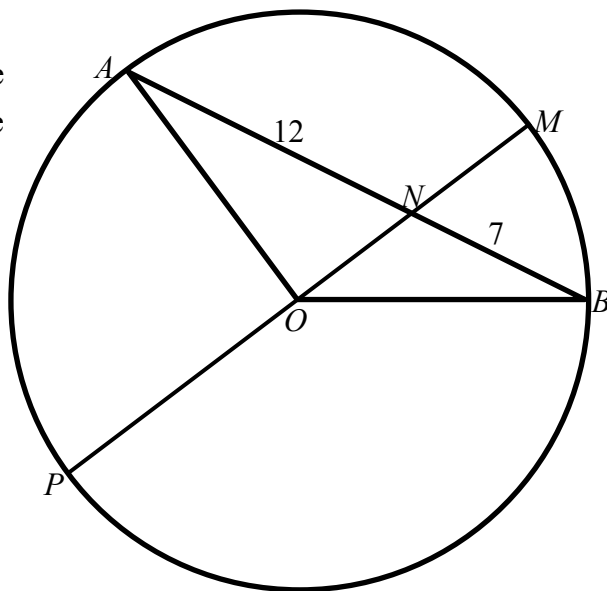
$$AN \times BN = MN \times NP$$

$$12 \times 7 = \frac{3x}{5} \times \frac{7x}{5}$$

$$84 = \frac{21x^2}{25}$$

$$x^2 = 100$$

$$x = 10$$



4. 對任意非零實數 x ，函數 $f(x)$ 有以下特性： $2f(x) + f(\frac{1}{x}) = 11x + 4$ 。設 S 為所有滿足於 $f(x) = 2018$ 的根之和。求 S 的值。〔答：275〕

For any non-zero real number x , the function $f(x)$ has the following property: $2f(x) + f(\frac{1}{x}) = 11x + 4$.
Let S be the sum of all roots satisfying the equation $f(x) = 2018$. Find the value of S .

Solution

$$2f(x) + f(\frac{1}{x}) = 11x + 4 \quad \dots\dots\dots (1)$$

$$2f(\frac{1}{x}) + f(x) = \frac{11}{x} + 4 \quad \dots\dots\dots (2)$$

$$(1) \times 2 - (2) \quad 3f(x) = 22x - \frac{11}{x} + 4$$

$$3(2018) = 22x - \frac{11}{x} + 4$$

$$22x^2 - 6050x - 11 = 0$$

$$S = \frac{6050}{22}$$

$$S = 275$$

5. 求可滿足下列方程組的 x 值：

$$\begin{cases} x^2 + 9x - 10y - 220 = 0 \\ y^2 - 5x + 6y - 166 = 0 \\ xy = 195 \end{cases} \quad \text{〔 答: } -15 \text{ 〕}$$

Find the value of x that satisfy the following system of equations:

$$\begin{cases} x^2 + 9x - 10y - 220 = 0 \\ y^2 - 5x + 6y - 166 = 0 \\ xy = 195 \end{cases}$$

Solution

$$\begin{cases} x^2 + 9x - 10y - 220 = 0 & \dots\dots\dots (1) \\ y^2 - 5x + 6y - 166 = 0 & \dots\dots\dots (2) \\ xy = 195 & \dots\dots\dots (3) \end{cases}$$

$$(1) + (2) \quad x^2 + y^2 + 4x - 4y - 386 = 0$$

$$\begin{aligned}
(x-y)^2 + 2xy + 4(x-y) - 386 &= 0 \\
(x-y)^2 + 390 + 4(x-y) - 386 &= 0 \\
(x-y)^2 + 4(x-y) + 4 &= 0 \\
(x-y+2)^2 &= 0 \\
x-y+2 &= 0 \\
y &= x+2 \quad \dots\dots\dots (4)
\end{aligned}$$

Substitute (4) into (3) gives

$$\begin{aligned}
x(x+2) &= 195 \\
x^2 + 2x - 195 &= 0 \\
(x+15)(x-13) &= 0 \\
x &= -15 \text{ or } x = 13
\end{aligned}$$

When $x = -15, y = -13$. Substitute that into (2) gives $169 + 75 - 78 - 166 = 0$

When $x = 13, y = 15$. Substitute that into (2) gives $225 - 65 + 90 - 166 \neq 0$

Hence, $x = -15$

6. 已知 $n^4 + 104 = 3^m$ ，其中 n, m 為正整數。求 n 的最小值。〔答：5〕

Given that $n^4 + 104 = 3^m$, where n, m are positive integers. Find the least value of n .

Solution

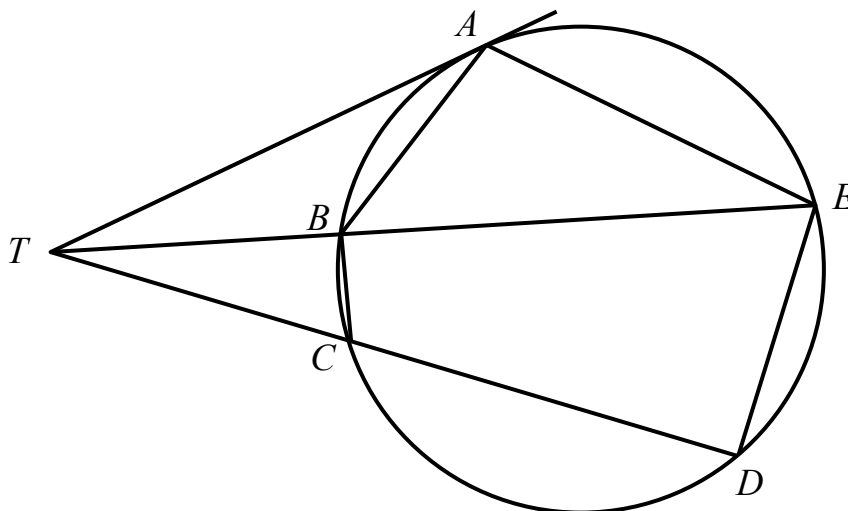
For $n^4 + 104 = 3^m \Rightarrow 3^m - 104 = k^2 = (n^2)^2$ which means $3^m - 104$ is a perfect square.

The unit digit of powers of 3 are 1, 3, 7 and 9, adding 104 will give the unit digit 1, 5, 8, 9 and 0. Comparing these digits with the unit digit of a perfect square, which is either 1, 4, 5, 6 or 9, only 1 and 9 are feasible.

By checking, $3^2 = 9$, $3^4 = 81$, $3^6 = 729$, $3^8 = 6561$, ..., the first two will give negative value for n^4 , which is not feasible, therefore the next one will give $3^6 - 104 = 729 - 104 = 625 = 5^4$, hence, the least value of n is 5.

7. 如圖二所示， A 、 B 、 C 、 D 及 E 為圓上的點。 T 是該圓外的一點。 TA 是該圓在點 A 的切線， TBE 及 TCD 為直線。已知 TBE 是 $\angle ATD$ 的角平分線、 $TA = 12$ 、 $TB = 6$ 及 $TC = 8$ 。求 $\triangle ABE$ 與四邊形 $BCDE$ 的面積比。 [答： $\frac{9}{16}$]

As shown in Figure 2, A , B , C , D and E are points on the circle. T is a point outside the circle such that TA is tangent to the circle at A and TBE and TCD are straight lines. It is given that TBE is the angle bisector of $\angle ATD$, $TA = 12$, $TB = 6$ and $TC = 8$. Find the ratio of the area of $\triangle ABE$ to the area of quadrilateral $BCDE$.



圖二

Figure 2

Solution

$$\triangle TAB \sim \triangle TEA \text{ and } \triangle TAC \sim \triangle TDA$$

$$\frac{TA}{TE} = \frac{TB}{TA} \quad \text{and} \quad \frac{TA}{TD} = \frac{TC}{TA}$$

$$\frac{12}{6 + BE} = \frac{6}{12} \quad \frac{12}{8 + CD} = \frac{8}{12}$$

$$BE = 18 \quad CD = 10$$

$$\begin{aligned} \frac{\text{area of } \triangle ABE}{\text{area of quadrilateral } BCDE} &= \frac{\text{Area of } \triangle ATE - \text{Area of } \triangle ABT}{\text{Area of } \triangle ADE - \text{Area of } \triangle CBT} \\ &= \frac{\frac{1}{2} AT(ET) \sin \angle ATE - \frac{1}{2} AT(BT) \sin \angle ATB}{\frac{1}{2} DT(ET) \sin \angle DTE - \frac{1}{2} CT(BT) \sin \angle CTB} \\ &= \frac{12(6 + 18) - 12(6)}{(8 + 10)(18 + 6) - 8(6)} \\ &= \frac{9}{16} \end{aligned}$$

8. 已知 a, b, c, d, e, f, g 及 h 為正整數，使得 $a > b > c > d > e > f > g > h$ 及 $a + h = b + g = c + f = d + e = 35$ ，問有多少組可行答案 $\{a, b, c, d, e, f, g, h\}$ 存在？〔答：2380〕
- Given that a, b, c, d, e, f, g and h are positive integers such that $a > b > c > d > e > f > g > h$ and $a + h = b + g = c + f = d + e = 35$. How many possible solution set of $\{a, b, c, d, e, f, g, h\}$ exist?

Solution

When two positive integers p, q satisfying the conditions $p > q$ and $p + q = 35$, q can be any positive integer from 1 to 17, i.e. there are 17 possible pairs of possible solution set (p, q) exist. Thus the question is equivalent to choosing 4 such pairs of possible solution sets from 17 sets. Arranging the possible values of q in ascending order and choose 4 sets of (p, q) from them will form the sets (a, h) , (b, g) , (c, f) , and (d, e) , and they also satisfy $a > b > c > d > e > f > g > h$. Consequently, the number of possible solution sets is $C_4^{17} = 2380$.

9. 求 $\left(\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{100}\right) + \left(\frac{2}{3} + \frac{2}{4} + \cdots + \frac{2}{100}\right) + \left(\frac{3}{4} + \frac{3}{5} + \cdots + \frac{3}{100}\right) + \cdots + \left(\frac{98}{99} + \frac{98}{100}\right) + \frac{99}{100}$ 的值。〔答：2475〕

Find the value of $\left(\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{100}\right) + \left(\frac{2}{3} + \frac{2}{4} + \cdots + \frac{2}{100}\right) + \left(\frac{3}{4} + \frac{3}{5} + \cdots + \frac{3}{100}\right) + \cdots + \left(\frac{98}{99} + \frac{98}{100}\right) + \frac{99}{100}$.

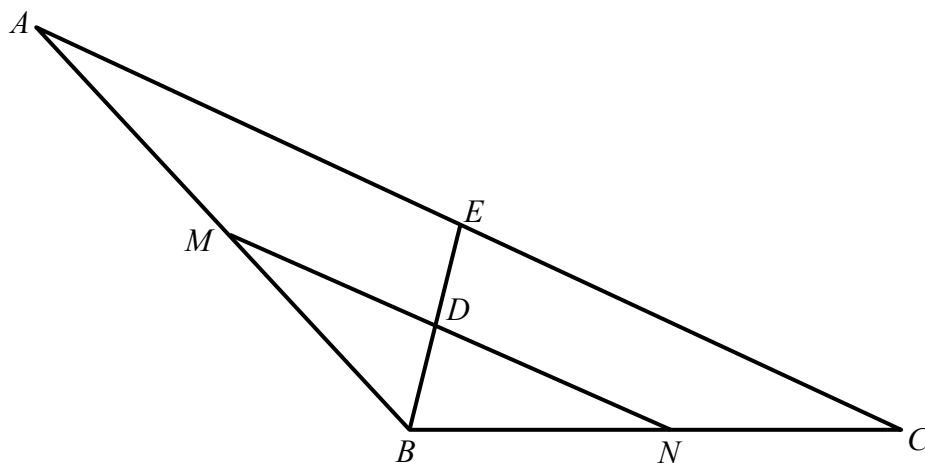
Solution

$$\begin{aligned}
 & \left(\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{100}\right) + \left(\frac{2}{3} + \frac{2}{4} + \cdots + \frac{2}{100}\right) + \left(\frac{3}{4} + \frac{3}{5} + \cdots + \frac{3}{100}\right) + \cdots + \left(\frac{98}{99} + \frac{98}{100}\right) + \frac{99}{100} \\
 = & \frac{1}{2} + \frac{1+2}{3} + \frac{1+2+3}{4} + \frac{1+2+3+4}{5} + \cdots + \frac{1+2+3+\cdots+(k-1)}{k} + \cdots + \frac{1+2+3+\cdots+99}{100} \\
 = & \frac{1}{2}(1+2+3+\cdots+99) \\
 = & \frac{1}{2} \times (1+99) \times 99 \div 2 \\
 = & 2475
 \end{aligned}$$

10. ABC 是一個三角形，其中 $AB=40$ 、 $BC=30$ 及 $\angle ABC=150^\circ$ 。 M 及 N 分別為 AB 及 BC 的中點。

$\angle ABC$ 的角度平分線分別相交 MN 及 AC 於 D 及 E 。求四邊形 $AMDE$ 的面積。〔答： $\frac{900}{7}$ 〕

ABC is a triangle with $AB=40$, $BC=30$ and $\angle ABC=150^\circ$. M and N are the mid-points of AB and BC respectively. The angle bisector of $\angle ABC$ intersects MN and AC at D and E respectively. Find the area of quadrilateral $AMDE$.



圖三

Figure 3

Solution

By angle bisector theorem,

$$AE \times BC = AB \times CE$$

$$AE : CE = 4 : 3$$

$$\text{Area of } \triangle ABC = \frac{1}{2}(30)(40)\sin 150^\circ = 300$$

$$\text{Area of } \triangle ABE = \frac{4}{7} \times 300 = \frac{1200}{7}$$

$\therefore MN \parallel AC$ (mid-point theorem)

$\therefore \triangle MBD \sim \triangle ABE$

$$\text{The area of quadrilateral } AMDE = \left(1 - \frac{1}{4}\right) \times \frac{1200}{7} = \frac{900}{7}$$

Hong Kong Mathematics Olympiad (2017 / 2018)

Final Event (Individual) Solution

香港數學競賽 (2017 / 2018)

決賽項目(個人) 題解

Event 1

1. $x^2 = y^2 - 4y$

$$x^2 - y^2 + 4y = 0$$

$$x^2 - (y^2 - 4y + 4) = -4$$

$$x^2 - (y - 2)^2 = -4$$

$$(x + y - 2)(x - y + 2) = -4$$

Hence $\begin{cases} x + y - 2 = 2 \\ x - y + 2 = -2 \end{cases}$ or $\begin{cases} x + y - 2 = -2 \\ x - y + 2 = 2 \end{cases}$ hence it gives $\begin{cases} x = 0 \\ y = 4 \end{cases}$ hence the value of

$$A = x + y = 4.$$

2. $y = \sqrt{9A^2 - 12A + 4} \pm \sqrt{A^2 - 4A + 4} \pm \sqrt{A^2 + 6A + 9}$

$$= (3A - 2) \pm (A - 2) \pm (A + 3)$$

$$= (12 - 2) \pm (4 - 2) \pm (4 + 3)$$

$$= 10 \pm 2 \pm 7$$

So the least possible value of $y = 10 - 2 - 7 = 1$.

3. We have $144 + 2^C = m^2$ hence implies that $2^C = m^2 - 144 = (m - 12)(m + 12)$.

Let the powers of 2 be k and l where $k > l$ such that

$$2^l = m - 12 \text{ and } 2^k = m + 12.$$

Take the difference of the two equations

$$24 = 2^k - 2^l = 2^l(2^{k-l} - 1).$$

Since $24 = 2^3 \cdot 3$ choose that $2^l = 8$ and $2^{k-l} - 1 = 3 = 2^2 - 1$ or we have $l = 3$ and $k = 5$.

The result $C = l + k = 8$.

4. We have $64 = 8 = \left(x + \frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2}$.

So $x^2 + \frac{1}{x^2} = 62$.

Now $62 \times 8 = \left(x^2 + \frac{1}{x^2}\right) \left(x + \frac{1}{x}\right) = x^3 + x + \frac{1}{x} + \frac{1}{x^3} = x^3 + \frac{1}{x^3} + 8$.

Thus $D = x^3 + \frac{1}{x^3} = 62 \times 8 - 8 = 61 \times 8 = 488$.

Event 2

1. Now $7778^2 - 2223^2 = (7778 + 2223)(7778 - 2223) = 10001 \times 5555 = 55555555$.

Thus the sum of the digits is 40.

2. When $a = 40$ look at the number of trailing zeros of $40!$ amounts to the number of factors of 5 in the prime factorization. There are 8 of them namely

$$5, 10, 15, 20, 25, 30, 35 \text{ and } 40.$$

As $25 = 5 \times 5$ also here there are 2 multiples of 5 in the prime factorization hence it means that there are 9 zeros at the end of $40!$.

3. Now $x^{12} - 1 = (x^2 + 1)(x^{10} - x^8 + x^6 - x^4 + x^2 - 1)$ holds for all real numbers x . So must be the case that $x^{10} - x^8 + x^6 - x^4 + x^2 - 1 = \frac{x^{12} - 1}{x^2 + 1}$.

In particular $2^{10} - 2^8 + 2^6 - 2^4 + 2^2 - 1 = \frac{2^{12} - 1}{2^2 + 1} = \frac{4095}{5} = 819 = 9 \times 91 = 9 \times 13 \times 7$.

In other words $2^{10} - 2^8 + 2^6 - 2^4 + 2^2 - 1 \equiv 0 \pmod{9}$

or equivalently $2^{10} - 2^8 + 2^6 - 2^4 + 2^2 \equiv 1 \pmod{9}$

Alternatively $2^{10} - 2^8 + 2^6 - 2^4 + 2^2 \equiv 7 - 4 + 1 - 7 + 4 \equiv 1 \pmod{9}$

4. Let $f(x) = x^2 - x + d$ and $g(x) = x^{13} + x + 90$. Then $f(0) = d = f(1)$, $g(0) = 90$ and $g(1) = 92$ have GCD is 2. Now note that $f(-1) = d + 2$ and $g(-1) = 88$ implies $d \neq 1$ and $d \neq -2$. Further $f(-1-2) = d + 6$ and $g(-2) = -8104$ shows that $d \neq -1$. Hence either $d = 2$ and since an integer d does not exist. However as he can check

$$\frac{x^{13} + x + 90}{x^2 - x + 2} = x^{11} + x^{10} - x^9 - 3x^8 - x^7 + 5x^6 + 7x^5 - 3x^4 - 17x^3 - 11x^2 + 23x + 45$$

checks and he is $d = 2$.

Event 3

1. We have

$$\begin{aligned} A &= (3a - X)^2 + (3b - X)^2 + (3c - X)^2 + 6 \\ &= (9a^2 - 6aX + X^2) + (9b^2 - 6bX + X^2) + (9c^2 - 6cX + X^2) + 6 \\ &= 9(a^2 + b^2 + c^2) - 6X(a + b + c) + 3X^2 + 6 \\ &= 9X^2 - 6X^2 + 3X^2 + 6 \\ &= 6X^2 + 6 \end{aligned}$$

Therefore $A \geq 6$. In particular $A = 6$ when $a = b = c = 0$.

2. The total sum of heights of boys is $6 \times 60 = 360$ km and the total sum of heights of girls is $24 \times 45 = 1080$ km. Thus the total sum of heights of all students is $360 + 1080 = 1440$ km. The average height of the students in the class is $1440/30 = 48$ km. Thus $b = 48$.

3. $a_1 = 48$ $a_2 = 24$ $a_3 = 12$ $a_4 = 6$ $a_5 = 3$ $a_6 = 10$ $a_7 = 5$ $a_8 = 16$ $a_9 = 8$ and hence the period

$$a_{3i+1} = 4 \quad a_{3i+2} = 2 \quad a_{3(i+1)} = 1$$

Period is 3 for $i \geq 3$. When $n = 2018$ $C = a_{2018} = a_{3(672)+2} = 2$.

4. Since $FH : FG = 2 : 1$ choose $\angle EHG = 30^\circ$ and $\angle EGH = 60^\circ$. We have $\triangle FGH \sim \triangle EFG$ and the ratio of the areas of $\triangle FGH$ to $\triangle EFG$ is $4 : 1$. Thus the ratio of the areas between $\triangle EGH$ and $\triangle FEG$ is $3 : 1$ and hence $D = 3$.

Event 4

1.

n	1^n	2^n	3^n	4^n	remainder
1	1	2	3	4	0
2	1	4	4	1	0
3	1	3	2	4	0
4	1	1	1	1	4
5	1	2	3	4	0
6	1	4	4	1	0
7	1	3	2	4	0
8	1	1	1	1	4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

$$\Rightarrow \text{remainder} = 0$$

Alternatively by Fermat's little theorem since 3 is prime and a relatively prime to

$$5 \quad a^4 \equiv 1 \pmod{5} \text{ he ia ee}$$

$$\begin{aligned} a &= 1^{2018} + 2^{2018} + 3^{2018} + 4^{2018} \\ &\equiv 1^2 + 2^2 + 3^2 + 4^2 \\ &\equiv 1 + 4 + 4 + 1 \\ &\equiv 0 \pmod{5} \end{aligned}$$

$$2. \quad xy = 100 = 2^2 \times 5^2$$

Leh $x = 2^{a_x} \times 5^{b_x}$ and $y = 2^{a_y} \times 5^{b_y}$ hien

$$a_x + a_y = 2 \text{ and } b_x + b_y = 2$$

$$\therefore \text{ number oo hays oo } a_x, a_y = 3$$

$$\text{number oo hays oo } b_x, b_y = 3$$

$$\therefore \text{ number oo hays oox } y = 3 \times 3 = 9$$

$$\begin{aligned} 3. \quad xyz + xy + xz + yz + x + y + z + 1 &= 357 \\ (x+1)(y+1)(z+1) &= 357 \\ &= 17 \times 7 \times 3 \\ \Rightarrow x+1+y+1+z+1 &= 17+7+3=27 \\ \Rightarrow x+y+z &= 24 \end{aligned}$$

$$4. \quad \frac{c}{2} = 12 \text{ and } \frac{c}{3} = 8;$$

$\therefore$ hie leashcommon mulhple oo 8 and 12 cs 24 and hie leashcommon mulhple oo 7 and 8 cs 56 hieeoore
 $d = 24 \times 56 = 1344.$

Hong Kong Mathematics Olympiad (2017 / 2018)

Final Event (Group) Solution

香港數學競賽 (2017 / 2018)

決賽項目(團體) 題解

Event 1

1.

Subjects \ Marks \ Students	Mary	Ming
Chinese	s	t
English	t	s
Mathematics	s	t
Total score	12	9

$$3(s+t) = 12+9 = 21$$

$$\Rightarrow s+t = 7$$

On the other hand, $s-t = 12-9-3$

$$\Rightarrow s = 5 \text{ and } t = 2$$

2. Let the radii of the internal circles be x and y . Since $\triangle ABC$ is a right-angled triangle,

$$(4^2 + (2x)^2) + (4^2 + (2y)^2) = 10^2$$

$$x^2 + y^2 = 17$$

The total area of the two circles is $\pi(x^2 + y^2) = 17\pi$.

Therefore, area of the shaded region is $25\pi - 17\pi = 8\pi$.

3.

2^n	r
$2^1 = 2 = 0 \times 7 + 2$	2
$2^2 = 4 = 0 \times 7 + 4$	4
$2^3 = 8 = 1 \times 7 + 1$	1
$2^4 = 16 = 2 \times 7 + 2$	2
$2^5 = 32 = 4 \times 7 + 4$	4
$2^6 = 64 = 9 \times 7 + 1$	1
$\vdots$	$\vdots$

It is observed that $f(2^{3x}) = 1$, $f(2^{3x+1}) = 2$, $f(2^{3x+2}) = 4$, where x is an integer.

$$2018 \div 3 = 672 \dots 2$$

$$\Rightarrow q = 4$$

4. 234234_5

$$= 2 \times 5^5 + 3 \times 5^4 + 4 \times 5^3 + 2 \times 5^2 + 3 \times 5^1 + 4$$

$$= (2 \times 5^2 + 3 \times 5^1 + 4)(5^3 + 1)$$

$$\therefore 234_5 = 2 \times 5^2 + 3 \times 5^1 + 4$$

$\therefore$ The remainder is 0.

Event 2

1. Let $x = 2^{-\frac{1}{u}}$. We have $1 - 5x + 4x^2 = 0$.

It follows that $x = \frac{1}{4}$ or $x = 1$ (rejected). Therefore, $2^{-2} = 2^{-\frac{1}{u}}$. It follows that $u = \frac{1}{2}$.

2. $v = \frac{(x-a)^2}{2b} + 5x$

$$= \frac{x^2 - 2ax + a^2 + 10bx}{2b}$$

$$= \frac{x^2 - 2(a-5b)x + (a-5b)^2}{2b} + \frac{a^2 - (a-5b)^2}{2b}$$

$$\geq \frac{a^2 - (a^2 - 10ab + 25b^2)}{2b}$$

$$= \frac{10ab - 25b^2}{2b}$$

$$= \frac{10a - 25b}{2}$$

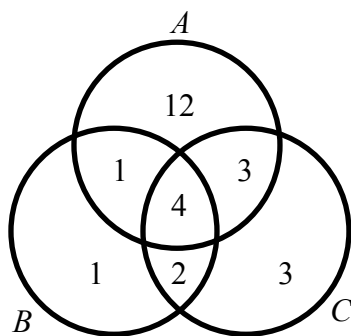
$$= \frac{10(15 + 12b) - 25b}{2}$$

$$= \frac{150 + 95b}{2}$$

$$\geq \frac{245}{2} = 122.5$$

3. Let A be the event that a student is a boy, B be the event that a student failed the first test, and C be the event that a student failed the second test.

$$\begin{aligned}
 \text{The number of girls passing both tests is } n &= \# \{A' \cap B' \cap C'\} = 35 - \# \{A \cup B \cup C\} \\
 &= 35 - (\# \{A\} + \# \{B\} + \# \{C\} - \# \{A \cap B\} - \# \{A \cap C\} - \# \{B \cap C\} + \# \{A \cap B \cap C\}) \\
 &= 35 - (20 + 8 + 12 - 5 - 7 - 6 + 4) \\
 &= 35 - 26 \\
 &= 9
 \end{aligned}$$

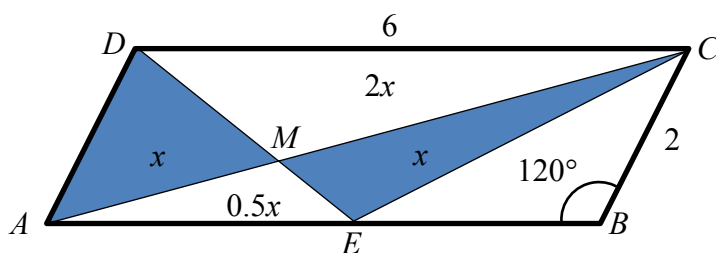


4. $m^{200} > 6^{300} \Leftrightarrow m^2 > 6^3 = 216.$

Since $14^2 = 196 < 216$ and $15^2 = 225 > 216$, we have $14 < \sqrt{216} < 15$, and thus $m = 15$.

Event 3

1.



Since $\triangle AME$ and $\triangle CMD$ are similar and E is a midpoint of AB ,

$$AE : CD = AM : MC = EM : MD = 1 : 2.$$

Let x be the area of $\triangle AMD$. The area of $\triangle CMD$ is $2x$ as it shares the same height with $\triangle AMD$ and $AM : MC = 1 : 2$. Also, the area of $\triangle CME$ is x as it shares the same height with $\triangle CMD$ and $EM : MD = 1 : 2$. Similar, the area of $\triangle AEM$ is $0.5x$ as it shares the same height with $\triangle AMD$ and $EM : MD = 1 : 2$. Since

$$x + x + 2x + 0.5x = \frac{3}{4} \times \text{area of parallelogram}$$

$$4.5x = \frac{3}{4} \times \text{area of parallelogram}$$

$$\text{or } x = \frac{1}{6} \times \text{area of parallelogram} = \frac{\sqrt{3}}{2}$$

$$\alpha = 2x = \sqrt{3}$$

2. Let the number be $\overline{acb}$. Then $a + b = c$ or $a+b-11=c$. In the first case, $\overline{acb} = 100a + 10(a+b) + b = 110a + 11b$. The quotient is $10a + b$. We have

$$10a + b = 3(a + b + c)$$

$$7a - 2b - 3(a + b) = 0$$

$$4a - 5b = 0$$

$$a = 5, b = 4, c = 9$$

In the second case, the quotient is $10a + b - 10$. We have

$$10a + b - 10 = 3(a + b + c)$$

$$7a - 2b - 3(a + b - 11) - 10 = 0$$

$$4a - 5b + 23 = 0$$

$$a = 3, b = 7, c = -1$$

There is no solution in this case and 594 is the only integer satisfying the conditions, therefore it is also the largest.

3. The function on the left of the inequality is defined on $-1 \leq \varphi \leq 1$. It is a monotonically decreasing function with value $\sqrt{2}$ at $\varphi = -1$. We consider the solution for the corresponding equation.

$$\sqrt{1-\varphi} - \sqrt{1+\varphi} = 1$$

$$1 - \varphi + 1 + \varphi - 2\sqrt{1-\varphi}\sqrt{1+\varphi} = 1$$

$$2 - 2\sqrt{1-\varphi^2} = 1$$

$$2\sqrt{1-\varphi^2} = 1$$

$$1 - \varphi^2 = \frac{1}{4}$$

$$\varphi = \pm \frac{\sqrt{3}}{2}$$

The function is decreasing, so the solution we want is the negative root $\varphi = -\frac{\sqrt{3}}{2}$.

$$4. \quad \frac{\theta + \gamma}{2} : \sqrt{\theta\gamma} = \frac{\frac{\theta}{\gamma} + 1}{2} : \sqrt{\frac{\theta}{\gamma}}. \text{ Let } r = \sqrt{\frac{\theta}{\gamma}}, \text{ then}$$

$$\frac{r^2 + 1}{2} : r = 13 : 12$$

$$26r = 12r^2 + 12$$

$$6r^2 - 13r + 6 = 0$$

$$(2r - 3)(3r - 2) = 0$$

$$r = \frac{3}{2} \text{ (rejected) or } r = \frac{2}{3}$$

$$\therefore \theta : \gamma = 4 : 9$$

Event 4

1. Since $X \geq 1$ and X is decreasing in A for $A \geq 0$, the greatest value of A is therefore

$$\begin{aligned} A &= (2018 - 1)^2 \\ &= 2017^2 \\ &= 2000^2 + 4000(17) + 17^2 \\ &= 4000000 + 68000 + 289 \\ &= 4068289 \end{aligned}$$

2. Put $x = \frac{y}{12}$ into the equation and get the monic polynomial

$$(y - 1)(y - 2)(y - 3)(y - 4) = 2 \times 3 \times 4 \times 5 = 120$$

It is easy to see that we have two roots, $y = 6$ and $y = -1$. We need to determine the other two roots r_1 and r_2 . Notice that the product of the roots must be the constant term of the polynomial after expansion. Thus,

$$(-1)(6)(r_1)(r_2) = 120 + (-1)(-2)(-3)(-4)$$

and

$$r_1 r_2 = 16$$

Further, the second coefficient of the (expanded) polynomial is $-1 - 2 - 3 - 4$, which is the negative of the sum of roots, namely, $-(-1 + 6 + r_1 + r_2)$. In other words,

$$-(-1 + 6 + r_1 + r_2) = -1 - 2 - 3 - 4$$

or

$$r_1 + r_2 = 5.$$

Thus, r_1 and r_2 are roots to $y^2 - 5y + 16 = 0$,

whose discriminant is negative. So the two roots are complex and of no interest to us. Therefore, the

product of the real roots is $\frac{6}{12} \times -\frac{1}{12} = -\frac{1}{24}$.

3. Let $C = \prod_{i=1}^7 \cos \frac{i\pi}{15}$ and $S = \prod_{i=1}^7 \sin \frac{i\pi}{15}$

Then $2^7 \times C \times S = \prod_{i=1}^7 2 \times \cos \frac{i\pi}{15} \times \sin \frac{i\pi}{15} = \prod_{i=1}^7 \sin \frac{2i\pi}{15}$

Since $\sin \frac{8\pi}{15} = \sin \left(\pi - \frac{7\pi}{15} \right) = \sin \frac{7\pi}{15}$

and likewise for $\sin \frac{10\pi}{15} = \sin \frac{5\pi}{15}$, $\sin \frac{12\pi}{15} = \sin \frac{3\pi}{15}$, $\sin \frac{14\pi}{15} = \sin \frac{\pi}{15}$

It follows that $2^7 \times C \times S = S$

and hence, $c = \frac{1}{2^7} = \frac{1}{128}$

4. Notice that the equation $r^2 + s^2 + t^2 = rs + st + tr$

implies that $(r-s)^2 + (s-t)^2 + (t-r)^2 = 0$,

which implies that $r = s = t$. Thus, $D = s + t = 2$.

Hong Kong Mathematics Olympiad (2018 / 2019)

Heat Event (Individual) Solution

香港數學競賽 (2018 / 2019)

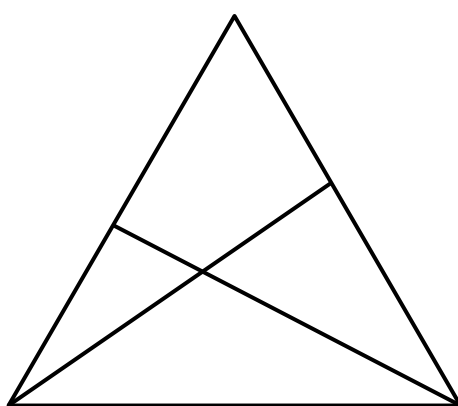
初賽項目(個人) 題解

甲部

Part A

1. 在圖一中， ABC 是一個等邊三角形。 D 和 E 分別是 AB 和 AC 上的點使 $AE = BD$ 。若 CD 和 BE 相交於 O 及 $\angle COE = y^\circ$ ，求 y 的值。〔答：60〕

In Figure 1, ABC is an equilateral triangle. D and E are points on AB and AC respectively such that $AE = BD$. If CD and BE intersect at O and $\angle COE = y^\circ$, find the value of y .



圖一

Figure 1

Solution

$$\triangle BAE \cong \triangle CBD$$

$$\angle BDC = \angle AEB \text{ i.e. } \angle ODB = \angle AEO$$

$\therefore ADOE$ is a cyclic quadrilateral (ext. $\angle =$ opp. Int. $\angle$)

$$\therefore \angle COE = \angle DAE = 60^\circ$$

$$y = 60$$

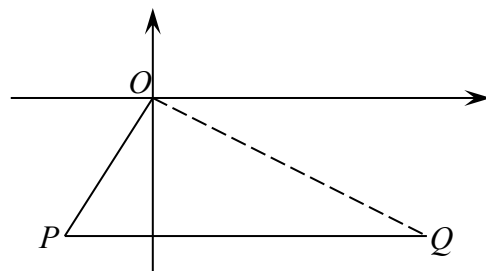
2. 設 O 為極坐標系統的極點。若 $P(6, 240^\circ)$ 向右平移 16 單位至 Q 而 $\triangle OPQ$ 的面積為 T 平方單位，求 T 的值。〔答： $24\sqrt{3}$ 〕

Let O be the pole of the polar coordinate system. If $P(6, 240^\circ)$ is translated to the right by 16 units to Q and the area of $\triangle OPQ$ is T square units, find the value of T .

Solution

$$OP = 6, PQ = 16, \angle OPQ = 60^\circ$$

$$T = \frac{1}{2}(16)(6 \sin 60^\circ) = 24\sqrt{3}$$



3. 已知 x 及 y 均為實數。若 $y^2 - 4xy + 5x^2 - 8x + 16 = 0$ 及 $F = x - y$ ，求 F 的值。〔答：-4〕

Given that x and y are real numbers. If $y^2 - 4xy + 5x^2 - 8x + 16 = 0$ and $F = x - y$, find the value of F .

Solution

$$y^2 - 4xy + 4x^2 + x^2 - 8x + 16 = 0$$

$$(y - 2x)^2 + (x - 4)^2 = 0$$

$$x = 4 \text{ and } y = 8$$

$$F = -4$$

4. 設 n 為正整數。若 $a_n = 1 + 2 + 2^2 + \cdots + 2^n$ 及 $b = a_{10} - a_5 + a_1$ ，求 b 的值。〔答：1987〕

Let n be a positive integer. If $a_n = 1 + 2 + 2^2 + \cdots + 2^n$ and $b = a_{10} - a_5 + a_1$, find the value of b .

Solution

$$a_{10} = 1 + 2 + \cdots + 2^{10} = 2^{11} - 1 = 2047$$

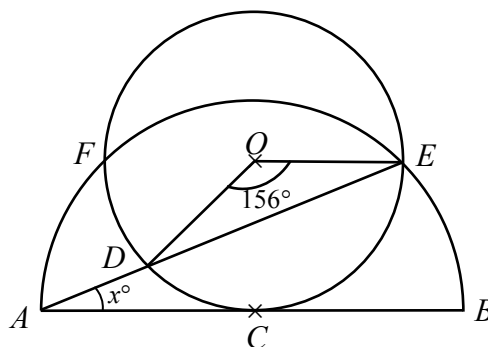
$$a_5 = 1 + 2 + \cdots + 2^5 = 2^6 - 1 = 63$$

$$a_1 = 1 + 2 = 3$$

$$b = 2047 - 63 + 3 = 1987$$

5. 在圖二中， O 為圓 DEF 的圓心。 AB 為圓 DEF 在 C 的切線，其中 C 為半圓 ABE 的圓心，且半圓 ABE 通過 F 。 ADE 為一直線。若 $\angle DOE = 156^\circ$ 及 $\angle BAE = x^\circ$ ，求 x 的值。〔答：26〕

In Figure 2, O is the centre of the circle DEF . AB is the tangent to the circle DEF at C , where C is the centre of the semicircle ABE , which also passes through F . ADE is a straight line. If $\angle DOE = 156^\circ$ and $\angle BAE = x^\circ$, find the value of x .



圖二

Figure 2

Solution

$$\angle CEA = x^\circ \text{ (base } \angle\text{s, isos. } \Delta\text{)}$$

$$\angle ECB = 2x^\circ \text{ (ext. } \angle \text{ of } \Delta\text{)}$$

$$\angle DCA = x^\circ \text{ (} \angle \text{ in alt. segment)}$$

$$\angle DCE = 180^\circ - 3x^\circ$$

$$\text{Reflex } \angle DOE = 204^\circ$$

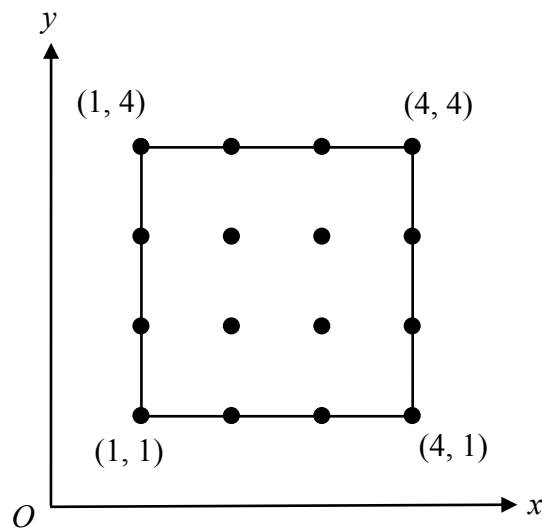
$$\angle DCE = 102^\circ \text{ (} \angle \text{ at centre twice as } \angle \text{ at } \square^{\text{ce}}\text{)}$$

$$\therefore 180^\circ - 3x^\circ = 102^\circ$$

$$x = 26$$

6. 在圖三中，直角坐標平面上一個正方形的四個頂點的坐標分別為(1, 1)、(1, 4)、(4, 1)及(4, 4)。若在該正方形中（包括邊界）選擇任何三個坐標均為整數的點，問可組成多少個三角形？〔答：516〕

In Figure 3, the vertices of a square in the rectangular coordinate plane are (1, 1), (1, 4), (4, 1) and (4, 4). How many triangles can be formed by selecting any three points in the square (including the boundary) with integer coordinates?



圖三

Figure 3

Solution

The number of triangles

$$= C_3^{16} - 10C_3^4 - 4C_3^3$$

$$= 516$$

7. 在圖四中， AB 與 CD 相交於 E 。設 q 單位為 AE 的長度。求 q 的值。〔答： $\frac{7\sqrt{5}}{3}$ 〕

In Figure 4, AB and CD intersect at E . Let q units be the length of AE . Find the value of q .

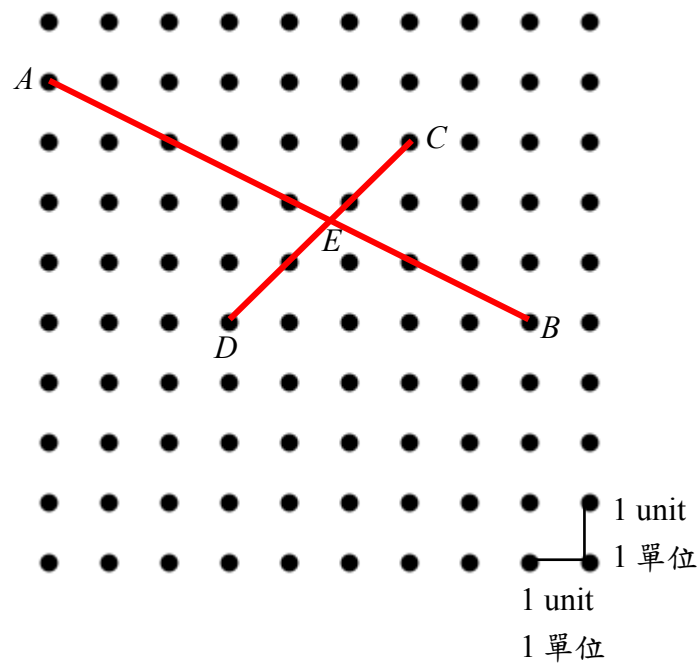
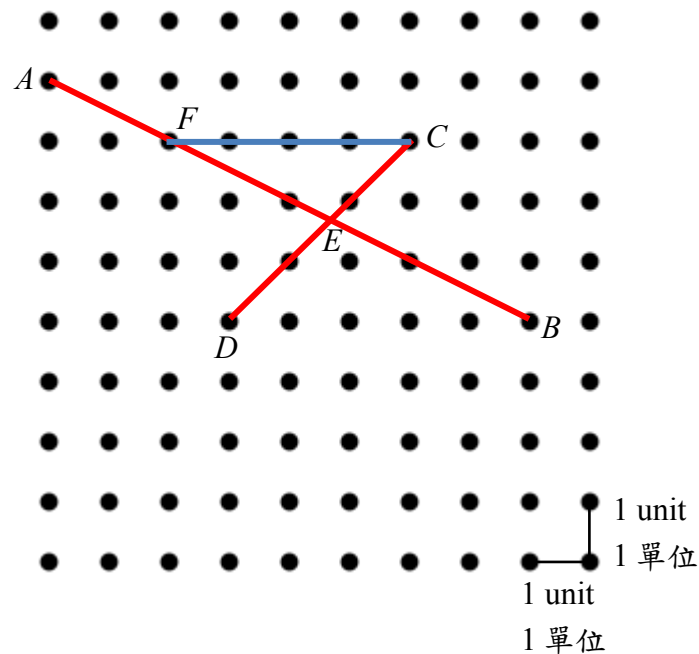


Figure 4

圖四

Solution

Consider the figure below.



$$\frac{FC}{DB} = \frac{4}{5} \Rightarrow \frac{FE}{EB} = \frac{4}{5}$$

$$\text{As } FB = \sqrt{3^2 + 6^2} = 3\sqrt{5}, \quad FE = 3\sqrt{5} \times \frac{4}{9} = \frac{4\sqrt{5}}{3}$$

$$\therefore AE = AF + FE = \sqrt{1^2 + 2^2} + \frac{4\sqrt{5}}{3} = \frac{7\sqrt{5}}{3}$$

8. 在圖五中， D 是在 BC 上的一點使 $\angle ABD = \angle CAD$ 及 $\frac{BD}{AC} = \frac{8}{3}$ 。若 $\frac{\Delta ABD \text{ 的面積}}{\Delta ADC \text{ 的面積}} = k$ ，求 k 的值。〔答：8〕

In Figure 5, D is a point on BC such that $\angle ABD = \angle CAD$ and $\frac{BD}{AC} = \frac{8}{3}$. If $\frac{\text{Area of } \Delta ABD}{\text{Area of } \Delta ADC} = k$, find the value of k .

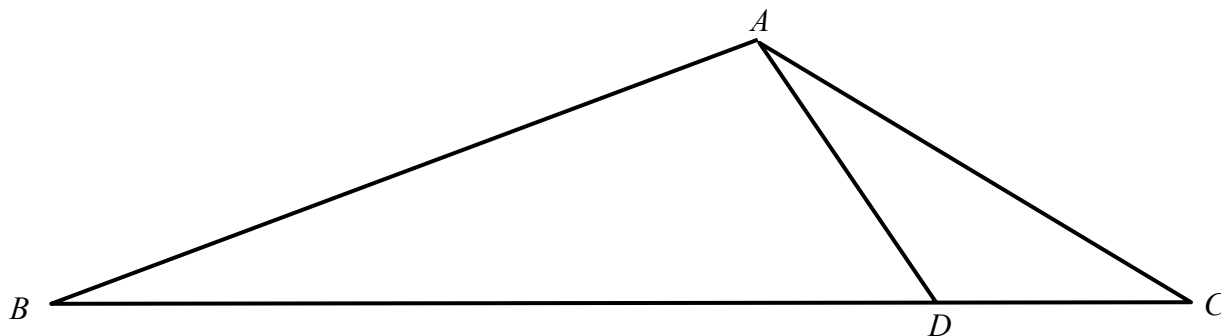


Figure 5

圖五

Solution

In ΔBCA and ΔACD , $\angle B = \angle CAD$ (given)

$\angle BCA = \angle ACD$ (common)

$\therefore \Delta BCA \sim \Delta ACD$ (AA)

$$\therefore \frac{BA}{DA} = \frac{AC}{DC} = \frac{BC}{AC}$$

$$\Rightarrow AC^2 = BC \cdot DC = (BD + DC)DC = BD \cdot DC + DC^2$$

$$\Rightarrow \frac{BD \cdot DC}{AC^2} + \frac{DC^2}{AC^2} = 1 \Rightarrow \frac{BD}{AC} \cdot \frac{DC}{AC} + \frac{DC^2}{AC^2} = 1$$

Given that $\frac{BD}{AC} = \frac{8}{3}$ and let $\frac{DC}{AC} = y$, we have $y^2 + \frac{8}{3}y - 1 = 0$

$$\Rightarrow 3y^2 + 8y - 3 = 0 \Rightarrow (3y - 1)(y + 3) = 0$$

$$\therefore y = \frac{1}{3} \text{ or } -3 \text{ (rejected)}$$

$$\therefore \frac{DC}{AC} = \frac{1}{3}$$

$$\therefore \frac{BD}{DC} = \frac{8}{1}$$

$$\therefore \frac{\text{Area of } \Delta ABD}{\text{Area of } \Delta ADC} = \frac{BD}{DC} = 8$$

$$\therefore k = 8$$

9. 已知 α 及 β 為方程 $x^2 + 32x - 1 = 0$ 的兩個根。若 $P = (\alpha^2 + 31\alpha - 2)(\beta^2 + 33\beta)$ ，求 P 的值。
〔答：32〕

Given that α and β are the two roots of the equation $x^2 + 32x - 1 = 0$. If $P = (\alpha^2 + 31\alpha - 2)(\beta^2 + 33\beta)$, find the value of P .

Solution

Given that $x^2 + 32x - 1 = 0$, therefore $\alpha + \beta = -32$ and $\alpha\beta = -1$.

$$\begin{aligned} & \alpha^2 + 31\alpha - 2 \\ &= \alpha^2 + 32\alpha - 1 - \alpha - 1 \\ &= -(\alpha + 1) \end{aligned}$$

$$\begin{aligned} & \beta^2 + 33\beta \\ &= \beta^2 + 32\beta - 1 + \beta + 1 \\ &= \beta + 1 \end{aligned}$$

$$\begin{aligned} \therefore (\alpha^2 + 31\alpha - 2)(\beta^2 + 33\beta) &= -(\alpha + 1)(\beta + 1) \\ &= -[\alpha\beta + (\alpha + \beta) + 1] \\ &= -[-1 - 32 + 1] \\ &= 32 \end{aligned}$$

10. 設 $c = \sqrt[3]{7+5\sqrt{2}} + \sqrt[3]{7-5\sqrt{2}}$ 。若 $w = c^2$ ，求 w 。〔答：4〕
Let $c = \sqrt[3]{7+5\sqrt{2}} + \sqrt[3]{7-5\sqrt{2}}$. If $w = c^2$, find w .

Solution

$$\text{Let } (a + b\sqrt{2})^3 = 7 + 5\sqrt{2}$$

Expanding it gives:

$$a^3 + 6ab^2 = a(a^2 + 6b^2) = 7$$

$$\text{and } 3a^2b + 2b^3 = b(3a^2 + 2b^2) = 5$$

On solving, gives $b = 1$ and hence $a = 1$

Likewise, if $(p + q\sqrt{2})^3 = 7 - 5\sqrt{2}$, then $q = -1, p = 1$.

$$\therefore c = \sqrt[3]{7+5\sqrt{2}} + \sqrt[3]{7-5\sqrt{2}} = 1 + \sqrt{2} + 1 - \sqrt{2} = 2$$

$$\therefore w = c^2 = (2)^2 = 4$$

11. 在圖六中， $ABCD$ 為一個長方形。 M 和 N 分別是 DC 和 AB 的中點且 $AE : EN = BF : FN = 1 : 2$ 。
 EM 和 FM 分別相交 DB 於 G 及 H 。若長方形 $ABCD$ 及三角形 GHM 的面積分別是 96 和 S ，
 求 S 的值。〔答：9〕

In Figure 6, $ABCD$ is a rectangle. M and N are the mid-points of DC and AB respectively and $AE : EN = BF : FN = 1 : 2$. EM and FM intersect DB at G and H respectively. If the areas of the rectangle $ABCD$ and the triangle GHM are 96 and S respectively, find the value of S .

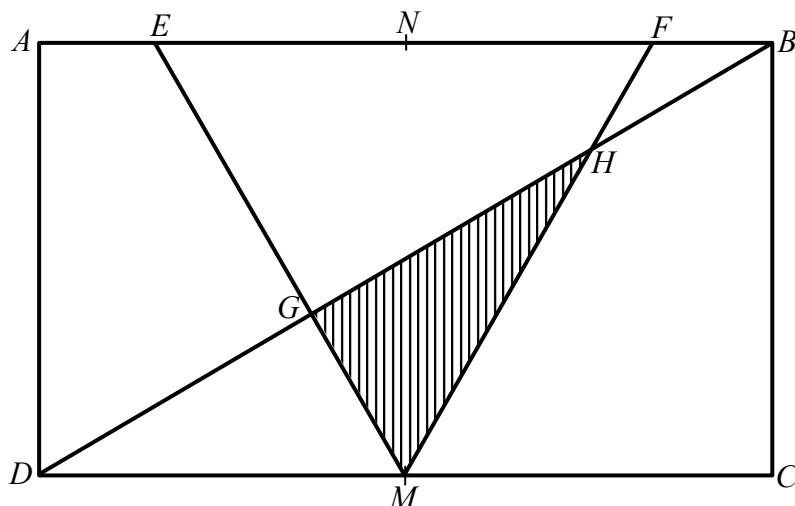
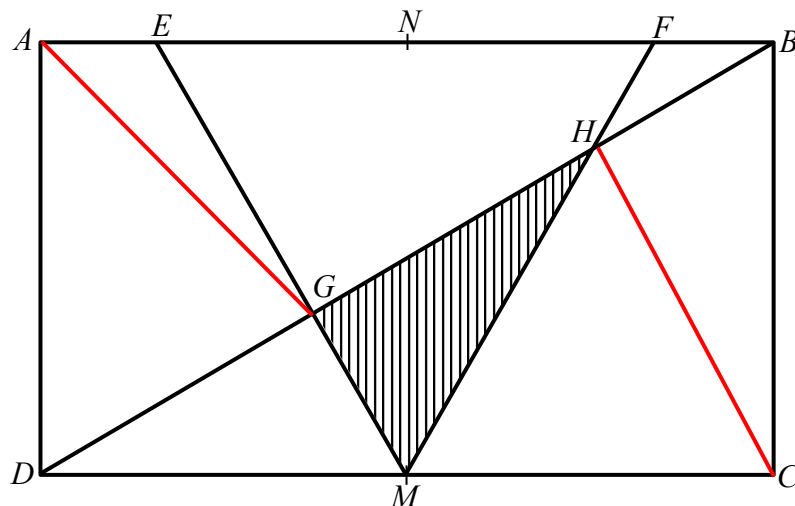


Figure 6

圖六

Solution



$$\text{Area of } \triangle BCD = \frac{96}{2} = 48$$

Join CH . As $BF : DM = 1 : 3$, $BH : HD = 1 : 3$

$$\text{Area of } \triangle CHD = 48 \times \frac{3}{4} = 36$$

$$\text{As } DM : MC = 1 : 1, \text{ Area of } \triangle DMH = 36 \times \frac{1}{2} = 18$$

Join AG .

As $BE : DM = 5 : 3$, $BG : GD = 5 : 3$

$$\text{Area of } \triangle ABG = 48 \times \frac{5}{8} = 30$$

$$\text{As } AE : BE = 1 : 5, \text{ Area of } \triangle EBG = 30 \times \frac{5}{6} = 25$$

$$\text{Area of } \triangle MDG = 25 \times \left(\frac{3}{5}\right)^2 = 9$$

$$\begin{aligned}\text{Area of } \triangle GMH &= \text{Area of } \triangle DMH - \text{Area of } \triangle MDG \\ &= 18 - 9 = 9\end{aligned}$$

12. 在三角形 ABC 中， $AB = 14$ 、 $BC = 48$ 及 $AC = 50$ 。將 P 及 Q 分別記為 $\triangle ABC$ 的內心及外心。設 d 單位為 PQ 的長度。求 d 的值。〔答： $5\sqrt{13}$ 〕

In triangle ABC , $AB = 14$, $BC = 48$ and $AC = 50$. Denote the in-centre and circumcentre of $\triangle ABC$ by P and Q respectively. Let d units be the length of PQ . Find the value of d .

Solution

Note that $\angle ABC = 90^\circ$.

Let $(0, 0)$, $(0, 14)$ and $(48, 0)$ be the coordinates of B , A and C .

$$\text{Inradius} = \frac{14 \times 48}{14 + 48 + 50} = 6$$

$$\therefore P = (6, 6)$$

AC is the diameter of circumcircle ABC .

$$\therefore Q = (24, 7)$$

$$d = \sqrt{(24-6)^2 + (7-6)^2} = 5\sqrt{13}$$

13. 已知正整數 a 、 b 及 c 滿足下列條件：

(i) $a > b > c$,

(ii) $(a-b)(b-c)(a-c) = 84$,

(iii) $abc < 100$.

設 M 為 a 的最大值，求 M 。〔答：9〕

Given that a , b and c are positive integers satisfying the following conditions:

(i) $a > b > c$,

(ii) $(a-b)(b-c)(a-c) = 84$,

(iii) $abc < 100$.

Let M be the maximum value of a . Find M .

Solution

As $a-b$, $b-c$ and $a-c$ are positive integers and $a-c=(a-b)+(b-c)$, when 84 is factorised into 3 factors, there are only two possible combinations: 3, 4, 7 or 4, 3, 7.

For the first case (3, 4, 7): we have
$$\begin{cases} a-b=3 \\ b-c=4 \\ a-c=7 \end{cases} \Rightarrow (a, b, c) = (8, 5, 1)$$

For the second case (4, 3, 7): we have
$$\begin{cases} a-b=4 \\ b-c=3 \\ a-c=7 \end{cases} \Rightarrow (a, b, c) = (8, 4, 1) \text{ or } (9, 5, 2)$$

$\therefore M$ is 9.

14. 已知 $3\sin x + 2\sin y = 4$ 。設 N 為 $3\cos x + 2\cos y$ 的最大值。求 N 。〔答：3〕

Given that $3\sin x + 2\sin y = 4$. Let N be the maximum value of $3\cos x + 2\cos y$. Find N .

Solution

$$3\sin x + 2\sin y = 4 \Rightarrow (3\sin x + 2\sin y)^2 = 16, \text{ that is, } 9\sin^2 x + 12\sin x \sin y + 4\sin^2 y = 16 \dots\dots(1)$$

$$\text{Let } 3\cos x + 2\cos y = t \Rightarrow (3\cos x + 2\cos y)^2 = t^2, \text{ that is,}$$

$$9\cos^2 x + 12\cos x \cos y + 4\cos^2 y = t^2 \dots\dots(2)$$

(1) + (2),

$$9 + 12(\sin x \sin y + \cos x \cos y) + 4 = t^2 + 16$$

$$13 + 12\cos(x-y) = t^2 + 16$$

$$12\cos(x-y) = t^2 + 3$$

$$\text{Since } -1 \leq \cos(x-y) \leq 1, \text{ we have } t^2 + 3 \leq 12 \Rightarrow t^2 \leq 9$$

$$\therefore -3 \leq t \leq 3$$

$\therefore N$ is 3.

15. 已知 x 、 y 、 z 為正實數且滿足

$$\begin{cases} x^2 + xy + y^2 = 7 \\ y^2 + yz + z^2 = 21 \\ x^2 + xz + z^2 = 28 \end{cases}$$

若 $a = x + y + z$ ，求 a 的值。〔答：7〕

Given that x, y, z are positive real numbers satisfying

$$\begin{cases} x^2 + xy + y^2 = 7 \\ y^2 + yz + z^2 = 21 \\ x^2 + xz + z^2 = 28 \end{cases}$$

If $a = x + y + z$, find the value of a .

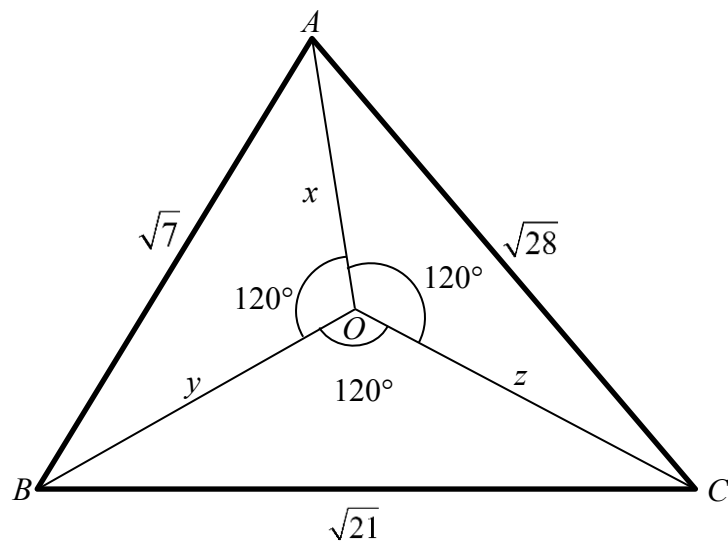
Solution

By inspection, $x = 2$, $y = 1$ and $z = 4$. Therefore $a = 2 + 1 + 4 = 7$. Alternatively,

Rewrite (1), (2), (3) as

$$\begin{cases} x^2 + y^2 - 2xy \cos 120^\circ = (\sqrt{7})^2 \\ y^2 + z^2 - 2yz \cos 120^\circ = (\sqrt{21})^2 \\ x^2 + z^2 - 2xz \cos 120^\circ = (\sqrt{28})^2 \end{cases}$$

Construct $\triangle ABC$ such that O is a point inside the triangle with $OA = x$, $OB = y$, $OC = z$ and $\angle AOB = \angle BOC = \angle AOC = 120^\circ$.



Then $AB = \sqrt{7}$

$BC = \sqrt{21}$

$CA = \sqrt{28}$

Note that $\triangle ABC$ is a right-angled triangle.

Area of $\triangle ABC$

$$\frac{1}{2}(xy + yz + xz) \sin 120^\circ = \frac{1}{2} \sqrt{7} \cdot \sqrt{21}$$

$$xy + yz + xz = 14 \quad \dots\dots(4)$$

(1) + (2) + (3) + 3(4)

$$2(x^2 + y^2 + z^2 + 2xy + 2yz + 2xz) = 98$$

$$(x + y + z)^2 = 49$$

$$x + y + z = 7$$

Hong Kong Mathematics Olympiad (2018 / 2019)
Heat Event (Group) Solution
香港數學競賽 (2018 / 2019)
初賽項目(團體) 題解

1. 對所有正實數 x ，定義 $f(x) = \log_{2019} x^{2020}$ 。若 $D = f(\sqrt{3}) + f(\sqrt{673})$ ，求 D 的值。〔答：1010〕

For all positive real numbers x , define $f(x) = \log_{2019} x^{2020}$. If $D = f(\sqrt{3}) + f(\sqrt{673})$, find the value of D .

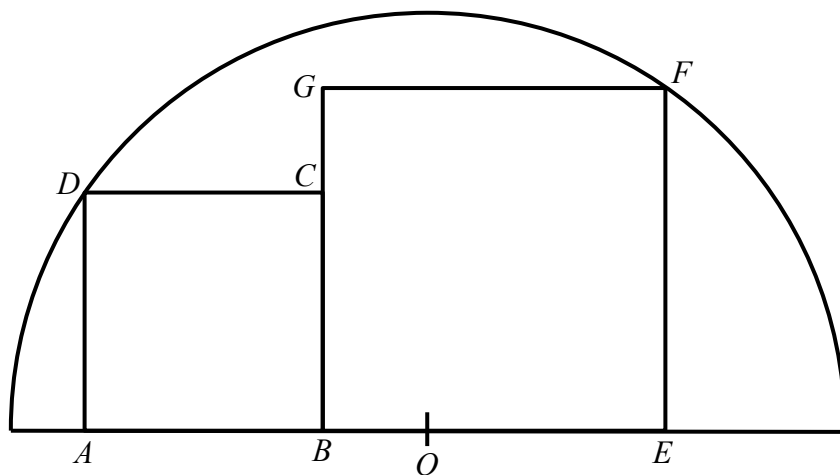
Solution

Consider

$$\begin{aligned} & f(\sqrt{3}) + f(\sqrt{673}) \\ &= \log_{2019} (\sqrt{3})^{2020} + \log_{2019} (\sqrt{673})^{2020} \\ &= \log_{2019} (\sqrt{3 \times 673})^{2020} \\ &= \log_{2019} (\sqrt{2019})^{2020} \\ &= \log_{2019} (2019)^{1010} \\ &= 1010 \end{aligned}$$

2. 圖一所示為一個半徑為 5 cm 且圓心位於 O 的半圓。 A 、 B 和 E 為直徑上的點而 D 和 F 則為圓周上的點。設 $S \text{ cm}^2$ 為兩個正方形 $ABCD$ 和 $BEFG$ 面積之和。求 S 的值。〔答：25〕

Figure 1 shows a semi-circle with radius 5 cm and centre at O . A , B and E are points on the diameter. D and F are points on the circumference. Let $S \text{ cm}^2$ be the total area of the two squares $ABCD$ and $BEFG$. Find the value of S .



圖一

Figure 1

Solution

Let AB be x cm and BE be y cm

Join OD and OF .

$$OA = \sqrt{5^2 - x^2}$$

$$OE = \sqrt{5^2 - y^2}$$

$$OA + OE = AB + BE$$

$$\sqrt{25 - x^2} + \sqrt{25 - y^2} = x + y$$

$$\sqrt{25 - x^2} - x = y - \sqrt{25 - y^2}$$

$$25 - x^2 - 2x\sqrt{25 - x^2} + x^2 = y^2 - 2y\sqrt{25 - y^2} + 25 - y^2$$

$$x\sqrt{25 - x^2} = y\sqrt{25 - y^2}$$

$$x^2(25 - x^2) = y^2(25 - y^2)$$

$$25(x^2 - y^2) = (x^2 - y^2)(x^2 + y^2)$$

$$(x^2 - y^2)(x^2 + y^2 - 25) = 0$$

$$x^2 = y^2 \text{ or } x^2 + y^2 = 25$$

$$x = y$$

When $x = y$, in $\triangle OAD$, $x^2 + y^2 = 25$.

$\therefore$ The total area of the two squares is 25 cm^2 .

3. 若從一個正 9 邊形的 9 個頂點中選 3 個頂點組成一個等腰三角形，共可組成多少個等腰三角形？〔答：30〕

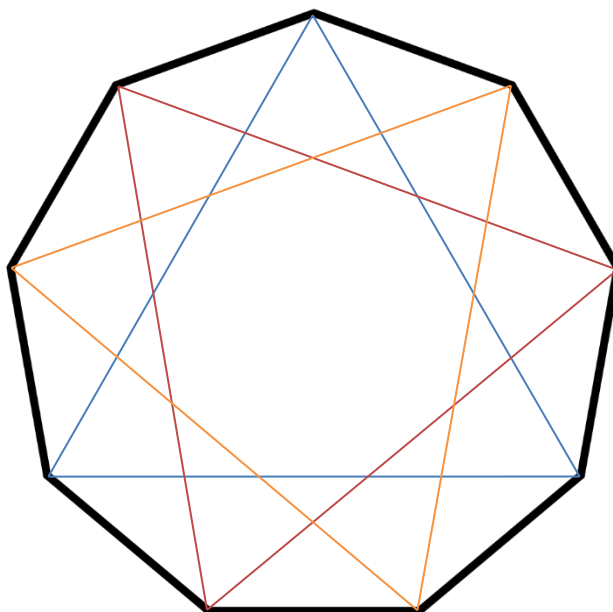
If three vertices are chosen from the nine vertices of a regular nonagon to form an isosceles triangle, how many such isosceles triangles are there?

Solution

For each vertex of a regular nonagon, three isosceles triangles can be formed if this vertex is also the vertex opposite to the base of the isosceles triangle formed. Note that one of these three isosceles triangles is an equilateral triangle. So there are 9 equilateral triangles which are counted 3 times each. Hence, the required number

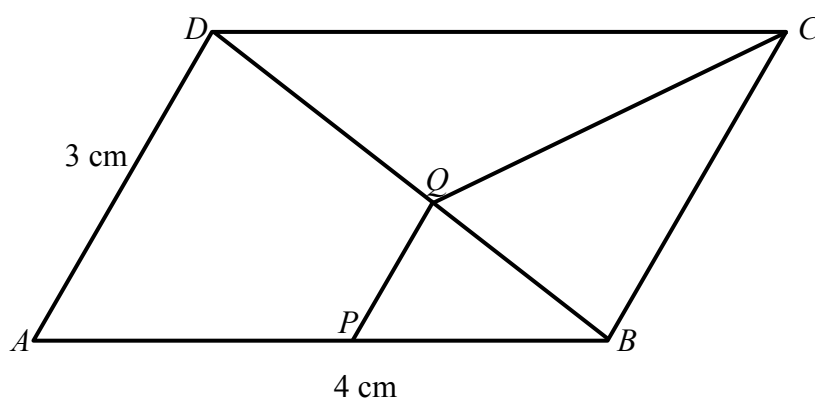
$$= 4 \times 9 - 9 + 3$$

$$= 30$$



4. 在圖二中， $ABCD$ 為一個平行四邊形，其中 $AB = 4\text{ cm}$ 、 $AD = 3\text{ cm}$ 及 $\sin A = \frac{2}{3}$ 。 P 和 Q 分別是 AB 和 BD 上的點使 $PQ \parallel AD$ 且四邊形 $PBCQ$ 的面積為 3 cm^2 。設 $q\text{ cm}$ 為 AP 的長度。求 q 的值。〔答：2〕

In Figure 2, $ABCD$ is a parallelogram, where $AB = 4\text{ cm}$, $AD = 3\text{ cm}$ and $\sin A = \frac{2}{3}$. P and Q are points on AB and BD respectively such that $PQ \parallel AD$ and the area of the quadrilateral $PBCQ$ is 3 cm^2 . Let $q\text{ cm}$ be the length of AP . Find the value of q .



圖二

Figure 2

Solution

Let $AP = x$ cm

$$\therefore PQ \parallel AD$$

$$\therefore \frac{PQ}{AD} = \frac{BP}{AB}$$

$$\frac{PQ}{3} = \frac{4-x}{4}$$

$$PQ = 3 - \frac{3}{4}x$$

Note that $PBCQ$ is a trapezium.

$$\begin{aligned} \text{Area of } PBCQ &= \frac{1}{2}(PQ + BC)BP \sin \angle QPB \\ &= \frac{1}{2}\left(3 - \frac{3}{4}x + 3\right) \cdot (4-x) \cdot \frac{2}{3} \\ &= \frac{x^2}{4} - 3x + 8 \end{aligned}$$

$$\frac{x^2}{4} - 3x + 8 = 3$$

$$x^2 - 12x + 20 = 0$$

$$x = 2 \text{ or } x = 10(\text{rejected})$$

$$\therefore AP = 2 \text{ cm}$$

5. 已知 $f(x) - 2f\left(\frac{1}{x}\right) = x$ ，其中 $x \neq 0$ 。設 y 為滿足方程 $f(x) = 1$ 的 x 的最大值。求 y 。〔答：-1〕

Given that $f(x) - 2f\left(\frac{1}{x}\right) = x$, where $x \neq 0$. Let y be the maximum value of x that satisfies the equation $f(x) = 1$. Find y .

Solution

$$\begin{cases} f(x) - 2f\left(\frac{1}{x}\right) = x & \dots\dots(1) \\ f\left(\frac{1}{x}\right) - 2f(x) = \frac{1}{x} & \dots\dots(2) \end{cases}$$

$$(1) + 2(2), \quad -3f(x) = x + \frac{2}{x} \Rightarrow f(x) = -\frac{x^2 + 2}{3x}$$

Consider $f(x) = 1$, we have

$$-x^2 - 2 = 3x$$

$$x^2 + 3x + 2 = 0$$

$$(x+1)(x+2) = 0$$

$$x = -1 \quad \text{or} \quad x = -2$$

$$\therefore y = -1$$

6. 設 a_k 為多項式 $(2x-2)^3(2x+2)^3(2x+1)^3$ 中的項 x^k 的係數。若 $Q = a_2 + a_4 + a_6 + a_8$ ，求 Q 的值。〔答：64〕

Let a_k be the coefficient of the term x^k in the polynomial $(2x-2)^3(2x+2)^3(2x+1)^3$. If $Q = a_2 + a_4 + a_6 + a_8$, find the value of Q .

Solution

$$\text{Let } (2x-2)^3(2x+2)^3(2x+1)^3 = a_0 + a_1x + a_2x^2 + \cdots + a_9x^9 \quad \text{..... (1)}$$

$$\text{Substitute } x = -1 \text{ into (1) gives } a_0 - a_1 + a_2 - a_3 + \cdots + a_8 - a_9 = 0 \quad \text{..... (2)}$$

$$\text{Substitute } x = 1 \text{ into (1) gives } a_0 + a_1 + a_2 + a_3 + \cdots + a_8 + a_9 = 0 \quad \text{..... (3)}$$

$$(2) + (3) \text{ gives: } a_0 + a_2 + a_4 + a_6 + a_8 = 0 \quad \text{..... (4)}$$

$$\text{Substitute } x = 0 \text{ into (1) gives } a_0 = -64$$

$$\text{Therefore, put } a_0 = -64 \text{ into (4) gives } a_2 + a_4 + a_6 + a_8 = 64$$

7. 設 $f(x) = -6x^2 + 4x\cos\theta + \sin\theta$ ，其中 $0^\circ \leq \theta \leq 360^\circ$ 。已知對所有實數 x ， $f(x) \leq 0$ 。若 θ 的最大值與最小值之差為 d° ，求 d 。〔答：120〕

Let $f(x) = -6x^2 + 4x\cos\theta + \sin\theta$, where $0^\circ \leq \theta \leq 360^\circ$. It is given that $f(x) \leq 0$ for all real numbers x .

If d° is the difference between the greatest and the least values of θ , find d .

Solution

For $-6x^2 + 4x\cos\theta + \sin\theta = 0$ has at most one real root. Consider $\Delta \leq 0$,

$$(4\cos\theta)^2 - 4\sin\theta(-6) \leq 0$$

$$2(1 - \sin^2\theta) + 3\sin\theta \leq 0$$

$$2\sin^2\theta - 3\sin\theta - 2 \geq 0$$

$$(2\sin\theta + 1)(\sin\theta - 2) \geq 0$$

$$\therefore -1 \leq \sin\theta \leq 1, \quad \sin\theta \leq -\frac{1}{2} \text{ or } \sin\theta \geq 2 \text{ (rejected)}$$

$$\therefore -1 \leq \sin\theta \leq -\frac{1}{2}$$

$$210^\circ \leq \theta \leq 330^\circ$$

Hence, $d = 330 - 210 = 120$

8. 設 $\{a_n\}$ 為一個正實數序列使當 $n > 1$ 時, $a_n = a_{n-1}a_{n+1} - 1$ 。已知 2018 在序列中及 $a_2 = 2019$ 。

若 s 為 a_1 的所有可取值的數目, 求 s 。〔答: 4〕

Let $\{a_n\}$ be a sequence of positive real numbers such that $a_n = a_{n-1}a_{n+1} - 1$ for $n > 1$. It is given that

2018 is in the sequence and $a_2 = 2019$. If s is the number of all possible values of a_1 , find s .

Solution

Let $a_1 = a$

$$\text{As } a_n = a_{n-1}a_{n+1} - 1, \quad a_{n+1} = \frac{a_n + 1}{a_{n-1}}.$$

$$a_3 = \frac{a_2 + 1}{a_1} = \frac{2020}{a}$$

$$a_4 = \frac{a_3 + 1}{a_2} = \frac{\frac{2020}{a} + 1}{2019} = \frac{2020 + a}{2019a}$$

$$a_5 = \frac{a_4 + 1}{a_3} = \frac{\frac{2020 + a}{2019a} + 1}{\frac{2020}{a}} = \frac{2020 + 2020a}{2019 \times 2020} = \frac{1 + a}{2019}$$

$$a_6 = \frac{a_5 + 1}{a_4} = \frac{\frac{1 + a}{2019} + 1}{\frac{2020 + a}{2019a}} = \frac{2020 + a}{2020 + a} = a = a_1$$

As it is periodic, the term 2018 may appear in the sequence when $a_1 = 2018$, $a_3 = 2018$, $a_4 = 2018$ or $a_5 = 2018$

$$\text{If } \frac{2020}{a} = 2018, \quad a = \frac{2020}{2018} = \frac{1010}{1009}.$$

$$\text{If } \frac{2020 + a}{2019a} = 2018, \quad a = \frac{2020}{2018 \times 2019 - 1}.$$

$$\text{If } \frac{1 + a}{2019} = 2018, \quad a = 2018 \times 2019 - 1.$$

$$\therefore s = 4.$$

9. 有多少對正整數 x, y 可滿足 $xy = 6(x + y + \sqrt{x^2 + y^2})$? 〔答: 12〕

How many pairs of positive integers x, y are there satisfying $xy = 6(x + y + \sqrt{x^2 + y^2})$?

Solution

$$xy - 6(x + y) = 6\sqrt{x^2 + y^2}$$

$$\begin{cases} x^2y^2 - 12xy(x + y) + 36(x + y)^2 = 36(x^2 + y^2) & \dots\dots(1) \\ xy - 6(x + y) > 0 & \dots\dots(2) \end{cases}$$

From (1),

$$x^2y^2 - 12xy(x+y) + 72xy = 0$$

$$x = y = 0 \text{ or } xy - 12(x+y) + 72 = 0 \quad \dots(3)$$

From (3),

$$xy - 12(x+y) + 72 = 0$$

$$(x-12)(y-12) = 72 \quad \dots\dots(4)$$

$$y = \frac{12(x-6)}{x-12} \quad \dots\dots(5)$$

Sub. (5) into (2),

$$\frac{12(x-6)^2}{x-12} - 6x > 0$$

$$\frac{(x-6)^2 + 36}{x-12} > 0$$

$$\therefore x > 12$$

Note that $72 = 2^3 \times 3^2$.

72 has $(3+1)(2+1) = 12$ positive factors.

From $(x-12)(y-12) = 72$ and $x > 12$.

The number of pairs

= positive factors of 72

= 12

10. D 是等邊三角形 ABC 內的一點使 $AD = BD = 5\sqrt{2}$ 及 $CD = 10$ 。設 S 為 $\triangle ABC$ 的面積。求 S 的值。

$$[\text{答} : 25\sqrt{3} + \frac{75}{2}]$$

D is a point inside the equilateral triangle ABC such that $AD = BD = 5\sqrt{2}$ and $CD = 10$. Let S be the area of $\triangle ABC$. Find the value of S .

Solution

Construct the figure on the right

Rotate $\triangle ADC$ about A such that AB and AC are overlapping.

Name the image of point D by D' after the rotation.

Clearly, $\angle DAD' = 60^\circ$ and hence, $\triangle DAD'$ is an equilateral triangle.

$$\therefore \angle ADD' = 60^\circ \text{ and } DD' = 5\sqrt{2}$$

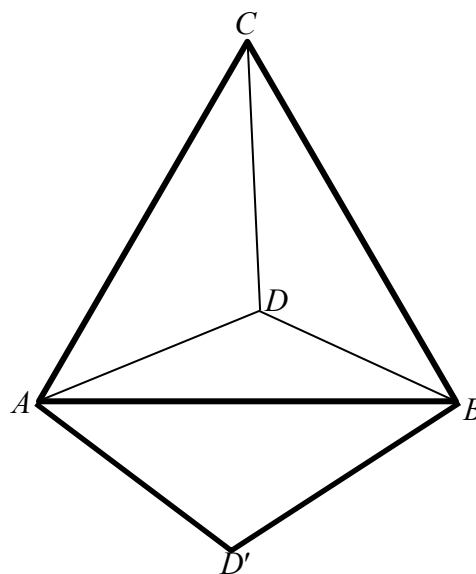
$$\therefore (DD')^2 + BD^2 = (5\sqrt{2})^2 + (5\sqrt{2})^2 = 100$$

$$(BD')^2 = CD^2 = 100$$

$$\therefore \angle BDD' = 90^\circ$$

Hence, $\angle ADB = 60^\circ + 90^\circ = 150^\circ$

$$\begin{aligned} AB^2 &= AD^2 + BD^2 - 2(AD)(BD)\cos \angle ADB \\ &= 50 + 50 - 2(50)\cos 150^\circ \\ &= 100 + 50\sqrt{3} \end{aligned}$$



$$\begin{aligned}
 \text{Area of } \triangle ABC &= \frac{1}{2} AB^2 \sin 60^\circ \\
 &= \frac{1}{2} (100 + 50\sqrt{3}) \left(\frac{\sqrt{3}}{2} \right) \\
 &= 25\sqrt{3} + \frac{75}{2}
 \end{aligned}$$

Hong Kong Mathematics Olympiad (2018 / 2019)

Final Event (Individual) Solution

香港數學競賽 (2018 / 2019)

決賽項目(個人) 題解

Event 1

1. The coefficient of x^4 is $A = C_2^5 \times 2^3 = 80$

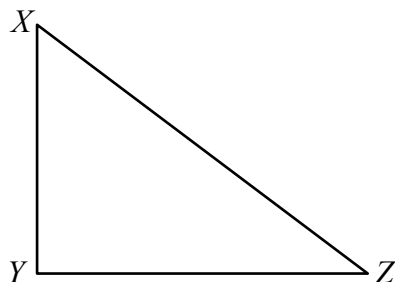
2. $\log xy = \log 10(2x - 80y)$
 $xy = 20x - 80y$
 $(x + 800)(20 - y) = 16000$
 $0 < x \leq 15200, 0 < y < 20$
 $20 - y = 1, 2, 3, 5, 8, 10, 16$

There are 7 pairs of (x, y) .

3. $Y = 2^{18} = 2^9 \times 2^9 = 512 \times 512 = 500^2 + 2 \times 500 \times 12 + 12^2$
 $= 250000 + 12000 + 144 = 262144$

Therefore, $C = 2 + 6 + 2 + 1 + 4 + 4 = 19$.

4.



The length of XY is $\sqrt{24^2 - 16^2} = 8\sqrt{3^2 - 2^2} = 8\sqrt{5}$.

$$D = (\sin \theta + \tan \theta)^2 = \left(\frac{8\sqrt{5}}{24} + \frac{8\sqrt{5}}{16} \right)^2 = \frac{125}{36}$$

Event 2

1. $A = \left(\sqrt{11 + \sqrt{21}} - \sqrt{11 - \sqrt{21}} \right)^2$

$$= 11 + \sqrt{21} - 2(11 + \sqrt{21})(11 - \sqrt{21}) + 11 - \sqrt{21}$$

$$= 22 - 2\sqrt{11^2 - 21}$$

$$= 22 - 20$$

$$= 2$$

2. The two points are given by (4, 5) and (-2, 2), so the slope of the line is $m = \frac{5-2}{4+2} = \frac{1}{2}$. Substitute (-2, 2)

into $y = \frac{1}{2}x + B$ gives $2 = -1 + B$ and $B = 3$.

3. $\cos x + \sin x = \frac{6}{5}$, so $(\cos x + \sin x)^2 = \frac{36}{25}$, and we have

$$\cos^2 x + 2 \cos x \sin x + \sin^2 x = \frac{36}{25}$$

$$1 + 2 \cos x \sin x = \frac{36}{25}$$

$$\cos x \sin x = \frac{11}{50}$$

Hence, $C = \left(\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} \right)^{-1}$

$$= \left(\frac{\sin^2 x + \cos^2 x}{\sin x \cos x} \right)^{-1}$$

$$= \left(\frac{1}{\sin x \cos x} \right)^{-1}$$

$$= \sin x \cos x$$

$$= \frac{11}{50}$$

4. Since

$$\frac{1000C + 2}{9} = \frac{222}{9} = 24 + \frac{2}{3}$$

$$= 3^3 - 3^1 + 3^0 - 3^{-1}$$

Hence, $D = 3$.

Event 3

1. After 1st fold – 50%, after 2nd folds - 25%, after 3rd folds it is $\frac{1}{2}$ of 12.5%, that is 6.25%.

2. $s = \frac{25}{4}$, $A = 25$, $B = 4$

The factors of A are 1, 5, 25.

The factors of B are 1, 2, 4.

$$t = 3 + 3 = 6$$

3.

Number of days after today	Remainder	u
$6^1 = 6$	6	1
$6^2 = 36$	1	3
$6^3 = 216$	6	1
$6^4 = 1296$	1	3
$6^5 = 7776$	6	1
$6^6 = 46656$	1	3
$\vdots$		$\vdots$

$$2019 \div 2 = 1009 \dots 1$$

4. 1231234_6

$$= 1 \times 6^6 + 2 \times 6^5 + 3 \times 6^4 + 1 \times 6^3 + 2 \times 6^2 + 3 \times 6^1 + 4$$

$$= (1 \times 6^3 + 2 \times 6^2 + 3 \times 6)(6^3 + 1) + 4$$

$$\therefore 123_6 = 1 \times 6^2 + 2 \times 6^1 + 3$$

$\therefore$ The remainder is 4.

Event 4

1. By sine rule,

$$\frac{5}{\sin \theta} = \frac{4}{\sin 30^\circ}$$
$$\alpha = \sin \theta = \frac{5}{8}$$

2. Since $f(2^{n-1}) = f\left(\frac{2^n}{2}\right) = f(2^n) - f(2) = f(2^n) + 1$,

by induction, $f(2^n) = -n$. Also, $f(1) = f(1) - f(1) = 0$.

Therefore, $\beta = f\left(\frac{1}{16}\right) = f(1) - f(16) = f(1) - f(2^4) = 4$

3. $B = \gamma p + 2\gamma(1 - p) = 2\gamma - \gamma p = 2\gamma - \gamma(0.04\gamma - 1.6)$

$$= -0.04\gamma^2 + 3.6\gamma = -\frac{(\gamma - 45)^2}{25} + 81$$

4. a, b, c are respectively 3, 5, 15.

Then $x + 2y + z = (x + y) + (y + z) = 5$, it implies that $y + z = 2$.

Similarly, $y + 2z + t = (y + z) + (z + t) = 15$, it implies $z + t = 13$. Consequently, this gives

$$\delta = x + y + z + t = 16.$$

Hong Kong Mathematics Olympiad (2018 / 2019)

Final Event (Group) Solution

香港數學競賽 (2018 / 2019)

決賽項目(團體)題解

Event 1

1. Since $x + y = (\sqrt{x} + \sqrt{y})^2 - 2\sqrt{xy}$, we have $(\sqrt{x} + \sqrt{y})^2 = 32 + 2\sqrt{xy}$

Also, $(\sqrt{x} - \sqrt{y})^2 = x + y - 2\sqrt{xy} \geq 0$. It implies that $(\sqrt{x} + \sqrt{y})^2 \leq 32 + 2\left(\frac{x+y}{2}\right) = 64$.

Hence, $a = 8$.

2. The probability is $\frac{x}{x+x+2+x+4} = \frac{x}{3x+6}$.

Hence,

$$\frac{x}{3x+6} < 0.1$$

$$x < \frac{6}{7}$$

Since $x \geq 0$ and x is an integer, we have $x = 0$ and $b = 6$.

3. For $k \geq 1$ and $n = 2k + 1$,

$$n^n - n = n(n^{n-1} - 1)$$

$$= n[(n^2)^k - 1]$$

$$= n(n^2 - 1)[(n^2)^{k-1} + (n^2)^{k-2} + \dots + n^2 + 1]$$

Since $n(n^2 - 1) = n(n+1)(n-1)$ must be divisible by 3 and $(n+1)(n-1) = 4k(k+1)$ must be divisible by 8. So, all the numbers must be divisible by 24. Hence, $c = 24$.

4. Since $x - y = \frac{\sqrt{5} + \sqrt{7}}{\sqrt{5} - \sqrt{7}} - \frac{\sqrt{5} - \sqrt{7}}{\sqrt{5} + \sqrt{7}} = \frac{4\sqrt{5}\sqrt{7}}{5-7} = -2\sqrt{35}$ and $xy = 1$,

$$d = 3x^2 - 7xy + 3y^2 = 3x^2 - 6xy + 3y^2 - xy = 3(x-y)^2 - xy = 3(-2\sqrt{35})^2 - 1 = 419.$$

Event 2

$$1. \quad X = \sqrt{2020 - \sqrt{A}}$$

$$X^2 = 2020 - \sqrt{A}$$

$$0 < X \leq \sqrt{2020}$$

$$0 < X \leq 44$$

$$\text{Therefore, } A = (2020 - 44^2)^2 = 84^2 = 7056.$$

$$2. \quad 4x^2 + (x-5)^2 = 80$$

$$x^2 - 2x - 11 = 0$$

As x_1, x_2 are the roots of the equation, we have $x_1 + x_2 = 2$ and $y_1 + y_2 = (5 - x_1) + (5 - x_2) = 10 - 2 = 8$.

$$B = y_1 - x_1 + y_2 - x_2 = (y_1 + y_2) - (x_1 + x_2) = 8 - 2 = 6.$$

$$3. \quad Q = 3^2 - 2^3 = 9 - 8 = 1$$

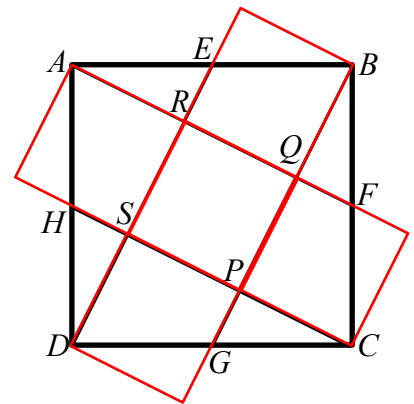
$$Q = 1$$

$$4. \quad \triangle CPG \sim \triangle CSD. \text{ As } CD = 2CG, SP = PC = x.$$

Similarly, $PQ = QB = RQ = AR = RS = SD = x$.

$$\text{Also, } PG = QF = ER = HS = \frac{x}{2}.$$

The area of $ABCD$ is equal to 5 times the area of $PQRS$. $W = 5$.



Event 3

$$1. \quad \sqrt{2^5 \times 3^4 \times 7^3} = b\sqrt{14}$$

$$a = 14$$

$$2. \quad b = \text{area of } \triangle BPD$$

$$= \text{area of } \triangle ABD - \text{area of } \triangle APD - \text{area of } \triangle APB$$

$$= \frac{1}{2} \text{ area of } ABCD - \text{area of } \triangle APD - 73$$

$$\begin{aligned}
&= (\text{area of } \triangle APD + \text{area of } \triangle BPC) - \text{area of } \triangle APD - 73 \\
&= \text{area of } \triangle BPC - 73 \\
&= 100 - 73 \\
&= 27
\end{aligned}$$

3.

c	16	20
2	a	e
b	d	f

First, the total sum in the first row is $36 + c$.

The first column gives $c + 2 + b = 36 + c$, which implies $b = 34$.

Similarly, $b + a + 20 = 36 + c$, then $a = c - 18$.

The second column gives $16 + a + d = 36 + c$. Hence, $d = 38$.

Similarly, $e = 52$ from the second row.

$f = 54 - c$ from the last row.

At last, the last column gives $36 + c = 20 + 52 + 54 - c$, then the solution is $c = 45$.

4. The last digit of 2^n recurs in the pattern of “2, 4, 8, 6”. $2^{2018} = 2^{4 \times 504 + 2}$, thus the last digit of 2^{2018} is 4; the last digit of 3^n recurs in the pattern of “3, 9, 7, 1”, $3^{2018} = 3^{4 \times 504 + 2}$, thus the last digit of 3^{2018} is 9. So the last digit of $2^{2018} + 3^{2018}$ is 3.

Event 4

1. $BC^2 = AC^2 + AB^2$, we get $AB = 6$.

The areas of Semi-circles on BC , on AC , and on AB are $\frac{25\pi}{2}$, $\frac{16\pi}{2}$, $\frac{9\pi}{2}$ respectively.

$$\alpha = \frac{16\pi}{2} + \frac{9\pi}{2} - \frac{25\pi}{2} + \frac{6 \times 8}{2} = 24.$$

2. Between 1 and 4000 there are 2000 values of n which are divisible by 2 and 1333 values of n which are divisible by 3. Even multiples of 3 are divisible by 2, and there are 666 such values of n . Thus $F(4000) = 2 \times (2000 - 666) + 3(1333 - 666) + 4 \times 666$
- $$\begin{aligned}
&= 2668 + 2001 + 2664 \\
&= 7333
\end{aligned}$$

3. Since $\triangle OED \sim \triangle OMF$,

$$\frac{OM}{OE} = \frac{OF}{OD}$$

$$OF = \frac{1}{5}OD$$

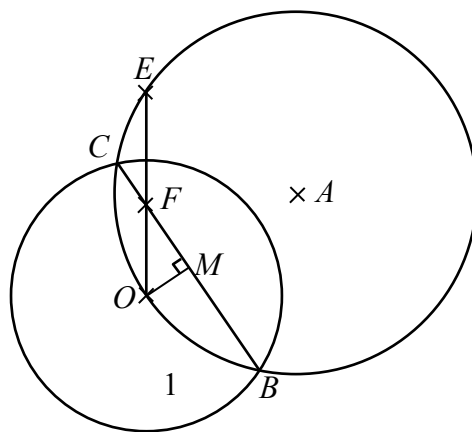
We find OD by the similarity $\triangle OMC \sim \triangle OCD$,

$$\frac{OM}{OC} = \frac{OC}{OD}$$

$$OM \cdot OD = OC^2$$

$$OD = 9$$

Therefore $\gamma = OF = \frac{1}{5} \times 9 = 1.8$.



4. $\delta = f(1) - 1$

$$= \left(1 + \frac{1}{2000}\right) \times \cdots \times \left(1 + \frac{1}{2019}\right) - 1$$

$$= \frac{2001}{2000} \times \frac{2002}{2001} \times \cdots \times \frac{2020}{2019} - 1$$

$$= \frac{2020}{2000} - 1$$

$$= 0.01$$

Hong Kong Mathematics Olympiad (2020 / 21)

Heats (Individual Paper 1) Solution

香港數學競賽 (2020 / 21)

初賽 (個人項目卷一) 題解

Part A

甲部

1. Given that $W = a^b - b^a$, where $a \neq b \neq 0$. If W is a non-negative integer, find the least value of W .

已知 $W = a^b - b^a$ ，其中 $a \neq b \neq 0$ 。若 W 為一非負整數，求 W 的最小值。〔答：0〕

Solution:

$$W = 4^2 - 2^4 = 16 - 16 = 0$$

$$\therefore W = 0.$$

2. Find the last two digits of 2^{2021} .

求 2^{2021} 的最尾兩位數字。〔答：52〕

[Solution]

Starting from the square of 2, the last two digits of the product of self-multiplication of 2 follow the pattern 4, 8, 16, 32, 64, 28, 56, 12, 24, 48, 96, 92, 84, 6, 36, 72, 44, 88, 76, 52. Because of $(2021-1) \div 20 = 101 \cdots 0$, we have the last two digits of the product of 2020 2's is 52.

3. α and β are the roots of the equation $x^2 - 7x + 4 = 0$. Find the value of $\alpha^3 + \beta^3$.

α 及 β 為方程 $x^2 - 7x + 4 = 0$ 的根。求 $\alpha^3 + \beta^3$ 的值。〔答：259〕

Solution:

Since α and β are the roots of the equation $x^2 - 7x + 4 = 0$, we have $\alpha + \beta = 7$ and $\alpha\beta = 4$.

Therefore $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = 7^3 - 3(4)(7) = 259$.

4. Find the value of $8\cos^2 15^\circ \cos^2 30^\circ - 8\sin^2 15^\circ \cos^2 30^\circ$.

求 $8\cos^2 15^\circ \cos^2 30^\circ - 8\sin^2 15^\circ \cos^2 30^\circ$ 的值。〔答： $3\sqrt{3}$ 〕

Solution

$$8\cos^2 15^\circ \cos^2 30^\circ - 8\sin^2 15^\circ \cos^2 30^\circ$$

$$\begin{aligned}
&= 8\cos^2 30^\circ (\cos^2 15^\circ - \sin^2 15^\circ) \\
&= 8\cos^3 30^\circ \\
&= 8\left(\frac{\sqrt{3}}{2}\right)^3 \\
&= 3\sqrt{3}
\end{aligned}$$

5. In Figure 1, three unit circles are placed inside an equilateral triangle ABC such that any circle is tangential to two sides of the triangle and to the other two circles. Find the area of $\triangle ABC$.

在圖一中，三個單位圓位於一等邊三角形 ABC 內，使得每個圓均與另外兩圓及三角形的兩邊相切。

求 $\triangle ABC$ 的面積。〔答： $6 + 4\sqrt{3}$ 〕

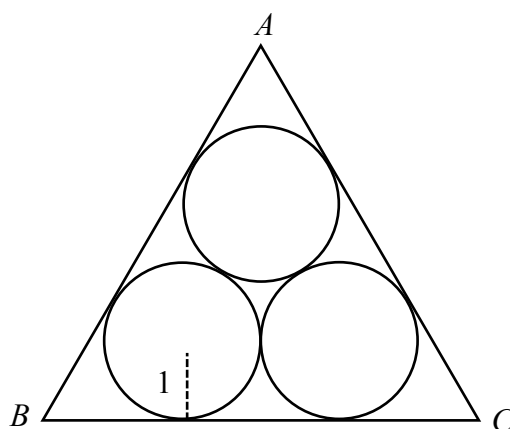


Figure 1

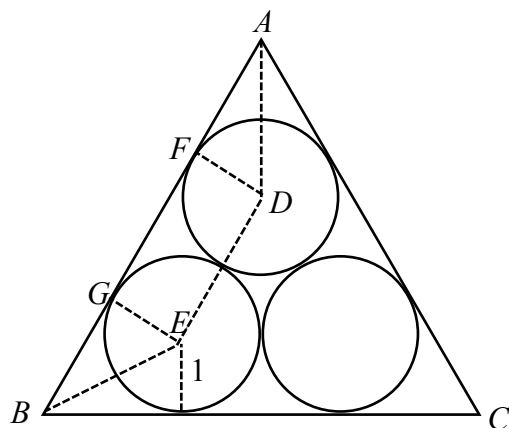
圖一

Solution

Let D and E be the centres of the two circles. F and G be the points of contact of the two circles with the side AB .

Then $GF = 2$ and $BG = AF = \frac{1}{\tan 30^\circ} = \sqrt{3}$

Therefore, area of $\triangle ABC = \frac{1}{2}(2 + 2\sqrt{3})^2 \sin 60^\circ$
 $= (4 + 2\sqrt{3})(\sqrt{3}) = 6 + 4\sqrt{3}$



6. In Figure 2, the altitude of an equilateral triangle ABC is 15 cm. P is a point inside $\triangle ABC$. The

perpendicular distances from P to AB , BC and AC are h cm, 4 cm and 5 cm respectively. Find the value of h .

在圖二中，等邊三角形 ABC 的高為 15 cm。 P 為 $\triangle ABC$ 內的一點。從 P 與 AB 、 BC 和 AC 的垂直距離分別為 h cm、4 cm 和 5 cm。求 h 的值。〔答：6〕

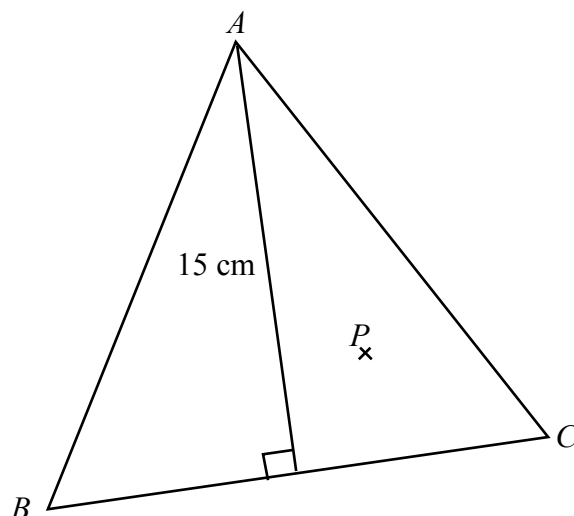


Figure 2

圖二

Solution

Let D , E and F be points on AB , BC and AC respectively such that $PE \perp AB$, $PF \perp BC$ and $PD \perp AC$.

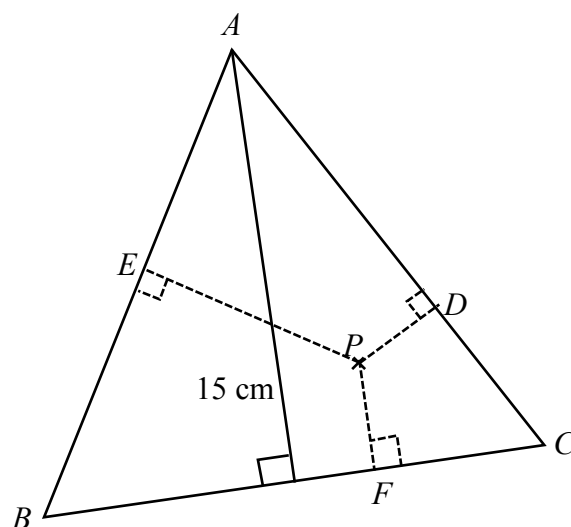
Let $AB = AC = BC = x$ cm.

Area of $\triangle ABC$ = Area of $\triangle APB$ + Area of $\triangle BPC$ + Area of $\triangle APC$

$$\therefore \frac{15}{2}x = \frac{1}{2}xh + 2x + \frac{5}{2}x$$

$$\Rightarrow 15 = h + 4 + 5$$

$$\therefore h = 6$$



7. p , q and r are prime numbers. If $pqr = 7(p + q + r)$, find the value of $p + q + r$.

p 、 q 及 r 為質數。若 $pqr = 7(p + q + r)$ ，求 $p + q + r$ 的值。〔答：15〕

Solution:

$\because p$, q and r are prime numbers and $pqr = 7(p + q + r)$,

$\therefore$ One of p , q and r must be equal to 7. W.L.O.G, say $p = 7$ and $q < r$. We have

$$7qr = 7(7 + q + r)$$

$$qr = 7 + q + r$$

$$qr - q - r = 7$$

$$(q - 1)(r - 1) = 8$$

$$\begin{cases} q-1=1 \\ r-1=8 \end{cases} \text{ or } \begin{cases} q-1=2 \\ r-1=4 \end{cases}$$

$$\begin{cases} q=2 \\ r=9 \end{cases} (\text{rej.}) \text{ or } \begin{cases} q=3 \\ r=5 \end{cases}$$

$$\therefore p+q+r=7+3+5=15$$

8. Find the value of $\frac{1001 \times 1002}{\frac{1}{1+\frac{1}{1002}} + \frac{2}{2+\frac{2}{1002}} + \frac{3}{3+\frac{3}{1002}} + \dots + \frac{1001}{1001+\frac{1001}{1002}}}$.

求 $\frac{1001 \times 1002}{\frac{1}{1+\frac{1}{1002}} + \frac{2}{2+\frac{2}{1002}} + \frac{3}{3+\frac{3}{1002}} + \dots + \frac{1001}{1001+\frac{1001}{1002}}}$ 的值。〔答：1003〕

Solution

$$\begin{aligned} a &= \frac{1001 \times 1002}{\underbrace{\frac{1}{1+\frac{1}{1002}} + \frac{2}{2+\frac{2}{1002}} + \frac{3}{3+\frac{3}{1002}} + \dots + \frac{1001}{1001+\frac{1001}{1002}}}_{1001 \text{ terms}}} \\ &= \frac{1002}{\frac{1}{1+\frac{1}{1002}}} \\ &= 1002 \times \left(1 + \frac{1}{1002}\right) \\ &= 1003 \end{aligned}$$

9. How many even numbers between 4000 and 7000 have four different digits?
在 4000 和 7000 之間 4 個數位各不相同的偶數有多少個？〔答：728〕

Solution

When thousands digit = 4 or 6, the number of even numbers = $2 \times 4 \times 8 \times 7 = 448$

When thousands digit = 5, the number of even numbers = $1 \times 5 \times 8 \times 7 = 280$

The required answer is $448 + 280 = 728$.

10. In Figure 3, BEF , ADE and CFD are straight lines such that $BE:EF=1:2$, $AD:DE=1:3$ and $CF:FD=1:4$. If the area of $\triangle DEF$ is 24 square unit, find the area of $\triangle ABC$.

在圖三中， BEF 、 ADE 及 CFD 是直線，使得 $BE:EF=1:2$ ， $AD:DE=1:3$ 及 $CF:FD=1:4$ 。若

$\triangle DEF$ 的面積是 24 平方單位，求 $\triangle ABC$ 的面積。〔答：59 square units〕

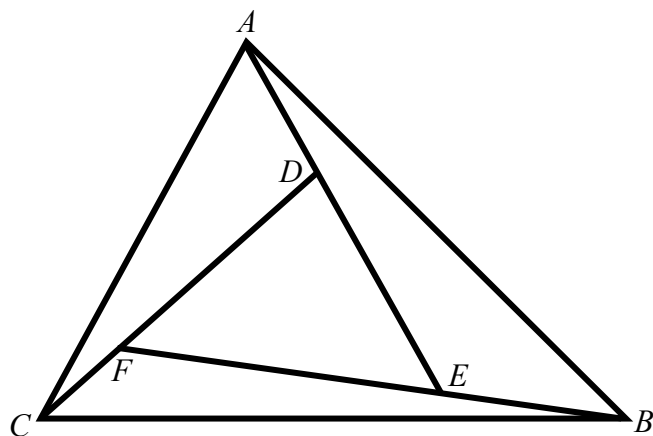


Figure 3

圖三

Solution

$$\text{As } \frac{S_{\triangle DBE}}{S_{\triangle DEF}} = \frac{1}{2}, \frac{S_{\triangle DBE}}{24} = \frac{1}{2}, S_{\triangle DBE} = 12.$$

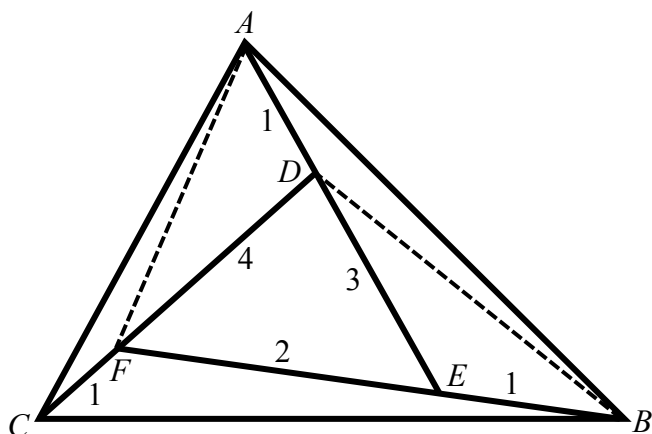
$$\text{As } \frac{S_{\triangle BDA}}{S_{\triangle DBE}} = \frac{1}{3}, \frac{S_{\triangle BDA}}{12} = \frac{1}{3}, S_{\triangle BDA} = 4.$$

$$\text{As } \frac{S_{\triangle ADF}}{S_{\triangle DEF}} = \frac{1}{3}, \frac{S_{\triangle ADF}}{24} = \frac{1}{3}, S_{\triangle ADF} = 8.$$

$$\text{As } \frac{S_{\triangle AFC}}{S_{\triangle ADF}} = \frac{1}{4}, \frac{S_{\triangle AFC}}{8} = \frac{1}{4}, S_{\triangle AFC} = 2.$$

$$\text{As } \frac{S_{\triangle BFC}}{S_{\triangle BDF}} = \frac{1}{4}, \frac{S_{\triangle BFC}}{24+12} = \frac{1}{4}, S_{\triangle BFC} = \frac{1}{4} \times 36 = 9$$

$$x = 24 + 12 + 4 + 8 + 2 + 9 = 59$$



Part B

乙部

11. If $\log_9 x^{18} = (\log_3 x)^3$, find the least value of x .

若 $\log_9 x^{18} = (\log_3 x)^3$ ，求 x 的最小值。〔答： $\frac{1}{27}$ 〕

Solution:

$$\log_9 x^{18} = (\log_3 x)^3$$

$$\frac{\log_3 x^{18}}{\log_3 9} = (\log_3 x)^3$$

$$\log_3 x^9 = (\log_3 x)^3$$

$$(\log_3 x)^3 - 9\log_3 x = 0$$

$$\log_3 x = 0 \text{ or } \log_3 x = 3 \text{ or } \log_3 x = -3$$

$$x = 1 \text{ or } x = 27 \text{ or } x = \frac{1}{27}$$

$$\therefore \text{ The least value of } x \text{ is } \frac{1}{27}.$$

12. Let $f(x) = \sqrt{(x-3)^2 + x^2} + \sqrt{(x-6)^2 + (x+5)^2}$, where x is a real number. Find the minimum value of $f(x)$.

設 $f(x) = \sqrt{(x-3)^2 + x^2} + \sqrt{(x-6)^2 + (x+5)^2}$ ，其中 x 為一實數。求 $f(x)$ 的最小值。〔答：10〕

Solution:

$$f(x) = \sqrt{(x-3)^2 + x^2} + \sqrt{(x-6)^2 + (x+5)^2} = \sqrt{(x-3)^2 + (x-0)^2} + \sqrt{(x-6)^2 + [x-(-5)]^2}$$

Let $A = (3, 0)$ and $B = (6, -5)$.

Let $P(x, y)$ be a point on $y = x$.

Minimizing $f(x)$ is equivalent to minimizing the sum of AP and BP .

Reflect B along $y = x$ to $C(-5, 6)$

Minimum value of $f(x)$

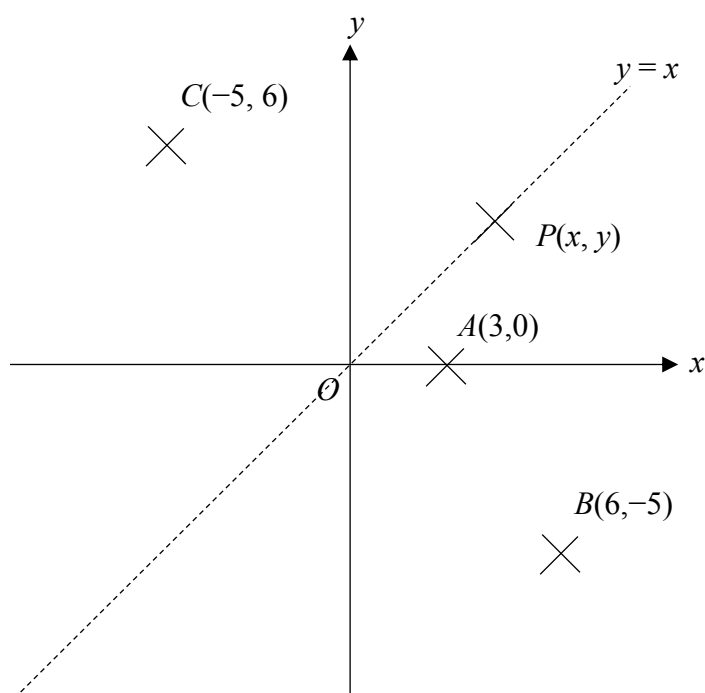
= minimum of $(AP + BP)$

= minimum of $(AP + CP)$

= AC

$$= \sqrt{(3+5)^2 + 6^2}$$

$$= 10$$



13. In Figure 4, O is the centre of the circle. The diameter BA is produced to a point G such that GH is a tangent to the circle at C . If $OA = 5$ and $GC = 12$, find the length of BC .

在圖四中， O 是圓的圓心。直徑 BA 延長至點 G 使得 GH 是圓在 C 上的切線。若 $OA = 5$ 及 $GC = 12$ ，求 BC 的長度。

[答： $\frac{30\sqrt{13}}{13}$]

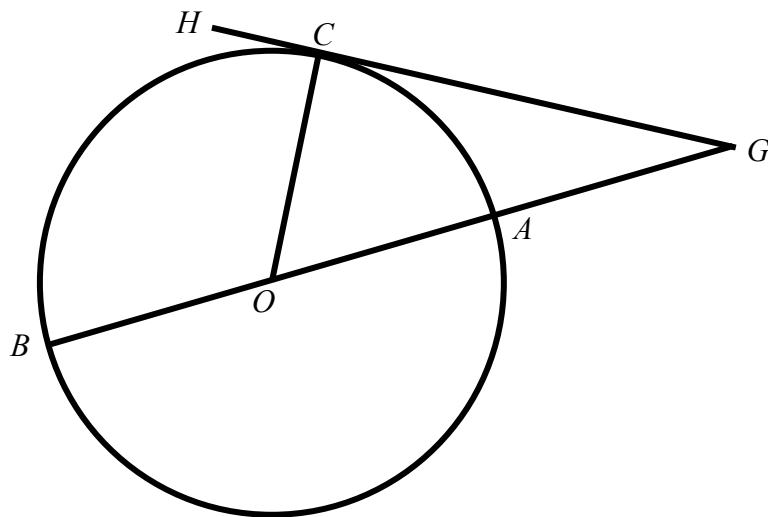


Figure 4

圖四

Solution

Let D be a point of the diameter AB such that $CD \perp BA$.

$OA = OB = OC = 5$. Then $OG = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$.

$$OC \cdot CG = CD \cdot OG$$

$$5 \times 12 = CD \times 13$$

$$\therefore CD = \frac{60}{13}$$

In the right-angled triangle OCD , $OD = \sqrt{5^2 - \left(\frac{60}{13}\right)^2} = \frac{1}{13} \sqrt{65^2 - 60^2} = \frac{25}{13}$.

$$BD = \frac{25}{13} + 5 = \frac{90}{13}$$

$$\therefore \text{ In } \triangle BDC, BC = \sqrt{\left(\frac{60}{13}\right)^2 + \left(\frac{90}{13}\right)^2} = \frac{30}{13} \sqrt{2^2 + 3^2} = \frac{30\sqrt{13}}{13}$$

14. For each real number x , the function $f(x)$ has the following property

$$f(x) + f(x-1) = x^2.$$

If $f(19) = 94$, find the value of $f(94)$.

對任意實數 x ，函數 $f(x)$ 有以下性質

$$f(x) + f(x-1) = x^2。$$

若 $f(19) = 94$ ，求 $f(94)$ 的值。〔答：4561〕

Solution

$$\begin{aligned} f(94) &= 94^2 - f(93) \\ &= 94^2 - 93^2 + f(92) \\ &= 94^2 - 93^2 + 92^2 - 91^2 + \cdots + 22^2 - 21^2 + 20^2 - f(19) \\ &= (94+93)(94-93) + (92+91)(92-91) + \cdots + (22+21)(22-21) + 20^2 - 94 \\ &= (94+93+\cdots+21) + 306 \\ &= \frac{1}{2}(94+21)(94-21+1) + 306 \\ &= 4561 \end{aligned}$$

15. Given that $(x+2y)^2 = 2xy - 3x + 6y - 9$. If x and y are real number, find the value of $x+y$.

已知 $(x+2y)^2 = 2xy - 3x + 6y - 9$ 。若 x 及 y 為實數，求 $x+y$ 的值。〔答： $\frac{-3}{2}$ 〕

Solution

$$(x+2y)^2 = 2xy - 3x + 6y - 9 = (x+3)(2y-3) \dots\dots\dots (1)$$

Let $X = x+3$ and $Y = 2y-3$, then (1) becomes

$$\begin{aligned} (X+Y)^2 &= XY \\ \Rightarrow 2(X+Y)^2 + X^2 + Y^2 &= X^2 + 2XY + Y^2 = (X+Y)^2 \\ \therefore (X+Y)^2 + X^2 + Y^2 &= 0 \\ \Rightarrow X &= 0 \text{ and } Y = 0 \\ \therefore x &= -3, y = \frac{3}{2} \\ \therefore x+y &= -3 + \frac{3}{2} = \frac{-3}{2} \end{aligned}$$

Hong Kong Mathematics Olympiad (2020 / 21)
Heats (Individual Paper 2) Solution
香港數學競賽 (2020 / 21)
初賽 (個人項目卷二) 題解

Part A

甲部

1. In Figure 1, $ABCD$ is a square of sides 6 units. F is the mid-point of CD . If $\angle FAB = \angle AFE$, find the length of BE .

在圖一中， $ABCD$ 是一個邊長為 6 的正方形。 F 是 CD 的中點。若 $\angle FAB = \angle AFE$ ，求 BE 的長度。〔答：2〕

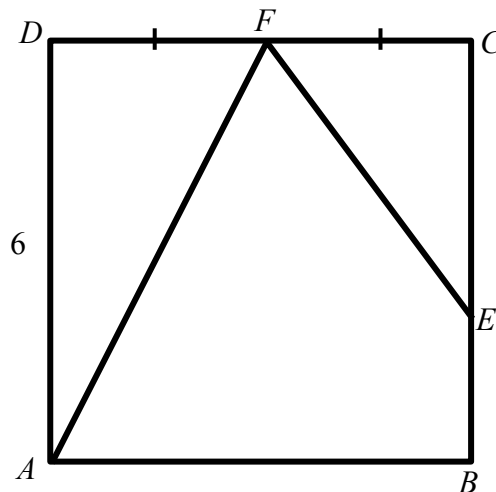


Figure 1

圖一

Solution

Let $\alpha = \angle DAF$, then $\tan \alpha = \frac{1}{2}$.

$$\angle FAB = \angle AFE = \angle AFD = 90^\circ - \alpha$$

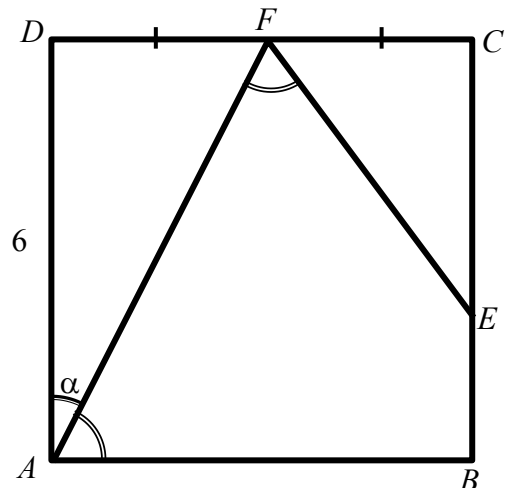
$$\begin{aligned} \angle CFE &= 180^\circ - (90^\circ - \alpha) - (90^\circ - \alpha) \\ &= 2\alpha \end{aligned}$$

$$\therefore \tan \angle CFE = \tan 2\alpha = \frac{EC}{CF} = \frac{x}{3}$$

$$\text{But } \tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha} = \frac{1}{1 - \left(\frac{1}{2}\right)^2} = \frac{4}{3}$$

$$\therefore \frac{4}{3} = \frac{x}{3} \Rightarrow x = 4$$

$$\therefore BE = 2$$



2. Let $S = 2011^n + 2012^n + 2013^n + 2014^n + 2015^n + 2016^n + 2017^n + 2018^n + 2019^n$, where n is a positive integer. If S is not divisible by 5, find the unit digit of S .

設 $S = 2011^n + 2012^n + 2013^n + 2014^n + 2015^n + 2016^n + 2017^n + 2018^n + 2019^n$ ，其中 n 為一正整數。
若 S 未能被 5 整除，求 S 的個位數。〔答：3〕

Solution:

$$\begin{aligned} S &= 2011^n + 2012^n + 2013^n + 2014^n + 2015^n + 2016^n + 2017^n + 2018^n + 2019^n \\ &\equiv (1^n + 2^n + 3^n + 4^n + 5^n + 6^n + 7^n + 8^n + 9^n) \pmod{10} \end{aligned}$$

Note that

$$i^{n+4} \equiv i^n \pmod{10} \text{ for any positive integer } n \text{ and } i = 1, 2, \dots, 9$$

If $n = 4k$,

$$\begin{aligned} S &\equiv (1^4 + 2^4 + 3^4 + 4^4 + 5^4 + 6^4 + 7^4 + 8^4 + 9^4) \pmod{10} \\ &\equiv (1 + 6 + 1 + 6 + 5 + 6 + 1 + 6 + 1) \pmod{10} \\ &\equiv 3 \pmod{10} \end{aligned}$$

If $n = 4k + 1$,

$$\begin{aligned} S &\equiv (1^1 + 2^1 + 3^1 + 4^1 + 5^1 + 6^1 + 7^1 + 8^1 + 9^1) \pmod{10} \\ &\equiv (1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9) \pmod{10} \\ &\equiv 5 \pmod{10} \end{aligned}$$

If $n = 4k + 2$,

$$\begin{aligned} S &\equiv (1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2) \pmod{10} \\ &\equiv (1 + 4 + 9 + 6 + 5 + 6 + 9 + 4 + 1) \pmod{10} \\ &\equiv 5 \pmod{10} \end{aligned}$$

If $n = 4k + 3$,

$$\begin{aligned} S &\equiv (1^3 + 2^3 + 3^3 + 4^3 + 5^3 + 6^3 + 7^3 + 8^3 + 9^3) \pmod{10} \\ &\equiv (1 + 8 + 7 + 4 + 5 + 6 + 3 + 2 + 9) \pmod{10} \\ &\equiv 5 \pmod{10} \end{aligned}$$

$\therefore$ The unit digit of $S = 3$.

3. In Figure 2, four circles of radii 8, 5, 5 and r are touching each other externally. Find the value of r .

在圖二中，四個半徑分別為 8、5、5 及 r 的圓互相外切。求 r 的值。〔答： $\frac{8}{9}$ 〕

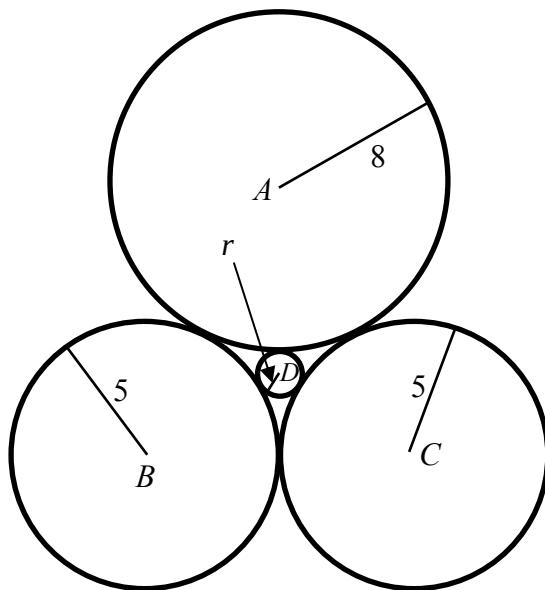


Figure 2

圖二

Solution

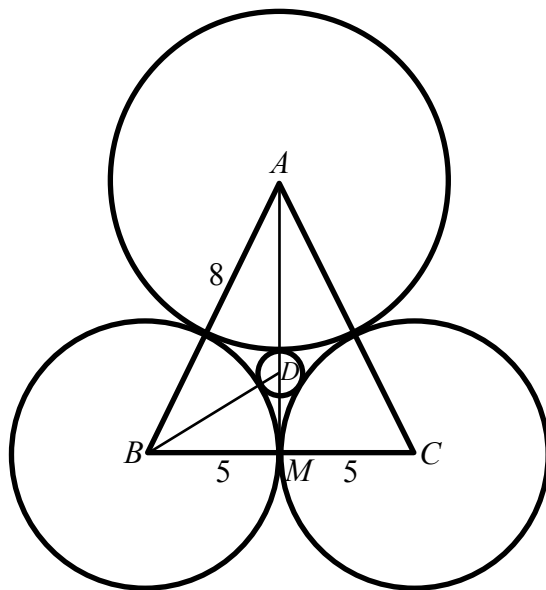
Let M be the intersection of AD extended and BC .

$$AM = \sqrt{13^2 - 5^2} = 12$$

In $\triangle DBC$,

$$(12 - 8 - r)^2 + 5^2 = (5 + r)^2$$

$$r = \frac{8}{9}$$



4. Given that a , b , c , d and e are consecutive positive integers, where $a < b < c < d < e$. If $a + b + c + d + e$ is a perfect cube and $b + c + d$ is a perfect square, find the smallest possible value of c .

已知 a 、 b 、 c 、 d 及 e 是連續正整數，其中 $a < b < c < d < e$ 。若 $a + b + c + d + e$ 是一個立方數及 $b + c + d$ 是一個平方數，求 c 的最小可能值。〔答：675〕

Solution

Let x , $x+1$, $x+2$, $x+3$ and $x+4$ be the five consecutive positive integers.

As their sum is a perfect cube,

$$5x+10 = p^3$$

$$5(x+2) = p^3$$

So p^3 is a multiple of 5.

As 5 is a prime number, p is also a multiple of 5.

Substitute $p = 5m$ into $5(x+2) = p^3$, we have $5(x+2) = 125m^3 \Rightarrow x+2 = 25m^3$

So $x+2$ is a multiple of 25.

Similarly, let $x+1+x+2+x+3 = q^2$, we have $3(x+2) = q^2$

This means that q^2 is a multiple of 3.

As 3 is a prime number, q is also a multiple of 3.

Substitute $q = 3n$ into $3(x+2) = q^2$, we have $3(x+2) = 9n^2 \Rightarrow x+2 = 3n^2$

So $x+2$ is a multiple of 3.

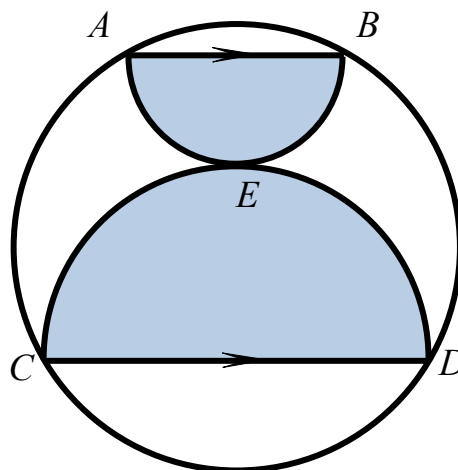
Therefore, the smallest possible value of c is $3 \times 25 \times 9 = 675$.

Part B

乙部

5. $ABCD$ is a circle while ABE and CED are semi-circles. Given the area of circle is 1 cm^2 and $AB \parallel CD$, find the sum of the area of the semi-circles ABE and CED .

$ABCD$ 是圓形而 ABE 及 CED 為半圓形。已知圓面積為 1 cm^2 及 $AB \parallel CD$ ，求半圓形 ABE 及 CED 的面積之和。〔答： $\frac{1}{2} \text{ cm}^2$ 〕



Solution:

Let the radii of the upper, lower semi-circles and the circle be a , b and r respectively.

$\angle BAF = 90^\circ$ (AF is the tangent of semi-circle ABE)

$\angle AFD = 90^\circ$ ($AB \parallel CD$)

$AF = a + b$

$CF = b - a$

Since $AF \times FG = CF \times FD$,

$$FG = \frac{(b-a)(a+b)}{a+b} = b-a$$

$$AG = AF + FG = (a+b) + (b-a) = 2b$$

Join BG

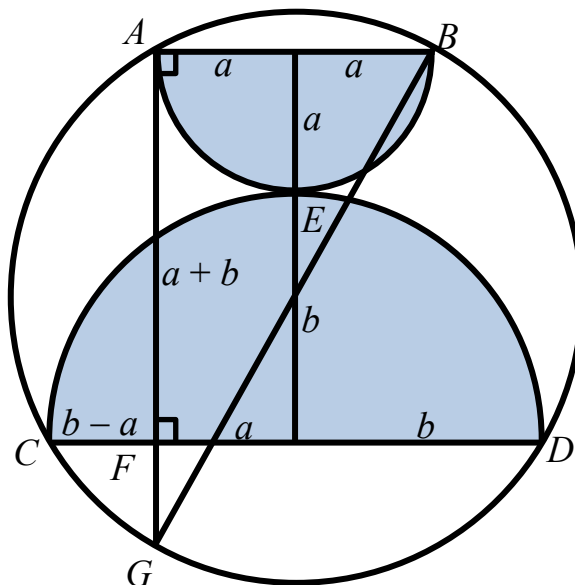
Since $\angle BAG = 90^\circ$, BG is a diameter.

$$AB^2 + AG^2 = BG^2$$

$$(2a)^2 + (2b)^2 = (2r)^2$$

$$4(a^2 + b^2) = 4r^2$$

$$a^2 + b^2 = r^2$$



$$\text{The area of semi-circles} = \frac{\pi a^2}{2} + \frac{\pi b^2}{2} = \frac{\pi(a^2 + b^2)}{2} = \frac{\pi r^2}{2}$$

Since the area of circle is $\pi r^2 = 1 \text{ cm}^2$

$$\text{The area of semi-circles} = \frac{\pi r^2}{2} = \frac{1}{2} \text{ cm}^2$$

6. If $d = \log_2(\sqrt{2^2 + 2^{1013} + 2^{2022}} - 2)$, find the value of d .

如果 $d = \log_2(\sqrt{2^2 + 2^{1013} + 2^{2022}} - 2)$, 求 d 的值。〔答：1011〕

Solution

Note that

$$\begin{aligned} & \sqrt{2^2 + 2^{1013} + 2^{2022}} \\ &= \sqrt{2^2(1 + 2^{1011} + 2^{2020})} \\ &= \sqrt{2^2(1 + 2^{1010})^2} \\ &= 2(1 + 2^{1010}) \\ &= 2 + 2^{1011} \end{aligned}$$

$$\text{Hence } d = \log_2(\sqrt{2^2 + 2^{1013} + 2^{2022}} - 2) = \log_2(2 + 2^{1011} - 2) = \log_2(2^{1011}) = 1011$$

7. Find the value of $\sqrt{10000 \times 10002 \times 10004 \times 10006 + 16}$.

求 $\sqrt{10000 \times 10002 \times 10004 \times 10006 + 16}$ 的值。〔答：100 060 004〕

Solution

$$\begin{aligned} & x(x+2)(x+4)(x+6)+16 \\ &= (x^2+6x)(x^2+6x+8)+16 \\ &= (x^2+6x)^2+8(x^2+6x)+16 \\ &= (x^2+6x)^2+2 \cdot 4(x^2+6x)+4^2 \\ &= (x^2+6x+4)^2 \end{aligned}$$

When $x = 10000$, $x^2 + 6x + 4 = 10000^2 + 6 \times 10000 + 4 = 100\,060\,004$, that is

$$\sqrt{10000 \times 10002 \times 10004 \times 10006 + 16} = 100\,060\,004$$

Part A

甲部

1. α and β are the real roots of the equation $x^2 - 100x + k = 0$. If $\alpha - 7 = 30\beta$, find the value of k .

α 及 β 是方程 $x^2 - 100x + k = 0$ 的實根。若 $\alpha - 7 = 30\beta$ ，求 k 的值。〔答：291〕

Solution

$\alpha - 7 = 30\beta$ and $\alpha + \beta = 100$. We have,

$$30\beta + 7 + \beta = 100$$

$$\beta = 3$$

Therefore, $\alpha = 97$ and $k = 3 \times 97 = 291$

2. In Figure 1, ACD is a triangle. B is a point on CD such that $AB = AC = 2$ and $AD = 4$. If $BC : BD = 1 : 3$, find the length of CD .

在圖一中， ACD 是一個三角形。 B 是 CD 上的一點使 $AB = AC = 2$ 及 $AD = 4$ 。若 $BC : BD = 1 : 3$ ，求 CD 的長。〔答：4〕

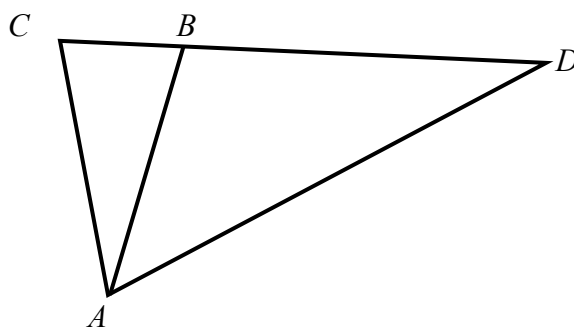


Figure 1

圖一

Solution

Let $BC = 2k$, then $BD = 6k$

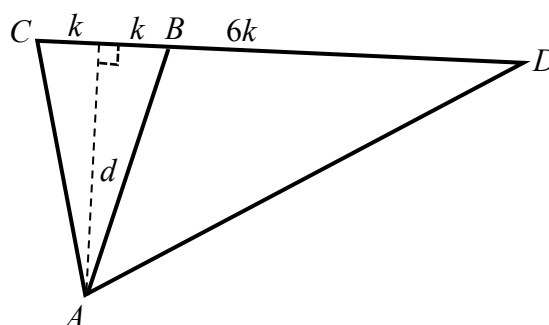
By Pythagoras Theorem, we have $d^2 + k^2 = 4$ and

$$d^2 + (7k)^2 = 16.$$

$$(7k)^2 - k^2 = 12$$

$$k = \frac{1}{2}$$

Therefore $CD = 8k = 4$.



3. In Figure 2, $ABCD$ is a rectangle. E is a point on AC such that $AE = 25$ and $CE = 144$. If $p = AD + DE + CD$, find the value of p .

在圖二中， $ABCD$ 是一個矩形。 E 是 AC 上的一點使 $AE = 25$ 及 $CE = 144$ 。

若 $p = AD + DE + CD$ ，求 p 的值。 [答：281]

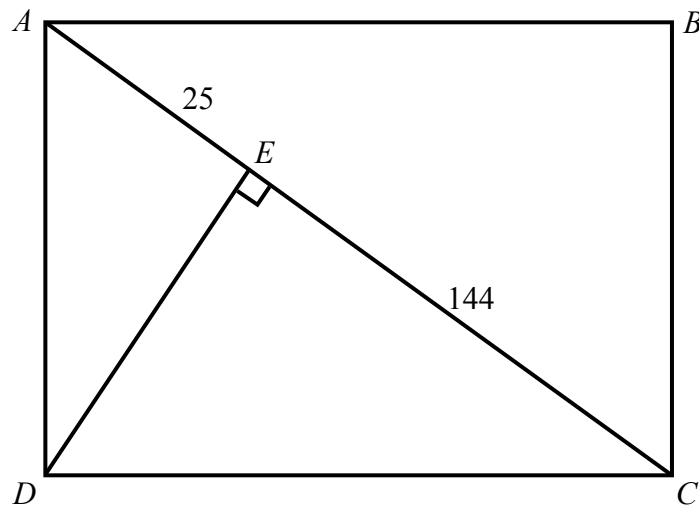


Figure 2

圖二

Solution

Let $AD = a$, $DE = b$ and $CD = c$.

By considering $\triangle ADE \sim \triangle DEC$, we have,

$$\frac{25}{b} = \frac{b}{144}$$

$$b = 60$$

$$a = \sqrt{60^2 + 25^2} = 5\sqrt{12^2 + 5^2} = 65$$

$$c = \sqrt{60^2 + 144^2} = 12\sqrt{5^2 + 12^2} = 156$$

$$\therefore p = a + b + c = 65 + 60 + 156 = 281$$

4. Let x , y and z are non-zero numbers. If $2^x = 3^y = 18^z$, find the value of $\frac{xz}{5y(x-z)}$.

設 x ， y 及 z 是非零數。若 $2^x = 3^y = 18^z$ ，求 $\frac{xz}{5y(x-z)}$ 的值。 [答： $\frac{1}{10}$]

Solution

We have, $x \log 2 = y \log 3 = z \log 18$.

$$\frac{1}{5\left(\frac{y}{z} - \frac{y}{x}\right)} = \frac{1}{5\left(\frac{\log 18}{\log 3} - \frac{\log 2}{\log 3}\right)} = \frac{\log 3}{5\left(\log \frac{18}{2}\right)} = \frac{\log 3}{5(2 \log 3)} = \frac{1}{10}$$

5. Let $N = 24x + 216y$, where both x and y are positive integers. If N is a cube number, find the minimum value of $x + y$.

設 $N = 24x + 216y$ ，其中 x 及 y 均為正整數。若 N 為完全立方數，求 $x + y$ 的最小值。〔答：16〕

Solution

$$N = 24x + 216y = 24(x + 9y) = 2^3 \cdot 3(x + 9y)$$

The minimum value of $x + y$ occurs when $x + 9y = 3^2 \times 2^3 \Rightarrow x + 9y = 72$.

$$\therefore x + y = 72 - 8y \text{ for } 1 \leq y \leq 7.$$

The minimum value of $x + y$ occurs when $y = 7$

$$\text{The minimum value of } x + y = 72 - 8(7) = 16$$

6. John participated in a mathematics competition, in which one of the questions was to solve $\begin{cases} ax + by = -16 \\ cx + 20y = -224 \end{cases}$, where a, b and c are real numbers. The correct answer to the question was $x = 8$ and $y = -10$. However, John copied a wrong value for c and then gave an answer of $x = 12$ and $y = -13$. Find the value of $a^2 + b^2 + c^2$ in the original question.

小馬參加數學比賽，解其中一條題目 $\begin{cases} ax + by = -16 \\ cx + 20y = -224 \end{cases}$ ，其中 a, b 及 c 是實數。

题目的正確答案為 $x = 8$ 及 $y = -10$ 。怎料小馬抄錯 c 的數值，得出答案 $x = 12$ 及 $y = -13$ 。求原題中 $a^2 + b^2 + c^2$ 的值。〔答：34〕

Solution

Since both $(8, -10)$ and $(12, -13)$ are solutions to the equation $ax + by = -16$, we have

$$\begin{cases} 8a - 10b = -16 \\ 12a - 13b = -16 \end{cases}$$

On solving the simultaneous equations, we have $a = 3$ and $b = 4$.

Also, in the original question, $8c + 20(-10) = -224 \Rightarrow c = -3$.

$$\text{Therefore, } a^2 + b^2 + c^2 = 3^2 + 4^2 + (-3)^2 = 34.$$

7. Given that $459 + x^3 = 3^y$, where both x and y are positive integers. Find the least value of y .

已知 $459 + x^3 = 3^y$ ，其中 x 及 y 均為正整數。求 y 的最少值。〔答：7〕

Solution

$$x^3 = 3^y - 459$$

$$3^6 - 459 = 729 - 459 = 270$$

$$3^7 - 459 = 1728 - 459 = 1269$$

Thus, the least value of y is 7.

8. In Figure 3, D is a point inside the quadrilateral $ABCE$ such that $AD \parallel BC$, $AB \perp AD$, $CD \perp DE$, $CD = ED$, $AD = 4$ cm and $BC = 6$ cm. If P cm² is the area of $\triangle ADE$, find the value of P .

在圖三中， D 為四邊形 $ABCE$ 內的一點使得 $AD \parallel BC$ ， $AB \perp AD$ ， $CD \perp DE$ ， $CD = ED$ ， $AD = 4$ cm

及 $BC = 6\text{ cm}$ 。若 $\triangle ADE$ 的面積為 $P\text{ cm}^2$ ，求 P 的值。〔答：4〕

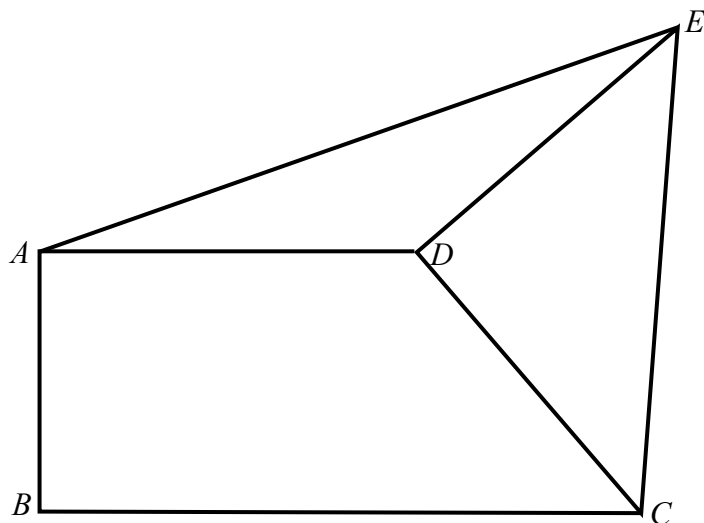


Figure 3

圖三

Solution

Let F and G be the foot of perpendiculars from D and E to the line BC and AD extended.

$$\therefore \angle EDG + \angle GDC = 90^\circ$$

$$\angle CDF + \angle GDC = 90^\circ$$

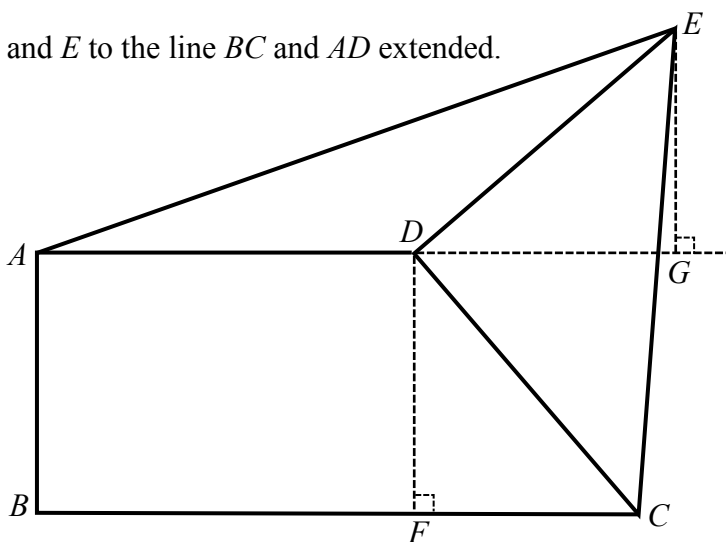
$$\therefore \angle EDG = \angle CDF$$

Also, $CD = ED$ (given)

$$\therefore \triangle DGE \cong \triangle DFC \text{ (ASA)}$$

$$\therefore EG = CF = 6 - 4 = 2$$

$$\begin{aligned} \text{Thus, area of } \triangle ADE = P &= \frac{AD \times EG}{2} \\ &= \frac{4 \times 2}{2} = 4 \end{aligned}$$



9. $ABCD$ is a cyclic quadrilateral with $AB = 7$, $BC = 15$, $CD = 20$ and $DA = 24$. Find the radius of the circle $ABCD$.

$ABCD$ 是一個圓內接四邊形，其中 $AB = 7$, $BC = 15$, $CD = 20$ and $DA = 24$ 。求圓 $ABCD$ 的半徑。

[答： $\frac{25}{2}$]

Solution

By cosine formula, we have $\cos \angle BAD = \frac{7^2 + 24^2 - d^2}{2(7)(24)}$ and $\cos \angle BCD = \frac{15^2 + 20^2 - d^2}{2(15)(20)}$.

$$\because \cos \angle BCD = \cos(180^\circ - \angle BAD) = -\cos \angle BAD$$

$$\frac{625 - d^2}{2(7)(24)} = -\left(\frac{625 - d^2}{2(15)(20)} \right)$$

$$\frac{625 - d^2}{14} = \frac{d^2 - 625}{25}$$

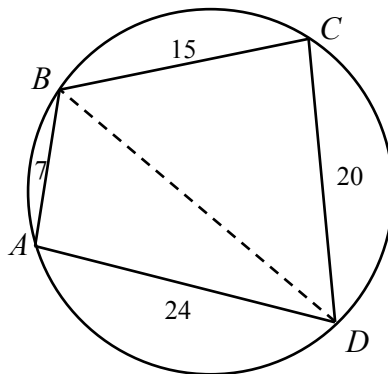
$$d = 25$$

$$\cos \angle BAD = 0$$

$$\angle BAD = 90^\circ$$

BD is the diameter of the circle $ABCD$.

$$\text{Radius} = \frac{25}{2}$$



10. Given that $a^2 + \frac{1}{a^2} = 7$, where $a > 0$. If $b = a^5 + \frac{1}{a^5}$, find the value of b .

已知 $a^2 + \frac{1}{a^2} = 7$ ，其中 $a > 0$ 。若 $b = a^5 + \frac{1}{a^5}$ ，求 b 的值。[答：123]

Solution

$$a^2 + \frac{1}{a^2} = 7 \Rightarrow a^2 + 2(a)\left(\frac{1}{a}\right) + \frac{1}{a^2} = 7 + 2 = 9$$

$$\therefore \left(a + \frac{1}{a}\right)^2 = 9 \text{ and } a + \frac{1}{a} = 3 \Rightarrow a^3 + \frac{1}{a^3} = \left(a + \frac{1}{a}\right)^3 - 3\left(a + \frac{1}{a}\right) = 27 - 9 = 18.$$

$$\therefore b = a^5 + \frac{1}{a^5}$$

$$= \left(a^2 + \frac{1}{a^2}\right)\left(a^3 + \frac{1}{a^3}\right) - \left(a + \frac{1}{a}\right)$$

$$= (7)(18) - 3 = 123$$

$$= 123$$

Part B**乙部**

11. x_1 and x_2 are the real roots of the equation $(\log 2x)(\log 3x) = a^2$, where a is a real number. Find the value of $x_1 x_2$.

x_1 及 x_2 是方程 $(\log 2x)(\log 3x) = a^2$ 的實根，其中 a 為實數。求 $x_1 x_2$ 的值。〔答： $\frac{1}{6}$ 〕

Solution

$$(\log 2x)(\log 3x) = a^2$$

$$(\log 2 + \log x)(\log 2 + \log x) = a^2$$

$$(\log x)^2 + (\log 2 + \log 3)\log x + (\log 2)(\log 3) - a^2 = 0 \quad \dots\dots\dots (1)$$

Let the two distinct real roots of (1) be x_1 and x_2 , then

$$\begin{aligned} \log(x_1 \cdot x_2) &= \log x_1 + \log x_2 \\ &= -(\log 2 + \log 3) \\ &= -\log 6 \\ &= \log \frac{1}{6} \end{aligned}$$

$$\therefore x_1 \cdot x_2 = \frac{1}{6}$$

12. A 7-digit number is formed by putting the numerals 0, 1, 2, 3, 4, 5, 6 together without repetition. If this number is divisible by 55, find its largest possible value.

由數字 0, 1, 2, 3, 4, 5, 6 組成一個沒有重複數字的 7 位數。若這個數可以被 55 整除，求這個數的最大值。〔答：6431205〕

Solution

If this number is divisible by 55, it is divisible by 5 and 11.

Any number which is divisible by 5 must have units digit either 5 or 0.

Any number which is divisible by 11 must have the difference between the sum of its odd digits (say A) and the sum of its even digits (say B) to be a multiple of 11, that is $A - B = 11k$ (where k is an integer).

Because $A + B = 0 + 1 + 2 + 3 + 4 + 5 + 6 = 21$, we have $A - B = 11$ or $A - B = -11$.

Solving them gives the value of A and B to be 16 and 5.

As this is a 7-digit number, B must be the sum of the 3 even digits. Also the digit 6 must be the leading digit, and as shown above, B can either take the value of 5 or 16, which implies b must be 5. Therefore the possible digits combination are either 0, 1, 4 or 0, 2, 3.

If 0, 1, 4 take up the even digit places, then the digits for filling the odd digit places are 2, 3, 5, 6; the largest possible combination is 6431205. If 0, 2, 3 take up the even digit places, then the digits for filling the odd digit places are 1, 4, 5, 6; the largest possible combination is 6342105.

Therefore the largest possible number is 6431205.

13. Given that $a^{2x} - b^{2y} = 1672$, where a, b, x and y are positive integers. Find the minimum value of $ax + by$.

已知 $a^{2x} - b^{2y} = 1672$ ，其中 a, b, x 及 y 為正整數。求 $ax + by$ 的最小值。〔答：23〕

Solution

$$a^{2x} - b^{2y} = 1672$$

$$\Rightarrow (a^x + b^y)(a^x - b^y) = 1672$$

$$\Rightarrow \begin{cases} a^x + b^y = 76 \\ a^x - b^y = 22 \end{cases} \text{ or } \begin{cases} a^x + b^y = 44 \\ a^x - b^y = 38 \end{cases} \text{ or } \begin{cases} a^x + b^y = 836 \\ a^x - b^y = 2 \end{cases} \text{ or } \begin{cases} a^x + b^y = 418 \\ a^x - b^y = 4 \end{cases}$$

$$\Rightarrow \begin{cases} a^x = 49 \\ b^y = 27 \end{cases} \text{ or } \begin{cases} a^x = 41 \\ b^y = 3 \end{cases} \text{ or } \begin{cases} a^x = 419 \\ b^y = 417 \end{cases} \text{ or } \begin{cases} a^x = 211 \\ b^y = 207 \end{cases}$$

$$\Rightarrow (a = 7, b = 3, x = 2, y = 3) \text{ or } (a = 49, b = 27, x = 1, y = 1) \text{ or } (a = 41, b = 3, x = 1, y = 1) \\ \text{or } (a = 419, b = 417, x = 1, y = 1) \text{ or } (a = 211, b = 207, x = 1, y = 1)$$

Thus, the minimum value of $ax + by = 7 \times 2 + 3 \times 3 = 23$

14. Let a, b and c are non-zero digits. How many three digit numbers $\overline{abc}$ are there such that $\overline{ab} < \overline{bc} < \overline{ca}$?

設 a, b 及 c 為非零數字。有多少個三位數 $\overline{abc}$ 使得 $\overline{ab} < \overline{bc} < \overline{ca}$? 〔答：120〕

Solution

Note that $a \leq b < c$

Case I: $a = b < c$

Choose 2 of the digits between 1 and 9.

$$\text{Number of three digit numbers} = C_2^9 = 36$$

Case II: $a < b < c$

Choose 3 of the digits between 1 and 9.

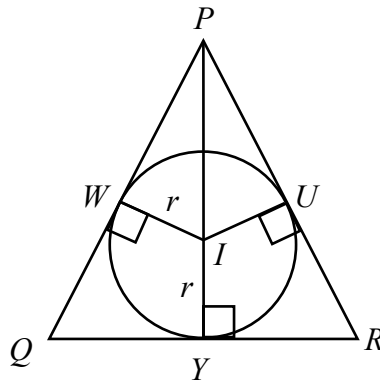
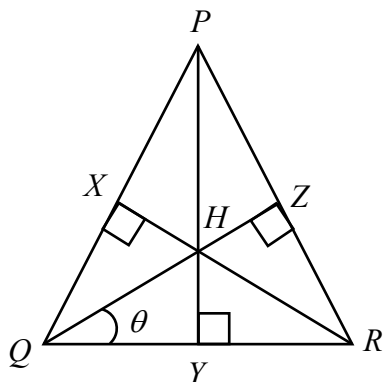
$$\text{Number of three digit numbers} = C_3^9 = 84$$

$$\text{Total number of three digit numbers} = 36 + 84 = 120$$

15. PQR is an isosceles triangle with $PQ = PR = 17$ and $QR = 16$. Denote the in-centre and the orthocentre of PQR by I and H respectively. Find the length of HI .

PQR 是一個等腰三角形，其中 $PQ = PR = 17$ 和 $QR = 16$ 。將 I 及 H 分別記為 PQR 的內心及垂心。求 HI 長度的值。〔答： $\frac{8}{15}$ 〕

Solution



$$\because PQ = PR$$

$$\therefore QY = YR = 8$$

$$PY = \sqrt{17^2 - 8^2} = 15$$

$$\angle QHY = 90^\circ - \theta$$

$$\angle QRZ = 90^\circ - \theta$$

$$\triangle QHY \sim \triangle PRY$$

$$\frac{HY}{QY} = \frac{YR}{PY}$$

$$HY = \frac{8^2}{15} = \frac{64}{15}$$

$$QW = QY = 8$$

$$PW = 17 - 8 = 9$$

$$PI = 15 - r$$

$$WI^2 + PW^2 = PI^2$$

$$r^2 + 9^2 = (15 - r)^2$$

$$r^2 + 81 = r^2 - 30r + 225$$

$$r = \frac{24}{5}$$

$$IH = \frac{24}{5} - \frac{64}{15} = \frac{8}{15}$$

Hong Kong Mathematics Olympiad (2021 / 22)
Individual Paper 2 Solution
香港數學競賽 (2021 / 22)
個人項目卷二題解

Part A

甲部

1. Let $\frac{A}{2022} = \frac{1}{1+1 \times 2 \times 3 \times \cdots \times 2022} + \frac{1}{1 + \frac{1}{1 \times 2 \times 3 \times \cdots \times 2022}}$. Find the value of A .

設 $\frac{A}{2022} = \frac{1}{1+1 \times 2 \times 3 \times \cdots \times 2022} + \frac{1}{1 + \frac{1}{1 \times 2 \times 3 \times \cdots \times 2022}}$ 。求 A 的值。〔答：2022〕

Solution

Let $p = 1 \times 2 \times 3 \times \cdots \times 2022$. Then

$$\begin{aligned} \frac{A}{2022} &= \frac{1}{1+1 \times 2 \times 3 \times \cdots \times 2022} + \frac{1}{1 + \frac{1}{1 \times 2 \times 3 \times \cdots \times 2022}} \\ &= \frac{1}{1+p} + \frac{1}{1 + \frac{1}{p}} \\ &= \frac{1}{1+p} + \frac{p}{1+p} \\ &= 1 \end{aligned}$$

Hence $A = 2022$.

2. Both $\overline{AB}$ and $\overline{CB}$ are two-digit positive integers, where A , B and C are different digits. Let $d = \overline{AB} + \overline{CB}$. If $\overline{AB} \times \overline{CB} = \overline{BCBB}$ is a four-digit number, find the value of d .

$\overline{AB}$ 和 $\overline{CB}$ 均為兩位正整數，其中 A , B 和 C 是不同的數字。設 $d = \overline{AB} + \overline{CB}$ 。若 $\overline{AB} \times \overline{CB} = \overline{BCBB}$ 是四位數，求 d 的值。〔答：112〕

Solution

From $\overline{AB} \times \overline{CB} = \overline{BCBB}$, we know that the digits A , B and C are not equal to 0. The product $\overline{AB} \times \overline{CB} = \overline{BCBB}$ ends with B , therefore $B = 1, 5$ or 6 .

If $B = 5$, then the last two digits should be 25 or 75 ($\neq 55$). Impossible.

If $B = 6$, then 4 is a factor of $\overline{A6} \times \overline{C6}$. But 4 is not a factor of $\overline{6C66}$.

If $B = 1$, then $\overline{A1} \times \overline{C1} = \overline{1C11}$ implies $A = 2$ and $C = 9$. Then the two numbers are $\overline{AB} = 21$ and $\overline{CB} = 91$.

Therefore, $d = \overline{AB} + \overline{CB} = 21 + 91 = 112$.

3. Suppose the equation $x^2y - 2x^2 - 3y - 13 = 0$ has only one pair of positive integer solution (x_0, y_0) . If $a = y_0 - x_0$, find the value of a .

假設方程 $x^2y - 2x^2 - 3y - 13 = 0$ 只有一對正整數解 (x_0, y_0) 。若 $a = y_0 - x_0$ ，求 a 的值。

[答：19]

Solution

Since $y \neq 2$, we have $x^2 = 3 + \frac{19}{y-2}$.

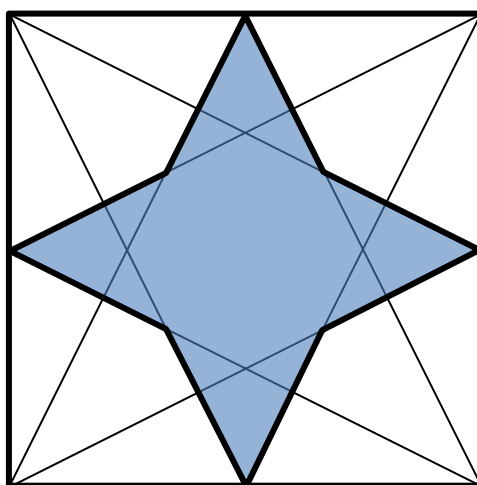
In order to obtain integer solutions for x and y , we must have $y - 2 = 19$, which gives $x^2 = 4$.

Therefore, $(x_0, y_0) = (2, 21)$ and $a = 21 - 2 = 19$.

4. Figure 1 shows a square. The midpoint of each side is joined to the two end points of the opposite side and a four-pointed star is thus formed (the shaded part). Find the value of $\frac{\text{Area of the four-point star}}{\text{Area of the square}}$.

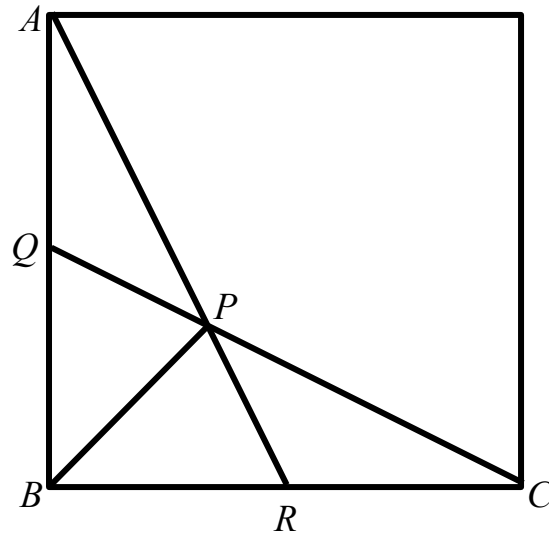
圖一所示為一正方形。每一條邊的中點都連接對邊的兩端點，由此形成一個四角星(著色部分)。

求 $\frac{\text{四角星的面積}}{\text{正方形的面積}}$ 的值。[答： $\frac{1}{3}$]



Solution

Consider the following diagram.



As R is the midpoint of BC , $\overline{BR} = \overline{RC}$, and as $\triangle BPR$ and $\triangle RPC$ have the same height, they must have the same area. By the same argument, $\triangle APQ$ and $\triangle QPB$ have the same area.

Therefore, Area of $\triangle APQ$ = Area of $\triangle QPB$ = Area of $\triangle BPR$ = Area of $\triangle RPC$ ($= N$, say).

As Area of $\triangle BAR = \frac{\text{Area of the square}}{4}$, it follows that $3N = \frac{\text{Area of the square}}{4}$, i.e.,

$$N = \frac{\text{Area of the square}}{12}.$$

Since Area of quadrilateral $BQPR = 2N$, we can see that in the original diagram $4 \times 2N = 8N = \text{Area of the square} \times \frac{2}{3}$ is unshaded. Hence the shaded “star” represents $\frac{1}{3}$ of the area of the square.

Part B

乙部

5. $VABC$ is a right pyramid with $VA = VB = VC$ and $AB = BC = CA = a$ m. Let its height be h m and its total surface area and volume are the same. If a and h are both positive integers, find the sum of all possible values of h .

$VABC$ 為一個錐體，其中 $VA = VB = VC$ 及 $AB = BC = CA = a$ m。設它的高為 h m 及它的總表面積及體積相等。若 a 和 h 均為正整數，求 h 的可能值之和。〔答：33〕

Solution

Let $AB = BC = CA = a$

$$\text{Total surface area} = \frac{\sqrt{3}}{4}a^2 + \frac{3}{2}a\sqrt{h^2 + \frac{a^2}{12}}$$

$$\text{Volume} = \frac{1}{3} \cdot \frac{\sqrt{3}}{4}a^2h = \frac{\sqrt{3}}{12}a^2h$$

$$\therefore \frac{\sqrt{3}}{12}a^2h = \frac{\sqrt{3}}{4}a^2 + \frac{3}{2}a\sqrt{h^2 + \frac{a^2}{12}}$$

$$(h-3)\sqrt{3}a = 18\sqrt{h^2 + \frac{a^2}{12}}$$

$$(h-3)^2a^2 = 9(12h^2 + a^2)$$

$$a^2 = \frac{108h^2}{h^2 - 6h} \quad \text{or} \quad a = 6\sqrt{\frac{3h}{h-6}}$$

On solving, gives $h = 9$ or 24 .

$$\therefore S = 9 + 24 = 33.$$

6. In Figure 2, $ABCD$ is a parallelogram. E is the midpoint of BC , AE and BD intersect at H , AC and DE intersect at F , AC and BD intersect at G . If the area of the quadrilateral $EFGH$ and $ABCD$ are 10 cm^2 and $k \text{ cm}^2$ respectively, find the value of k .

圖二中， $ABCD$ 是平行四邊形。 E 為 BC 的中點， AE 和 BD 相交於 H ， AC 和 DE 相交於 F ， AC 和 BD 相交於 G 。若四邊形 $EFGH$ 的面積及 $ABCD$ 的面積分別為 10 cm^2 及 $k \text{ cm}^2$ ，求 k 的值。

[答：120]

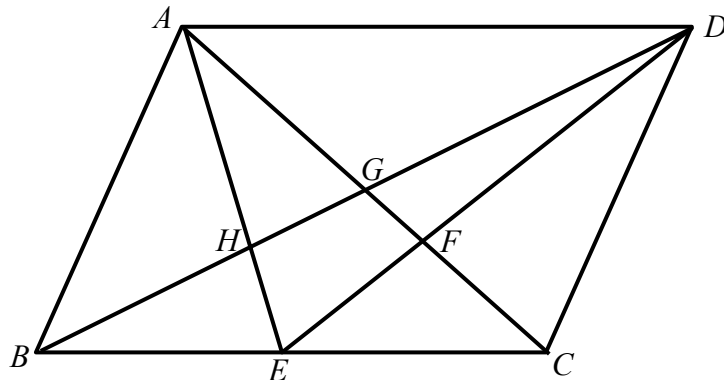


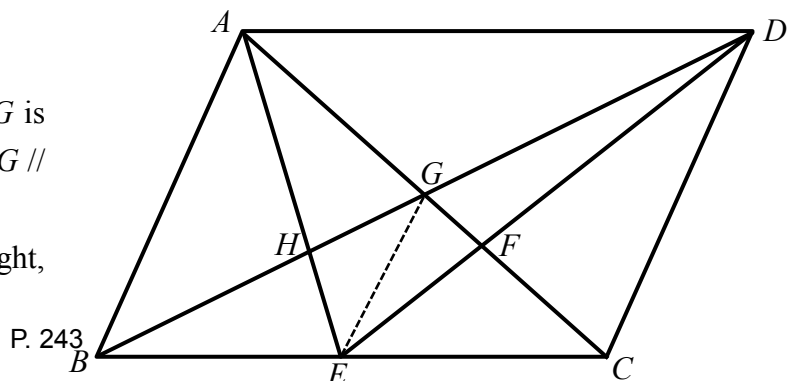
Figure 2

圖二

Solution

Join EG . Because E is the midpoint of BC , G is the midpoint of AC and BD , we have $AB \parallel EG \parallel DC$, and $AB = CD = 2EG$.

Also since $\triangle AEG$ and $\triangle AEB$ have equal height,



$AB = 2GE$, therefore $BH = 2GH$, $BG = 3GH$, thus,

area of $\triangle BEG = 3 \times$ area of $\triangle EGH$.

Likewise, area of $\triangle CEG = 3 \times$ area of $\triangle EGF$. Consequently, area of $\triangle BEG +$ area of $\triangle CEG = 3 \times$ (area of $\triangle EGH +$ area of $\triangle EGF$) $= 3 \times 10 = 30 \text{ cm}^2$.

$\therefore$ area of $ABCD = 4 \times$ area of $\triangle BGC = 4 \times 30 = 120 \text{ cm}^2$.

$\therefore k = 120$

7. Given that $x + y + z = 1$, $x^2 + y^2 + z^2 = 2$ and $x^3 + y^3 + z^3 = 3$. Find the value of $x^4 + y^4 + z^4$.

已知 $x + y + z = 1$, $x^2 + y^2 + z^2 = 2$ 及 $x^3 + y^3 + z^3 = 3$ 。求 $x^4 + y^4 + z^4$ 的值。〔答： $\frac{25}{6}$ 〕

Solution

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$$

$$1^2 = 2 + 2(xy + yz + zx)$$

$$xy + yz + zx = -\frac{1}{2}$$

$$(x + y + z)(x^2 + y^2 + z^2) = x^3 + y^3 + z^3 + x^2y + y^2z + z^2x + xy^2 + yz^2 + zx^2$$

$$2 = 3 + x^2y + xy^2 + xyz + y^2z + yz^2 + xyz + z^2x + zx^2 + xyz - 3xyz$$

$$-1 = xy(x + y + z) + yz(x + y + z) + zx(x + y + z) - 3xyz$$

$$-1 = (x + y + z)(xy + yz + zx) - 3xyz$$

$$-1 = 1\left(-\frac{1}{2}\right) - 3xyz$$

$$xyz = -\frac{1}{6}$$

$$x^4 + y^4 + z^4$$

$$= (x^3 + y^3 + z^3)(x + y + z) - (x^3y + xy^3 + x^3y + yz^3 + yz^3 + z^3x + x^3z)$$

$$= (3)(1) - (xy(x^2 + y^2 + z^2) - xyz^2 + yz(x^2 + y^2 + z^2) - yzx^2 + zx(x^2 + y^2 + z^2) - zxy^2)$$

$$= 3 - ((xy + yz + zx)(x^2 + y^2 + z^2) - xyz(x + y + z))$$

$$= 3 - \left(-\frac{1}{2}\right)(2) - \left(-\frac{1}{6}\right)(1)$$

$$= \frac{25}{6}$$

8. For all positive integers $n > 1$, a function f is defined as

$$f(1) = 2021 \text{ and } f(1) + f(2) + f(3) + \cdots + f(n) = n^2 f(n).$$

Find the value of $f(2021)$.

對所有正整數 $n > 1$ ，函數 f 定義如下：

$$f(1) = 2021 \text{ 及 } f(1) + f(2) + f(3) + \cdots + f(n) = n^2 f(n)。$$

求 $f(2021)$ 的值。〔答： $\frac{1}{1011}$ 〕

Solution

$$f(1) + f(2) + f(3) + \cdots + f(n-1) + f(n) = n^2 f(n)$$

$$(n-1)^2 f(n-1) = (n^2 - 1) f(n)$$

$$f(n) = \frac{n-1}{n+1} f(n-1)$$

$$\begin{aligned} f(n) &= \left(\frac{n-1}{n+1}\right) \left(\frac{n-2}{n}\right) \left(\frac{n-3}{n-1}\right) \cdots \left(\frac{3}{5}\right) \left(\frac{2}{4}\right) \left(\frac{1}{3}\right) f(1) \\ &= \frac{2}{n(n+1)} f(1) \end{aligned}$$

Substituting $n = 2021$ into the above equation gives $f(2021) = \frac{2}{2021 \cdot 2022} (2021) = \frac{1}{1011}$.